INTERSECTIONS ON REGULAR SURFACES

by P. Deligne

In the first part of this exposé, after some preliminaries from intersection theory (contained in [5] $\cup$ [7]), we give Artin's proof of a special case of a theorem of Raynaud [9] (1.13), and an a priori proof of a theorem which Néron [8] had verified case by case (1.15). The theorems of [9] used in [2], and hence in IX, follow fairly easily from (1.13).

In the second part, continuing in the same vein, I explain how the "Italians" [1], [3] calculated intersection numbers on surfaces. The main results, 2.9 and 2.11, may also be found in Northcott [12] and [13]. To conclude, in 2.27, 2.30, and 2.34 I give a conjectural relation between the dimension of the versal formal moduli scheme of a singularity of a reduced curve, the invariant δ of the singularity, and an invariant measuring its symmetry.

This exposé will not be used in the remainder of the seminar.

1. Intersection numbers on arithmetic surfaces

1.0. Throughout this section, let X be a regular scheme of dimension 2, let Y be a reduced divisor on X, and let k be a subfield of $H^0(Y, \underline{O}_Y)$. We assume that Y is a curve (EGA II 7.4.1) proper over k. Here are three examples of this situation.

Example 1.1. Let (S, η, s) be a trait and let $f : X \longrightarrow S$ be a regular, proper, and flat S-scheme whose fibres are purely one-dimensional. We take $Y = (f^{-1}(s))_{\text{red}}$ and $k = k(s)$. From 1.10 onward we shall restrict ourselves to this case.

Example 1.2. Let X_o be the spectrum of a normal local ring of dimension 2, let s be the closed point of X_o, and let $f : X \longrightarrow X_o$ be a regular connected X_o-scheme, proper over X_o, distinct from X_o but agreeing with X_o over $X_o \setminus \{s\}$. We take $Y = (f^{-1}(s))_{\text{red}}$ and $k = k(s)$.

Example 1.3. We take X to be a complete regular surface over a field k, and Y to be a reduced curve on X.

Recall the following lemma (SGA 6 IV, Ex. 1.9).

Lemma 1.4. Let Z_o be a closed subscheme of a Noetherian scheme Z. Then the inclusion of Z_o in Z induces an isomorphism from the Grothendieck group of the category of coherent $\underline{O}_{Z_o}$-modules to the Grothendieck group of the category of coherent sheaves on Z with support in Z_o.

This lemma allows us to define the Euler–Poincaré characteristic $\chi(\mathcal{F})$ of a coherent sheaf $\mathcal{F}$ on X supported on Y so that:

(i) If $\mathcal{F}$ is a coherent sheaf of $\underline{O}_Y$-modules, then

$$\chi(\mathcal{F}) = \dim_k H^0(\mathcal{F}) - \dim_k H^1(\mathcal{F}). \tag{1.4.1}$$

(ii) χ is additive with respect to short exact sequences.

If the cohomology sheaves of a complex K of $\underline{O}_X$-modules are coherent sheaves supported on Y, we set

$$\chi(K) = \sum_i (-1)^i \chi(\underline{H}^i(K)). \tag{1.4.2}$$

If $\mathcal{F}$ is supported at a point y of Y, then

$$\chi(\mathcal{F}) = \text{length}_{\underline{O}_y}(\mathcal{F}_y)[k(y) : k]. \tag{1.4.3}$$

Under the hypotheses of (1.1), (1.2), or (1.3), one has

$$\chi(\mathcal{F}) = \text{length}\, f_*\mathcal{F} - \text{length}\, R^1 f_*\mathcal{F}. \tag{1.4.4}$$

Let D be a divisor ≥ 0 supported on Y, and let $\mathcal{L}$ be an invertible sheaf on D. There is then a unique way to define the degree $\deg_D(\mathcal{L})$ such that

$$\begin{cases} \deg_D(\mathcal{L}' \otimes \mathcal{L}'') = \deg_D(\mathcal{L}') + \deg_D(\mathcal{L}''), \\ \text{for } \mathcal{L} \subset \underline{O}, \text{ one has } \deg_D(\mathcal{L}) = -\chi(\underline{O}/\mathcal{L}). \end{cases} \tag{1.4.5}$$

The Riemann–Roch formula on D is then

$$\chi(\mathcal{L}) = \chi(\underline{O}_D) + \deg_D(\mathcal{L}). \tag{1.4.6}$$

If $D = \sum n_i D_i$ is the decomposition of D into reduced irreducible divisors, then

$$\deg_D(\mathcal{L}) = \sum n_i \deg_{D_i}(\mathcal{L}). \tag{1.4.7}$$

Definition 1.5. Let D and E be two divisors ≥ 0 on X, and suppose that D is supported on Y. The intersection number of D and E (relative to the field k (1.0)) is defined by

$$(D, E) = \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) = \chi(\underline{O}_D \otimes \underline{O}_E) - \chi(\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E)).$$

Indeed, under the hypotheses made here, $\mathrm{Tor}^2(\underline{O}_D, \underline{O}_E) = 0$.

Proposition 1.6. Under the hypotheses of 1.5, one has:

(i) The intersection number (D, E) is additive in the divisors D and E; this allows us, by linearity, to define the intersection number of two divisors which are not necessarily positive.

(ii) $(D, E) = \deg_D(\underline{O}(E))$.

(iii) If D and E meet at only finitely many points, then $(D, E) = \chi(\underline{O}_D \otimes \underline{O}_E) \geq 0$, and equality can occur only if D and E are disjoint.

(iv) If D and E are supported on Y, then $(D, E) = (E, D)$.

(v) If N is the conormal sheaf of D, then

$$D^2 = (D, D) = -\deg_D(N).$$

(vi) If E, not necessarily positive, is the divisor of a meromorphic function on X, then $(D, E) = 0$.

Proof. The exact sequence

$$0 \longrightarrow \underline{O}(-E) \longrightarrow \underline{O}_X \longrightarrow \underline{O}_E \longrightarrow 0$$

gives, by (1.4.6),

$$\begin{aligned} (D, E) &\overset{\mathrm{def}}{=} \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_E) \\ &= \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}_X) - \chi(\underline{O}_D \overset{\mathbb{L}}{\otimes} \underline{O}(-E)) \\ &= \chi(\underline{O}_D) - \chi(\underline{O}(-E)|D) = -\deg_D(\underline{O}(-E)) = \deg_D(\underline{O}(E)). \end{aligned}$$

This proves (ii), (vi), and biadditivity, by (1.4.5) and (1.4.7). Under the hypotheses of (iii), one has $\mathrm{Tor}^1(\underline{O}_D, \underline{O}_E) = 0$, which gives the conclusion. Under the hypotheses of (v), one has

$$\underline{O}_D \otimes \underline{O}_D = \underline{O}_D \quad \text{and} \quad \mathrm{Tor}^1(\underline{O}_D, \underline{O}_D) \simeq N \quad \text{(SGA 6 VII 2.5)},$$

hence

$$(D, D) = \chi(\underline{O}_D) - \chi(N) = -\deg_D(N).$$

Finally, (iv) follows from the symmetry of the tensor product.

Notice that (ii), (iv), and (vi) remain true without assuming the divisors positive.

Proposition 1.7. Let $(Y_i)_{1 \leq i \leq k}$ be the irreducible components of Y, f a meromorphic function on X, and

$$\mathrm{div}(f) = \sum_i n_i Y_i + Z$$

its divisor, with $|Z| \cap Y$ finite. Then, putting $D_i = n_i Y_i$, one has, for $(a_i) \in \mathbb{Q}^k$:

$$\left(\sum_i a_i D_i\right)^2 = -\sum_{i<j} (a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).$$

Proof (after [7]). By 1.6 (v), (vi),

$$\begin{aligned}\left(\sum_i a_i D_i\right)^2 &= \sum_i a_i \left(D_i, \sum_j a_j D_j\right) = \sum_i a_i \left(D_i, \sum_j a_j D_j - a_i \operatorname{div} f\right) \\ &= \sum_i a_i \left(D_i, \sum_{j \neq i} (a_j - a_i) D_j - a_i Z\right) \\ &= \sum_{i \neq j} a_i (a_j - a_i)(D_i, D_j) - \sum_i a_i^2 (D_i, Z) \\ &= -\sum_{i<j} (a_i - a_j)^2 (D_i, D_j) - \sum_i a_i^2 (D_i, Z).\end{aligned}$$

Corollary 1.8. Under the hypotheses of Example 1.1, take for f a uniformizer π of S. Here $Z = 0$ and

$$\left(\sum_i k_i Y_i\right)^2 = -\sum_{i<j} \frac{1}{n_i n_j} (k_i n_j - k_j n_i)^2 (Y_i, Y_j);$$

thus the intersection form is negative, and if Y is connected, its kernel consists precisely of the multiples of $\operatorname{div}(\pi)$.

Corollary 1.9 (Mumford [7]). Under the hypotheses of Example 1.2, take for f a nonzero element of the maximal ideal of the local ring $\underline{O}_s$. Here $Z > 0$, so the intersection form is negative definite.

1.10. For the remainder of this section we restrict ourselves to the case considered in Example 1.1.

Lichtenbaum [5] proved that, in this case, X is automatically projective over S, so that there is no difficulty in defining the complex $Rf^!\underline{O}_S$ ([4] III 8.7, p. 190). Since f has relative dimension 1 and X and S are regular (Gorenstein would suffice), there exists ([4] V 8.3 and V 9.1) an invertible sheaf $\omega_{X/S}$ on X, unique up to unique isomorphism, such that

(1.10.1) $$Rf^!\underline{O}_S \simeq \omega_{X/S}[1] \quad .$$

We denote by K any divisor (effective or not) such that $\omega_{X/S} \simeq \underline{O}(K)$.

Let D be a divisor ≥ 0 supported on Y, and let N be its conormal sheaf:

$$\begin{tikzcd} D \arrow[rr, "i"] \arrow[dr, "g"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

One has

(1.10.2) $$Rg^!\underline{O}_S = Ri^!Rf^!\underline{O}_S = Ri^!\omega_{X/S}[1] = \omega_{X/S}\big|_D \otimes N^\vee[0].$$

By the duality theorem ([4] V 11.1, p. 210),

(1.10.3) $$Rg_*Rg^!\underline{O}_S \simeq \operatorname{RHom}(Rg_*\underline{O}_D, \underline{O}_S) \quad .$$

Moreover, for every $\underline{O}_S$-module M supported at the closed point,

$$(1.10.4) \qquad \chi\big(\mathrm{RHom}(M, \underline{O}_S)\big) = -\chi(M),$$

so that here, using (1.4.4) and (1.10.3), we obtain

$$\chi(Rg^!\underline{O}_S) = -\chi(\underline{O}_D) \quad .$$

Since, by (1.4.6),

$$\chi(Rg^!\underline{O}_S) = \chi(\underline{O}_D) + \deg_D(Rg^!\underline{O}_S),$$

we conclude from (1.10.2) that

$$-2\chi(\underline{O}_D) = \deg_D(Rg^!\underline{O}_S) = \deg_D(\omega_{X/S}|_D \otimes N^\vee) = \deg_D(\omega_{X/S}) - \deg_D(N),$$

which, by 1.6(ii) and (v), may be rewritten as follows.

Proposition 1.11. Under the hypotheses of (1.1), for every divisor $D \geq 0$ supported on Y, one has

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D+K) \quad .$$

1.12. We denote by d the greatest common divisor of the geometric multiplicities of the irreducible components of $X_s = f^{-1}(s)$; that is, the greatest common divisor of the lengths of the local rings at the generic points of the geometric special fibre of f. If X_η has a rational point, or equivalently if $f : X \longrightarrow S$ has a section x, then $d = 1$; more precisely, f is smooth at $x(s)$. Indeed, x is a closed immersion of regular schemes, hence a regular immersion, and EGA IV 17.12.1 applies.

Here is Artin's proof of a special case of a theorem of Raynaud [9], which only assumes that d is prime to the characteristic p of k.

Theorem 1.13. With the preceding notation, if Y is geometrically connected and $d = 1$, then X is cohomologically flat over S in dimension 0 (EGA III 7.8.1); more precisely (EGA III 7.6.9(ii)), one has

$$k \xrightarrow{\sim} H^0(X_s, \underline{O}_{X_s}).$$

It is easy to reduce to the case in which S is strictly local. Since Y is geometrically connected, $H^0(Y, \underline{O}_Y)$ is a purely inseparable extension of k; and since $(d,p) = 1$, at least one irreducible component of Y is

geometrically reduced. It follows readily that $H^0(Y, \underline{O}_Y) = k$.

Let D be a divisor on X such that

$$Y \leq D < X_s \quad .$$

Since $d = 1$, D is not a rational multiple of X_s; hence, by 1.8,

$$(D, X_s - D) = -(D, D) > 0 \quad .$$

The divisor Y therefore has an irreducible component E such that

$$0 < E \leq X_s - D \quad \text{and} \quad (D, E) > 0 \quad .$$

Consider the diagram

$$\begin{tikzcd}
0 \arrow[r] & \underline{O}(-D-E) \arrow[r] \arrow[d, "\varphi"] & \underline{O} \arrow[r] \arrow[d, equal] & \underline{O}_{D+E} \arrow[r] \arrow[d, "\psi"] & 0 \\
0 \arrow[r] & \underline{O}(-D) \arrow[r] & \underline{O} \arrow[r] & \underline{O}_D \arrow[r] & 0
\end{tikzcd}$$

The snake lemma gives an isomorphism between $\mathrm{Ker}(\psi)$ and $\mathrm{Coker}(\varphi)$. This cokernel is precisely the restriction of $\underline{O}(-D)$ to E, whence an exact sequence

$$0 \longrightarrow \underline{O}(-D)|_E \longrightarrow \underline{O}_{D+E} \longrightarrow \underline{O}_D \longrightarrow 0 \quad .$$

By construction, $\deg_E(\underline{O}(-D)|_E) = -(D,E) < 0$, so that $H^0(\underline{O}(-D)|_E) = 0$ and the restriction map

$$0 \longrightarrow H^0(\underline{O}_{D+E}) \longrightarrow H^0(\underline{O}_D)$$

is injective. This allows us, by induction, to construct a sequence of divisors

$$Y = D_0 \leq D_1 \leq \cdots \leq D_n = X_s$$

such that $H^0(\underline{O}_{D_i})$ injects into $H^0(\underline{O}_{D_{i-1}})$. Thus $H^0(\underline{O}_{X_s}) = H^0(\underline{O}_Y) = k$, which completes the proof of 1.13.

The following fact is proved in [5] or [10].

Theorem 1.14 (Castelnuovo, Enriques, ...). *Under the hypotheses of* (1.1), *let* D *be a reduced irreducible divisor contained in* Y. *Suppose that* $H^1(D, \underline{O}_D) = 0$ *and that, on putting* $k' = H^0(D, \underline{O}_D)$, *one has*

$$(D,D) = -[k' : k] \quad .$$

Then D *is a projective line over* k', *and there exist a regular* S*-scheme* X_1, *a closed point* x_1 *of* X_1 *with residue field* k', *and an isomorphism from* X *to the* S*-scheme obtained from* X_1 *by blowing up* x_1, *under which* D *is identified with the fibre over* x_1.

A divisor D having these properties is called an *exceptional curve of the first kind*.

I cannot resist the pleasure of proving the following proposition of Néron [8], Proposition 1′, p. 90.

Proposition 1.15. *Suppose that* X_η *is an elliptic curve (in particular, that it has a section), and that* X *contains no exceptional curve of the first kind. Then, with the notation of* (1.10.1), *the canonical map*

$$f^* f_* \omega_{X/S} \xrightarrow{\sim} \omega_{X/S}$$

is universally an isomorphism.

Since $d = 1$ (1.12), by (1.13) it is enough to prove that $\omega_{X/S}$ is isomorphic to $\underline{O}_X$. Since $\omega_{X/S}$ is trivial on the elliptic curve X_η, we know a priori that

$$\omega_{X/S} \simeq \underline{O}\Big(\sum_i n_i Y_i\Big),$$

where the Y_i are the irreducible components of Y; and since $\mathrm{div}(\pi) \sim 0$, we may assume the n_i positive. If $\sum_i n_i Y_i$ is proportional to $\mathrm{div}(\pi)$, it is an integral multiple of $\mathrm{div}(\pi)$ and we are done. Otherwise,

$$\Big(\sum_i n_i Y_i\Big)^2 < 0,$$

there is an i such that

$$\Big(Y_i, \sum_j n_j Y_j\Big) < 0,$$

and the following lemma applies.

Lemma 1.16. Under the hypotheses of 1.1, suppose that Y is connected and that either Y is reducible or $\chi(\underline{O}_Y) \leq 0$. Then an irreducible component Y_i of Y is an exceptional curve of the first kind if and only if $(Y_i, K) < 0$.

Let $k' = H^0(Y_i, \underline{O}_{Y_i})$. The Riemann–Roch formula 1.11 gives

$$\chi(\underline{O}_{Y_i}) = [k' : k]\big(1 - \dim_{k'} H^1(Y_i, \underline{O}_{Y_i})\big) = -\frac{1}{2}\big((Y_i, Y_i) + (Y_i, K)\big) \quad .$$

If Y_i is exceptional of the first kind, it follows that $(Y_i, K) = -[k' : k] < 0$. Conversely, if $(Y_i, K) < 0$, then $\chi(\underline{O}_{Y_i}) > 0$, which implies $\chi(\underline{O}_{Y_i}) = [k' : k]$ and that Y is reducible. Hence, by 1.8, $(Y_i, Y_i) < 0$. The negative integers (Y_i, Y_i) and (Y_i, K) are divisible by $[k' : k]$, and therefore necessarily

$$(Y_i, Y_i) = (Y_i, K) = -[k' : k] \quad .$$

Exercise. (a) Under the hypotheses of (1.1), suppose that X_η is a principal homogeneous space under an elliptic curve and that X contains no exceptional curve of the first kind. Show that on X the Riemann–Roch formula (1.11) takes the form

$$\chi(\underline{O}_D) = -\frac{1}{2}(D, D).$$

(b) Suppose in addition that the residue field $k(s)$ is algebraically closed. Show that the diagram of irreducible components of X_s is a multiple of one of the diagrams considered by Néron [8], pp. 124–125, fourth column (cases a through c8). The diagram determines the genera of the reduced irreducible components, their multiplicities in X_s, and the nature and disposition of the singularities of $(X_s)_{\mathrm{red}}$.

First treat the case in which X_s is irreducible. Once this case has been excluded, prove that every reduced irreducible component is a projective line of self-intersection -2. If $X_s = \sum_i n_i D_i$ is the decomposition of X_s into reduced irreducible divisors, rewrite this condition as

$$(*) \qquad n_i = \frac{1}{2} \sum_{j \neq i} n_j (D_i, D_j) \quad .$$

On the other hand, X_s is connected, so that:

(**) For any i and j, there exists a chain $i = i_1, i_2, \ldots, i_n = j$ with $(D_{i_k}, D_{i_{k+1}}) \neq 0$.

Now find all solutions in integers of $(*) + (**)$. First show that, except in Néron's cases b2 and c2, the intersection numbers (D_i, D_j) are either zero or one.

2. Italian intersections

Let S' and S be two smooth projective surfaces over a field k, and let $f : S' \longrightarrow S$ be a birational morphism. If D and E are two divisors on S, then

$$(D, E) = (f^*D, f^*E),$$

as one sees, for example, by moving D and E.

It is therefore natural to seek a "birational" expression for the intersection number. I propose to explain how the Italians proceeded. My principal source of information is [1], which contains a long bibliography. See also Manin [6] and Northcott [12], [13].

(2.0) Let S be the spectrum of a regular local ring of dimension 2, let s be the closed point of S, and let k be a subfield of $k(s)$ over which $k(s)$ is finite. Intersection numbers and Euler–Poincaré characteristics will be computed over k.

In this section, an S-scheme $p : S' \longrightarrow S$ will be called a birational transform of S if it is regular and connected, and if p is proper and induces an isomorphism $S' \setminus p^{-1}(s) \longrightarrow S \setminus \{s\}$. Everything said about S will also apply to the localizations of its birational transforms at a closed point.

(2.1) Let $f : (S'', p'') \longrightarrow (S', p')$ be a morphism of S-schemes between birational transforms of S. It is known [5] that f has a unique factorization

$$f = f_n \cdots f_1, \qquad f_i : S_{i+1} \longrightarrow S_i,$$

such that S_{i+1} is obtained from S_i by blowing up a finite set X_i of closed points, and $f_i(X_{i+1}) \subset X_i$. The set X_i is the set of points of S_i at which the projection of S'' onto S_i is not an isomorphism.

In particular, f induces an isomorphism over the complement of a finite subset of S' containing the generic points of the irreducible components of $p'^{-1}(s)$. Hence f^{-1} defines an injection

from the set $\mathrm{Irr}(p'^{-1}(s))$ of irreducible components of $p'^{-1}(s)$ into $\mathrm{Irr}(p''^{-1}(s))$.

The birational transforms of S have no S-automorphisms. We write I for the set of S-isomorphism classes of birational transforms of S, ordered by domination. The set I is filtered to the right and has least element S.

Definition 2.2. The set S^* of points of S infinitely near to s is the direct-limit set

$$S^* = \varinjlim_{I} \mathrm{Irr}(p^{-1}(s)) \quad .$$

If an element $t \in S^*$ is represented on a birational transform S' of S by a complete reduced irreducible curve T, the field $H^0(T, \underline{O}_T)$ depends only on t, not on the choice of S'. It is called the residue field $k(t)$ of t.

Let S_1 be the blow-up of S along s. The projective line s^*, the inverse image of $\{s\}$ in S_1, is called the curvetta of s and defines an element s^* of S^* with residue field $k(s)$.

Let S' be a birational transform of S, let t be a closed point of S', and let S'_t be the spectrum of the local ring of S' at t. There is an evident injection $(S'_t)^* \hookrightarrow S^*$; in particular, t defines an element t^* of S^* with the same residue field. We write $(S')^*$ for the union in S^* of the $(S'_t)^*$ as t ranges over the closed points of S'.

The following proposition follows from the description 2.1 of the birational transforms of S.

Proposition 2.3. For every $t \in S^*$, there is a unique sequence of birational transforms $(S_i)_{0 \leq i \leq n+1}$ of S and closed points $t_i \in S_i$ $(0 \leq i \leq n)$ such that

(i) $S_0 = S$, and S_{i+1} is the blow-up of S_i along $\{t_i\}$;

(ii) $t_{i+1} \in t_i^*$ $(0 \leq i < n)$, and $t = t_n^*$.

Moreover, for every birational transform (S', p') of S, the element t belongs to the image of $\mathrm{Irr}(p'^{-1}(s))$ in S^* if and only if S' dominates S_{n+1}.

The integer n is called the order of t. We write $S(t)$ (respectively $S_+(t)$) for the birational transform S_n (respectively S_{n+1}) of S. By abuse of notation, we also write t for the closed point t_n of S_n, and t^* for its inverse image t_n^* in S_{n+1}.

We equip S^* with the following order:

$$t_1 \prec t_2 \quad \Longleftrightarrow \quad S_+(t_2) \text{ dominates } S_+(t_1) \quad .$$

If $t_1 \prec t_2$, then $k(t_2)$ is an extension of $k(t_1)$. The ordered set S^* has least element s^*, and every bounded subset of S^* is finite and totally ordered (S^* is a "tree rooted at s^*"). Moreover, the order n of $t \in S^*$ is the distance in the tree S^* from s^* to t.

The following proposition makes the description 2.1 of the birational transforms of S explicit.

Proposition 2.4. The function assigning to a birational transform $p : S' \to S$ the subset $\mathrm{Irr}(p^{-1}(s))$ of S^* induces a bijection between I (2.1) and the finite subsets of S^* which, together with each element,

contain all its predecessors.

Let D be a divisor ≥ 0 on S, and let $t \in S^*$. We say that t belongs to D if the closed point t of $S(t)$ (2.3) belongs to the strict transform (the schematic closure of $D - \{s\}$) of D on $S(t)$.

Definition 2.5. Let $t, t' \in S^*$. We say that t' is proximate to t if $t' \in S_+(t)^*$ (2.2) and belongs to the curvetta t^* of t (2.3).

If t' is an immediate successor of t, then t' is proximate to t. If in a sequence $(t_i)_{0 \leq i \leq n}$ every t_{i+1} is an immediate successor of t_i, and if t_n is proximate to t_0, then every t_i with $0 < i \leq n$ is proximate to t_0, while t_n is not proximate to t_i for $0 < i \leq n-1$.

(2.6) Let A be a Noetherian local ring of dimension n, let T be its spectrum, let m be its maximal ideal, let $k = A/m$ be its residue field, and let $P = \sum_{0 \leq i < n} a_i X^i$ be its Hilbert–Samuel polynomial:

$$P(r) = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Recall that the multiplicity μ_s of the closed point s in T is defined by

$$a_{n-1} = \frac{\mu_s}{(n-1)!} \quad .$$

When $n = 1$, this simply says

$$\mu_s = \dim_k(m^r/m^{r+1}) \qquad \text{for sufficiently large } r.$$

Geometrically, μ_s may be defined as the degree, in the projective space $\mathbf{P}_k(m/m^2)$, of the special fibre of the blow-up of T along the closed subscheme $\{s\}$.

Recall that if Y and Z are two closed subschemes of a scheme X, then the strict transform of Y in the blow-up $\widetilde{X}$ of X along Z—the schematic closure of $Y \setminus Z$ in $\widetilde{X}$—is precisely the blow-up of Y along $Y \cap Z$.

Suppose that T is a closed subscheme of a regular scheme U. Let $p : U_1 \longrightarrow U$ be the blow-up of U along $\{s\}$, let s^* be the inverse image of s in U_1, and let T' be the strict transform of T in U_1. The preceding discussion shows that the multiplicity of T at s is the degree, in the projective space s^*, of the subscheme $s^* \cap T'$.

Suppose further that T is a divisor on U. Let T^* be its total transform in U_1 (the inverse-image divisor), and define ν by $T^* = \nu s^* + T'$. If X is a line in s^*, and Euler–Poincaré characteristics are calculated as alternating sums of dimensions over k, then

$$(X, s^*) = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{s^*}) = \deg_X \underline{O}(s^*) = -1, \tag{2.6.1}$$

and we have seen that

$$(X, T') = \chi(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T'}) = \deg_{s^*}(s^* \cap T') = \mu_s \quad . \tag{2.6.2}$$

The projection formula

$$Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_{T^*}) = Rp_*(\underline{O}_X \overset{\mathbb{L}}{\otimes} Lp^*\underline{O}_T) = Rp_*\underline{O}_X \overset{\mathbb{L}}{\otimes} \underline{O}_T$$

and its corollary

$$(X, T^*) = \nu(X, s^*) + (X, T') = 0$$

give, by (2.6.1) and (2.6.2), the identity

$$\mu_s = \nu \quad . \tag{2.6.3}$$

Proposition 2.7. Let T be a divisor ≥ 0 on a regular scheme U, s a closed point of U, U_1 the blow-up of U along $\{s\}$, s^* the inverse image of s in U_1, T' the strict transform of T on U_1, and T^* its total transform.

The following three quantities are equal:

(i) the multiplicity, in Samuel's sense, of the point s on T;

(ii) the degree, in the projective space s^*, of $s^* \cap T'$;

(iii) the multiplicity of s^* in the divisor T^*.

2.8. Return to the hypotheses of 2.0. Let D be a divisor ≥ 0 on S and let $t \in S^*$. The multiplicity $\mu_t(D)$ of D at t is, by definition, the multiplicity at t of the strict transform of D on $S(t)$ (2.3). The support of the function $t \mapsto \mu_t(D)$ is the set of points infinitely near to s that belong to D (2.4). When D is regular, the numbers $\mu_t(D)[k(t) : k(s)]$ are zero or one: indeed, the strict transforms D' of D are isomorphic to D, and one applies the definition (2.6) of multiplicity.

Intersection numbers can be calculated in terms of multiplicities.

Theorem 2.9. If the divisors D and E meet only at s, then

$$(D, E) = \sum_{t \in S^*} \mu_t(D)\mu_t(E)[k(t) : k],$$

where the sum is over the points infinitely near to s which belong to both D and E. Proof. Let $p : S' \longrightarrow S$ be a birational transform of S. We denote the total transforms of D and E by D^* and E^*, and their strict transforms by D' and E'.

Lemma 2.9.1. (i) The total transform D^* is characterized by:

(a) $D^* - D'$ is supported on $p^{-1}(s)$;

(b) if X is supported on $p^{-1}(s)$, then $(D^*, X) = 0$.

(ii) $(D, E) = (D^*, E^*) = (D^*, E') = (D', E^*)$.

Assertion (i)(a) is immediate, and (i)(b) follows from 1.6(vi). That (a) and (b) determine D^* follows from 1.9. By construction, the cokernel of the map $\underline{O}_E \longrightarrow p_*\underline{O}_{E'}$ is finite; hence

$$\begin{aligned}(D^*, E') &= \chi(Lp^*\underline{O}_D \otimes \underline{O}_{E'}) \\ &= \chi(\underline{O}_D \otimes Rp_*\underline{O}_{E'}) = \chi(\underline{O}_D \otimes \underline{O}_E) = (D, E),\end{aligned}$$

which, by (i), implies (ii).

Take for S' the blow-up of S at s. By (2.9.1), (2.6.3), and 2.7,

$$\begin{aligned}(D, E) &= (D^*, E') = (D' + \mu_s(D)s^*, E') \\ &= (D', E') + \mu_s(D)(s^*, E') \\ &= (D', E') + \mu_s(D)\mu_s(E)[k(s) : k].\end{aligned} \tag{2.9.2}$$

Theorem 2.9 now follows by induction on (D, E), through repeated application of (2.9.2).

(2.10) Let A be a reduced Noetherian local ring of dimension one, let s be the closed point of $\mathrm{Spec}(A)$, let $\widetilde{A}$ be the normalization of A, and let k be a field over which $k(s)$ is finite. Suppose that $\widetilde{A}$ is finite over A (equivalently, that the completion $\widehat{A}$ is reduced). The δ-invariant of the singularity of $\mathrm{Spec}(A)$ at s is the length over k of the A-module $\widetilde{A}/A$.

Theorem 2.11. Let D be a reduced curve on S. Then:

(i) for every $t \in S^*$, with the notation of 2.8, one has

$$\mu_t(D) = \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \mu_{t'}(D)[k(t') : k(t)] \quad .$$

(ii) Suppose that the normalization $\widetilde{D}$ of D is finite over D, and let $\delta = \chi(\underline{O}_{\widetilde{D}}/\underline{O}_D)$ be the invariant of the singularity of D at s. For all but finitely many $t \in S^*$, $\mu_t(D)$ is zero or one, and

$$\delta = \sum_{t \in S^*} \frac{1}{2} \mu_t(D)\big(\mu_t(D) - 1\big)[k(t) : k] \quad .$$

Let $p : S_1 \longrightarrow S$ be the blow-up of S along $\{s\}$, let D^* be the total transform of D, and let D' be its strict transform. By (2.7),

$$D^* = D' + \mu_s(D) s^* \tag{2.11.1}$$

and

$$\mu_s(D) = (s^*, D')\,[k(s) : k]^{-1} \quad . \tag{2.11.2}$$

The curvetta s^* is nonsingular. Thus, by 2.8 and Definition 2.5, formula 2.9, applied to the divisors s^* and D' on S_1, becomes

$$(s^*, D') = \sum_{\substack{t' \text{ proximate} \\ \text{to } s}} \mu_{t'}(D)[k(t') : k] \quad . \tag{2.11.3}$$

Assertion (i) for $s = t$ follows from (2.11.2) and (2.11.3); the general case is obtained by applying this result to the localization of $S(t)$ at t (2.3).

Let m be the maximal ideal defining $\{s\}$, and let $n \geq 0$. It is known (SGA VII 3.5) that

$$p_* \underline{O}_{S_1}(-ns^*) = m^n \quad \text{and} \quad R^i p_* \underline{O}_{S_1}(-ns^*) = 0 \quad (i > 0). \tag{2.11.4}$$

It follows from this and from the exact sequence

$$0 \longrightarrow \underline{O}_{S_1}(-ns^*) \longrightarrow \underline{O}_{S_1} \longrightarrow \underline{O}_{ns^*} \longrightarrow 0$$

that

$$p_* \underline{O}_{ns^*} = \underline{O}_S / m^n \quad \text{and} \quad R^i p_* \underline{O}_{ns^*} = 0 \quad (i > 0), \tag{2.11.5}$$

and that

$$\chi(\underline{O}_{ns^*}) = \chi(\underline{O}_S / m^n) = \frac{n(n+1)}{2}[k(s) : k] \quad . \tag{2.11.6}$$

Put $n = 0$ in (2.11.4). For every complex K of coherent sheaves on S, the projection formula gives

$$K \xrightarrow{\sim} Rp_* Lp^* K \qquad \big(\simeq K \overset{\mathbb{L}}{\otimes} Rp_* \underline{O}_{S_1}\big).$$

In particular, since $Lp^* \underline{O}_D = p^* \underline{O}_D = \underline{O}_{D^*}$, one has

$$p_* \underline{O}_{D^*} = \underline{O}_D \quad \text{and} \quad R^i p_* \underline{O}_{D^*} = 0 \quad (i > 0). \tag{2.11.7}$$

Put $E = \mu_s(D) s^*$. By (2.11.1), (2.9.1)(i)(b), and (2.6.1),

$$\begin{aligned} \chi(\underline{O}_{D' \cap E}) &= (D', E) = (D^* - E, E) = -(E, E) \\ &= -\mu_s(D)^2 (s^*, s^*) = \mu_s(D)^2 [k(s) : k]. \end{aligned} \tag{2.11.8}$$

The exact sequence

$$0 \longrightarrow \underline{O}_{D^*} \longrightarrow \underline{O}_{D'} \oplus \underline{O}_E \longrightarrow \underline{O}_{D' \cap E} \longrightarrow 0$$

gives, by direct image and (2.11.7), the exact sequence

$$0 \longrightarrow \underline{O}_D \longrightarrow p_*\underline{O}_{D'} \oplus p_*\underline{O}_E \longrightarrow p_*\underline{O}_{D' \cap E} \longrightarrow 0,$$

and hence

$$0 \longrightarrow p_*\underline{O}_E \longrightarrow p_*\underline{O}_{D' \cap E} \longrightarrow p_*\underline{O}_{D'}/\underline{O}_D \longrightarrow 0 \quad .$$

Consequently, by (2.11.6) and (2.11.8),

$$\begin{aligned} \chi(p_*\underline{O}_{D'}/\underline{O}_D) &= \mu_s(D)^2[k(s):k] - \frac{1}{2}\mu_s(D)\big(\mu_s(D)+1\big)[k(s):k] \\ &= \frac{1}{2}\mu_s(D)\big(\mu_s(D)-1\big)[k(s):k] \quad . \end{aligned} \tag{2.11.9}$$

The schemes D and D' are reduced, differ only over s, and D' is finite over D. They therefore have the same normalization, and the δ-invariant of the singularity of D at s is the sum of $\chi(p_*\underline{O}_{D'}/\underline{O}_D)$ and the invariants δ of the singularities of D' at the points of $D' \cap s^*$. We prove 2.11(ii) by induction on δ. If $\mu_s(D) > 1$, the induction hypothesis applies to D', and 2.11(ii) follows from (2.11.9). If $\mu_s(D) = 1$, then $(D', s^*) = 1$ and $D' \cap s^*$ is a regular divisor on D'; by EGA IV 19.1.1, D' is regular. Formula (2.11.9) then shows that D is itself regular, and 2.11(ii) follows from 2.8.

2.12. Let $t \in S^*$, and let S' be a birational transform of S which dominates $S_+(t)$ (2.3). The <u>total transform</u> of t on S', again denoted by t^*, is the total transform on S' of the divisor t^* on $S_+(t)$.

Let $\nu : S^* \longrightarrow \mathbf{N}$ be a function of finite support, and suppose that, for every $t \in S^*$,

$$\nu(t) \geq \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \nu(t')[k(t'):k(t)] \tag{2.12.1}$$

(the proximate-point inequality). Let $f : S' \longrightarrow S$ be a birational transform of S which dominates $S_+(t)$ for each t in the support of ν, and let

$$\nu^* = \sum_{t \in S^*} \nu(t) t^*$$

be the corresponding divisor on S'.

We write $I(\nu)$ for the ideal of $\underline{O}_S$ obtained by direct image under f of the ideal of $\underline{O}_{S'}$ defining ν^*:

$$I(\nu) = f_*\underline{O}_{S'}(-\nu^*) \quad .$$

By (2.11.4), this ideal does not depend on the choice of S'. The ideals of S obtained in this way are those which Zariski [11] calls "complete."

<u>Theorem</u> 2.13. <u>Under the preceding hypotheses, one has</u>

$$\chi(\underline{O}_S/I(\nu)) = \sum_{t \in S^*} \frac{\nu(t)(\nu(t)+1)}{2}[k(t):k],$$

<u>and</u>

$$R^i f_* \underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0.$$

In old French, this is expressed briefly by saying: "requiring a plane curve to pass n times through a point t, whether proper or infinitely near, imposes $n(n+1)/2$ conditions, generally independent, on the coefficients of its equation."

We first prove the following lemma.

Lemma 2.14. Let $p : S_1 \longrightarrow S$ be the blow-up of S along $\{s\}$; let T be a closed subscheme of S_1 with ideal q; let n be an integer; let $J = \underline{O}_{S_1}(-s^*)$ be the ideal defining the divisor s^*; put $Y = \operatorname{Spec}(\underline{O}_{S_1}/J^n q)$; and let D be a regular divisor on S, with strict transform D' on S_1. Suppose that $T \cap s^*$ and $T \cap D'$ are finite.

(i) If $\operatorname{length}_{k(s)}(T \cap s^*) \leq n+1$, then

$$R^i p_*(qJ^n) = 0 \quad \text{for } i > 0,$$

and, if T is Artinian,

$$\operatorname{length}_{k(s)}\big(\underline{O}_S/p_*(qJ^n)\big) = \frac{n(n+1)}{2} + \operatorname{length}_{k(s)}(T).$$

(ii) If $\operatorname{length}_{k(s)}(T \cap s^*) \leq n$, then

$$\operatorname{length}_{k(s)}\Big(D \cap \operatorname{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = n + \operatorname{length}_{k(s)}(D' \cap T) \quad .$$

Consider the exact sequence on S_1

$$0 \longrightarrow J^n q \longrightarrow J^n \xrightarrow{u} J^n \otimes \underline{O}_T \longrightarrow 0 \quad . \tag{2.14.1}$$

To prove that $R^i p_*(J^n q) = 0$ for $i > 0$, it is enough, by (2.11.4), to show that $p_*(u)$ is surjective. By the theorem on formal functions, it suffices to prove that the maps

$$u_k : p_*(J^n \otimes \underline{O}/J^k) \longrightarrow p_*(J^n \otimes \underline{O}_T \otimes \underline{O}/J^k)$$

are surjective.

Write Gr_J for the associated graded object of the J-adic filtration. The module J^m/J^{m+1} is isomorphic to $\underline{O}_{s^*}(m)$; hence $R^1 p_*(J^m/J^{m+1}) = 0$, and the map

$$p_*(J^n/J^{n+k+1}) \longrightarrow p_*(J^n/J^{n+k})$$

is surjective. Therefore, to prove by induction on k that the maps u_k are surjective, it suffices to prove the surjectivity of

$$v_k : p_*\big(J^n \otimes \operatorname{Gr}_J^k(\underline{O})\big) \longrightarrow p_*\big(J^n \otimes \operatorname{Gr}_J^k(\underline{O}_T)\big) \qquad (k \geq 0).$$

The module $\operatorname{Gr}_J^k(\underline{O}_T)$ is a quotient of $J^k/J^{k+1} \otimes \underline{O}_T$, and thus has length at most $n+1$ over $k(s)$. The kernel N of the surjective map

$$v_k' : J^n \otimes \operatorname{Gr}_J^k(\underline{O}) \simeq \underline{O}_{s^*}(n+k) \longrightarrow J^n \otimes \operatorname{Gr}_J^k(\underline{O}_T)$$

is therefore an invertible sheaf on the projective line s^* of degree ≥ -1. Hence $R^1 p_* N = 0$, and $v_k = p_*(v_k')$ is surjective.

If T is Artinian, (2.14.1) now gives

$$\operatorname{length}_{k(s)}\big(p_* J^n / p_*(J^n q)\big) = \operatorname{length}_{k(s)}(\underline{O}_T),$$

and 2.14(i) follows from (2.11.4) and (2.11.6).

We now prove (ii). Since D is regular, its strict transform D' is isomorphic to D and meets s^* with length one. The assumption $\mathrm{length}_{k(s)}(T \cap s^*) \leq n$ therefore implies

$$\mathrm{length}_{k(s)}\big((D' \cup T) \cap s^*\big) \leq n+1,$$

so that $D' \cup T$ satisfies the hypothesis of (i).

The projection p induces an isomorphism between D and D'. Under this isomorphism, the ideal of $\underline{O}_D$ defining the subscheme

$$D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)$$

corresponds to the ideal of D' generated by the global sections of qJ^n on S_1. This ideal is contained in the ideal of D' defining $D' \cap Y$. It is in fact equal to it, because every section of $\underline{O}_{D' \cup Y}$ extends to a section of $\underline{O}_{S_1}$: if $I_{D' \cup Y}$ is the ideal defining $D' \cup Y$, the obstruction to extension lies in $R^1 p_*(I_{D' \cup Y})$. One has

$$\begin{aligned} I_{D' \cup Y} &= \underline{O}(-D') \cap qJ^n \\ &= (\underline{O}(-D') \cap J^n) \cap qJ^n \\ &= \underline{O}(-D')J^n \cap qJ^n \\ &= (\underline{O}(-D') \cap q)J^n, \end{aligned}$$

and by (i), applied to $D' \cup T$, there is no obstruction.

Thus

$$\mathrm{length}_{k(s)}\Big(D \cap \mathrm{Spec}\big(\underline{O}_S/p_*(qJ^n)\big)\Big) = \mathrm{length}_{k(s)}(D' \cap Y) \quad . \tag{2.14.2}$$

The isomorphism

$$J^n \otimes \big(\underline{O}_{D'}/q\underline{O}_{D'}\big) \xrightarrow{\sim} \underline{O}_{D'}J^n/\underline{O}_{D'}qJ^n$$

gives the exact sequence

$$0 \longrightarrow J^n \otimes \underline{O}_{D' \cap T} \longrightarrow \underline{O}_{D' \cap Y} \longrightarrow \underline{O}_{D' \cap ns^*} \longrightarrow 0,$$

and hence

$$\mathrm{length}_{k(s)}(D' \cap Y) = n(D', s^*) + \mathrm{length}_{k(s)}(D' \cap T). \tag{2.14.3}$$

Since $(D', s^*) = 1$, assertion 2.14(ii) follows from (2.14.2) and (2.14.3).

We now prove, by induction on the number of elements in the support of ν, the assertions of 2.13 and the following assertion:

(2.14.4) if D is a regular divisor on S, then $\quad \chi\Big(D \cap \mathrm{Spec}(\underline{O}_S/I(\nu))\Big) = \sum\limits_{\substack{t \in S^* \\ t \text{ on } D}} \nu(t)[k(t):k].$

If $\nu = 0$, then $\nu^* = 0$, $I(\nu) = \underline{O}_S$, and 2.13 and (2.14.4) are immediate (the second assertion of 2.13 follows by repeated application of (2.11.4) with $n = 0$). Otherwise $\nu(s) > 0$, and there is an S-morphism $g : S' \longrightarrow S_1$.

For each closed point a of S_1, let S_{1a} be the localization of S_1 at a, let ν_a be the restriction of ν to S_{1a}^*, let $S'_a = S' \times_{S_1} S_{1a}$, let $I(\nu_a) = g_*\underline{O}_{S'_a}(-\nu_a^*)$, and put

$$T_a = \mathrm{Spec}(\underline{O}_{S_1}/I(\nu_a)), \qquad T = \coprod_a T_a,$$

where the union is finite and disjoint. Let q be the ideal of T. With the notation of 2.14, one has

$$g_*\underline{O}_{S'}(-\nu^*) = qJ^{\nu(s)}. \tag{2.14.5}$$

The induction hypothesis applies to (S_{1a}, S'_a, ν_a). Hence

$$R^i g_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0, \tag{2.14.6}$$

$$\chi(T) = \sum_{t\in S^*\setminus\{s\}} \frac{\nu(t)(\nu(t)+1)}{2}[k(t):k], \tag{2.14.7}$$

and, by (2.14.4) applied to the divisor s^* on S_1 and by (2.12.1),

$$\text{length}_{k(s)}(T\cap s^*) = \sum_{\substack{t \text{ proximate} \\ \text{to } s}} \nu(t)[k(t):k(s)] \le \nu(s).$$

This inequality allows us to apply 2.14. From (2.14.5) we obtain

$$R^i p_* g_*\underline{O}_{S'}(-\nu^*) = 0 \quad \text{for } i > 0$$

and

$$\text{length}_{k(s)}\big(\underline{O}_S/p_*g_*\underline{O}_{S'}(-\nu^*)\big) = \frac{\nu(s)(\nu(s)+1)}{2} + \text{length}_{k(s)}(T).$$

The assertions of 2.13 therefore follow from (2.14.6) and (2.14.7). Finally, 2.14(ii) gives

$$\text{length}_{k(s)}\Big(D\cap \text{Spec}(\underline{O}_S/I(\nu))\Big) = \nu(s) + \text{length}_{k(s)}(D'\cap T).$$

Applying (2.14.4) to D' gives

$$\begin{aligned}\chi\Big(D\cap \text{Spec}(\underline{O}_S/I(\nu))\Big) &= \nu(s)[k(s):k] + \sum_{\substack{t \text{ on } D \\ t\neq s}} \nu(t)[k(t):k] \\ &= \sum_{\substack{t\in S^* \\ t \text{ on } D}} \nu(t)[k(t):k].\end{aligned}$$

This completes the proof of 2.13. (2.15) Fragments of the preceding discussion generalize to higher dimension. Let S be the spectrum of a regular local ring of dimension n, with closed point s, where $k(s)$ is a finite extension of k. We may again consider the set I of schemes $f: S' \longrightarrow S$ obtained from S by successive blow-ups of closed points, and put, as in 2.1,

$$S^* = \varinjlim_{S'\in I} \text{Irr}\big(f^{-1}(s)\big).$$

The set S^* will again be called the set of <u>points of S infinitely near to s</u>.

If X is a closed subscheme of S and $t \in S^*$, the multiplicity of X at t is defined as in 2.8.

Let $f: S' \longrightarrow S$ be a composite of blow-up morphisms at closed points, let D be a divisor on S, and let $D^* = f^*D$ be its total transform on S'.

Let E be a curve on S, let E' be its strict transform on S', and suppose that $D\cap E$ is supported at s. The projection formula again gives

$$(D, E) = (D^*, E'). \tag{2.16}$$

Proceeding as in 2.9, one proves with the aid of (2.16) that

$$(D, E) = \sum_{t \in S^*} \mu_t(D)\mu_t(E)[k(t) : k] \quad . \tag{2.17}$$

Define proximate points as in 2.5. Proceeding as in 2.11, one verifies that

$$\mu_t(E) = \sum_{\substack{t' \text{ proximate} \\ \text{to } t}} \mu_{t'}(E)[k(t') : k] \quad . \tag{2.18}$$

On the other hand, formula 2.11(ii) is false in general (see the examples in 2.25).

2.19. Let S be a regular projective surface over a field k. Write S^* for the disjoint sum, taken over the closed points of S,

$$S^* = \coprod_{s \in S} \operatorname{Spec}(\underline{O}_{s,S})^*,$$

and call S^* the set of points infinitely near to the points of S.

Let I be the set of surfaces obtained from S by successive blow-ups of closed points. The set I is cofinal in the set of regular models of the function field of S, ordered by domination. For $i \in I$, let S_i denote the corresponding surface, equipped with $p_i : S_i \longrightarrow S$. We write $\operatorname{Div}^{\pm}(S_i)$ for the group of divisors, effective or not, on S_i.

Definition 2.20. (i) The group of "curves" on S is the inductive limit

$$\varinjlim_{i \in I} \operatorname{Div}^{\pm}(S_i).$$

(ii) The category of "invertible sheaves" on S is the inductive limit
(SGA 4 VI 6) of the categories of invertible sheaves on the $(S_i)_{i \in I}$.
These objects are birational invariants of S.

Example 2.21. (i) Every "curve" C on S defines an "invertible sheaf" $\underline{O}(C)$ on S.

(ii) Every divisor C on S defines a "curve" on S, denoted by C^*.

(iii) If $t \in S^*$, define the surface $S_+(t)$ and the divisor t^* on $S_+(t)$ as in 2.3; the "curve" defined by this divisor will again be denoted by t^*.

(2.22) One verifies easily that a "curve" C (respectively an "invertible sheaf" $\mathcal{L}$) can be written in one and only one way in the form

$$C = C_0^* - \sum_{t \in S^*} i_t t^* \qquad \text{respectively} \qquad \mathcal{L} = \mathcal{L}_0 \otimes \underline{O}\left(-\sum_{t \in S^*} i_t t^*\right),$$

where C_0 (respectively $\mathcal{L}_0$) is a divisor (respectively an invertible sheaf) on S, and where the integers i_t are almost all zero. We say that C (respectively $\mathcal{L}$) is the curve C_0 (respectively the invertible sheaf $\mathcal{L}_0$) endowed with the virtual multiplicities i_t.

The completion $\operatorname{Div}^{\pm}(S) \times \mathbb{Z}^{S^*}$ of the group $\operatorname{Div}^{\pm}(S) \times \mathbb{Z}^{(S^*)}$ of "curves" on S is again a birational invariant of S; it is called the group of "generalized curves" on S. On S, a "generalized curve" C is represented by a divisor C_0 endowed with a possibly infinite collection of virtual multiplicities.

"Generalized invertible sheaves" on S are defined in the same way.

Example 2.23. Suppose that k is perfect. Then the invertible sheaf $\Omega^2_{S/k}$, endowed with virtual multiplicity -1 at every point $t \in S^*$, is a birational invariant of S; denote it by ${\Omega^2_{S/k}}^*$.

Example 2.24. Let $f : S_1 \longrightarrow S_2$ be a dominant morphism of regular projective surfaces over k. The inverse image of a "curve" or an "invertible sheaf" on S_2 is defined without difficulty by passing to a model of S_2 on which this "curve" or "invertible sheaf" is an ordinary divisor or invertible sheaf. By passage to the limit (I have not checked whether this passage to the limit is possible), one similarly defines the inverse image of a "generalized curve" or a "generalized invertible sheaf." Suppose that k is perfect and that f is generically étale. The <u>different</u> $\mathfrak{d}(f)$ of f is then the effective divisor on S_1 characterized by

$$f^*\Omega^2_{S_2/k}(\mathfrak{d}(f)) \xrightarrow{\sim} \Omega^2_{S_1/k} \quad .$$

This different is not a birational invariant of f; to obtain a birational invariant one must consider the generalized coincidence curve

$$\mathfrak{d}^*(f) = \mathfrak{d}(f) + \sum_{t \in S_1^*} t^* - f^*\left(\sum_{t \in S_2^*} t^*\right),$$

or, briefly,

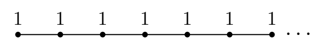
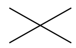
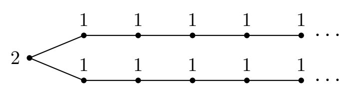
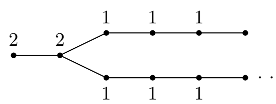
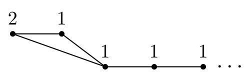
$$\mathfrak{d}^*(f) = {\Omega^2_{S_1/k}}^* - f^*{\Omega^2_{S_2/k}}^* \quad .$$

This "generalized curve" is supported on the closed subset of S_1 where f is not étale.

(2.25) <u>Example</u>.

Let k be an algebraically closed field and let E_n be affine n-space over k. Let C be a reduced curve in E_n having a singularity at the origin. With this singularity we associate its <u>Enriques diagram</u> (cf. [3]): its vertices are the points of C infinitely near the origin (2.4 and 2.15), and each vertex is joined by a line to the vertices proximate to it (2.5 and 2.15). This diagram determines the multiplicities of these points on C (2.11 and 2.18) and, when $n = 2$, the invariant δ (2.10 and 2.11) of the singularity of C at the origin.

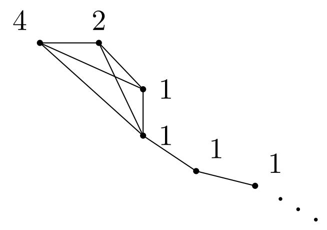
For the simplest plane singularities one obtains the following table. A number next to a vertex is its multiplicity.

Sketch	*Equation*	δ	*Enriques diagram*
	$y = 0$	0	
	$xy = 0$	1	
	$y(y - x^2) = 0$	2	
	$y^2 = x^3$	1	

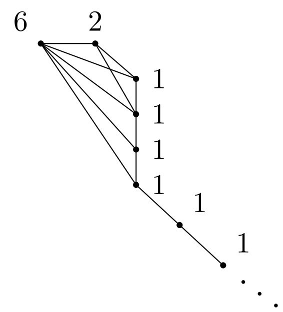
If $n > 2$, the invariant δ is no longer determined by this diagram. For example, the union of three concurrent coplanar lines (respectively, noncoplanar lines) has invariant $\delta = 3$ (respectively, $\delta = 2$), although the Enriques diagram is the same. I do not know whether this phenomenon also occurs for unibranch curves.

Here are two examples of diagrams in the case $n = 3$ which cannot occur for plane curves (cf. 2.5). We give, in order, the equations of the curve, the equations of the pure transform after blowing up the origin, parametric equations of the curve, the invariant δ, and the Enriques diagram.

(2.25.1)
$$\left\{\begin{array}{l} \begin{cases} y^2 = xz^4, \\ z^3 = x^2, \end{cases} \qquad \begin{cases} (y/z)^2 = (x/z)^3, \\ \qquad z = (x/z)^2, \end{cases} \\[2ex] \qquad \begin{cases} x = u^6, \\ y = u^7, \\ z = u^4, \end{cases} \qquad \delta = 5 \end{array}\right.$$



$$\left\{\begin{array}{l} \begin{cases} y^2 z = x^3, \\ \quad z^3 = y^2, \end{cases} \qquad \begin{cases} (y/z)^2 = (x/z)^3, \\ \qquad\quad z = (y/z)^2, \end{cases} \\ \qquad\quad \begin{cases} x = u^8, \\ y = u^9, \qquad \delta = 10 \\ z = u^6, \end{cases} \end{array}\right. \tag{2.25.2}$$



2.26. Theorem 2.27 below was stated as a false conjecture in the first version of this exposé (circulated by the IHES). It owes much to conversations with D. S. Rim and includes some of his results (D. S. Rim, "Torsion Differentials and Deformations," Theorem 2.7(i)).

Let C be a reduced projective curve over an algebraically closed field k, and let $s \in C(k)$. We shall write

A_s	for the local ring of C at s (or its completion, which would make no difference below);
$\widetilde{A}_s$	for the normalization of A_s;
$\delta(s)$	for $\dim_k(\widetilde{A}_s/A_s)$;
$m_0(s)$	for the dimension of the formal versal scheme T of deformations of $\operatorname{Spec}(A_s)$.

For an irreducible component E of T, put likewise $m_0^E(s) = \dim(E)$.

The module $\widetilde{\tau}$ of derivations of $\widetilde{A}_s$ and the module τ of derivations of A_s are torsion-free. If A_s is integral, they embed in the module of derivations of the fraction field of A_s; in general, they embed in the module of derivations of the total quotient ring of A_s (a product of fields). Moreover, through these embeddings, τ and $\widetilde{\tau}$ are commensurable: both $\tau/(\tau \cap \widetilde{\tau})$ and $\widetilde{\tau}/(\tau \cap \widetilde{\tau})$ are finite-dimensional over k. Set

$$m_1(s) = [\widetilde{\tau} : \tau] \underset{\text{def}}{=} \dim_k \frac{\widetilde{\tau}}{\tau \cap \widetilde{\tau}} - \dim_k \frac{\tau}{\tau \cap \widetilde{\tau}}.$$

Theorem 2.27. Let E be an irreducible component of the formal versal scheme T of deformations of A_s. If the deformation of A_s over the generic point e of E is smooth—that is, if the image of the formal non-smoothness subscheme (defined by a Jacobian ideal: VI 5), which is finite over T, does not contain e—then

$$3\delta(s) = m_0^E(s) + m_1(s).$$

The validity of 2.27 depends only on the completion of A_s. Replacing (C, s) by a pair (C', s) formally isomorphic to it at s, we may suppose that the following conditions hold:

(a) s is the only singular point of C;

(b) C has no infinitesimal automorphisms.

Condition (b) ensures that the deformation functor of C is pro-representable. It is even effectively pro-representable, since there is no obstruction to lifting a prescribed ample invertible sheaf on C. Let $f : X_0 \longrightarrow T_0$ be the universal deformation of C, and let t_0 be the closed point of T_0, so that $C \xrightarrow{\sim} X_{0t_0}$.

Lemma 2.28. There exist a scheme T_1 of finite type over k, a point $t_1 \in T_1(k)$, and a proper flat morphism $f : X_1 \longrightarrow T_1$, whose fiber at t_1 is C, such that for every $u \in T_1(k)$ the completion of T_1 at u is a formal moduli scheme for the fiber X_{1u}.

Here is a direct proof of this corollary of general results of Artin. Let $\mathcal{L}$ be a very ample invertible sheaf on C such that $H^1(C, \mathcal{L}) = 0$, and put $\mathbb{P} = \mathbb{P}(H^0(C, \mathcal{L}))$. We have an embedding $v : C \hookrightarrow \mathbb{P}$. Let T_2 be the Hilbert scheme parametrizing the subvarieties of $\mathbb{P}$ having the same Hilbert polynomial as $v(C) \subset \mathbb{P}$, and let $t_2 \in T_2$ correspond to $v(C)$. The completion $\widehat{T}_2$ of T_2 at t_2 is smooth over T_0. The vertical tangent bundle of $\widehat{T}_2/T_0$ is induced by the subbundle L of the tangent bundle of T_2 which is the image of $\mathrm{Lie}(\underline{\mathrm{Aut}}(\mathbb{P}))$ under the action of $\underline{\mathrm{Aut}}(\mathbb{P})$ on T_2.

In a neighborhood of t_2, let $f_1, \ldots, f_n$, with $n = \dim \underline{\mathrm{Aut}}(\mathbb{P})$, be equations defining a subscheme T_1' of T_2 which passes through t_2 and whose completion at t_2 is étale over T_0. Take for T_1 a neighborhood of t_2 in T_1' on which the tangent bundle is transverse to L.

2.29. The morphism of functors

$$(\text{deformations of } C) \longrightarrow (\text{deformations of } A_s)$$

corresponds to a morphism $p : T_0 \longrightarrow T$. By VI 2.10, p is smooth. Let t be the closed point of T. The Zariski tangent space of $p^{-1}(t)$ at t_0 classifies the locally trivial deformations of C over $k[\varepsilon]$, where $\varepsilon^2 = 0$. Putting $E_0 = p^{-1}(E)$, we obtain

$$\dim(E_0) = \dim(E) + \dim H^1(C, T^1_{C/k}). \tag{2.29.1}$$

Since X_0/T_0 is flat, all its fibers have the same arithmetic genus p, where $1 - p = \chi(\underline{O})$. By hypothesis, the fiber at the generic point of E_0 is smooth. The formal moduli scheme of a smooth proper curve being smooth of dimension $3p - 3$, Lemma 2.28 gives

$$\dim(E_0) = 3p - 3. \tag{2.29.2}$$

Let $\widetilde{C}$ be the normalization of C, of arithmetic genus $\widetilde{p} = p - \delta(s)$. We have

$$\begin{aligned} \dim H^1(C, T^1_{C/k}) &= -\chi(C, T^1_{C/k}) \\ &= -\chi(\widetilde{C}, T^1_{\widetilde{C}/k}) + m_1(s) \\ &= -[3(p - \delta(s)) - 3] + m_1(s). \end{aligned} \tag{2.29.3}$$

Formulas (2.29.1), (2.29.2), and (2.29.3) prove 2.27.

Conjecture 2.30. Every reduced proper curve has a smooth deformation.

This conjecture implies, by 2.27 and 2.28, that the formal versal scheme of deformations of A_s is always equidimensional, of dimension $3\delta(s) - m_1(s)$.

2.31. Let C be a reduced proper curve over k, of arithmetic genus p. We shall write

M_0 for the dimension of the formal versal scheme T of deformations of C;

M_1 for the dimension of the Lie algebra of $\underline{\mathrm{Aut}}(C)$.

For an irreducible component E of T, write M_0^E for the dimension of E. The global analogue of 2.27 is the following.

Proposition 2.32. If the deformation of C over the generic point of E is smooth, then

$$3p-3 = M_0^E - M_1.$$

Let $X \subset C(k)$ be a finite set of smooth points of C such that every vector field on C which vanishes at every point $x \in X$ is zero. The deformation functor of (C, X) is pro-representable and satisfies a variant of 2.28. The formal moduli scheme of (C, X) is smooth over T, of relative dimension $\#X - M_1$, and the analogue of 2.28 gives

$$M_0^E + (\#X - M_1) = 3p - 3 + \#X,$$

which proves 2.32.

Suppose now that k has characteristic zero. Then $M_1 = \dim \underline{\mathrm{Aut}}(C)$. Here is an analogous interpretation of $m_1(s)$.

Lemma 2.33. If k has characteristic zero, every derivation of A_s extends to a derivation of $\widetilde{A}_s$.

If D is a derivation of A_s, then

$$\exp(Dt) = \sum_{n \geq 0} \frac{D^n t^n}{n!}$$

is an automorphism of the completion at $(s, 0)$ of $A_s \otimes_k k[t]$. By transport of structure, this automorphism acts on the normalization $(\widetilde{A}_s \otimes_k k[t])^\wedge$ of that ring. Reducing modulo t^2, we obtain an automorphism $1 + Dt$ of $\widetilde{A}_s^\wedge \otimes_k k[t]/(t^2)$, and D thereby extends to a derivation of $\widetilde{A}_s^\wedge$. Since it also extends to the fraction field of A_s (to the total quotient ring when A_s is not integral), it extends to $\widetilde{A}_s$.

Suppose that s is singular. A similar argument then shows that the vector field on $\mathrm{Spec}(\widetilde{A}_s)$ defined by D vanishes at every point above s.

Let c be an ideal of $\widetilde{A}_s$ contained in the conductor of $\widetilde{A}_s/A_s$. Put

$$\widetilde{G}_c = \underline{\mathrm{Aut}}(\widetilde{A}_s/c),$$

and let G_c be the subgroup of $\widetilde{G}_c$ which stabilizes the subalgebra A_s/c. If $d \subset c$, then $\widetilde{G}_d$ stabilizes c/d and maps onto $\widetilde{G}_c$. Notice that $g \in \widetilde{G}_c$ is determined by $g(t)$ for a uniformizer t, and that $g(t)$ may be any uniformizer. Consequently the pointed homogeneous space $\widetilde{G}_c/G_c$ is independent of c; denote it by $\widetilde{G}/G$.

Let $\widetilde{\tau}_0$ be the $\widetilde{A}_s$-module of vector fields on $\mathrm{Spec}(\widetilde{A}_s)$ which vanish at the points above s. When s is singular, one has

$$\dim_k(\widetilde{\tau}/\widetilde{\tau}_0) = \text{the number of branches},$$

and

$$\widetilde{\tau}_0/\tau \xrightarrow{\sim} \text{the tangent space at the origin of } \widetilde{G}/G.$$

Proposition 2.34. In characteristic zero, when s is singular, the integer $m_1(s)$ is the sum of the number of branches and the dimension of $\widetilde{G}/G$.

Bibliography

[1] P. Defrise, *Étude locale des correspondances rationnelles entre surfaces algébriques*, thèse d'agrégation, Brussels, 1949.

[2] P. Deligne and D. Mumford, "The irreducibility of the space of curves of a given genus," *Publ. Math. IHES* 36 (1969), 75–110.

[3] Enriques–Chisini, *Lezioni sulla teoria geometrica delle equazioni e delle funzioni algebriche*, Zanichelli, Bologna, 1924.

[4] R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, 1966.

[5] S. Lichtenbaum, "Curves over discrete valuation rings," *Amer. J. Math.* 15, no. 2 (April 1968), 380–405.

[6] I. Manin, "Rational surfaces over perfect fields," *Publ. Math. IHES*, no. 30, 55–114.

[7] D. Mumford, "The topology of normal singularities of an algebraic surface and a criterion for simplicity," *Publ. Math. IHES*, no. 9, 5–22.

[8] A. Néron, *Modèles minimaux des variétés abéliennes sur les corps locaux et globaux*, Publ. Math., no. 21.

[9] M. Raynaud, "Spécialisation du foncteur de Picard. Critères numériques de représentabilité," *C. R. Acad. Sci. Paris* 264 (June 1967), 1001–1004.

[10] Saint-Donat, *Sur un théorème de G. Castelnuovo et F. Enriques*, thèse de troisième cycle, Lyon.

[11] O. Zariski, "On polynomial ideals defined by infinitely near base points," *Amer. J. Math.* 60 (1938), 151.

[12] D. G. Northcott, "On the genus formula for plane curves," *J. London Math. Soc.* 30 (1955), 376–382.

[13] D. G. Northcott, "Abstract dilatations and infinitely near points," *Proc. Cambridge Philos. Soc.* 52 (1956), 178–197.

The notion of an "infinitely near point" is discussed in Lipman's appendix to Chapter II of the following work.

[14] O. Zariski, *Algebraic Surfaces*, Ergebnisse 61, Springer-Verlag.

SGA 7

EXPOSÉ XI

COHOMOLOGY OF COMPLETE INTERSECTIONS

by P. DELIGNE

1. Cohomology of projective space and complete intersections

Let X be a nonsingular complex projective variety, purely of dimension n, and let $\underline{O}(1)$ be an ample invertible sheaf on X. By Akizuki and Nakano [1], Kodaira's vanishing theorem [4] implies that

$$H^i(X, \Omega^j_X(-1)) = 0 \qquad \text{for } i + j < n. \tag{*}$$

Again by [1], this result implies the "weak Lefschetz theorem" in complex cohomology. We propose to prove, by brute force, that if X is a projective space over a field of arbitrary characteristic, then (*) remains true. This will allow us, following [1], to compare the Hodge and de Rham cohomology of complete intersections over a field of arbitrary characteristic with that of projective space.

We shall then spell out some consequences of the results obtained. In ğ2 we give qualitative results deduced from Hirzebruch's numerical formula (2.2). I wish to thank N. Katz for his help in the preparation of that section.

1.0. Let $\underline{E}$ be a locally free module of rank $r+1$ on a scheme S. We assume that $r > 0$. Recall that $\mathbb{P}(\underline{E})$ denotes the relative projective space over S of hyperplanes of $\underline{E}$. Let $p : \mathbb{P}(\underline{E}) \to S$ be the projection, and let

$$\eta \in H^0\big(S, R^1 p_* \Omega^1_{\mathbb{P}(\underline{E})/S}\big)$$

be the first Chern class, in the Hodge sense, of the invertible sheaf $\underline{O}(1)$.

Recall that η is the image of the class of $\underline{O}(1)$ in $H^0(S, R^1 p_* \underline{O}^*)$ under the logarithmic differential

$$df/f : \underline{O}^* \longrightarrow \Omega^1_{\mathbb{P}(\underline{E})/S}.$$

For every sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, put

$$\underline{F}(n) = \underline{F} \otimes \underline{O}(1)^{\otimes n} \qquad (n \in \mathbb{Z}).$$

Theorem 1.1. (i) The sheaves $R^i p_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ are locally free, and their formation is compatible with arbitrary base change.

(ii) For $0 \le i \le r$, the section η^i of $R^i p_*(\Omega^i_{\mathbb{P}(\underline{E})/S})$ defines an isomorphism from $\underline{O}_S$ onto this sheaf. Moreover,

$$R^i p_* \Omega^j_{\mathbb{P}(\underline{E})/S} = 0 \qquad \underline{\text{if}}\ i \neq j\ \ \underline{\text{or}}\ \ i > r.$$

(iii) If $n \neq 0$, then, outside the following two cases, $R^i p_*(\Omega^j_{\mathbb{P}(\underline{E})/S}(n))$ vanishes:

$$i = 0 \quad \underline{\text{and}} \quad n > j, \tag{iii$_a$}$$
$$i = r \quad \underline{\text{and}} \quad n < j - r. \tag{iii$_b$}$$

Let

$$\mathbb{V}(\underline{E}) = \operatorname{Spec}\big(\operatorname{Sym}^* \underline{E}\big)$$

be the affine space defined by $\underline{E}$, let e be its zero section, and put $\mathbb{V}(\underline{E})^* = \mathbb{V}(\underline{E}) \setminus e(S)$. Let a and a' be the projections of $\mathbb{V}(\underline{E})$ and $\mathbb{V}(\underline{E})^*$ onto S, and let q be the projection of $\mathbb{V}(\underline{E})^*$ onto $\mathbb{P}(\underline{E})$:

$$\begin{tikzcd}
\mathbb{V}(\underline{E}) \arrow[dr, "a"'] & \mathbb{V}(\underline{E})^* \arrow[l, hook'] \arrow[r, "q"] \arrow[d, "a'"] & \mathbb{P}(\underline{E}) \arrow[dl, "p"] \\
& S \arrow[ul, bend left, "e"] &
\end{tikzcd}.$$

The punctured affine space $\mathbb{V}(\underline{O}(1))^*$ over $\mathbb{P}(\underline{E})$ is canonically isomorphic to $\mathbb{V}(\underline{E})^*$. Consequently, for every quasi-coherent sheaf $\underline{F}$ of $\underline{O}$-modules on $\mathbb{P}(\underline{E})$, there is a canonical isomorphism

$$q_* q^* \underline{F} \simeq \bigoplus_{n\in\mathbb{Z}} \underline{F}(n). \tag{1.1.1}$$

This grading of $q_* q^* \underline{F}$ can also be obtained from the action on this sheaf of the group $\mathbb{G}_m$ of homotheties of $\mathbb{V}(\underline{E})$.

Relative vector fields on $\mathbb{V}(\underline{E})$ are identified with S-morphisms from $\mathbb{V}(\underline{E})$ to itself, the target being identified with the tangent space at the origin. The vector field X defined by the identity map of $\mathbb{V}(\underline{E})$ is invariant under homotheties. It is tangent to the fibers of q and defines a trivialization

$$X : \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow \underline{O}_{\mathbb{V}(\underline{E})^*}. \tag{1.1.2}$$

If the x_i form a basis of $\underline{E}$, then

$$X = \sum_i x_i \, dx^i.$$

Since $\Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})}$ has rank one, the exact sequence

$$0 \longrightarrow q^*\Omega^1_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \longrightarrow 0$$

induces exact sequences

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \longrightarrow \Omega^1_{\mathbb{V}(\underline{E})^*/\mathbb{P}(\underline{E})} \otimes q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0.$$

Using (1.1.2), these become

$$0 \longrightarrow q^*\Omega^i_{\mathbb{P}(\underline{E})/S} \longrightarrow \Omega^i_{\mathbb{V}(\underline{E})^*/S} \xrightarrow{\iota_X} q^*\Omega^{i-1}_{\mathbb{P}(\underline{E})/S} \longrightarrow 0. \tag{1.1.3}$$

Canonically,

$$\Omega^i_{\mathbb{V}(\underline{E})/S} = a^* \bigwedge^i \underline{E}. \tag{1.1.4}$$

The short exact sequences (1.1.3) are therefore obtained by splitting the following long exact sequence on $\mathbb{V}(\underline{E})^*$:

$$\cdots \longrightarrow a'^* \bigwedge^i \underline{E} \xrightarrow{\iota_X} a'^* \bigwedge^{i-1} \underline{E} \xrightarrow{\iota_X} a'^* \bigwedge^{i-2} \underline{E} \longrightarrow \cdots . \tag{1.1.5}$$

This long exact sequence is simply the restriction to $\mathbb{V}(\underline{E})^*$ of the Koszul resolution of $\underline{O}_{e(S)}$ on $\mathbb{V}(\underline{E})$.

Push the long exact sequence (1.1.5) forward to $\mathbb{P}(\underline{E})$ by the affine morphism q. By (1.1.1) and the formula $q_* a'^* = q_* q^* p^*$, one obtains a long exact sequence

$$\cdots \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^i \underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^{i-1} \underline{E}(n) \longrightarrow \cdots . \tag{1.1.6}$$

It splits into short exact sequences, the images of (1.1.3),

$$0 \longrightarrow \bigoplus_{n\in\mathbb{Z}} \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow \bigoplus_{n\in\mathbb{Z}} p^* \bigwedge^i \underline{E}(n) \xrightarrow{\iota_X} \bigoplus_{n\in\mathbb{Z}} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0. \tag{1.1.7}$$

The exact sequences (1.1.7) are direct sums of the exact sequences

$$0 \longrightarrow \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow p^* \bigwedge^i \underline{E}(n-i) \xrightarrow{\iota_X} \Omega^{i-1}_{\mathbb{P}(\underline{E})/S}(n) \longrightarrow 0. \tag{1.1.8}$$

Recall (EGA III, 2.1.15, or FAC) that the sheaves $R^i p_* \underline{O}(k)$ are locally free and their formation is compatible with arbitrary base change, and that

$$R^i p_* \underline{O}(k) = 0 \quad \text{if } i \neq 0, r, \quad \text{or if } i = 0, k < 0, \quad \text{or if } i = r, k > -r-1, \tag{1.1.9}$$

and hence that $R^i p_* p^* \bigwedge^j \underline{E}(k)$ vanishes under the same conditions.

The exact sequences (1.1.8) give a resolution

$$0 \longrightarrow \Omega^i_{\mathbb{P}(\underline{E})/S}(n) \xrightarrow{E} p^* \bigwedge^i \underline{E}(n-i) \xrightarrow{\iota_X} \cdots \longrightarrow p^* \bigwedge^0 \underline{E}(n) \longrightarrow 0 \tag{1.1.10}$$

of $\Omega^i_{\mathbb{P}(\underline{E})/S}(n)$. The spectral sequence defined by this resolution degenerates ($E_1 = E_\infty$) for degree reasons by (1.1.9); for $0 < j < r$, it gives

$$R^j p_* \Omega^i_{\mathbb{P}(\underline{E})/S}(n) = \mathcal{H}\left(\cdots \xrightarrow{\iota_X} p_* \bigwedge^{i-j} \underline{E}(n-i+j) \xrightarrow{\iota_X} \cdots \right). \tag{1.1.11}$$

Since $r \geq 1$, a depth argument shows that

$$\bigoplus_{n\in\mathbb{Z}} p_* p^* \bigwedge^i \underline{E}(n) = p_* q_* q^* p^* \bigwedge^i \underline{E} = a'_* a'^* \bigwedge^i \underline{E} = a_* a^* \bigwedge^i \underline{E}. \tag{1.1.12}$$

Thus $p_* p^* \bigwedge^i \underline{E}(n)$ is the homogeneous part of weight $n+i$ in $a_* a^* \bigwedge^i \underline{E}$, namely $\mathrm{Sym}^n(\underline{E}) \otimes \bigwedge^i \underline{E}$. Remembering that (1.1.5) comes from the Koszul resolution, (1.1.11) shows that, for

$0 < j < r$, the sheaf $R^j p_* \Omega^i_{\mathbb{P}(\underline{E})/S}(n)$ can be nonzero only when $i - j = 0$ and $n - i + j = 0$, that is, only when $n = 0$ and $i = j$. In that case, moreover, $R^i p_* \Omega^i_{\mathbb{P}(\underline{E})/S}$ is isomorphic to $\underline{O}_S$.

For $j = 0$, (1.1.10) also gives

$$p_* \Omega^i_{\mathbb{P}(\underline{E})/S}(n) = \ker \left(\iota_X : p_* \bigwedge^{i} \underline{E}(n-i) \longrightarrow p_* \bigwedge^{i-1} \underline{E}(n-i+1) \right), \tag{1.1.13}$$

and the same argument shows that ι_X is injective when $n - i \leq 0$.

By (1.1.8), with $n = 0$ and $i = r + 1$, one has

$$\Omega^r_{\mathbb{P}(\underline{E})/S} \simeq p^* \bigwedge^{r+1} \underline{E}(-r-1).$$

When S is the spectrum of a field, the vanishing assertions in 1.1(ii) and (iii) now follow from the preceding argument for $j \neq r$, and follow from it by duality ([2], V, 11.2, p. 211) for $j = r$. In particular, for any j and n there is at most one i for which $R^i p_* \Omega^j_{\mathbb{P}(\underline{E})/S}(n) \neq 0$. This proves (i), since $\Omega^j_{\mathbb{P}(\underline{E})/S}$ is flat over S (the hardest part is locating the reference in EGA III; we suggest 6.10.5, 7.4.1, and 7.5.5, followed by a passage to the limit).

Knowing that $R^i p_* \Omega^i_{\mathbb{P}(\underline{E})/S}$ is an invertible sheaf for $0 \leq i \leq r$, it remains only to prove that, when S is the spectrum of a field, $\eta^i \neq 0$. Indeed, η^r is the cohomology class of a rational point (the intersection of r hyperplanes), hence a basis of $H^r(\mathbb{P}^r, \Omega^r)$ (FGA, Exposé 149, no. 4).

1.2. The Euler–Poincaré characteristics of the $\Omega^i(n)$ may be calculated directly from (1.1.10) or (1.1.13). The rather unappetizing result is

$$\chi\big(\mathbb{P}^r_k, \Omega^j_{\mathbb{P}^r_k}(n)\big) = \sum_{0 \leq i \leq j} (-1)^{i-j} \binom{r+1}{i} \binom{r+n-i}{r}.$$

<u>Proposition</u> 1.3. <u>Let $f : X \to S$ be a proper smooth morphism, let $\underline{W}$ be a locally free sheaf of rank c on X, let $\underline{F}$ be a coherent sheaf on X, let $d > 0$ be an integer, and let $s : \underline{W} \to \underline{O}_X$ be a section of $\underline{W}^\vee$. Assume that S is noetherian and that:</u>

(a) <u>The closed subscheme H of X defined by the vanishing of s is smooth over S.</u>

(b) <u>Locally on X, if the s_i are the coordinates of s in a basis of $\underline{W}^\vee$, the s_i form both an $\underline{O}_X$-regular sequence and an $\underline{F}$-regular sequence.</u>

(c) <u>For every sequence of integers k_ℓ, not all zero, one has</u>

$$R^i f_* \left(\bigotimes_\ell \bigwedge^{k_\ell} \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F} \right) = 0 \qquad \underline{\text{for }} i + j < d.$$

<u>Then:</u>

$$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \xrightarrow{\sim} R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \underline{\text{for }} i + j < d - c, \tag{i}$$

$$R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F} \right) \hookrightarrow R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F} \right) \qquad \underline{\text{for }} i + j = d - c. \tag{ii}$$

The conormal sheaf $N_{H/X}$ of H in X is isomorphic to the restriction of $\underline{W}$ to H, so the exact sequence

$$0 \longrightarrow N_{H/X} \longrightarrow \Omega^1_{X/S} \otimes \underline{O}_H \longrightarrow \Omega^1_{H/S} \longrightarrow 0 \tag{1.3.1}$$

defines a decreasing filtration on $\Omega^j_{X/S} \otimes \underline{O}_H$ whose successive quotients are

$$\mathrm{Gr}^k\big(\Omega^j_{X/S} \otimes \underline{O}_H\big) \simeq \bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S}. \tag{1.3.2}$$

It follows from (b) that one also has

$$\mathrm{Gr}^k\big(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\big) = \bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S} \otimes \underline{F}. \tag{1.3.3}$$

We prove (i) and (ii) by induction on j, simultaneously for all $\underline{F}$ satisfying (b) and (c). The assertions are empty for $j < 0$.

Suppose, then, that (i) and (ii) have been proved for $j' < j$. The induction hypothesis applies to the sheaves $\bigwedge^k \underline{W} \otimes \underline{F}$ and, by (c), gives

$$R^i f_* \left(\bigwedge^k \underline{W} \otimes \Omega^{j-k}_{H/S} \otimes \underline{F}\right) = 0 \quad \text{for } k > 0 \text{ and } i + j < d - c + k. \tag{1.3.4}$$

The filtration of $\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}$ considered above defines a spectral sequence

$$E_1^{p,i-p} = R^i f_* \left(\bigwedge^p \underline{W} \otimes \Omega^{j-p}_{H/S} \otimes \underline{F}\right) \Longrightarrow R^i f_* \left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right).$$

By (1.3.4), $E_1^{p,i-p} = 0$ when $p > 0$ and $i + j \leq d - c$; hence

$$\begin{cases} R^i f_* \left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right) \xrightarrow{\sim} R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j < d - c, \\ R^i f_* \left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right) \hookrightarrow R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j = d - c. \end{cases} \tag{1.3.5}$$

By hypothesis (b), with a suitable differential, the sheaves $\bigwedge^k \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F}$ form a left Koszul resolution of $\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}$. This filtration defines a spectral sequence

$$E_1^{-p,q} = R^q f_* \left(\bigwedge^p \underline{W} \otimes \Omega^j_{X/S} \otimes \underline{F}\right) \Longrightarrow R^{q-p} f_* \left(\Omega^j_{X/S} \otimes \underline{O}_H \otimes \underline{F}\right). \tag{1.3.6}$$

By hypothesis (c), $E_1^{-p,q} = 0$ for $p > 0$ and $q + j < d$, and trivially for $p \notin [0, c]$. Consequently,

$$\begin{cases} R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F}\right) \xrightarrow{\sim} R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j < d - c, \\ R^i f_* \left(\Omega^j_{X/S} \otimes \underline{F}\right) \hookrightarrow R^i f_* \left(\Omega^j_{H/S} \otimes \underline{F}\right) & \text{if } i + j = d - c. \end{cases} \tag{1.3.7}$$

Assertions (i) and (ii) now follow from (1.3.5) and (1.3.7).

1.4. Let $\underline{E}$ be a locally free sheaf of rank $r + 1$ on a scheme S, let $\mathbb{P}(\underline{E})$ be the corresponding projective bundle, let $d \leq r$ be an integer, and let $\underline{a} = (a_1, \ldots, a_d)$ be a sequence of integers. A relative complete intersection of multidegree $\underline{a}$ in $\mathbb{P}(\underline{E})$ is a closed subscheme which has codimension d in every fiber and which, locally on S for the étale topology, is the intersection of d hypersurfaces of respective degrees $a_1, \ldots, a_d$.

By 1.1, hypotheses (b) and (c) of 1.3 hold for $X/S = \mathbb{P}(\underline{E})/S$, $\underline{F} = \underline{O}_X$,

$$\underline{W} = \bigoplus_i \underline{O}(-a_i),$$

and a section s of $\underline{W}^\vee$ defining a relative complete intersection $Y \subset \mathbb{P}(\underline{E})$ of multidegree $\underline{a}$.

Theorem 1.5. Under the hypotheses of 1.4, let $n = r - d$ be the relative dimension of Y, and let $\eta \in H^0(S, R^1 f_* \Omega^1_{Y/S})$ be the first Chern class of $\underline{O}(1)$ in the diagram

$$\begin{tikzcd} Y \arrow[rr, hook] \arrow[dr, "f"'] & & \mathbb{P}(\underline{E}) \arrow[dl, "g"] \\ & S & \end{tikzcd} .$$

If Y is smooth over S, then:

(i) The sheaves $R^i f_* \Omega^j_{Y/S}$ are locally free, and their formation is compatible with arbitrary base change.

(ii) The spectral sequence
$$E_1^{p,q} = R^q f_* \Omega^p_{Y/S} \Longrightarrow R^{p+q} f_* (\Omega^*_{Y/S}),$$
relating Hodge and de Rham cohomology degenerates ($E_1 = E_\infty$); its abutment is locally free, and its formation is compatible with arbitrary base change.

(iii)$_a$ One has
$$R^i f_* \Omega^j_{Y/S} = 0 \qquad \text{if } i \neq j \text{ and } i + j \neq n.$$

(iii)$_b$ η^i is a basis of $R^i f_* \Omega^i_{Y/S}$ if $2i < n$.

(iii)$_c$ If $n < 2i \leq 2n$, and if α_i is the basis of $R^i f_* \Omega^i_{Y/S}$ dual to the basis η^{n-i} of $R^{n-i} f_* \Omega^{n-i}_{Y/S}$, then
$$\eta^i = \left(\prod_{h=1}^d a_h \right) \alpha_i.$$

(iii)$_d$ If $n = 2m$, then $\eta^m \neq 0$ (provided $S \neq \varnothing$), and the quotient $R^m f_* \Omega^m_{Y/S} / \underline{O}_S \eta^m$ is locally free.

It is enough to prove 1.5 in a "universal" case, which permits us to assume that S is smooth over $\mathrm{Spec}(\mathbb{Z})$. In this case (ii) follows from (i), since it holds over $\mathbb{C}$, and therefore over every field of characteristic zero, by Hodge theory. Moreover, since S is reduced, (i) follows from (iii), assumed proved when S is the spectrum of a field. Indeed, by (iii), for each j the functions

$$s \longmapsto \dim_{k(s)} H^i(Y_s, \Omega^j_{Y_s})$$

are constant on S, except perhaps when $i = j$; hence they are all constant, because their alternating sum is constant (EGA III, 7.9.2). The conclusion then follows from EGA III, 6.10.5, 7.6.9, and 7.7.5.

By 1.4 and 1.3, one has

$$(1.5.1) \qquad \begin{cases} R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \xrightarrow{\sim} R^i f_* \Omega^j_{Y/S} & \text{for } i + j < n, \\ R^i g_* \Omega^j_{\mathbb{P}(\underline{E})/S} \hookrightarrow R^i f_* \Omega^j_{Y/S} & \text{for } i + j = n. \end{cases}$$

When S is the spectrum of a field, assertions (iii)$_a$, (iii)$_b$, and (iii)$_d$ follow from (1.5.1) and Serre duality ([2], V, 11.2, p. 211). The cohomology class

η^n is the class of a zero-cycle of degree $\prod_{h=1}^d a_h$ (the degree of the complete intersection), so

$$\eta^n = \left(\prod_{h=1}^d a_h\right) \alpha_n.$$

By definition, $\eta^{n-i}\alpha_i = \alpha_n$, and (iii)$_c$ follows.

Thus (i) and (ii) hold; in the universal case, the assertions in (iii) follow from (i) and from (iii) in the case already proved, where S is the spectrum of a field.

Theorem 1.6. Keep the hypotheses and notation of 1.4 and 1.5, assume that Y is smooth, let ℓ be a prime invertible in all residue fields of S, and let $\eta_\ell \in H^0(S, R^2 f_* \mathbb{Z}_\ell(1))$ denote the ℓ-adic first Chern class of $\underline{O}(1)$ (SGA 5, VII). Then:

(i) $R^i f_* \mathbb{Z}_\ell = 0$ if i is odd and $i \neq n$.

(ii) If $2i < n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has η_ℓ^i as a basis.

(iii) If $n < 2i \leq 2n$, then $R^{2i} f_* \mathbb{Z}_\ell(i)$ is isomorphic to $\mathbb{Z}_\ell$ and has

$$\eta_\ell^i \Big/ \prod_{h=1}^d a_h$$

as a basis.

(iv) The $\mathbb{Z}_\ell$-sheaf $R^n f_* \mathbb{Z}_\ell$ is constructible, twisted constant, and torsion-free. If n is even, then $\eta_\ell^{n/2} \neq 0$ (provided $S \neq \varnothing$), and $R^n f_* \mathbb{Z}_\ell(n/2)/\mathbb{Z}_\ell \eta_\ell^{n/2}$ is torsion-free.

The $\mathbb{Z}_\ell$-sheaves $R^i f_* \mathbb{Z}_\ell$ are constructible and twisted constant, and their formation is compatible with arbitrary base change, by SGA 4, XVI, 2.2. It therefore suffices to prove 1.6 when S is the spectrum of an algebraically closed field. Repeated application of SGA 4, XIV, 3.3 then shows, for every m, that the canonical map

$$H^q_Y\big(\mathbb{P}(\underline{E}), \mathbb{Z}/\ell^m\big) \longrightarrow H^q(Y, \mathbb{Z}/\ell^m)$$

is bijective for $q > r + d$ and surjective for $q = r + d$. From this and Poincaré duality (SGA 4, XVIII), one deduces (SGA 5, VII) that the restriction map

$$H^q\big(\mathbb{P}(\underline{E}), \mathbb{Z}_\ell\big) \longrightarrow H^q(Y, \mathbb{Z}_\ell)$$

is bijective for $q < n$, and injective with torsion-free cokernel for $q = n$. This proves (i) for $i < n$, as well as (ii) and (iv). As in 1.5, one checks that η_ℓ^n is $\prod_{h=1}^d a_h$ times the canonical class; (i) and (iii) then follow by Poincaré duality (SGA 4, XVIII).

1.7. If Y is a complete intersection of dimension at least three in a projective space $\mathbb{P}^r(k)$ over a field k, then $\mathrm{Pic}(Y)$ is known to be isomorphic to $\mathbb{Z}$ and to be generated by $\underline{O}(1)$ (SGA 2, XII, 3.7). In dimension two, when Y is smooth, one can still prove the following.

Theorem 1.8. Let Y be a smooth complete-intersection surface in the projective space $\mathbb{P}^r(k)$ over an algebraically closed field k. Then $\underline{\mathrm{Pic}}^\tau(Y) = 0$, and the class of $\underline{O}(1)$ is not divisible in the Néron–Severi group $\mathrm{NS}(Y)$.

Since $H^1(Y, \underline{O}) = 0$ by 1.5(iii)$_a$, one knows that $\underline{\mathrm{Pic}}^0(Y) = 0$. The vanishing of $\underline{\mathrm{Pic}}^\tau(Y)$ then follows from the familiar lemma below, the fact that $H^2(Y, \mathbb{Z}_\ell(1))$ is torsion-free by 1.6(iv), and the fact that $H^0(Y, \Omega^1_Y) = 0$ by 1.5(iii)$_a$.

Lemma 1.9. Let Y be a complete algebraic variety over an algebraically closed field k.

(i) If ℓ is prime to the characteristic of k, then the ℓ-torsion of $\mathrm{NS}(Y)$ identifies with the ℓ-torsion of $H^2(Y, \mathbb{Z}_\ell(1))$.

(ii) If k has nonzero characteristic p and Y is smooth, then the kernel of multiplication by p in $\mathrm{Pic}(Y)$ identifies with the subgroup of $H^0(Y,\Omega^1_Y)$ consisting of locally logarithmic differentials.

The Kummer exact sequences

$$0 \longrightarrow \mu_{\ell^n} \longrightarrow \mathbb{G}_m \xrightarrow{x\mapsto x^{\ell^n}} \mathbb{G}_m \longrightarrow 0$$

give exact sequences

$$0 \longrightarrow \mathrm{Pic}(Y)/\ell^n\,\mathrm{Pic}(Y) \longrightarrow H^2(Y,\mu_{\ell^n}) \longrightarrow H^2(Y,\mathbb{G}_m)_{\ell^n} \longrightarrow 0,$$

where

$$\mathrm{Pic}(Y)/\ell^n\,\mathrm{Pic}(Y) = \mathrm{NS}(Y)\otimes\mathbb{Z}/\ell^n.$$

Passing to the inverse limit gives

$$0 \longrightarrow \mathrm{NS}(Y)\otimes\mathbb{Z}_\ell \longrightarrow H^2(Y,\mathbb{Z}_\ell(1)) \longrightarrow \mathrm{Hom}\big(\mathbb{Q}_\ell/\mathbb{Z}_\ell, H^2(Y,\mathbb{G}_m)\big) \longrightarrow 0.$$

The cokernel is torsion-free because $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is divisible. This proves (i).

To prove (ii), it is enough to observe that the exact sequence

$$0 \longrightarrow \underline{O}^*_Y \xrightarrow{x\mapsto x^p} \underline{O}^*_Y \xrightarrow{df/f} \Omega^1_Y$$

defines a short exact sequence

$$0 \longrightarrow \underline{O}^*_Y \longrightarrow \underline{O}^*_Y \longrightarrow \Omega^1_{Y,\text{loc log}} \longrightarrow 0,$$

and then to take the corresponding long exact cohomology sequence. This argument is due to Cartier, who proves moreover that, if C denotes the Cartier operator on $H^0(Y,\Omega^1_Y)$, then the locally logarithmic differentials form the kernel of $C-\mathrm{Id}$.

Suppose, for a contradiction, that the class of $\underline{O}(1)$ in $\mathrm{NS}(Y)$ has the form nu, with $n>1$. If ℓ divides n and is prime to the characteristic, then in ℓ-adic cohomology one again has

$$\eta_\ell = n\,c_1(u),$$

contradicting 1.6(iv).

Finally, if the characteristic p of k is nonzero and p divides n, then in Hodge cohomology one again has

$$\eta = n\,c_1(u) = 0,$$

contradicting $1.5(\mathrm{iii})_d$ and completing the proof.

2. Numerical results

2.1. Let $\underline{a}=(a_i)_{1\le i\le d}$ be a sequence of integers ≥ 1, and let $V_n(\underline{a})$ denote a smooth complete intersection of dimension n and multidegree $\underline{a}$ in a projective space $\mathbb{P}^{n+d}$ over a field k. Put

$$h^{p,q}(\underline{a}) = \dim_k H^q\big(V_{p+q}(\underline{a}),\Omega^p\big)$$

and

$$h^{p,q}_0(\underline{a}) = h^{p,q}(\underline{a}) - \delta_{p,q}.$$

Thus the $h^{p,q}_0(\underline{a})$ are the Hodge numbers of the middle-dimensional primitive cohomology of a complete intersection of multidegree $\underline{a}$. They depend only on $\underline{a}$, p, and q, by 1.5. Set

$$
(2.1.1) \qquad H(\underline{a}) = \sum_{p,q\geq 0} h_0^{p,q}(\underline{a}) y^p z^q \in \mathbb{Z}[[y,z]].
$$

For $d=1$ and $\underline{a}=(a)$, put $h^{p,q}(a) = h^{p,q}(\underline{a})$ and $H(a) = H(\underline{a})$. For $d=0$, put $\sup(a_i) = 1$. The following formula is the special case $k=0$ of formula (2), p. 160, in Hirzebruch [3]:

$$
(2.2) \qquad \sum_{n=0}^{\infty}\sum_{p} \chi\big(V_n(\underline{a}),\Omega^p\big) y^p z^{n+d} = \frac{1}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + y(1-z)^{a_i}}.
$$

We shall transform this into the following formula.
<u>Theorem</u> 2.3 (Hirzebruch). <u>With the notation of 2.1, one has</u>

$$
H(\underline{a}) = \frac{1}{(1+y)(1+z)} \left[\prod_{i=1}^{d} \frac{(1+y)^{a_i} - (1+z)^{a_i}}{-(1+y)^{a_i} z + (1+z)^{a_i} y} - 1 \right].
$$

By 1.5,

$$
\chi\big(V_n(\underline{a}),\Omega^p\big) = (-1)^p + (-1)^{n-p} h_0^{p,n-p}(\underline{a}).
$$

Consequently, the left-hand side I of (2.2) can be rewritten as

$$
\begin{aligned}
I &= \sum_{n=0}^{\infty} \sum_{p+q=n} \left((-1)^p + (-1)^q h_0^{p,q}(\underline{a})\right) y^p z^{p+q+d} \\
&= \sum_{p,q\geq 0} (-1)^p (yz)^p z^q z^d + \sum_{p,q\geq 0} (-1)^q h_0^{p,q}(\underline{a}) (yz)^p z^q z^d \\
&= z^d \left[\frac{1}{(1+yz)(1-z)} + H(\underline{a})(yz,-z) \right].
\end{aligned}
$$

The right-hand side II of (2.2) can be rewritten as

$$
II = \frac{z^d}{(1+yz)(1-z)} \prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + yz(1-z)^{a_i}}.
$$

After cancelling z^d in $I = II$, one obtains

$$
H(\underline{a})(yz,-z) = \frac{1}{(1+yz)(1-z)} \left[\prod_{i=1}^{d} \frac{(1+yz)^{a_i} - (1-z)^{a_i}}{(1+yz)^{a_i} z + (1-z)^{a_i} yz} - 1 \right],
$$

and 2.3 follows by a change of variables.
<u>Corollary</u> 2.4. <u>With the notation of 2.1:</u>

(i) <u>For $d=1$, one has</u>

$$
\begin{aligned}
H(a) &= -\frac{(1+y)^{a-1} - (1+z)^{a-1}}{(1+y)^a z - (1+z)^a y} \\
&= \frac{\displaystyle\sum_{i,j\geq 0} \binom{a-1}{i+j+1} y^i z^j}{1 - \displaystyle\sum_{i,j\geq 1} \binom{a}{i+j} y^i z^j}.
\end{aligned}
$$

(ii) Writing $|P|$ for the number of elements of a subset P of $[1,d]$, one has

$$H(\underline{a}) = \sum_{\substack{P\subset[1,d]\\ P\neq\varnothing}} (1+y)^{|P|-1}(1+z)^{|P|-1}\prod_{i\in P} H(a_i).$$

By 2.3,

$$\begin{aligned}
H(a) &= -\frac{1}{(1+y)(1+z)}\left[\frac{(1+y)^a-(1+z)^a}{(1+y)^a z-(1+z)^a y}+1\right]\\
&= -\frac{1}{(1+y)(1+z)}\left[\frac{(1+y)^a(1+z)-(1+z)^a(1+y)}{(1+y)^a z-(1+z)^a y}\right]\\
&= -\frac{(1+y)^{a-1}-(1+z)^{a-1}}{(1+y)^a z-(1+z)^a y}.
\end{aligned}$$

Let N and D be the numerator and denominator in this last expression, so that $H(a) = -N/D$.

One has

$$N = \sum_k \binom{a-1}{k}(y^k-z^k) = (y-z)\sum_{i,j\geq 0}\binom{a-1}{i+j+1}y^i z^j$$

and

$$\begin{aligned}
D = \sum_k \binom{a}{k}(y^k z - y z^k) &= (z-y) + yz\sum_{k\geq 1}\binom{a}{k}(y^{k-1}-z^{k-1})\\
&= (y-z)\left(-1+yz\sum_{i,j\geq 0}\binom{a}{i+j+2}y^i z^j\right)\\
&= (y-z)\left(-1+\sum_{i,j\geq 1}\binom{a}{i+j}y^i z^j\right).
\end{aligned}$$

This proves (i).

Now invert the formula in 2.3 when $d=1$. One finds

$$-\frac{(1+y)^a-(1+z)^a}{(1+y)^a z-(1+z)^a y} = 1+(1+y)(1+z)H(a).$$

Substituting this identity into 2.3 gives

$$\begin{aligned}
H(\underline{a}) &= \frac{1}{(1+y)(1+z)}\left[\prod_{i=1}^{d}(1+(1+y)(1+z)H(a_i))-1\right]\\
&= \frac{1}{(1+y)(1+z)}\left[\sum_{\substack{P\subset[1,d]\\ P\neq\varnothing}}(1+y)^{|P|}(1+z)^{|P|}\prod_{i\in P}H(a_i)\right],
\end{aligned}$$

which proves (ii). Theorem 2.5. With the notation of 2.1:

(i) If $p+q=p'+q'$ and $p\leq p'\leq q'\leq q$, then

$$h_0^{p,q}(\underline{a}) \leq h_0^{p',q'}(\underline{a}).$$

(ii) If $p \leq q$, then $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$p > \left\lfloor \frac{p+q+d-\sum_i a_i}{\sup_i(a_i)} \right\rfloor .$$

Here $\lfloor\ \rfloor$ denotes the integer part of the quotient.

Let (*) be the following property of a formal power series $P = \sum a_{p,q} y^p z^q$:

(*) the $a_{p,q}$ are nonnegative, $a_{p,q} = a_{q,p}$, and whenever $p+q = p'+q'$ and $p \leq p' \leq q' \leq q$, one has $a_{p,q} \leq a_{p',q'}$.

Assertion (i) follows at once from the formulas in 2.4, the identity $(1-x)^{-1} = \sum_{n\geq 0} x^n$, and the following lemma, whose proof is left to the reader.

Lemma 2.5.1. The set of formal power series satisfying (*) is stable under addition and multiplication.

We first prove (ii) in the special case of hypersurfaces ($d = 1$). For $p \leq q$, the desired formula can then be rewritten as

$$h_0^{p,q}(a) \neq 0 \quad \Longleftrightarrow \quad p \geq \left\lfloor \frac{p+q+1}{a} \right\rfloor \tag{2.5.2}$$

or, equivalently, $(p+1)a \geq p+q+2$. From the second formula in 2.4(i), one sees that $h_0^{p,q}(a)$ is nonzero if and only if there exist pairs (e,f) and (c_i,d_i) such that

$$\begin{cases} (e,f) \geq (0,0), & e+f \leq a-2, \\ (c_i,d_i) \geq (1,1), & c_i+d_i \leq a, \\ (p,q) = (e,f) + \sum\limits_i (c_i,d_i). & \end{cases} \tag{2.5.3}$$

For fixed $p+q$, the smallest possible p is obtained by taking $e = 0$ and every $c_i = 1$; it is the number of indices i. Thus it is the lower bound of those p for which

$$p \geq \frac{(p+q)-(a-2)}{a}, \qquad \text{i.e.} \qquad (p+1)a \geq p+q+2.$$

We now pass to the general case. If $\underline{a}'$ is obtained from $\underline{a}$ by deleting the a_i equal to one, then $H(\underline{a}) = H(\underline{a}')$. In view of the form of 2.5(ii), we may therefore assume that every $a_i > 1$. If $d = 0$, then $h_0^{p,q} = 0$ and the assertion is correct; we lose no generality by assuming $d > 0$ and $a_d = \sup_i(a_i)$.

By 2.4(ii) and (2.5.2), the condition $h_0^{p,q}(\underline{a}) \neq 0$ is equivalent to the existence of a nonempty subset P of $[1,d]$, a pair (e,f), and pairs $(c_i,d_i)_{i\in P}$ such that

$$\begin{cases} (|P|-1, |P|-1) \geq (e,f) \geq (0,0), & \\ c_i, d_i \geq \left\lfloor \dfrac{c_i+d_i+1}{a_i} \right\rfloor , & (c_i,d_i) \geq (0,0), \\ (p,q) = (e,f) + \sum\limits_{i\in P} (c_i,d_i). & \end{cases} \tag{2.5.4}$$

For fixed $n = p+q$, seek the smallest possible value of p. Put $n_i = c_i + d_i$. For such a p, conditions (2.5.4) become

$$
(2.5.5) \qquad \begin{cases} a_i = 2 \Longrightarrow n_i \text{ is even} & \text{(otherwise (2.5.4) is impossible)}, \\ c_i \geq \left\lfloor \dfrac{n_i+1}{a_i} \right\rfloor, & c_i \leq n_i, \\ (|P|-1, |P|-1) \geq (e,f) \geq (0,0), \\ n = e + f + \sum_{i \in P} n_i, \\ p = e + \sum_{i \in P} c_i. \end{cases}
$$

This description first shows that the smallest p is obtained for $P = [1, d]$, to which case we henceforth restrict. If

$$
(2.5.6) \qquad \sum_i (a_i - 2) + (d-1) \geq n,
$$

then a solution exists with $p = c_i = e = 0$. Otherwise, if $a_d \neq 2$, the best choice is $n_i = a_i - 2$ for $i \neq d$, so that $c_i = 0$ for $i \neq d$, $e = 0$, $f = d-1$, and

$$
n_d = n - (d-1) - \sum_{i \neq d} (a_i - 2).
$$

Then

$$
(2.5.7) \qquad p = \left\lfloor \frac{n - (d-1) - \sum_{i \neq d}(a_i - 2) + 1}{a_d} \right\rfloor,
$$

and hence, for $p \leq q$, one has $h_0^{p,q}(\underline{a}) \neq 0$ if and only if

$$
(2.5.8) \qquad p \geq \left\lfloor \frac{n - \sum_{i \neq d}(a_i - 1) + 1}{a_d} \right\rfloor.
$$

Formula (2.5.8) reduces to (2.5.2) for $d = 1$ and is trivially true when (2.5.6) holds. If $d \geq 2$, (2.5.6) fails, and all the a_i equal two, the best choice is instead

$n_i = 0$ for $i \neq d$, $e = 0$, and $f = d-1$ or $d-2$, chosen congruent to n modulo two, with $n_d = n - f$. If $n \equiv d - 1 \pmod 2$, this proves (2.5.7) as above. Otherwise, observe that

$$
\left\lfloor \frac{n - (d-2) + 1}{2} \right\rfloor = \left\lfloor \frac{n - (d-1) + 1}{2} \right\rfloor,
$$

so (2.5.7), and hence (2.5.8), remains valid when $a_d = 2$.

It remains to verify that (2.5.8) is equivalent to 2.5(ii). Indeed,

$$
\left\lfloor \frac{n - \sum_{i \neq d}(a_i - 1) + 1}{a_d} \right\rfloor = \left\lfloor \frac{n + d - \sum_{i \neq d} a_i}{a_d} \right\rfloor = \left\lfloor \frac{p + q + d - \sum_i a_i}{a_d} \right\rfloor + 1.
$$

2.6. If X is a complete intersection of dimension n in $\mathbb{P}^r$, it follows at once from (2.5.8) that the primitive part of $H^n(X)$ vanishes only when X is an odd-dimensional quadric.

Definition 2.7. If X is a proper smooth algebraic variety over $\mathbb{C}$, the Hodge level of $H^n(X)$ is the supremum of the numbers $q - p$ for which $p + q = n$ and $h^{p,q}(X) \neq 0$.

The Hodge level ℓ is therefore congruent to n modulo two, and is at most

$$
\min\bigl(n, 2\dim(X) - n\bigr).
$$

It is nonnegative unless $H^n(X)=0$, in which case it is $-\infty$. [1]

Corollary 2.8. Retain the notation of 2.1, put $k=\mathbb{C}$, and suppose that $a_d=\sup_i(a_i)$. Let ℓ be an integer with $0\le\ell<n$, congruent to n modulo two. The Hodge level of $H^n(V_n(\underline{a}))$ is at most ℓ if and only if

$$\sum_{i\neq d}(a_i-1)\le(\ell+1)-(a_d-2)\frac{n-\ell}{2}.$$

By (2.5.8), if the Hodge level is nonnegative, it is the maximum of zero and

$$n-2\left\lfloor\frac{n-\sum_{i\neq d}(a_i-1)+1}{a_d}\right\rfloor.$$

The inequality

$$n-2\left\lfloor\frac{n-\sum_{i\neq d}(a_i-1)+1}{a_d}\right\rfloor\le\ell$$

may be rewritten as

$$\frac{n-\ell}{2}\le\left\lfloor\frac{n-\sum_{i\neq d}(a_i-1)+1}{a_d}\right\rfloor,$$

or equivalently as

$$\sum_{i\neq d}(a_i-1)\le 1+n-a_d\frac{n-\ell}{2}=\ell+1-(a_d-2)\frac{n-\ell}{2}.$$

2.9. Using formula 2.8, it is easy, for small ℓ, to enumerate the multidegrees of complete intersections in $\mathbb{P}^r$ of dimension $n>1$ whose H^n has level ℓ. The table given in the first version of this exposé was, however, incorrect; the table below is extracted from the more complete one given by M. Rapoport in his article "Complément à l'article de P. Deligne *La conjecture de Weil pour les surfaces K3*", *Inventiones Math.* 15, 227–236 (1972). For $\ell\le 1$, the complete intersections $V_n(\underline{a})$ of dimension n and multidegree $\underline{a}=(a_1,\dots,a_d)$, with $n>\ell$, $d\ge 1$, and $a_i\ge 2$, having Hodge level ℓ, are the following:

$$\begin{array}{ll}
\ell=-\infty & V_{2k+1}(2)\qquad (k\ge 0),\\
\ell=0 & V_{2k}(2)\ (k\ge 1),\quad V_{2k}(2,2)\ (k\ge 1),\quad V_2(3),\\
\ell=1 & V_{2k+1}(2,2)\ (k\ge 1),\quad V_{2k+1}(2,2,2)\ (k\ge 1),\\
 & V_3(3),\quad V_5(3),\quad V_3(3,2),\quad V_3(4).
\end{array}$$

2.10. For small $\underline{a}$, the generating functions of the $h_0^{p,q}(\underline{a})$ are, by 2.4,

$$\begin{aligned}
H(2) &= \frac{1}{1-yz},\\
H(3) &= \frac{2+y+z}{1-3yz-y^2z-yz^2},\\
H(2,2) &= \frac{y+z}{(1-yz)^2}+\frac{2}{(1-yz)^2}+\frac{1}{1-yz},
\end{aligned}$$

so that

[1] The printed source reads $\min(n,\dim(X)-n)$. The factor 2 is required by the Hodge ranges $0\le p,q\le\dim(X)$; the English text corrects this evident source typo.

$$
\begin{aligned}
h_0^{p,p}(2) &= 1 \qquad , \\
h_0^{1,1}(3) &= 6 \qquad , \\
h_0^{p,p}(2,2) &= 2p+3 \quad .
\end{aligned}
$$

BIBLIOGRAPHY

[1] Y. Akizuki and S. Nakano, "Note on Kodaira–Spencer's proof of Lefschetz theorems", *Proc. Jap. Acad.* 30, 266–272 (1954).

[2] R. Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, 1966.

[3] F. Hirzebruch, *Topological Methods in Algebraic Geometry*, 3rd ed., Springer-Verlag, 1966.

[4] K. Kodaira, "On a differential geometric method in the theory of analytic stacks", *Proc. Nat. Acad. Sci. USA* 39, 1268–1273 (1953).

S G A 7

E X P O S É XII

QUADRICS

by P. DELIGNE

0.0. The present exposé collects several well-known results about quadrics which we shall need in order to prove the Picard–Lefschetz formula.

0.1. Let m be a section of an $\mathcal{O}_S$-module M on a scheme S. We shall say that m is nowhere zero on S if, for every $s \in S$, the image of m in the $k(s)$-vector space M_s obtained from M by the base change $\{s\} \longrightarrow S$ is nonzero. We use the same terminology when $S = \operatorname{Spec}(A)$ and m is an element of an A-module M.

1. Quadratic forms

1.1. Let $S = \operatorname{Spec}(A)$ be an affine scheme, let V be a locally free A-module of rank n, and let Q be a quadratic form on V (Bourbaki, *Algèbre*, Ch. 9, §3, no. 4). Recall that Q is called nondegenerate if the associated bilinear form

$$\Phi(x, y) = Q(x + y) - Q(x) - Q(y)$$

is nondegenerate, that is, if it induces an isomorphism from V to its dual V^*. If 2 is not invertible in A, this can occur only when n is even (loc. cit.).

The quadratic forms on V form a projective A-module; a quadratic form Q is "nowhere zero" on S (0.1) if and only if the elements $Q(v)$ generate the unit ideal of A.

The notation $\mathbb{P}(V)$ is used in the sense of EGA; recall that if A is a field, then $\mathbb{P}(V)$ is the projective space whose set of rational points is the quotient $(V^* \setminus \{0\})/A^*$.

The quadratic form Q identifies with a section of the invertible sheaf $\mathcal{O}(2)$ on the projective space $\mathbb{P}(V^*)$ over S. When Q is nowhere zero, the ("homogeneous") equation $Q = 0$ defines a subscheme of $\mathbb{P}(V^*)$, flat and purely of relative dimension $n-2$ over S. This scheme is called the quadric defined by Q. The form Q is called ordinary if it is nowhere zero on S and the quadric that it defines is smooth over S.

The form Q is ordinary if and only if every form obtained from it by extending the scalars from A to a field is ordinary. If A is a field, then:

a) if n is even or $\operatorname{char}(A) \neq 2$, then Q is ordinary if and only if Q is nondegenerate;

b) if n is odd and $\operatorname{char}(A) = 2$, then Q is ordinary if and only if the kernel N of the associated (alternating) bilinear form Φ has dimension one and Q does not vanish on N.

Proposition 1.2. If Q is ordinary and $n = 2m$ (respectively, $n = 2m + 1$), then, locally for the étale topology on S, V has a basis $e_1, \ldots, e_n$ such that

$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m}$$

(respectively,

$$Q\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{m} x_i x_{i+m} + \lambda x_{2m+1}^2, \qquad (\lambda \text{ invertible}).$$

[2]

We prove 1.2 by induction on m. If $m = 0$, the assertion is clear. If $m > 0$, the quadric with equation $Q = 0$ is smooth over S and has nonempty geometric fibers; it therefore has sections locally for the étale topology. Locally, such a section is defined by an element $e \in V$, nowhere zero on $\mathrm{Spec}(A)$, with $Q(e) = 0$. At no point of S does e belong to the kernel of Φ; hence, locally, one can find f' such that $\Phi(e, f') = 1$. If

$$f = -Q(f')e + f',$$

then

$$Q(e) = Q(f) = 0, \qquad \Phi(e, f) = 1.$$

[3]

If V_2 is the orthogonal complement of the submodule V_1 generated by e and f, then $V = V_1 \oplus V_2$, and the conclusion follows by applying the induction hypothesis to V_2.

1.3. For the definition and principal properties of the Clifford algebra of Q, we refer to Bourbaki, *Algèbre*, Ch. 9, §9. We denote by ρ the canonical map from V to $C(Q)$.

Lemma 1.3.1. Under the hypotheses of 1.1, for every $\lambda \in A$ there exists a unique homomorphism

$$[\lambda]\colon C^+(\lambda Q) \longrightarrow C^+(Q)$$

such that, for $x, y \in V$,

$$[\lambda]\big(\rho(x)\rho(y)\big) = \lambda\rho(x)\rho(y).$$

This homomorphism is induced on passing to the quotient by the endomorphism $[\lambda]$ of the even part $T^+(V)$ of the tensor algebra of V, which acts as λ^r on $T^{2r}(V)$. Indeed,

$$[\lambda]\big(x \otimes x - \lambda Q(x)\big) = \lambda\big(x \otimes x - Q(x)\big).$$

Suppose that Q is nowhere zero on S. Lemma 1.3.1 then defines a transitive system of isomorphisms between the algebras $C^+(\lambda Q)$ for $\lambda \in A^*$.

Consequently, the algebra $C^+(Q)$ depends (up to unique isomorphism) only on V and on Q up to multiplication by a factor; in other words, $C^+(Q)$ depends only on V and on the subscheme of $\mathbb{P}(V^*)$ with equation $Q = 0$. Every homothety with invertible ratio is an automorphism of the pair $(V, X \subset \mathbb{P}(V^*))$; one checks that these automorphisms act trivially on $C^+(V, X)$. It follows that the algebra $C^+(V, X)$ depends only on the pair of schemes $X \subset \mathbb{P}(V^*)$ over A. We shall denote it by $C^+(X, \mathbb{P}(V^*))$. 1.4. Let V be a locally free A-module of rank $2m > 0$, and let Q be a nondegenerate quadratic form on V. Suppose given a decomposition $V = W_1 \oplus W_2$ into two totally singular subspaces. The form $Q(w_1 + w_2)$ then places W_1 and W_2 in perfect duality. Recall that, for $f \in W_2$ (identified with W_1^*), contraction by f is the endomorphism of degree -1 of $\bigwedge W_1$ defined by

$$f \lrcorner (e_1 \wedge \cdots \wedge e_p) = \sum_{i=1}^{p} (-1)^{i-1} \langle e_i, f\rangle e_1 \wedge \cdots \wedge \widehat{e_i} \wedge \cdots \wedge e_p.$$

[4]

[2] In both displayed sums the printed source has upper limit $m-1$. That leaves a two-dimensional radical and contradicts the statement; the standard ordinary normal form, and the induction that follows, require the upper limit m.

[3] The final term in the printed definition of f is f; it is the evident source typo f', as is confirmed by the displayed conclusion.

[4] The printed formula has $(-1)^i$ and begins the remaining wedge with an undefined e_0. The English restores the standard contraction formula $(-1)^{i-1} e_1 \wedge \cdots \widehat{e_i} \cdots \wedge e_p$, required by the Clifford action displayed immediately below. The reading was checked in a targeted high-detail enlargement of folio 65.

Endow the A-module $\bigwedge W_1$ with the $\mathbb{Z}/2$-grading for which

$$(\bigwedge W_1)^+ = \bigoplus_i \bigwedge^{2i} W_1, \qquad (\bigwedge W_1)^- = \bigoplus_i \bigwedge^{2i+1} W_1.$$

The algebra $\mathrm{End}(\bigwedge W_1)$ is then $\mathbb{Z}/2$-graded: its even (respectively, odd) endomorphisms are those that preserve the grading (respectively, have degree one modulo two).

We refer to Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, for the proof of the existence of a unique isomorphism s of $\mathbb{Z}/2$-graded algebras from $C(Q)$ to $\mathrm{End}(\bigwedge W_1)$ such that

$$s\big(\rho((e,0))\big) = e \wedge (-), \qquad s\big(\rho((0,f))\big) = f \lrcorner (-).$$

Temporarily denote by $e(W_1, W_2)$ the element of $C^+(Q)$ for which $s(e(W_1, W_2))$ is the identity on $(\bigwedge W_1)^+$ and zero on $(\bigwedge W_1)^-$. The center $Z(Q)$ of $C^+(Q)$ is isomorphic to $A \oplus A$, this decomposition being defined by the idempotents $e(W_1, W_2)$ and $1 - e(W_2, W_1)$. Moreover, $C^+(Q)$ is a matrix algebra over $Z(Q)$.

From this and 1.2 one deduces, as in loc. cit.:

Proposition 1.5. Under the hypotheses of 1.1, if $n = 2m > 0$ and Q is nondegenerate, then the center $Z(Q)$ of $C^+(Q)$ is an étale algebra, locally free of rank two over A, and $C^+(Q)$ is an Azumaya algebra over $Z(Q)$.

1.6. Let V and Q be as in 1.4, and let W be a totally singular direct summand of V, locally free of rank m. Let W_1' be a complement of W, so that $V = W \oplus W_1'$. Every complement W_2' of W identifies with the graph of a map $u : W_1' \to W$. The form Φ places W and W_1' in duality, so u is uniquely determined by the bilinear form on W_1'

$$B(x, y) = \Phi(x, u(y)),$$

and W_2' is totally singular if and only if

$$B(x, x) = -Q(x)$$

on W_1'. Since such forms B always exist, W has totally singular complements. Having chosen one of them, say W', the set of totally singular complements W_0' of W identifies with the set of alternating forms on W', or equivalently with the set of sections of the affine space $\mathbb{E}(\bigwedge^2 W')$ over S.

For each section x of $\mathbb{E}(\bigwedge^2 W')$, let $W_0'(x)$ be the corresponding complement of W and let $e(x) = e(W, W_0'(x))$. The function $x \mapsto e(x)$ is compatible with every extension of scalars and hence defines a morphism of S-schemes

$$e : \mathbb{E}(\bigwedge^2 W') \longrightarrow \mathrm{Spec}(Z(Q)).$$

The S-scheme $\mathbb{E}(\bigwedge^2 W')$ is flat with geometrically connected fibers, whereas $\mathrm{Spec}(Z(Q))$ is étale over S. The morphism e therefore factors through S, hence through a section of $\mathrm{Spec}(Z(Q))$ over S; equivalently, $e(x)$ is independent of x. [5] This justifies the following notation.

Notation 1.7. For W as above and any totally singular complement W' of W, denote by $e(W)$ the idempotent $e(W, W') \in Z(Q)$.

1.8. We saw in 1.3 that, for invertible λ, $C^+(Q)$ identifies with $C^+(\lambda Q)$. A continuity argument analogous to the preceding one shows that this identification carries

$$e(W) \in Z(Q) \qquad \text{to} \qquad e(W) \in Z(\lambda Q).$$

[5] The printed sentence says that e factors through a section of the affine space $\mathbb{E}(\bigwedge^2 W')$. The connected-to-étale argument instead gives the factorization through S and a section of $\mathrm{Spec}(Z(Q))$, exactly as the following conclusion requires.

[6]

1.9. Recall that if B and C are two $\mathbb{Z}/2$-graded algebras, then multiplication in their graded tensor product $B \otimes' C$ is defined by the Koszul rule

$$(b_1 \otimes c_1)(b_2 \otimes c_2) = (-1)^{\deg(c_1)\deg(b_2)} b_1 b_2 \otimes c_1 c_2$$

when the b_i and c_i are homogeneous.

If (V_1, Q_1) and (V_2, Q_2) are two quadratic modules, then

$$C(Q_1 \oplus Q_2) = C(Q_1) \otimes' C(Q_2)$$

(Bourbaki, loc. cit.).

If P and Q are two finite-type, locally free, $\mathbb{Z}/2$-graded A-modules, we identify the $\mathbb{Z}/2$-graded algebras $\operatorname{End}(P) \otimes' \operatorname{End}(Q)$ and $\operatorname{End}(P \otimes Q)$ by setting, for homogeneous u, v, p, q,

$$(u \otimes v)(p \otimes q) = (-1)^{\deg(v)\deg(p)} u(p) \otimes v(q).$$

Lemma 1.10. Let (V, Q) be the quadratic module which is the sum of two quadratic modules (V_1, Q_1) and (V_2, Q_2), where V_i is locally free of rank $2m_i$ and Q_i is nondegenerate. Let $V_i = W_i \oplus W_i'$ be a decomposition of V_i into two totally singular subspaces $(i = 1, 2)$. Then the diagram

$$\begin{tikzcd}
C(Q) \arrow[rr, "\sim"] \arrow[d, "\wr"] & & C(Q_1) \otimes' C(Q_2) \arrow[d, equal] \\
\operatorname{End}(\bigwedge(W_1 \oplus W_2)) \arrow[r, phantom, "\simeq"] & \operatorname{End}(\bigwedge W_1 \otimes \bigwedge W_2) \arrow[r, phantom, "\simeq"] & \operatorname{End}(\bigwedge W_1) \otimes' \operatorname{End}(\bigwedge W_2)
\end{tikzcd}$$

is commutative.

The verification is left to the reader. One deduces at once from this lemma that

$$e(W_1 \oplus W_2) = e(W_1)e(W_2) + (1 - e(W_1))(1 - e(W_2)). \tag{1.10.1}$$

1.11. The following construction is borrowed from Micali and Villamayor. If $p : Z \longrightarrow S$ is a double étale covering of S, then Z may be regarded as a $\mathbb{Z}/2$-torsor over S. Denote the addition of torsors by $*$; for a section s of Z, let $e(s)$ be the idempotent of $p_*\mathcal{O}_Z$ that equals one on s and zero elsewhere. With the notation of 1.10, there is a unique isomorphism

$$\operatorname{Spec}(Z(Q_1)) * \operatorname{Spec}(Z(Q_2)) \longrightarrow \operatorname{Spec}(Z(Q)), \tag{1.11.1}$$

whose underlying addition morphism is

$$+ : \operatorname{Spec}(Z(Q_1)) \times_S \operatorname{Spec}(Z(Q_2)) \longrightarrow \operatorname{Spec}(Z(Q)),$$

and which, after every base change, satisfies

$$e(s + t) = e(s)e(t) + (1 - e(s))(1 - e(t)).$$

If $s(W)$ is the section of $\operatorname{Spec}(Z(Q))$ such that $e(s(W)) = e(W)$, then formula (1.10.1) becomes

$$s(W_1 \oplus W_2) = s(W_1) + s(W_2).$$

[6] The printed cross-reference is 1.6; the scalar-change identification is the construction of 1.3.1.

Proposition 1.12. Let A be a field, let V be a vector space of dimension $2m > 0$ over A, let Q be a nondegenerate quadratic form on V, and let W_1, W_2 be two totally isotropic subspaces of V of dimension m. Then $e(W_1) = e(W_2)$ if and only if $\dim(W_1/(W_1 \cap W_2))$ is even.[7]

(Cf. Bourbaki, *Algèbre*, Ch. 9, §6, Exercise 28.) Let $e_1, \ldots, e_m, f_1, \ldots, f_m$ be a basis of V such that

a) $Q(\sum_i x_i e_i + \sum_i y_i f_i) = \sum_i x_i y_i$;

b) W_1 is generated by the e_i;

c) W_2 is generated by $(e_i)_{1 \leq i \leq s}$ and $(f_i)_{s < i \leq m}$.

Then V is the sum of the $V_i = Ae_i + Af_i$, and

$$W_\alpha = \bigoplus_i (W_\alpha \cap V_i), \qquad \alpha = 1, 2.$$

The addition formula 1.11 therefore reduces the proof to the case $\dim V = 2$. If $W_1 = W_2$, the assertion is clear. Otherwise, if $W_1 = Ae$ and $W_2 = Af$ with $\Phi(e, f) = 1$, then $e(W_1) = fe$ and $e(W_2) = ef = 1 - fe$.

2. Quadrics

2.1. Let k be an algebraically closed field and V a vector space of dimension $n+2$ over k. A smooth quadric of dimension n over k is a k-scheme X isomorphic to the subscheme of $\mathbb{P}(V^*)$ defined by the equation $Q = 0$, for some ordinary quadratic form Q over k.

Remark 2.2. For $n = 0, 1, 2$, X is a smooth quadric of dimension n over k if and only if

$$\begin{aligned} n = 0 &\ : \ X \simeq \operatorname{Spec}(k) \amalg \operatorname{Spec}(k), \\ n = 1 &\ : \ X \simeq \mathbb{P}^1_k, \\ n = 2 &\ : \ X \simeq \mathbb{P}^1_k \times \mathbb{P}^1_k. \end{aligned}$$

Lemma 2.3. With the notation of 2.1, suppose that X is defined by $Q = 0$.

(i) $\Omega^n_X \simeq \mathcal{O}_X(-n)$.

(ii)

$$\operatorname{Pic}(X) \simeq \begin{cases} 0, & n = 0, \\ \mathbb{Z}, & n = 1;\ \mathcal{O}_X(1) \text{ is the square of a generator}, \\ \mathbb{Z} \oplus \mathbb{Z}, & n = 2, \\ \mathbb{Z}, & n \geq 3;\ \mathcal{O}_X(1) \text{ is a generator}. \end{cases}$$

(iii) $H^i(X, \mathcal{O}_X) = 0$ for $i > 0$, and $H^1(X, \mathcal{O}_X(1)) = 0$.

(iv) The natural map

$$H^0\big(\mathbb{P}(V^*), \mathcal{O}_{\mathbb{P}(V^*)}(1)\big) = V^* \xrightarrow{\sim} H^0(X, \mathcal{O}_X(1))$$

is an isomorphism; and, for $n > 0$, $H^0(X, \mathcal{O}_X) = k$.

Assertion (i) follows from the fact that the conormal sheaf of X in $\mathbb{P}(V^*)$ is $\mathcal{O}_X(-2)$ and that the canonical sheaf of $\mathbb{P}^r$ is $\mathcal{O}(-r-1)$. Assertion (ii) follows from 2.2 when $n \leq 2$, and from SGA 2, XII, 3.7 when $n \geq 3$. Finally, (iii) and (iv) follow from FAC, no. 78.

Definition 2.4. A smooth quadric of dimension n over a scheme S is an S-scheme $f : X \longrightarrow S$, proper and smooth over S, whose geometric fibres are smooth quadrics.

[7] The printed proposition says that V has dimension $2m$ "over V"; the base field introduced in the same sentence is A.

For $n = 0$, a smooth quadric of dimension zero over S is simply a double étale covering of S. For $n = 1$, it is simply a Severi–Brauer variety of dimension one over S.

2.5. Let $\rho : X \longrightarrow S$ be a smooth quadric of dimension n over S. By 2.3(i), the inverse of the invertible sheaf $\Omega^n_{X/S}$ is ample.

By 2.3(ii), the Picard scheme $\mathrm{Pic}_{X/S}$ is smooth over S (because $H^2(X, \mathcal{O}_X) = 0$), unramified over S (because $H^1(X, \mathcal{O}_X) = 0$), and essentially proper over S (because X/S is smooth). Thus, locally on S for the étale topology, it is a constant discrete group having the structure described in 2.3(ii).[8]

In particular, if X has a section, there is an invertible sheaf L on X such that $L^{\otimes(-n)}$ and $\Omega^n_{X/S}$ have the same image in $\mathrm{Pic}_{X/S}$; any two such sheaves are locally isomorphic on S. By 2.3(iv) and [1], 0.5, p. 19, $\rho_* L$ is locally free and its formation commutes with every base change. Hence $\rho_* L$ is locally free of rank $n+2$, the map $\rho^*\rho_* L \longrightarrow L$ is surjective, and the canonical map

$$X \longrightarrow \mathbb{P}(\rho_* L)$$

is a closed immersion. It identifies X with a relative divisor of degree two in $\mathbb{P}(\rho_* L)$; locally on S, X is the quadric in $\mathbb{P}(\rho_* L)$ defined by an ordinary quadratic form on $(\rho_* L)^*$, unique up to an invertible factor.
[9]

2.6. Suppose that $n > 0$. If L_1 and L_2 are two invertible sheaves of the type considered above, then locally on S there is an isomorphism $u : L_1 \xrightarrow{\sim} L_2$. It is unique up to an invertible factor, since $\rho_* \mathcal{O}_X = \mathcal{O}_S$ by 2.3(iv). Consequently all the isomorphisms $u : L_1 \xrightarrow{\sim} L_2$ induce the same isomorphism

$$\mathbb{P}(u) : \mathbb{P}(\rho_* L_1) \xrightarrow{\sim} \mathbb{P}(\rho_* L_2).$$

Thus $\mathbb{P}(\rho_* L_1)$ is canonically isomorphic to $\mathbb{P}(\rho_* L_2)$ and depends only on X/S, not on the choice of L. We denote it by $P(X)$.

The projective space $P(X)$ is defined locally on S for the étale topology. By descent it defines a Severi–Brauer variety over S, again denoted $P(X)$, in which X is a relative divisor of degree two.

Locally on S, the pair $X \hookrightarrow P(X)$ defines a Clifford algebra $C^+(X, P(X))$ (1.3). Descent yields an algebra over S, which we denote simply by $C^+(X)$.

2.7. Suppose that $n = 2m > 0$. The centre Z of $C^+(X)$ is then an $\mathcal{O}_S$-algebra defining a double étale covering $Z(X)$ of S.

A generator of X is a subscheme D of X which is a linear subspace of dimension m in $P(X)$.

Locally on S, when $P(X) \simeq \mathbb{P}(V^*)$ and X is defined by $Q = 0$, a generator D of X is, by definition, identified with a locally direct-summand submodule $W(D)$ of V, totally singular and of rank $m+1$. By 1.8, the idempotent $e(W(D))$ in the centre of $C^+(X)$ depends only on $D \subset X$. Let $e(D)$ denote the corresponding section of $Z(X)$, namely the one on which $e(W(D))$ has value one.

The generators of X/S are the S-points of a scheme $\mathrm{Gen}(X)$ projective over S, and e defines a morphism of S-schemes

$$e : \mathrm{Gen}(X) \longrightarrow Z(X). \tag{2.7.1}$$

When $n = 0$, we set $Z(X) = X$, call a section of X a generator of X, and define e to be the identity.

Proposition 2.8. Under the hypotheses of 2.7, $p : \mathrm{Gen}(X) \longrightarrow S$ is projective and smooth, and (2.7.1) is its Stein factorization.

[8] At both occurrences in this sentence the print cites 2.3(iii); the Picard-group computation is 2.3(ii).

[9] After introducing the projection as ρ, the print changes its symbol to p in the direct-image formulas and cites 2.3(iii) where the required base-change statement is 2.3(iv). The notation and cross-reference have been regularized here.

The question is local for the étale topology on S. Locally, if D is a generator, then by 1.6 the generators D' disjoint from D are identified with the S-points of a suitable affine space over S. In this way $\mathrm{Gen}(X)$ is covered by open subschemes smooth over S.

To prove that the geometric fibres of e are geometrically connected, it is enough to consider the case in which S is the spectrum of an algebraically closed field k and X is defined by $Q = 0$ in $\mathbb{P}(V^*)$.

Let W_1 and W_2 be two totally isotropic subspaces of rank $m+1$ in V such that $e(W_1) = e(W_2)$, that is, such that $W_1/(W_1 \cap W_2)$ has even dimension (1.12). Proceeding as in 1.12, one obtains an orthogonal decomposition

$$V = V_0 \oplus \bigoplus_i V_i$$

such that

a) $W_\alpha = (W_\alpha \cap V_0) \oplus \bigoplus_i (W_\alpha \cap V_i)$ for $\alpha = 1, 2$;

b) $W_1 \cap V_0 = W_2 \cap V_0$;

c) $\dim V_i = 4$ and $V_i = (W_1 \cap V_i) \oplus (W_2 \cap V_i)$.

It remains to prove that $W_1 \cap V_i$ and $W_2 \cap V_i$ belong to the same connected family of maximal totally singular subspaces of V_i. In other words, the assertion reduces to the case of quadrics of dimension two. Such a quadric is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, and its generators are the subschemes $\{t\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{t\}$; the proposition is then evident.

3. Cohomology of quadrics

3.1. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S. We write

$$\eta \in H^0\big(S, R^2 p_* \mathbb{Z}_\ell(1)\big)$$

for the "cohomology class of a hyperplane section." Locally on S for the étale topology, $P(X) \simeq \mathbb{P}(V^*)$; by definition, η is then the image in $H^0(S, R^2 p_* \mathbb{Z}_\ell(1))$ of the first Chern class

$$c_1(\mathcal{O}(1)) \in H^2(X, \mathbb{Z}_\ell(1)).$$

[10]

3.2. Suppose that $n = 2m$. A generator D of X then defines a cohomology class $c\ell'(D) \in H^n(X, \mathbb{Z}_\ell(m))$, whose image in $H^0(S, R^n p_* \mathbb{Z}_\ell(m))$ is denoted $c\ell(D)$. By 2.8, the latter depends only on the section $e(D)$ of $Z(X)$. We therefore obtain a morphism of étale sheaves on S

$$c\ell : Z(X) \longrightarrow R^n p_* \mathbb{Z}_\ell(m). \tag{3.2.1}$$

Theorem 3.3. Let $p : X \longrightarrow S$ be a smooth quadric of dimension n over S, and let ℓ be a prime number invertible on S.

(i) $R^{2i+1} p_* \mathbb{Z}_\ell = 0$.

(ii) If $0 \le 2i < n$ (respectively, if $n < 2i \le 2n$), then $R^{2i} p_* \mathbb{Z}_\ell(i)$ is canonically isomorphic to the constant sheaf $\mathbb{Z}_\ell$, and is generated by η^i (respectively, by $\eta^i/2$).

(iii) If $n = 2m$, the map deduced from (3.2.1),

$$\mathbb{Z}_\ell^{Z(X)} \longrightarrow R^n p_* \mathbb{Z}_\ell(m), \tag{3.3.1}$$

[10] The displayed definition in the print has $R^1 p_*$, whereas the following sentence and the degree of $c_1(\mathcal{O}(1))$ both give $R^2 p_*$.

is an isomorphism. If α and β are two disjoint sections of $Z(X)$ (assumed isomorphic to $S \amalg S$, as is locally the case), then

$$
\begin{array}{ll}
\text{(a)} & \eta^m = c\ell(\alpha) + c\ell(\beta), \\[1ex]
\text{(b)} & \begin{cases} \mathrm{Tr}(c\ell(\alpha)^2) = \mathrm{Tr}(c\ell(\beta)^2) = 1, & c\ell(\alpha)c\ell(\beta) = 0, \quad m \text{ even}, \\ c\ell(\alpha)^2 = c\ell(\beta)^2 = 0, & \mathrm{Tr}(c\ell(\alpha)c\ell(\beta)) = 1, \quad m \text{ odd}, \end{cases} \\[1ex]
\text{(c)} & \eta\big(c\ell(\alpha) - c\ell(\beta)\big) = 0.
\end{array}
$$

[11]

Assertions (i) and (ii) are recalled only for reference (XI, 1.6, supplemented by XI, 2.6 when $2i+1 = n$ in (i)).

It is enough to verify (iii) when S is the spectrum of an algebraically closed field k (one could even take $S = \mathrm{Spec}(\mathbb{C})$).

Formula (a). We may suppose that X is the quadric in $\mathbb{P}^{2m+1}(k)$ with equation

$$\sum_{i=0}^{m} x_i x_{i+m+1} = 0.$$

The class η^m is the cohomology class of the plane section defined by

$$x_i = 0 \qquad (0 \le i < m),$$

and this cycle is the sum of the generators D_1, D_2 defined by

$$
\begin{array}{llll}
D_1: & x_i = 0 & (0 \le i \le m), & \\
D_2: & x_i = 0 & (0 \le i < m), & x_{2m+1} = 0.
\end{array}
$$

By 1.12, $e(D_1) \neq e(D_2)$, and hence

$$\eta^m = c\ell(D_1) + c\ell(D_2) = c\ell(\alpha) + c\ell(\beta).$$

[12]

Formula (b). If generators D_1, D_2 are disjoint (respectively, meet transversally in one point), then $c\ell(D_1)c\ell(D_2) = 0$ (respectively, $\mathrm{Tr}(c\ell(D_1)c\ell(D_2)) = 1$). By 1.12, when m is even we have $e(D_1) \neq e(D_2)$ (respectively, $e(D_1) = e(D_2)$); when m is odd we have $e(D_1) = e(D_2)$ (respectively, $e(D_1) \neq e(D_2)$). Formula (b) follows.

Formula (c). This is clear when $n = 0$. Otherwise, by (ii), it is enough to prove

$$\eta^m\big(c\ell(\alpha) - c\ell(\beta)\big) = 0.$$

Indeed,

$$\mathrm{Tr}(\eta^m(c\ell(\alpha) - c\ell(\beta))) = \mathrm{Tr}\big(c\ell(\alpha)^2 - c\ell(\beta)^2\big) = 0.$$

Verification. When m is even,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\big) = 1 + 1;$$

when m is odd,

$$2 = \mathrm{Tr}(\eta^{2m}) = \mathrm{Tr}\big(c\ell(\alpha)^2 + 2c\ell(\alpha)c\ell(\beta) + c\ell(\beta)^2\big) = 2.$$

[11] The print repeats the tag (3.2.1) on the map in part (iii), but the proof itself refers to this map as (3.3.1).

[12] The printed ambient projective space is $\mathbb{P}^{m+1}$, although the displayed nondegenerate equation has $2m+2$ coordinates. The required ambient space is $\mathbb{P}^{2m+1}$.

[13]

That (3.3.1) is an isomorphism now follows from the fact that the two bilinear forms in two variables

$$B(x,y) = x_1y_2 + x_2y_1, \qquad B(x,y) = x_1y_1 + x_2y_2$$

have discriminant ± 1.

Verification 3.4. Let X be a smooth quadric of dimension n over a finite field $\mathbb{F}_q$. In view of 3.3, the Lefschetz trace formula gives the classical formula

$$\#X(\mathbb{F}_q) = \begin{cases} \displaystyle\sum_{i=0}^{n} q^i, & n \text{ odd}, \\ \displaystyle\sum_{i=0}^{n} q^i + \varepsilon q^m, & n = 2m, \quad \varepsilon = \pm 1. \end{cases}$$

When n is even, $\varepsilon = 1$ if X is split, that is, if it has a rational generator, and $\varepsilon = -1$ otherwise.

3.5. Let $f : X \longrightarrow S$ be a smooth quadric of dimension $n = 2m$. The primitive part of $R^n f_*\mathbb{Z}_\ell(m)$ is the orthogonal complement of η^m; with the notation of 3.3, it is generated by $c\ell(\alpha) - c\ell(\beta)$. The primitive quotient of $R^n f_*\mathbb{Z}_\ell(m)$ is the quotient by $\mathbb{Z}_\ell\eta^m$. The primitive part and primitive quotient of $R^n f_*\mathbb{Z}_\ell$ are obtained by tensoring with $\mathbb{Z}_\ell(-m)$.

3.6. Let V be a locally free vector bundle of rank $n+2$ on S, and let X be the smooth quadric over S defined by an ordinary quadratic form Q on V. Let H moreover be a hyperplane in $\mathbb{P}(V^*)$ meeting X transversally, and put $X^o = X - H$. The schemes X and $Y = X \cap H$ are smooth quadrics over S, of dimensions n and $n-1$. We may therefore use 3.3 to compute the cohomology of $X^o = X - Y$.

(3.6.1)
$$\begin{tikzcd} X^o \arrow[r, hook] \arrow[dr, "f"'] & X \arrow[d, "p"] & Y \arrow[l, hook'] \arrow[dl, "q"] \\ & S & \end{tikzcd}.$$

If $n = 0$, then $Y = \varnothing$ and $X = X^o$. Suppose that $n > 0$. We have two long exact sequences dual to one another:

(3.6.2)
$$\cdots \xrightarrow{\partial} R^i f_! \mathbb{Z}_\ell \longrightarrow R^i p_*\mathbb{Z}_\ell \xrightarrow{r_i} R^i q_*\mathbb{Z}_\ell \longrightarrow \cdots$$

and

(3.6.3)
$$\cdots \xrightarrow{\partial} R^{i-2} q_*\mathbb{Z}_\ell(-1) \longrightarrow R^i p_*\mathbb{Z}_\ell \longrightarrow R^i f_*\mathbb{Z}_\ell \longrightarrow \cdots .$$

For n even, the restriction homomorphism r_i is an isomorphism for $i \neq n, 2n$; for n odd, it is an isomorphism for $i \neq n-1$. If $n = 2m$ and α is the cohomology class of a generator, then

$$r_n(\alpha) = \tfrac{1}{2}\eta^m,$$

so r_n is surjective and its kernel is the primitive part of $R^n p_*\mathbb{Z}_\ell$. If $n = 2m+1$, then $r_{2m}(\eta^m) = \eta^m$, so r_{2m} is injective and its cokernel is the primitive quotient of $R^{2m} q_*\mathbb{Z}_\ell$. It follows that $R^i f_!\mathbb{Z}_\ell = 0$ for $i \neq n, 2n$, and that $R^i f_!\mathbb{Z}_\ell$ is a twisted constant free $\mathbb{Z}_\ell$-sheaf of rank one for $i = n, 2n$. By duality, or from (3.6.3), one likewise shows that $R^i f_*\mathbb{Z}_\ell = 0$ for $i \neq 0, n$, that $f_*\mathbb{Z}_\ell = \mathbb{Z}_\ell$, and that $R^n f_*\mathbb{Z}_\ell$ has rank one. [14]

Suppose that $n = 2m$ and retain the notation of 3.3.

[13] In these two top-degree intersection computations the print has η^2 in place of η^{2m}, and once prints $c\ell(\alpha^2)$ in place of $c\ell(\alpha)^2$.

[14] This paragraph prints $r = 2m+1$ instead of $n = 2m+1$ and cites 3.5.3 instead of the displayed sequence (3.6.3).

$$\begin{tikzcd}
0 \arrow[r] & R^n f_!\mathbb{Z}_\ell(m) \arrow[r] \arrow[d, "\varphi"] & R^n p_*\mathbb{Z}_\ell(m) \arrow[r] \arrow[d, equal] & R^n q_*\mathbb{Z}_\ell(m) \arrow[r] & 0 \\
0 & R^n f_*\mathbb{Z}_\ell(m) \arrow[l] & R^n p_*\mathbb{Z}_\ell(m) \arrow[l] & R^{n-2} q_*\mathbb{Z}_\ell(m-1) \arrow[l] & 0. \arrow[l]
\end{tikzcd}$$

The sheaf $R^n f_!\mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, whose images in $R^n p_*\mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$. By 3.3(iii)(b),

$$\mathrm{Tr}(\delta^2) = (c\ell(\alpha) - c\ell(\beta))^2 = (-1)^m \cdot 2,$$

and $\pm\varphi(\delta)$ is twice the natural dual generators $\pm\delta'$ of $R^n f_*\mathbb{Z}_\ell(m)$, namely the opposite images of $c\ell(\alpha)$ and $c\ell(\beta)$.

Suppose now that $n = 2m+1$ and apply the notation of 3.3 to Y.

$$\begin{tikzcd}
0 \arrow[r] & R^{2m} p_*\mathbb{Z}_\ell(m) \arrow[r] & R^{2m} q_*\mathbb{Z}_\ell(m) \arrow[r, "\partial"] & R^n f_!\mathbb{Z}_\ell(m) \arrow[r] \arrow[d, "\varphi"] & 0 \\
& R^{2m+2} p_*\mathbb{Z}_\ell(m+1) & R^{2m} q_*\mathbb{Z}_\ell(m) \arrow[l] & R^n f_*\mathbb{Z}_\ell(m+1) \arrow[l] & 0. \arrow[l]
\end{tikzcd}$$

[15]

The sheaf $R^n f_!\mathbb{Z}_\ell(m)$ then has two natural opposite generators $\pm\delta$, namely $\partial c\ell(\alpha)$ and $\partial c\ell(\beta)$. We have

$$\delta\,\varphi(\delta) = \delta^2 = (\partial c\ell(\alpha))^2 = \partial\big(c\ell(\alpha)\,\partial c\ell(\alpha)\big) = 0,$$

and therefore $\varphi(\delta) = 0$. On the other hand, $R^n f_*\mathbb{Z}_\ell(m+1)$ has two natural opposite generators $\pm\delta'$, whose images in $R^{2m} q_*\mathbb{Z}_\ell(m)$ are $\pm(c\ell(\alpha) - c\ell(\beta))$, and $\mathrm{Tr}(\delta\delta') = \pm 1$.

These results are assembled in the following table.

Table 3.7. Cohomology of affine quadrics of dimension n, $X^o = X - Y$.

A. The cohomology is torsion-free. The nonzero Betti numbers are

$$\begin{aligned}
\text{ordinary cohomology:}\quad & b_0 = b_n = 1,\\
\text{cohomology with compact supports:}\quad & b_{2n} = b_n = 1,
\end{aligned}$$

except that $b_0 = 2$ when $n = 0$.

B. If $n = 2m > 0$, then

$$\begin{aligned}
R^n f_!\mathbb{Z}_\ell(m) &= \text{the primitive part of } R^{2m} p_*\mathbb{Z}_\ell(m),\\
R^n f_*\mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} p_*\mathbb{Z}_\ell(m).
\end{aligned}$$

If $n = 2m+1$, then

$$\begin{aligned}
R^n f_!\mathbb{Z}_\ell(m) &= \text{the primitive quotient of } R^{2m} q_*\mathbb{Z}_\ell(m),\\
R^n f_*\mathbb{Z}_\ell(m+1) &= \text{the primitive part of } R^{2m} q_*\mathbb{Z}_\ell(m).
\end{aligned}$$

Locally, these groups have natural generators defined up to sign; δ denotes those for cohomology with compact supports, and δ' those for ordinary cohomology.

C. The map

$$\varphi : R^n f_!\mathbb{Z}_\ell \longrightarrow R^n f_*\mathbb{Z}_\ell$$

[15] The leftmost term in the printed lower row has twist (m); duality and the other three terms require the displayed twist $(m+1)$.

is zero when n is odd, and sends $\pm\delta$ to $\pm 2\delta'$ when $n > 0$ is even. Moreover,

$$\mathrm{Tr}(\delta\delta') = \pm 1, \qquad \mathrm{Tr}(\delta^2) = \begin{cases} 0, & n = 2m+1, \\ (-1)^m \cdot 2, & n = 2m. \end{cases}$$

Verification 3.8. The complex affine quadric S in $\mathbb{C}^n$ with equation $\sum_i z_i^2 = 1$ is diffeomorphic to the tangent bundle of the real sphere $S \cap \mathbb{R}^n$. This agrees with A and C.

BIBLIOGRAPHY

References

[1] D. Mumford, *Geometric Invariant Theory*, Ergebnisse **34**, Springer-Verlag, 1965.

SGA 7 — EXPOSÉ XIII

The Formalism of Vanishing Cycles

by P. Deligne

Source contents

Introduction

In this exposé we take up again the formalism introduced in no. 2 of Grothendieck's Exposé I. The only new point is the introduction of the algebraic analogue of variation in the classical theory.

The topological picture I have in mind is as follows. Let X be an analytic space, let $D \subset \mathbb{C}$ be a disc centred at 0, put $D^* = D - \{0\}$, and let $f : X \longrightarrow D$ be a proper morphism (the nonproper case will be considered later). Replace D by a sufficiently small disc centred at 0, replace X by the inverse image of this small disc, and continue to denote the result by $f : X \longrightarrow D$. According to the complex-analytic variant of Thom's isotopy theorem, $f^{-1}(D^*)$ is a topological fibre bundle over D^*. I do not know whether one can always construct a pair (Γ, r) of the kind described below, but it seems plausible that one can.

(a) Γ is a locally coherent system of local trivializations of the bundle $f^{-1}(D^*)$ over D^*. In other words, Γ defines a $\pi_1(D^*)$-equivariant trivialization of the inverse image of X on a universal covering $\widetilde{D}^*$ of D^*.

(b) r is a retraction of X onto the special fibre $X_0 = f^{-1}(0)$, compatible with Γ: if, on a connected open subset U of D^*, Γ identifies $X_U = f^{-1}(U)$ with $X_1 \times U$, then $r = r(x_1, u)$ is independent of u.

Let (X, f, Γ, r) be given. Let $t \in D$ and $X_t = f^{-1}(t)$ be the "general fibre." The bundle $\varphi^* X$, obtained from X/D by pullback along

$$\varphi : [0,1] \longrightarrow D^*, \qquad x \longmapsto \exp(2\pi i x)\, t,$$

is trivialized by Γ. The monodromy $T : X_t \longrightarrow X_t$ is the homeomorphism

$$X_t = (\varphi^* X)_0 \underset{\Gamma}{\xrightarrow{\sim}} (\varphi^* X)_1 = X_t.$$

Let $r_t : X_t \longrightarrow X_0$ be the restriction of r to X_t. Then $r_t T = r_t$. One reconstructs (X, f) from (X_t, X_0, T, r_t) as follows:

(a) identify the ends $X_t \times \{0\}$ and $X_t \times \{1\}$ of $X_t \times [0,1]$ using T, obtaining $f' : X' \longrightarrow S^1$;

(b) r_t defines $r' : X' \longrightarrow X_0$, and (X, f) is identified with the map of mapping cones

$$\operatorname{cone}(f') : \operatorname{cone}(r' : X' \longrightarrow X_0) \longrightarrow D = \operatorname{cone}(S^1 \longrightarrow \mathrm{pt}).$$

Put $\Psi = (r, f) : X \longrightarrow X_0 \times D$. The higher Leray direct images $R^i\Psi_*\mathbb{Z}$ satisfy the following condition:

(*) The space D^* is covered by open subsets U such that the restriction of the sheaf F to $X_0 \times U$ is pulled back from a sheaf on X_0.

Construction 0.1. The sheaves F on $X_0 \times D$ satisfying (*) are identified with systems (F_0, F_t, T, α) in which

a) F_0 is a sheaf on X_0;

b) F_t is a sheaf on X_0, endowed with an action of the generator T of $\pi_1(D^*, t)$;

c) α is a morphism from F_0 to the sheaf of T-invariants of F_t.

Let G be a sheaf on $X_0 \times D^*$ satisfying (*), and let $\widetilde{D}^*$ be the universal covering of D^*, with base point t. The pullback of G to $X_0 \times \widetilde{D}^*$ is, in one and only one way, pulled back from a sheaf G_t on X_0. The sheaf G_t is endowed with an action of $\pi_1(D^*, t)$, and G is obtained from $\mathrm{pr}_1^* G_t$ on $X_0 \times \widetilde{D}^*$ by taking the quotient by $\pi_1(D^*, t)$.

A sheaf F on $X_0 \times D$ is determined by

1) its restriction F_0 to X_0;

2) its restriction G to $X_0 \times D^*$;

3) a morphism $\alpha : F_0 \longrightarrow i^* j_* G$, where $i : X_0 \hookrightarrow X_0 \times D$ and $j : X_0 \times D^* \hookrightarrow X_0 \times D$.

If F satisfies (*), then G is described by (G_t, T) on X_0, and the asserted description of F follows.

For the sheaves $R^i\Psi_*\mathbb{Z}$ we have

$$(R^i\Psi_*\mathbb{Z})_0 = \begin{cases} \mathbb{Z}, & i = 0, \\ 0, & i > 0, \end{cases}$$

whereas, for $i > 0$, the $(R^i\Psi_*\mathbb{Z})_t$ deserve the name sheaves of vanishing cycles (cf. the discussion at the beginning of paragraph 2 of Exposé I).

This discussion is useful only after it is refined by passing to the derived category: the $R^i\Psi_*\mathbb{Z}$ are obtained from an object of $D^+(X_0 \times D)$, and indeed from an object $R\Psi_*\mathbb{Z}$ of the derived category of the category of abelian sheaves on $X_0 \times D$ satisfying (*). This object determines not only the sheaves of vanishing cycles and their monodromy, but also the classical "variation" morphisms (cf. 1.4 and XIV, 3.1).

We shall define its analogue in algebraic geometry. If S is a strictly henselian trait and $f : X \longrightarrow S$ is a morphism of schemes, we shall define an object $R\Psi(\mathbb{Z}/\ell^n)$ of the derived category of the category of abelian sheaves on the product topos $(X_0)_{\mathrm{et}} \times S_{\mathrm{et}}$. A sheaf on this topos may be interpreted as a system (F_0, F_t, T, α) in which

a) F_0 is a sheaf on X_0;

b) F_t is a sheaf on X_0, endowed with a continuous action T of the inertia group I;

c) α is a morphism from F_0 to the sheaf of I-invariants of F_t.

We shall have to work in a slightly more general framework than that described above:

a) The construction of $R\Psi(\mathbb{Z}/\ell^n)$ does not require f to be proper. For a topological interpretation, one would repeat the preceding construction after replacing X by a neighbourhood of the special fibre X_0.

b) To construct $R\Psi(\mathbb{Z}/\ell^n)$ and prove its properties, one must consider not only the constant sheaf $\mathbb{Z}/\ell^n$, but also general sheaves and even complexes of sheaves.

c) When the residue field k of the henselian trait S is not assumed separably closed, the product topos $(X_0)_{\text{et}} \times S_{\text{et}}$ must be replaced by the fibred product topos

$$(X_0)_{\text{et}} \times_{\text{Spec}(k)_{\text{et}}} S_{\text{et}}.$$

The reader who dislikes 2-fibred products of topoi may take the Galois description of this topos in 1.2.4 as its definition.

0.2. Notation and terminology.

(0.2.1) trait: the spectrum of a discrete valuation ring.

(0.2.2) henselian: for a scheme, the spectrum of a henselian local ring.

(0.2.3) strictly henselian: for a ring, henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.2.4) geometric point of S: a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ with image $x \in S$, where $k(\bar{x}) =_{\text{def}} \bar{k}$ is a separable closure of $\kappa(x)$. We say that $\bar{x}$ is localized at x. By abuse of language, we shall sometimes also call a morphism $\bar{x} : \text{Spec}(\bar{k}) \longrightarrow S$ a geometric point when $\bar{k}$ is a separably closed extension of $\kappa(x)$; such a morphism factors uniquely through a geometric point of S in the proper sense.

(0.2.5) $s, \eta, \bar{s}, \bar{\eta}, \bar{S}$: If $S = \text{Spec}(V)$ is a henselian trait, we shall normally denote by s the closed point of S, by η its generic point, by $\bar{\eta}$ a generic geometric point of S (localized at η), and by $\bar{s}$ the corresponding geometric point localized at s (I, 0.0.3). Let $\bar{S}$ denote the spectrum of the normalization of V in $k(\bar{\eta})$; its residue field is an inseparable extension of $k(\bar{s})$. Write $\text{Gal}(\bar{\eta}/\eta)$ for the Galois group of $k(\bar{\eta})$ over $k(\eta)$, and similarly for s. There is an exact sequence

$$0 \longrightarrow I \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Gal}(\bar{s}/s) \longrightarrow 0,$$

where I is the inertia group.

(0.2.6) sheaf: on a scheme X, always means a sheaf on the étale site X_{et} of X.

(0.2.7) prime to: a torsion sheaf F on a scheme S (or a torsion abelian group F) is prime to the residue characteristics of S if, for every $s \in S$, multiplication by the characteristic exponent of $k(s)$ is an automorphism of F.

1. Sheaves of sets

The sheaves of vanishing cycles will be defined as values of derived functors of the left exact functor Ψ defined in no. 3 for every sheaf of sets.

1.1. Galois sheaves.

1.1.1. Let Y be a scheme over a field k with separable closure $\bar{k}$, and put $\bar{Y} = Y \otimes_k \bar{k}$. Let G be a profinite group and let

$$u : G \longrightarrow \text{Gal}(\bar{k}/k)$$

be a continuous homomorphism from G to the Galois group of $\bar{k}$ over k. The Galois group $\text{Gal}(\bar{k}/k)$ acts on the left on $\bar{Y}$ by transport of structure. The group G acts on $\bar{Y}$ through u.

Let $\mathcal{F}$ be a sheaf of sets on (the étale site of) $\bar{Y}$. An action of G on $\mathcal{F}$ compatible with the action of G on $\bar{Y}$ is an action of G by automorphisms of the pair $(\bar{Y}, \mathcal{F})$ which induces the given action on $\bar{Y}$. In other words, it is a system of isomorphisms

$$\sigma(g) : u(g)_* \mathcal{F} \longrightarrow \mathcal{F}$$

satisfying $\sigma(gh) = \sigma(g)\,\sigma(h)$. If G acts on $\mathcal{F}$, then for every U étale over Y, with $\bar{U} = U \otimes_k \bar{k}$ étale over $\bar{Y}$, the group G acts on $\mathcal{F}(\bar{U})$.

Definition 1.1.2. We say that G acts continuously on $\mathcal{F}$ if, for every quasi-compact U étale over Y, the action of G on the discrete set $\mathcal{F}(\bar{U})$ is continuous.

It would amount to the same thing to consider only affine U.

If $\mathcal{F}$ is a sheaf of sets on Y, with inverse image $\bar{\mathcal{F}}$ on $\bar{Y}$, then $\mathrm{Gal}(\bar{k}/k)$ acts on $\bar{\mathcal{F}}$ by transport of structure, compatibly with its action on $\bar{Y}$.

Reminder 1.1.3.

(i) The action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{\mathcal{F}}$ is continuous.

(ii) The functor

$$\mathcal{F} \longmapsto \bar{\mathcal{F}}$$

where $\bar{\mathcal{F}}$ is endowed with the action of $\mathrm{Gal}(\bar{k}/k)$, is an equivalence from the category of sheaves of sets on Y to the category of sheaves of sets on $\bar{Y}$ endowed with a continuous action of $\mathrm{Gal}(\bar{k}/k)$ compatible with the action of $\mathrm{Gal}(\bar{k}/k)$ on $\bar{Y}$.

Write $\bar{k}$ as the direct limit of finite Galois extensions k_i. Let

$$\varphi_i : Y_i = Y \otimes_k k_i \longrightarrow Y, \qquad \bar{\varphi} : \bar{Y} \longrightarrow Y$$

be the projections. For U étale over Y, assumed affine, put $U_i = U \otimes_k k_i$. Then

$$\bar{\mathcal{F}}(\bar{U}) = \varinjlim_i (\varphi_i^* \mathcal{F})(U_i).$$

On $(\varphi_i^* \mathcal{F})(U_i)$, the group $\mathrm{Gal}(\bar{k}/k)$ acts through $\mathrm{Gal}(k_i/k)$, which proves (i).

The functor in (ii) has as right adjoint the functor

$$\mathcal{G} \longmapsto (\bar{\varphi}_* \mathcal{G})^{\mathrm{Gal}(\bar{k}/k)}.$$

For every $\mathcal{F}$ on Y, one has

$$\bar{\varphi}_* \bar{\varphi}^* \mathcal{F} = \varinjlim_i \varphi_{i*} \varphi_i^* \mathcal{F}.$$

Thus, to prove that

$$\mathcal{F} \xrightarrow{\sim} (\bar{\varphi}_* \bar{\varphi}^* \mathcal{F})^{\mathrm{Gal}(\bar{k}/k)},$$

it is enough to prove, for every i, that

$$\mathcal{F} \xrightarrow{\sim} (\varphi_{i*} \varphi_i^* \mathcal{F})^{\mathrm{Gal}(k_i/k)}.$$

It is even enough to prove that this arrow becomes an isomorphism after extension of scalars from k to k_i; then $\mathrm{Spec}(k_i)$ becomes a sum of copies of $\mathrm{Spec}(k)$ permuted simply transitively by $\mathrm{Gal}(k_i/k)$, and the assertion becomes trivial.

The exact functor in (ii) is therefore fully faithful. To conclude, it remains to verify that the arrow

$$\bar{\varphi}^* (\bar{\varphi}_* \mathcal{G})^{\mathrm{Gal}(\bar{k}/k)} \longrightarrow \mathcal{G}$$

is an epimorphism. Let $\bar{\varphi}_i : \bar{Y} \longrightarrow Y_i$. The hypothesis implies that

$$\varinjlim_i \bar{\varphi}_i^* (\bar{\varphi}_{i*} \mathcal{G})^{\mathrm{Gal}(\bar{k}/k_i)} \longrightarrow \mathcal{G}$$

is an epimorphism.

We have

$$\bar{\varphi}^*(\bar{\varphi}_*\mathcal{G})^{\mathrm{Gal}(\bar{k}/k)} = \bar{\varphi}_i^*\left(\varphi_i^*\varphi_{i*}\left(\left(\bar{\varphi}_{i*}\mathcal{G}\right)^{\mathrm{Gal}(\bar{k}/k_i)}\right)^{\mathrm{Gal}(k_i/k)}\right),$$

and the conclusion follows by observing, by the same Galois-descent argument as above, that for $\mathcal{G}'$ on Y_i, endowed with an action of $\mathrm{Gal}(k_i/k)$ compatible with its action on Y_i, one has

$$\varphi_i^*(\varphi_{i*}\mathcal{G}')^{\mathrm{Gal}(k_i/k)} \xrightarrow{\sim} \mathcal{G}'.$$

1.2. The topos $Y \times_s S$

1.2.1.

Let S be a henselian local scheme with closed point $i : s \hookrightarrow S$. The functor

$$S' \longmapsto S' \times_S s = i^* S'$$

is an equivalence from the category of finite étale S-schemes S' to the category of finite étale s-schemes s' (spectra of étale $k(s)$-algebras). The inverse functor sp^*, from the étale site of s to the étale site of S, is a morphism of sites

$$\mathrm{sp} : S \longrightarrow s,$$

the specialization morphism (cf. SGA 4, VIII, 7). If s' is étale over s, one says that $\mathrm{sp}^*(s')$ is obtained from S by extension of the residue field. For a sheaf F on S, there is a functorial isomorphism

(1.2.1.1) $$\mathrm{sp}_* F = i^* F.$$

For every open subscheme of S, $j : U \hookrightarrow S$, or for every point of S, $j : \eta \hookrightarrow S$, we shall again write sp for the composite morphism $\mathrm{sp} \circ j$. For a sheaf F on U (respectively on η), one has

(1.2.1.2) $$\mathrm{sp}_* F = i^* j_* F.$$

1.2.2.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}, I$ be as in 0.2.5.

$$s \overset{i}{\hookrightarrow} S \overset{j}{\hookleftarrow} \eta \quad .$$

Here is the Galois-theoretic interpretation of sheaves on s, η, and S, and of the specialization morphism.

(a) Sheaves F on s (respectively on η) are identified, by $F \mapsto F_{\bar{s}}$ (respectively $F \mapsto F_{\bar{\eta}}$), with sets endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$) (SGA 4, VIII, 2.1, or 1.1.3).

(b) A sheaf F on S defines sheaves $F_s = i^*F$ and $F_\eta = j^*F$ on s and η, and an adjunction arrow $F \longrightarrow j_*j^*F$ inducing

$$\varphi : F_s = i^*F \longrightarrow i^*j_*j^*F = \mathrm{sp}_* F_\eta.$$

It is known (SGA 4, IV, 9.5.4) that the functor

$$F \longmapsto (F_s, F_\eta, \varphi)$$

is an equivalence from the category of sheaves on S to the category of triples (F_s, F_η, φ), where F_s is a sheaf on s, F_η is a sheaf on η, and $\varphi : F_s \longrightarrow i^*j_*F_\eta$.

Under the equivalence in (a), the functor i^*j_* is identified with the functor of "invariants under I". Thus sheaves on S are identified with triples consisting of a set $F_{\bar{s}}$ endowed with an action of $\mathrm{Gal}(\bar{s}/s)$, a set $F_{\bar{\eta}}$ endowed with an action of $\mathrm{Gal}(\bar{\eta}/\eta)$, and an equivariant morphism $\varphi : F_{\bar{s}} \longrightarrow F_{\bar{\eta}}$.

(c) Under (a) and (b), the morphisms sp, j, i, and $\mathrm{sp} = \mathrm{sp} \circ j$ are expressed as follows:

$$
\begin{array}{lll}
\mathrm{sp} : S \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = (F_{\bar{s}}, F_{\bar{s}}, \mathrm{Id}), \\
& \mathrm{sp}_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
j : \eta \longrightarrow S : & j^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{\eta}}, \\
& j_*(F_{\bar{\eta}}) & = (F_{\bar{\eta}}^I, F_{\bar{\eta}}, \text{inclusion}), \\
i : s \longrightarrow S : & i^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) & = F_{\bar{s}}, \\
& i_*(F_{\bar{s}}) & = (F_{\bar{s}}, \{0\}, \text{projection}), \\
\mathrm{sp} = \mathrm{sp} \circ j : \eta \longrightarrow s : & \mathrm{sp}^*(F_{\bar{s}}) & = F_{\bar{s}}, \\
& \mathrm{sp}_*(F_{\bar{\eta}}) & = F_{\bar{\eta}}^I.
\end{array}
$$

1.2.3.

Let S be a henselian trait as in 1.2.2, and let Y be a scheme over s. We shall have to work with the 2-fiber product $Y \times_s S$ of the étale topoi of Y and S over the étale topos of s. By Giraud [1], this 2-fiber product exists, and it is readily seen to admit the Galois-theoretic description given below. Those who dislike 2-fiber products may take this description as the definition.

Construction 1.2.4.

Sheaves F on $Y \times_s S$ are identified with triples (F_s, F_η, φ) in which:

(a) F_s is a sheaf on Y, or equivalently a sheaf $F_{\bar{s}}$ on $\bar{Y} = Y \otimes_s \bar{s}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ compatible with the action of $\mathrm{Gal}(\bar{s}/s)$ on $\bar{Y}$;

(b) F_η is a sheaf $F_{\bar{\eta}}$ on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$ compatible with the action (via 1.1.1) of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $\bar{Y}$;

(c) φ is an equivariant morphism from $F_{\bar{s}}$ to $F_{\bar{\eta}}$.

Similarly, a sheaf F_η on $Y \times_s \eta$ is an object as in 1.2.4(b). There is a diagram of topoi

(1.2.4.1)
$$
\begin{tikzcd}
Y \arrow[d] & \arrow[l, "\mathrm{sp}"'] & Y \arrow[r, hook, "i"] \arrow[d] & Y \times_s S \arrow[d] & Y \times_s \eta \arrow[l, hook', "j"'] \arrow[d] \\
s & & s \arrow[r, hook, "i"] & S & \eta \arrow[l, hook', "j"']
\end{tikzcd}
.
$$

The topos $Y \times_s S$ is the union of the open subtopos $Y \times_s \eta$ and its closed complement Y. The formulas in 1.2.2(c) remain valid for $\mathrm{sp} : Y \times_s S \longrightarrow Y$ (also denoted pr_1), $j : Y \times_s \eta \hookrightarrow Y \times_s S$, $i : Y \hookrightarrow Y \times_s S$, and $\mathrm{sp} : Y \times_s \eta \longrightarrow Y$ (also denoted pr_1).

1.2.4.2. If 1.2.4 is taken as the definition of $Y \times_s S$, one must show by a direct construction that, up to unique isomorphism, $Y \times_s S$ does not depend on the choice of a separable closure $k(\bar{\eta})$ of $k(\eta)$. This is done by interpreting the F_η of 1.2.4(b) as a functor which, to each separable closure $k(\bar{\eta})$ of $k(\eta)$ defining a separable closure $k(\bar{s})$ of $k(s)$, associates a sheaf on $Y \otimes_{k(s)} k(\bar{s})$; the F_η of 1.2.4(b) is the restriction of this functor to $\{\bar{\eta}\}$.

1.2.5.

Every geometric point $\bar{x}$ of $Y_{\bar{s}}$ defines a geometric point $(\bar{x}, \bar{\eta})$ of $Y \times_s \eta$, whose corresponding fiber functor is

$$F_\eta \longmapsto (F_{\bar{\eta}})_{\bar{x}}.$$

It defines geometric points $(\bar{x}, \bar{\eta})$ and $(\bar{x}, \bar{s})$ of $Y \times_s S$ by

$$F_{(\bar{x},\bar{\eta})} = (F_{\bar{\eta}})_{\bar{x}} \qquad \text{and} \qquad F_{(\bar{x},\bar{s})} = (F_{\bar{s}})_{\bar{x}}.$$

One verifies that all the points of the topoi $Y \times_s \eta$ and $Y \times_s S$ are obtained in this way. We shall not use this. One also verifies at once that these sets of points are conservative.

*__Remark 1.2.6.__

The construction in 1.2.5 defines an equivalence from the category $\mathrm{Point}(Y \times_s \eta)$ (respectively $\mathrm{Point}(Y \times_s S)$) of points of $Y \times_s \eta$ (respectively of $Y \times_s S$) to the 2-fiber product

$$\mathrm{Point}(Y) \times_{\mathrm{Point}(s)} \mathrm{Point}(\eta).$$

This equivalence also follows from the fact that $Y \times_s \eta$ (respectively $Y \times_s S$) is a 2-fiber product of topoi.*

1.2.7.

By the functoriality of 2-fiber products, diagram (1.2.4.1) is functorial in Y and S. In particular:

a) Every morphism $f : Y \longrightarrow Y'$ of schemes over s induces a morphism of topoi from $Y \times_s S$ to $Y' \times_s S$. One has

(1.2.7.1) $$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)),$$

and, when f is quasi-compact,

(1.2.7.2) $$f_*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_*F_{\bar{s}}, f_*F_{\bar{\eta}}, f_*(\varphi)).$$

Analogous formulas hold for $f : Y \times_s \eta \longrightarrow Y' \times_s \eta$.

b) Every surjective morphism of henselian traits $f : S' \longrightarrow S$ induces a morphism of topoi from $Y_{s'} \times_{s'} S'$ to $Y \times_s S$. Again writing f for the natural morphism $Y_{s'} = Y \otimes_{k(s)} k(s') \longrightarrow Y$, one has

(1.2.7.3) $$f^*(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^*F_{\bar{s}}, f^*F_{\bar{\eta}}, f^*(\varphi)).$$

When f is finite, f_* is expressed as an induced representation.

1.2.8.

Let $f : Y \longrightarrow Y'$ be a separated morphism, locally of finite type, between s-schemes. Define a "direct image with proper support" functor

$$f_! : (\text{sheaves of pointed sets on } Y \times_s S) \longrightarrow (\text{sheaves of pointed sets on } Y' \times_s S)$$

by the formula

(1.2.8.1) $$f_!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f_!F_{\bar{s}}, f_!F_{\bar{\eta}}, f_!(\varphi)).$$

When f is a locally closed immersion, $f_!$ is called the extension-by-zero functor.

An analogous functor is defined for $Y \times_s \eta$.

1.2.9.

Let $f : Y \hookrightarrow Y'$ be a locally closed immersion of s-schemes. The functor $f_!$ has a right adjoint $f^!$, "sections with support in Y", satisfying

$$f^!(F_{\bar{s}}, F_{\bar{\eta}}, \varphi) = (f^! F_{\bar{s}}, f^! F_{\bar{\eta}}, f^!(\varphi)). \tag{1.2.9.1}$$

Suppose only that f is quasi-finite. The functor $f_!$ from abelian sheaves on $Y \times_s S$ to abelian sheaves on $Y' \times_s S$ still has a right adjoint $f^!$, given by (1.2.9.1) (cf. SGA 4, XVIII, 3.1.8). If f is étale, then $f^! = f^*$.

Analogous functors are defined for $Y \times_s \eta$.

1.3. The functor Ψ

1.3.1.

Let S be a henselian trait as in 0.2.5, and let $p : X \longrightarrow S$. We propose to define a left exact functor Ψ from sheaves on X to sheaves on the topos $X_s \times_s S$ constructed in 1.2. In general, this functor is not the direct-image functor of a morphism of topoi.

We shall also define

$$\Psi_\eta : (\text{sheaves on } X_\eta) \longrightarrow (\text{sheaves on } X_s \times_s \eta).$$

1.3.2.

Put $\bar{X} = X \times_S \bar{S}$. There is a Cartesian diagram

$$\begin{tikzcd} X_{\bar{s}} \arrow[r, hook, "\bar{i}"] \arrow[d] & \bar{X} \arrow[d] & X_{\bar{\eta}} \arrow[l, hook', "\bar{j}"'] \arrow[d] \\ X_s \arrow[r, hook] & X & X_\eta \arrow[l, hook'] \end{tikzcd} \tag{1.3.2.1}$$

over the diagram

$$\begin{tikzcd} \bar{s} \arrow[r, hook] \arrow[d] & \bar{S} \arrow[d] & \bar{\eta} \arrow[l, hook'] \arrow[d] \\ s \arrow[r, hook] & S & \eta \arrow[l, hook'] \end{tikzcd} .$$

Let F be a sheaf on X_η, with inverse image $F_{\bar{\eta}}$ on $X_{\bar{\eta}}$. Put

$$\Psi_\eta(F) = \bar{i}^* \bar{j}_* F_{\bar{\eta}}. \tag{1.3.2.2}$$

By transport of structure, this sheaf is endowed with a continuous action of $\mathrm{Gal}(\bar{\eta}/\eta)$, compatible with the action of $\mathrm{Gal}(\bar{\eta}/\eta)$ on $X_{\bar{s}}$. The functor

$$\Psi_\eta : (X_{\eta,\mathrm{et}})^\sim \longrightarrow (X_s \times_s \eta)^\sim \tag{1.3.2.3}$$

is left exact.

1.3.3.

Let F be a sheaf on X, and let F_η and F_s be its restrictions to X_η and X_s. Put

$$(\Psi(F))_\eta = \Psi_\eta(F_\eta) \tag{1.3.3.1}$$

and

$$(\Psi(F))_s = F_s. \tag{1.3.3.2}$$

Let $\bar{F}$ be the inverse image of F on $\bar{X}$, and let φ' be the adjunction morphism

$$\varphi' : \bar{F} \longrightarrow \bar{j}_*\bar{j}^*\bar{F}.$$

This morphism induces

$$\varphi : F_{\bar{s}} = \bar{i}^*\bar{F} \longrightarrow \bar{i}^*\bar{j}_*\bar{j}^*\bar{F} = (\Psi_\eta(F_\eta))_{\bar{\eta}}.$$

The triple

$$\Psi(F) = ((\Psi(F))_s, (\Psi(F))_\eta, \varphi)$$

is a sheaf on $X_s \times_s S$. The functor

$$\Psi : (X_{\mathrm{et}})^\sim \longrightarrow (X_s \times_s S)^\sim \tag{1.3.3.3}$$

is left exact.

1.3.4.

When there is no risk of confusion, we shall write

$$\Psi_\eta(F) = (\Psi(F))_\eta, \qquad \Psi_{\bar{\eta}}(F) = (\Psi(F))_{\bar{\eta}}, \qquad \text{and} \qquad \Psi_{\bar{\eta}}(F) = (\Psi_\eta(F))_{\bar{\eta}}.$$

1.3.5.

For the remainder of this section, we list functoriality properties of Ψ (cf. 1.2.7–1.2.9). They will be useful chiefly in their derived form (2.1).

Let $f : X \longrightarrow X'$ be a morphism of S-schemes, giving a diagram

$$\begin{tikzcd} X_{\bar{s}} \arrow[r, hook, "\bar{i}"] \arrow[d, "f"] & \bar{X} \arrow[d, "f"] & X_{\bar{\eta}} \arrow[l, hook', "\bar{j}"'] \arrow[d, "f"] \\ X'_{\bar{s}} \arrow[r, hook, "\bar{i}"] & \bar{X}' & X'_{\bar{\eta}} \arrow[l, hook', "\bar{j}"'] \end{tikzcd} \tag{1.3.5.1}$$

(the printed diagram inadvertently omits the bar from its top-middle $\bar{X}$). The ambiguity in the notation for the arrows simplifies the formulas and causes no confusion: only one interpretation makes sense of the composites below. Likewise, we write Ψ for the functor (1.3.3.3) for both X and X'.

The morphisms of functors described below for Ψ have variants for Ψ_η, which we generally omit to state explicitly.

1.3.6.

The base-change morphism $\bar{i}^*f_* \longrightarrow f_*\bar{i}^*$ induces a morphism of functors

$$\Psi f_* \longrightarrow f_*\Psi, \tag{1.3.6.1}$$

which is an isomorphism when f is proper (SGA 4, XII, 5.1(i)).

The essential case is $X' = S$. One obtains a morphism

$$f_* \longrightarrow f_*\Psi \tag{1.3.6.2}$$

whose essential component is

$$\Gamma(X_{\bar{\eta}}, F) \longrightarrow \Gamma(X_{\bar{s}}, \Psi_{\bar{\eta}}(F)); \tag{1.3.6.3}$$

this morphism is an isomorphism when f is proper.

1.3.7.

The base-change morphism $f^*\bar{j}_* \longrightarrow \bar{j}_* f^*$ induces a morphism of functors

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.7.1}$$

1.3.8.

Suppose that f is quasi-finite. The functors $f_!$ commute with base change; the morphism $f_!\bar{j}_* \longrightarrow \bar{j}_* f_!$ induces a morphism of functors

$$f_!\Psi \longrightarrow \Psi f_!. \tag{1.3.8.1}$$

When f is finite, this is the inverse of (1.3.6.1).

1.3.9.

The situation is dual for $f^!$ (we suppose that f is quasi-finite when working with abelian sheaves, or that it is an immersion when working with sheaves of pointed sets). This functor commutes with direct images, and the morphism $\bar{i}^* f^! \longrightarrow f^! \bar{i}^*$ induces a morphism of functors

$$\Psi f^! \longrightarrow f^!\Psi. \tag{1.3.9.1}$$

When f is étale, this is the inverse of (1.3.7.1).

1.3.10.

Finally, a base change $S' \xrightarrow{f} S$ gives a base-change morphism

$$f^*\Psi \longrightarrow \Psi f^*. \tag{1.3.10.1}$$

1.4. The variation

1.4.1.

Let S be a henselian trait as in 0.2.5, and let Y be a scheme over s. Let Λ be a ring. More generally, one could take for Λ a sheaf of rings on Y. For every object K of the derived category $D(Y \times_s S, \Lambda)$ of the category of sheaves of Λ-modules on $Y \times_s S$, we propose to define a <u>variation</u> morphism.

1.4.2.

A complex of sheaves of Λ-modules K on $Y \times_s S$ is identified with a triple (K_s, K_η, φ) in which

a) K_s (respectively K_η) is a complex of sheaves $K_{\bar{s}}$ (respectively $K_{\bar{\eta}}$) on $\bar{Y}$, endowed with a continuous action of $\mathrm{Gal}(\bar{s}/s)$ (respectively $\mathrm{Gal}(\bar{\eta}/\eta)$);

b) φ is an equivariant morphism $\varphi : K_{\bar{s}} \longrightarrow K_{\bar{\eta}}$.

Here and below we retain the scheme Y fixed in 1.4.1; the printed opening sentence of 1.4.2 inadvertently writes $X \times_s S$.

Such a complex K is always homotopic to a complex $K' = (K'_s, K'_\eta, \varphi')$ such that φ' is injective and the exact sequence

$$0 \longrightarrow K'_{\bar{s}} \longrightarrow K'_{\bar{\eta}} \longrightarrow \mathrm{coker}(\varphi') \longrightarrow 0 \tag{1.4.2.1}$$

is split degree by degree. It is enough to take $K'_s = K_s$ and to take for K'_η the direct sum of K_η and the cone of $\mathrm{sp}^* K_s$. Put

$$\Phi(K) = \mathrm{coker}(\varphi').$$

The exact sequence (1.4.2.1) defines a distinguished triangle

(1.4.2.2)
$$
\begin{tikzcd}
 & \Phi(K) \arrow[dl] & \\
\mathrm{sp}^* K_s \arrow[rr] & & K_\eta \arrow[ul]
\end{tikzcd}
$$

in the category $K(Y \times_s \eta, \Lambda)$ of complexes of sheaves of Λ-modules on $Y \times_s \eta$, up to homotopy. This triangle depends only on $K \in \mathrm{Ob}\, K(Y \times_s S, \Lambda)$.

The construction passes to the derived category and associates with every object K of $D(Y \times_s S, \Lambda)$ a distinguished triangle of $D(Y \times_s \eta, \Lambda)$.

Triangle (1.4.2.2) is useful chiefly through the long exact hypercohomology sequence that it induces:

$$\cdots \longrightarrow H^i(\bar{Y}, K_{\bar{s}}) \longrightarrow H^i(\bar{Y}, K_{\bar{\eta}}) \longrightarrow H^i(\bar{Y}, \Phi(K)_{\bar{\eta}}) \longrightarrow \cdots .$$

1.4.3.

The inertia group I acts trivially on $K'_{\bar{s}}$. Hence, for every $\sigma \in I$, the endomorphism $\sigma - 1$ of $K'_{\bar{\eta}}$ (notation (1.4.2.1)) factors through a morphism

$$\mathrm{Var}(\sigma) : \mathrm{coker}(\varphi') \longrightarrow K'_{\bar{\eta}}.$$

This construction passes to the derived category: for $K \in \mathrm{Ob}\, D(Y \times_s S, \Lambda)$ and $\sigma \in I$, it defines the <u>variation</u>

(1.4.3.1) $$\mathrm{Var}(\sigma) : \Phi(K)_{\bar{\eta}} \longrightarrow K_{\bar{\eta}}.$$

Writing q for the arrow from K_η to $\Phi(K)$, one has

(1.4.3.2) $$\sigma = 1 + \mathrm{Var}(\sigma) q \qquad \text{on } K_{\bar{\eta}},$$
(1.4.3.3) $$\sigma = 1 + q\, \mathrm{Var}(\sigma) \qquad \text{on } \Phi(K)_{\bar{\eta}},$$

and

(1.4.3.4) $$\mathrm{Var}(\sigma\tau) = \mathrm{Var}(\sigma) q\, \mathrm{Var}(\tau) + \mathrm{Var}(\sigma) + \mathrm{Var}(\tau).$$

2. Vanishing cycles

2.1. The functor $R\Psi$

2.1.1.

Let S be a henselian trait as in 0.2.5, and let Λ be a torsion ring whose torsion is prime to the residual characteristic p of S (0.2.7). More generally, one could take for Λ a torsion sheaf whose torsion is prime to the residual characteristics; the most important case is $\Lambda = \mathbb{Z}/\ell^n$ with ℓ an invertible prime in $\mathcal{O}_S$.

Let X be a scheme over S. The functor Ψ (respectively Ψ_η) carries sheaves of Λ-modules on X (respectively on X_η) to sheaves of Λ-modules on $X_s \times_s S$ (respectively on $X_s \times_s \eta$). Let $R\Psi$ and $R\Psi_\eta$ be the derived functors

$$R\Psi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s S, \Lambda),$$

$$R\Psi_\eta : D^+(X_\eta, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda),$$

and let $R\Phi$ be the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X, \Lambda) \longrightarrow D^+(X_s \times_s \eta, \Lambda).$$

2.1.2.

Consider the commutative diagram (1.3.2.1)

(1.3.2.1)
$$\begin{tikzcd} X_{\bar{s}} \arrow[r, hook, "\bar{i}"] \arrow[d] & \bar{X} \arrow[d] & X_{\bar{\eta}} \arrow[l, hook', "\bar{j}"'] \arrow[d] \\ X_s \arrow[r, hook, "i"] & X & X_\eta \arrow[l, hook', "j"'] \end{tikzcd}$$

If F is an injective sheaf of modules on X, its restriction to the open subscheme X_η is still injective. If F_η is injective on X_η, its inverse image $F_{\bar{\eta}}$ is acyclic. Thus the derived forms of (1.3.3.1) and (1.3.3.2) are

(2.1.2.1)
$$R\Psi(K)_s = i^*K \qquad (K \in \operatorname{Ob} D^+(X, \Lambda)),$$

(2.1.2.2)
$$R\Psi(K)_\eta = R\Psi_\eta(K_\eta) \qquad (K \in \operatorname{Ob} D^+(X, \Lambda)),$$

and

(2.1.2.3)
$$R\Psi_\eta(K)_{\bar{\eta}} = \bar{i}^* R\bar{j}_* K_{\bar{\eta}} \qquad (K \in \operatorname{Ob} D^+(X_\eta, \Lambda)).$$

This allows us to write, without conflict,

$$R\Psi_\eta(K) = (R\Psi(K))_\eta, \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi(K))_{\bar{\eta}}, \qquad \text{and} \qquad R\Psi_{\bar{\eta}}(K) = (R\Psi_\eta(K))_{\bar{\eta}}.$$

With this notation, the distinguished triangle (1.4.2.2) becomes

(2.1.2.4)
$$\begin{tikzcd} & R\Phi(K) \arrow[dl] & \\ \mathrm{sp}^* i^* K \arrow[rr] & & R\Psi_\eta(K_\eta) \arrow[ul] \end{tikzcd} .$$

The printed bottom-right term has K_s; the defining identity (2.1.2.2) requires K_η, as written here.

For $\sigma \in I$, there is a <u>variation</u> morphism

(2.1.2.5)
$$\operatorname{Var}(\sigma) : R\Phi(K)_{\bar{\eta}} \longrightarrow R\Psi_{\bar{\eta}}(K).$$

The sheaves $R^i\Phi(\Lambda)$, or more generally $R^i\Phi(K)$, are the <u>sheaves of vanishing cycles</u>.

2.1.3.

Let $(\bar{x}, \bar{\eta})$ be the geometric point of $X_s \times_s \eta$ defined by a geometric point $\bar{x}$ of $X_{\bar{s}}$ (1.2.5). We compute the fiber of $R\Psi_\eta(K_\eta)$ at $(\bar{x}, \bar{\eta})$ using (2.1.2.3) (cf. I, 2.3). The strict henselization $X_{(\bar{x})}$ of X at $\bar{x}$ is a scheme over the strict henselization S_{nr} of S, with geometric generic point η_{nr}.

Proposition 2.1.4.

<u>One has</u>
$$(R\Psi(K_\eta))_{(\bar{x}, \bar{\eta})} = R\Gamma\big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\big)\,;$$
<u>in particular,</u>
$$R^i\Psi(K_\eta)_{(\bar{x}, \bar{\eta})} = \mathbb{H}^i\big(X_{(\bar{x})} \times_{\eta_{\mathrm{nr}}} \bar{\eta}, K\big)\,.$$

Apply SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms). This gives the following restatement.

Restatement 2.1.5.

If f is smooth and F is a locally constant sheaf, then

$$R\Phi(F) = 0.$$

More generally, if $K \in \operatorname{Ob} D^+(X, \Lambda)$, then $R\Phi(K) = 0$ wherever f is smooth and the sheaves $\mathcal{H}^i(K)$ are locally constant. See also 2.1.11.

2.1.6.

Let $f : Y \longrightarrow Y'$ be a morphism of schemes over s. The functors f^*, f_*, $f^!$, and $f_!$ (1.2.7–1.2.9) have natural analogues in the derived category.

a) The functor f^* is exact and therefore derives trivially.

b) The functor f_* has a right derived functor Rf_*. When f is quasi-compact, one has

$$(Rf_*K)_{\bar{s}} = Rf_{\bar{s}*}(K_{\bar{s}}) \tag{2.1.6.1}$$

and

$$(Rf_*K)_{\bar{\eta}} = Rf_{\bar{s}*}(K_{\bar{\eta}}). \tag{2.1.6.2}$$

c) When f is quasi-finite, the functor $f_!$ is exact and therefore derives trivially. In general, the derived functor of $f_!$ is pathological, and one proceeds as in SGA 4, XVII. Suppose that f is separated and of finite type and that Y' is noetherian. There is then [2] a factorization $f = \widetilde{f}\, j$, where j is an open immersion and $\widetilde{f}$ is proper, and one defines

$$Rf_! = R\widetilde{f}_* \circ j_!.$$

The arguments of SGA 4, XVII show that $Rf_!$ is independent of the choice of the factorization $\widetilde{f}\, j$. The analogues of (2.1.6.1) and (2.1.6.2) hold.

d) We shall use $Rf^!$ only for a quasi-finite morphism, in which case it is the derived functor of $f^!$. Again, the analogues of (2.1.6.1) and (2.1.6.2) hold.

2.1.7.

We now review the functoriality properties of $R\Psi$, that is, we derive 1.3.5–1.3.10. Let $f : X \longrightarrow X'$ be a morphism of S-schemes, and retain the notation of 1.3.5.

The morphism (1.3.6.1) defines

$$R\Psi\, Rf_* \longrightarrow Rf_*\, R\Psi. \tag{2.1.7.1}$$

By the proper base-change theorem (SGA 4, XII, 5.1), this morphism is an isomorphism if f is proper.

The morphism (1.3.7.1) defines

$$f^* R\Psi \longrightarrow R\Psi f^*. \tag{2.1.7.2}$$

By the smooth base-change theorem (SGA 4, XVI, 1.2), this morphism is an isomorphism if f is smooth.

The isomorphism (2.1.7.1) for f proper and the morphism (1.3.8.1) define a morphism

$$Rf_! R\Psi \longrightarrow R\Psi Rf_!, \tag{2.1.7.3}$$

which, when f is proper, is the inverse of (2.1.7.1). The print mistakenly refers here to the later special case (2.1.8.1).

When f is quasi-finite, (1.3.9.1) derives to

$$R\Psi Rf^! \longrightarrow Rf^! R\Psi, \tag{2.1.7.4}$$

which, when f is étale, is the inverse of (2.1.7.2). The printed reference to (2.1.7.1) is incompatible with the two functors involved.

Let $f : S' \longrightarrow S$ be a change of trait, and put $X' = X \times_S S'$:

$$\begin{tikzcd} X' \arrow[r, "f"] \arrow[d, "p"'] & X \arrow[d, "p"] \\ S' \arrow[r, "f"] & S \end{tikzcd}$$

(for the ambiguity in the notation, see 1.3.5). Suppose that f is surjective. The base-change arrow (1.3.10.1) derives to

$$f^* R\Psi \longrightarrow R\Psi f^*. \tag{2.1.7.5}$$

The same statements hold for $R\Psi_\eta$.

2.1.8.

The essential case of (2.1.7.1) and (2.1.7.3) is $X' = S$. For $K \in \operatorname{Ob} D^+(X_{\bar\eta}, \Lambda)$ one then obtains morphisms

$$Rf_* \longrightarrow Rf_* R\Psi \tag{2.1.8.1}$$

and

$$Rf_! R\Psi \longrightarrow Rf_!, \tag{2.1.8.2}$$

hence

$$\mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \tag{2.1.8.3}$$

and

$$\mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \longrightarrow \mathbb{H}^i_c(X_{\bar\eta}, K), \tag{2.1.8.4}$$

and the commutativity of the diagrams

$$\begin{tikzcd} (R^i f_* K)_{\bar s} \arrow[r, "\mathrm{sp}"] \arrow[d] & \mathbb{H}^i(X_{\bar\eta}, K) \arrow[d] \\ \mathbb{H}^i(X_{\bar s}, K) \arrow[r] & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \end{tikzcd} \tag{2.1.8.5}$$

and

$$\begin{tikzcd} \mathbb{H}^i_c(X_{\bar s}, K) \arrow[rr, "\mathrm{sp}"] \arrow[dr] & & \mathbb{H}^i_c(X_{\bar\eta}, K) \\ & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \arrow[ur] & \end{tikzcd} . \tag{2.1.8.6}$$

For $K \in \operatorname{Ob} D^+(X_\eta, \Lambda)$ and $\sigma \in I$, the following diagrams are commutative:

$$
(2.1.8.7) \qquad
\begin{tikzcd}
\mathbb{H}^i(X_{\bar\eta}, K) \arrow[rr, dash, "\sigma-1"] \arrow[d] & & \mathbb{H}^i(X_{\bar\eta}, K) \arrow[d] \\
\mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K)) \arrow[r] & \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \arrow[r, "\mathrm{Var}(\sigma)"] & \mathbb{H}^i(X_{\bar s}, R\Psi_{\bar\eta}(K))
\end{tikzcd}
$$

and

$$
(2.1.8.8) \qquad
\begin{tikzcd}
\mathbb{H}^i_c(X_{\bar\eta}, K) \arrow[rr, "\sigma-1"] & & \mathbb{H}^i_c(X_{\bar\eta}, K) \\
\mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \arrow[r] \arrow[u] & \mathbb{H}^i_c(X_{\bar s}, R\Phi_{\bar\eta}(K)) \arrow[r, "\mathrm{Var}(\sigma)"] & \mathbb{H}^i_c(X_{\bar s}, R\Psi_{\bar\eta}(K)) \arrow[u]
\end{tikzcd}
.
$$

In (2.1.8.8) the English restores the bar omitted from the printed bottom-left $X_{\bar s}$ and the missing closing parenthesis in the bottom-middle term. In (2.1.8.7), however, the upper connector really is printed as a rule without an arrowhead, whereas (2.1.8.8) has an arrow; this source-visible distinction is retained.

When f is proper, (2.1.8.1) and (2.1.8.2), as well as (2.1.8.3) and (2.1.8.4), are mutually inverse isomorphisms. The distinguished triangle (2.1.2.4) gives a long exact sequence

$$
(2.1.8.9) \qquad \cdots \longrightarrow \mathbb{H}^i(X_{\bar s}, K) \longrightarrow \mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \mathbb{H}^i(X_{\bar s}, R\Phi_{\bar\eta}(K)) \longrightarrow \cdots .
$$

Thus the vanishing complex $R\Phi(K)$ expresses the difference between the cohomologies of the geometric fibers $X_{\bar s}$ and $X_{\bar\eta}$, with coefficients in the inverse image of K. By (2.1.8.7), the variation $\mathrm{Var}(\sigma)$ determines the action of the inertia group I on $\mathbb{H}^i(X_{\bar\eta}, K)$. These facts underlie all applications of the method of vanishing cycles.

The properness hypothesis imposed above on f may be replaced by regularity-at-infinity conditions on f and K. For example, one has the following result (for $K \in \operatorname{Ob} D(X_{\bar\eta}, \Lambda)$).

Proposition 2.1.9.

Suppose that f admits a factorization $f = f_1 k$

$$
X \xhookrightarrow{\quad k \quad} X_1 \xrightarrow{\quad f_1 \quad} S,
$$

where f_1 is proper and k identifies X with the complement in X_1 of a relative normal-crossings divisor D (thus, by hypothesis, f_1 is smooth on a neighborhood of D). If, on a neighborhood U of D, the sheaves $\mathcal{H}^i(K)$ are locally constant on $U - (D \cup X_0)$ and tamely ramified along the normal-crossings divisor $D \cup X_0$, then (2.1.8.1) and (2.1.8.2) are isomorphisms.

The hypercohomology spectral sequence for K shows that it is enough to treat the case in which K consists of a sheaf F_η in degree zero. After replacing S by its normalization in a finite extension (which is permissible, as one sees directly from the definitions) and applying Abhyankar's lemma, we may suppose that F_η is the restriction to X_η of a sheaf F on X, locally constant on $U \cap X$ and tamely ramified along D. With the notation of 2.1.2, let $\overline{F}$ be the inverse image of F on $\overline{X}$, and write $\bar f : \overline{X} \longrightarrow \overline{S}$.

To prove that (2.1.8.1) is an isomorphism, it is enough to prove that

$$
R\bar f_*\big(R\bar j_* \bar j^* \overline{F}\big)
$$

commutes with base change by $\bar s \longrightarrow \overline{S}$. In the distinguished triangle

$$
\longrightarrow \overline{F} \longrightarrow R\bar j_* \bar j^* \overline{F} \longrightarrow \Delta \longrightarrow,
$$

the sheaves $\mathcal{H}^i(\Delta)$ are supported on $X_{\bar s}$; for Δ, the base-change statement in question is tautological. It therefore remains to prove that $R\bar f_*(\overline{F})$ commutes with base change by $\bar s \longrightarrow \overline{S}$, or, in view of the base-change theorem for the proper morphism f_1, that $Rk_*(\overline{F})$ commutes with base change $X_{1\bar{s}} \longrightarrow X_1$. This is Lemma 2.1.10 below.

To prove that (2.1.8.2) is an isomorphism, it is enough to prove, for the sheaf F, that extension by zero $k_!$ commutes with the functor $R\bar{j}_*$; that is, on a neighborhood of D,

$$R\Phi(k_! F) = 0 \qquad \text{on } X_{1\bar{s}}.$$

This is Lemma 2.1.11 below.

Lemma 2.1.10.

Let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Formation of Rj_*F commutes with arbitrary base change $T \longrightarrow S$.

The question is local on X for the étale topology. We may therefore suppose that there is a morphism $g : X' \longrightarrow X$, where X' is the normalization of X in a finite étale surjective covering $g' : U' \longrightarrow U$, tamely ramified along D, such that g'^*F is constant on U'. By Abhyankar's lemma, we may further suppose that X' is smooth over S and that U' is the complement in X' of a relative normal-crossings divisor D'.

The sheaf F injects into $g'_*g'^*F$, and the cokernel again satisfies the hypotheses of the lemma. Iterating this construction (the same g continues to work) gives a resolution of F by sheaves of the form g'_*G, where G is constant and g' is as above. It therefore suffices to prove the assertion for such a sheaf, or, equivalently, for $(X'/S, D', G)$ with G constant on U'. This is a special case of SGA 5, II, 4.4.

Lemma 2.1.11.

Let S be a henselian trait, let $f : X \longrightarrow S$ be smooth, let $D \subset X$ be a relative normal-crossings divisor, put $U = X - D \hookrightarrow^j X$, and let F be a sheaf of Λ-modules on U which is locally constant and tamely ramified along D. Then

$$R\Phi(j_! F) = 0.$$

The dévissage just used reduces us to the case in which F is constant, with value G. We may suppose that D is the union of smooth divisors $(D_i)_{i \in I}$. For $J \subset I$, let D_J be the intersection of the D_i for $i \in J$. The sheaf F has a resolution $F^\bullet$ whose terms are sums of sheaves G_J, constant on D_J and extended by zero to X. It is therefore enough to prove the lemma for one such sheaf, and this follows from (2.1.5), applied to D_J/S.

I conjecture that (2.1.7.5) is an isomorphism when S and S' are excellent. Here is a partial result (see also 2.4.2).

Proposition 2.1.12.

The morphism (2.1.7.5) is an isomorphism in the following cases:

(a) S is excellent and S' is its completion;
(b) S and S' have equal characteristic zero.

In case (b), we reduce to the situation in which S and S' are strictly local and a uniformizer of S is still a uniformizer of S' (extend the scalars without changing $\bar{S}' \longrightarrow \bar{S}$). In both cases (a) and (b), writing an overbar for normalization in a separable closure of the fraction field, one then has

$$\bar{S}' \simeq S' \times_S \bar{S}.$$

Returning to the definitions, it is enough to prove that $S' \longrightarrow S$ is universally locally acyclic (apply SGA 4, XVI, 1.1). This follows from I, 0.5.1 (based on Artin's version of Néron desingularization) and SGA 4, XV, 2.1 (the local-acyclicity theorem for smooth morphisms).

2.1.13.

Suppose that $f : X \longrightarrow S$ is of finite type. The cohomological dimension (SGA 4, XVII, 1.2.9) of $R\Psi$ is then at most $\dim(X_\eta)$, as follows from 2.1.4 and SGA 4, XIV, 3.2 (cf. I, 4.2(i)). Thus $R\Psi$ extends to the entire derived category,

$$R\Psi : D(X, \Lambda) \longrightarrow D(X_s \times_s S, \Lambda),$$

and it carries D^- into D^-. Let $L \in \operatorname{Ob} D^-(\Lambda)$ be a bounded-above complex of right Λ-modules, and let $K \in \operatorname{Ob} D^-(X, \Lambda)$. There is then an isomorphism

$$L \overset{\mathbf{L}}{\otimes}_\Lambda R\Psi(K) \xrightarrow{\sim} R\Psi\big(L \overset{\mathbf{L}}{\otimes}_\Lambda K\big). \tag{2.1.13.1}$$

To verify (2.1.13.1), first replace L by a complex of free Λ-modules. Since $R\Psi$ is triangulated and has finite cohomological dimension, it is enough to verify the assertion after replacing L by one of its terms. Passage to the filtered colimit reduces us to the case in which L is a finite free module, and then to the tautological case $L = \Lambda$.

It follows from (2.1.13.1), by the usual argument (cf. SGA 4, XVII, 5.2.11), that if $K \in \operatorname{Ob} D(X_\eta, \Lambda)$ has finite Tor-dimension at most d, where $d \in \mathbb{Z}$ (SGA 4, XVII, 4.1.9), then $R\Psi_\eta(K)$ likewise has Tor-dimension at most d.

2.2. A compatibility (trace morphisms)

2.2.1.

Let S be a henselian trait, and let $u : X_s \longrightarrow Y_s$ be a separated, quasi-finite, flat morphism of finite presentation between s-schemes. For every abelian sheaf F on Y_s, there is a trace morphism

$$\operatorname{Tr}_u : u_! u^* F \longrightarrow F$$

(SGA 4, XVII, 6.2.3). If F is an abelian sheaf on $Y_s \times_s S$ or on $Y_s \times_s \eta$, this gives an analogous morphism $\operatorname{Tr}_u : u_! u^* F \longrightarrow F$.

Now let $u : X \longrightarrow Y$ be a separated, quasi-finite, flat morphism of finite presentation between S-schemes. We again write u for the induced morphism

$$X_s \times_s S \longrightarrow Y_s \times_s S,$$

and we use Ψ for the functors of 1.3.3, both for X and for Y.

Proposition 2.2.2.

For every abelian sheaf F on Y, the following diagram is commutative:

$$\begin{tikzcd} u_! u^* \Psi F \arrow[r, "(1.3.7.1)"] \arrow[d, "\operatorname{Tr}_u"'] & u_! \Psi u^* F \arrow[r, "(1.3.8.1)"] & \Psi u_! u^* F \arrow[d, "\operatorname{Tr}_u"] \\ \Psi F \arrow[rr, equal] & & \Psi F. \end{tikzcd}$$

The verification is left to the reader. There is a variant for Ψ_η and for F on Y_η.

2.2.3.

Let Λ be as in 2.1.1, and let $u : X \longrightarrow Y$ be a quasi-finite flat morphism between separated, finite-type, flat S-schemes:

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

The print refers here to 2.2.1, but the coefficient ring Λ is fixed in 2.1.1. We again write f and g for the morphisms

$$X_s \times_s S \longrightarrow S, \qquad Y_s \times_s S \longrightarrow S,$$

and suppose that $\dim Y_\eta \leq n$. For every $K \in \operatorname{Ob} D(S, \Lambda)$, there is a trace morphism (SGA 4, XVIII, 2.9)

$$\mathrm{Tr}_f : Rf_! f^* K(n)[2n] \longrightarrow K.$$

Corollary 2.2.4.

The following diagram is commutative:

$$\begin{tikzcd} Rg_! u_! u^* R\Psi\, g^* K(n)[2n] \arrow[r, "(1.3.7.1)"] \arrow[d, "\mathrm{Tr}_u"'] & Rf_! R\Psi\, f^* K(n)[2n] \arrow[r, "(2.1.10.1)"] & Rf_! f^* K(n)[2n] \arrow[d, "\mathrm{Tr}_f"] \\ Rg_! R\Psi\, g^* K(n)[2n] \arrow[r, "(2.1.10.1)"] & Rg_! g^* K(n)[2n] \arrow[r, "\mathrm{Tr}_g"] & K \end{tikzcd}$$

If u is étale, (1.3.7.1) is an isomorphism.

With the arguments suppressed, the diagram (2.2.4) expands as follows:

$$\begin{tikzcd} Rg_! u_! u^* R\Psi\, g^* \arrow[r] \arrow[d, "\mathrm{Tr}_u"'] \arrow[dr, phantom, "(1)"] & Rg_! R\Psi\, u_! u^* g^* \arrow[r] \arrow[d, "\mathrm{Tr}_u"'] \arrow[dr, phantom, "(2)"] & Rg_! u_! u^* g^* \arrow[r, "\sim"] \arrow[d, "\mathrm{Tr}_u"'] \arrow[dr, phantom, "(3)"] & Rf_! f^* \arrow[d, "\mathrm{Tr}_f"] \\ Rg_! R\Psi\, g^* \arrow[r, equal] & Rg_! R\Psi\, g^* \arrow[r] & Rg_! g^* \arrow[r, "\mathrm{Tr}_g"] & K. \end{tikzcd}$$

The commutativity of (1) follows from 2.2.2; that of (2) is the functoriality of (2.1.10.2); and that of (3) is the transitivity of the trace (SGA 4, XVIII, 2.9, Variant 3).

2.2.4.1.

There is an analogous result for Ψ_η and $K \in \operatorname{Ob} D(\eta, \Lambda)$.

2.2.5.

Let Φ be a subset of Y_s which is proper over s, and write Rg_Φ for the "direct image with support in Φ" functor. Let Tr_g denote the composite

$$\begin{aligned} \mathrm{Tr}_g : Rg_\Phi R\Psi g^* K(n)[2n] &\longrightarrow Rg_! R\Psi g^* K(n)[2n] \\ &\longrightarrow Rg_! g^* K(n)[2n] \longrightarrow K, \end{aligned} \tag{2.2.5.1}$$

and similarly for Ψ_η and $K \in D(\eta, \Lambda)$. The print has Rf_Φ as the first functor in (2.2.5.1); the immediately preceding definition and the remainder of the composite require Rg_Φ, which is used here.

If Φ is a finite set of closed points, then

$$Rg_\Phi \circ u_! = Rf_{u^{-1}\Phi},$$

and hence there is a trace morphism

$$\mathrm{Tr}_u : Rf_{u^{-1}\Phi}u^* = Rg_\Phi u_! u^* \longrightarrow Rg_\Phi.$$

By 2.2.4, the following diagram is commutative:

$$\begin{tikzcd}
Rf_{u^{-1}\Phi}u^* R\Psi\, g^*K(n)[2n] \arrow[r] \arrow[d, "\mathrm{Tr}_u"'] & Rf_{u^{-1}\Phi}R\Psi\, f^*K(n)[2n] \arrow[r, "\mathrm{Tr}_f"] & K \arrow[d, equal] \\
Rg_\Phi R\Psi\, g^*K(n)[2n] \arrow[rr, "\mathrm{Tr}_g"'] & & K.
\end{tikzcd}$$

Suppose that u is étale, let $x \in X_s$, and suppose that x and $y = u(x)$ have the same residue field. Then

$$Rf_{\{x\}} \xrightarrow{\sim} Rg_{\{y\}}.$$

The print has $\{x\}$ also on the right, but the support for g lies in Y_s and is $\{y\}$, as the following diagram confirms. This isomorphism, given by Tr_u, expresses the étale locality of cohomology with support at x. The diagram

(2.2.5.2)
$$\begin{tikzcd}
Rf_{\{x\}}R\Psi\, f^*K(n)[2n] \arrow[r, "\mathrm{Tr}_f"] \arrow[d, equal] & K \arrow[d, equal] \\
Rg_{\{y\}}R\Psi\, g^*K(n)[2n] \arrow[r, "\mathrm{Tr}_g"] & K
\end{tikzcd}$$

is commutative.

Scholium 2.2.6.

The compatibility (2.2.5.2) says that the trace morphism (2.2.5.1), for $\Phi = \{y\}$, is a local invariant: it depends only on the henselization of the S-scheme X at x. The same holds for $R\Psi_\eta$. The print's references to (2.3.5.2) and (2.3.5.1) are the immediately preceding (2.2.5.2) and (2.2.5.1); likewise, $\Phi \subset Y_s$ makes $\{y\}$, rather than the printed $\{x\}$, the relevant support.

2.3. Finiteness theorem (in equal characteristic zero)

Theorem 2.3.1.

Let S be a henselian trait as in 0.2.5, let Λ be a noetherian torsion ring, and let $f : X \longrightarrow S$ be a morphism of finite type. Suppose that S has equal characteristic zero. Then, for every constructible sheaf of Λ-modules F on X_η, the sheaves $R^i\Psi_{\bar\eta}(F)$ on $X_{\bar s}$ are constructible.

We begin by repeating the opening of the proof of SGA 4, XIV, 1.10. Apply SGA 4, IX, 2.14(ii). There is a resolution $F^\bullet$ of F

$$\begin{tikzcd}
 & F \arrow[d] & & \\
0 \arrow[r] & F^0 \arrow[r] & F^1 \arrow[r] & \cdots
\end{tikzcd}$$

by sheaves of the form $\bigoplus_i p'_{i*}G_i$, where $p'_i : Y_{i\eta} \longrightarrow X_\eta$ is finite and G_i is a constructible constant sheaf. We may suppose that $Y_{i\eta}$ is reduced; let $p_i : Y_i \longrightarrow X$ be the normalization of X in $Y_{i\eta}$. Using the spectral sequence of the resolution $F^\bullet$ and applying (2.1.7.1) in the elementary case of the finite morphism p_i, we see that it suffices to prove the theorem for each pair (Y_i, G_i).

We next prove the theorem when F is the constant sheaf defined by a Λ-module M. Recall that Λ is a $\mathbb{Z}/n$-algebra for a suitable n. The module M may therefore be dévissé into modules M_i over $\Lambda/\ell_i\Lambda$, where each $\ell_i \mid n$ is a suitable prime. This reduces us to the case in which Λ is a $\mathbb{Z}/\ell$-algebra, with ℓ prime. Formula (2.1.13.1) then gives

$$M \otimes_{\mathbb{Z}/\ell} R^i\Psi_{\bar\eta}(\mathbb{Z}/\ell) \xrightarrow{\sim} R^i\Psi_{\bar\eta}(M).$$

It remains to treat the constant sheaf $F = \mathbb{Z}/\ell$.

Choose a descending sequence of closed subsets

$$X_\eta = X'_0 \supset \cdots \supset X'_n$$

such that $(X'_i - X'_{i+1})_{\mathrm{red}}$ is smooth over η, and let $X_i = \overline{X'_i}$ be the closure in X. The constant sheaf $\mathbb{Z}/\ell$ is dévissé into the constant sheaves $\mathbb{Z}/\ell$ on the strata $X'_i - X'_{i+1}$, extended by zero. Thus:

Reduction 1. It is enough to prove Theorem 2.3.1 for a sheaf $j_!\mathbb{Z}/\ell$, where $j : U \hookrightarrow X_\eta$ is the inclusion of a smooth open subset.

By resolution of singularities there is a morphism $g : X_1 \longrightarrow X$ such that

(a) g is proper and induces an isomorphism $g^{-1}(U) \xrightarrow{\sim} U$;
(b) X_1 is regular and $X_1 - g^{-1}(U)$ is a normal-crossings divisor in X_1, which is a union of regular divisors D_α.

Let $j_1 : g^{-1}(U) \hookrightarrow X_1$. Then

$$Rg_* j_{1!}\mathbb{Z}/\ell = j_!\mathbb{Z}/\ell.$$

Applying (2.1.7.1) and the finiteness theorem SGA 4, XIV, 1.1, we see that it suffices to prove Theorem 2.3.1 for $(X_1, j_{1!}\mathbb{Z}/\ell)$.

There is a resolution

$$0 \longrightarrow j_{1!}\mathbb{Z}/\ell \longrightarrow \mathbb{Z}/\ell \longrightarrow \bigoplus_\alpha (\mathbb{Z}/\ell)_{D_\alpha} \longrightarrow \bigoplus_{\alpha,\beta} (\mathbb{Z}/\ell)_{D_\alpha \cap D_\beta} \longrightarrow \cdots,$$

and hence:

Reduction 2. It is enough to prove Theorem 2.3.1 for the constant sheaf $\mathbb{Z}/\ell$ on a regular S-scheme X such that X_s is a normal-crossings divisor in X.

In this final case one uses the explicit computation I, 3.3, which in turn uses the purity theorem SGA 4, XIX, 3.2 and the characteristic-zero hypothesis.

2.4. Isolated singularities

2.4.1.

Throughout this section, for simplicity, we suppose that S is a strictly henselian trait. Let Λ be a noetherian torsion ring whose torsion is prime to the residual characteristic of S, let $f : X \longrightarrow S$ be a morphism of finite type, and let F be a constructible sheaf of Λ-modules on X. A point $x \in X$ is called a smooth point of (X, F) if f is smooth at x and F is locally constant in a neighborhood of x. Let Σ denote the set of points where (X, F) is not smooth, and let Σ_s be its special fiber. We know that $R\Phi(F) = 0$ outside Σ_s.

Theorem 2.4.2.

At every isolated point x of Σ_s, the modules $R^i\Phi_{\bar{\eta}}(F)_x$ are Λ-modules of finite type and are invariant under change of trait (via 2.1.7.1).

In equal characteristic zero, these assertions were proved in 2.1.12 and 2.3.1 without assuming that x is isolated.

Let x be an isolated point of Σ_s. The theorem is local in a neighborhood of x, so we may suppose that f is proper. We then have the long exact sequence (2.1.8.9)

$$\cdots \longrightarrow H^i(X_s, F) \longrightarrow H^i(X_{\bar{\eta}}, F) \longrightarrow \mathbb{H}^i\big(X_s, R\Phi_{\bar{\eta}}(F)\big) \longrightarrow \cdots .$$

In this sequence, $H^i(X_s, F)$ and $H^i(X_{\bar{\eta}}, F)$ are invariant under change of trait and are Λ-modules of finite type (SGA 4, XIV, 1.1). The modules $\mathbb{H}^i(X_s, R\Phi_{\bar{\eta}}(F))$ therefore have the same properties. Recall the following lemma.

Lemma 2.4.3.

Let (T, Λ) be a ringed topos, let Z be a closed subtopos of T, and let $i : Z \hookrightarrow T$ be the inclusion. Then the functor

$$i_* : D(Z, \Lambda) \longrightarrow D(T, \Lambda)$$

is an equivalence from $D(Z, \Lambda)$ onto the full subcategory of $D(T, \Lambda)$ consisting of the complexes K for which $\mathcal{H}^i(K)$ is supported on Z.

Indeed, for such a complex K, the morphism $K \longrightarrow i_*i^*K$ is a quasi-isomorphism.

We may consequently regard $R\Phi_{\bar{\eta}}(F)$ as an object of

$$D^b(\Sigma_s, \Lambda) = D^b(\Sigma_s - \{x\}, \Lambda) \oplus D^b(\{x\}, \Lambda).$$

Passing to a direct summand, we find that

$$\mathbb{H}^i\big(\{x\}, R\Phi_{\bar{\eta}}(F)\big) = R^i\Phi_{\bar{\eta}}(F)_x$$

has the properties stated in 2.4.2. This completes the proof.

Remark 2.4.4.

The same proof shows that, if K is constructible in $D^+(X, \Lambda)$ and x is isolated in the support of $R\Phi_{\bar{\eta}}(K)$, then the cohomology modules of the stalk $R\Phi_{\bar{\eta}}(K)_x$ are of finite type over Λ. If, after a change of trait, x remains an isolated point of the support, these modules are invariant under that change of trait.

2.4.5.

Let $x \in X_s$ be an isolated point where (X, F) is not smooth. More generally, let $K \in \operatorname{Ob} D^b(X, \Lambda)$, and let x be an isolated point of the support of $R\Phi(K)$. Let $i : \{x\} \hookrightarrow X_s$ be the inclusion. The complex $i_*i^*R\Phi(K)$, which is isomorphic to $R\Phi(K)$ in a neighborhood of x, is a direct summand of $R\Phi(K)$ by 2.4.3. Thus, for $\sigma \in I$, the variation morphism induces

$$\operatorname{Var}(\sigma) : i_*i^*R\Phi_{\bar{\eta}}(K) \longrightarrow R\Psi_{\bar{\eta}}(K),$$

that is,

$$\operatorname{Var}(\sigma) : \big(R\Phi_{\bar{\eta}}(K)\big)_x \longrightarrow R\Gamma_{\{x\}}\big(R\Psi_{\bar{\eta}}(K)\big). \tag{2.4.5.1}$$

The printed $R\Phi(F)$ in the direct-summand sentence is $R\Phi(K)$ in this general formulation.

2.4.6.

Suppose that the support Σ of $R\Phi(K)$ consists of finitely many isolated points. For every $\sigma \in I$, the endomorphism $\sigma - 1$ of $R\Psi_{\bar\eta}(K)$ is then the composite displayed in the following diagram:
(2.4.6.1)

$$\begin{tikzcd}
R\Psi_{\bar\eta}(K) \arrow[r] \arrow[d, "\sigma-1"'] & R\Phi_{\bar\eta}(K) = \bigoplus_{x\in\Sigma} i_{x*}\big(R\Phi_{\bar\eta}(K)\big)_x \arrow[d, "\bigoplus i_{x*}\operatorname{Var}(\sigma)"'] \\
R\Psi_{\bar\eta}(K) & \bigoplus_{x\in\Sigma} i_{x*}R\Gamma_{\{x\}}R\Psi_{\bar\eta}(K) \arrow[l]
\end{tikzcd}$$

If f is proper, applying H^i to (2.4.6.1) shows that the endomorphism $\sigma - 1$ of $\mathbb{H}^i(X_{\bar\eta}, K)$ is the composite
(2.4.6.2)

$$\begin{tikzcd}
\mathbb{H}^i(X_{\bar\eta}, K) \arrow[r, "\sim"] \arrow[d, "\sigma-1"'] & \mathbb{H}^i\big(X_s, R\Psi_{\bar\eta}(K)\big) \arrow[r] & \mathbb{H}^i\big(X_s, R\Phi_{\bar\eta}(K)\big) = \bigoplus_{x\in\Sigma} R^i\Phi_{\bar\eta}(K)_x \arrow[d, "\bigoplus \operatorname{Var}(\sigma)"'] \\
\mathbb{H}^i(X_{\bar\eta}, K) & \mathbb{H}^i\big(X_s, R\Psi_{\bar\eta}(K)\big) \arrow[l, "\sim"'] & \bigoplus_{x\in\Sigma} \mathbb{H}^i_{\{x\}}\big(X_s, R\Psi_{\bar\eta}(K)\big) \arrow[l]
\end{tikzcd}$$

This describes the action of I on $\mathbb{H}^i(X_{\bar\eta}, K)$; its essential ingredients are the variation morphisms (2.4.5.1), which are local on X. We also have the long exact sequence

$$\longrightarrow \mathbb{H}^i(X_s, K) \longrightarrow \mathbb{H}^i(X_{\bar\eta}, K) \longrightarrow \bigoplus_{x\in\Sigma} R^i\Phi_{\bar\eta}(K)_x \longrightarrow \cdots . \tag{2.4.6.3}$$

The same argument applies when f is not necessarily proper, provided that (X, K) satisfies 2.1.9. The diagram (2.4.6.5) below defines a morphism

$$\operatorname{Var}(\sigma) : R\Gamma(X_{\bar\eta}, K) \longrightarrow R\Gamma_c(X_{\bar\eta}, K) \tag{2.4.6.4}$$

such that the endomorphism $\sigma - 1$ of $R\Gamma(X_{\bar\eta}, K)$ (respectively, $R\Gamma_c(X_{\bar\eta}, K)$) is the composite $\alpha \operatorname{Var}(\sigma)$ (respectively, $\operatorname{Var}(\sigma)\alpha$), where

$$\alpha : R\Gamma_c(X_{\bar\eta}, K) \longrightarrow R\Gamma(X_{\bar\eta}, K).$$

(2.4.6.5)

$$\begin{tikzcd}
R\Gamma(X_{\bar\eta},K) \arrow[r, "\sim"] \arrow[d, "\mathrm{Var}(\sigma)"'] & R\Gamma\big(X_s,R\Psi_{\bar\eta}(K)\big) \arrow[r] & \displaystyle\bigoplus_{x\in\Sigma}\big(R\Phi_{\bar\eta}(K)\big)_x \arrow[d, "\oplus"'] \\
R\Gamma_c(X_{\bar\eta},K) \arrow[r, "\sim"'] & R\Gamma_c\big(X_s,R\Psi_{\bar\eta}(K)\big) & \displaystyle\bigoplus_{x\in\Sigma}R\Gamma_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big) \arrow[l]
\end{tikzcd}\quad .$$

In this case we also have the long exact sequences

(2.4.6.6)

$$\begin{tikzcd}
\cdots \arrow[r] & \mathbb{H}^i_c(X_s,K) \arrow[r] \arrow[d] & \mathbb{H}^i_c(X_{\bar\eta},K) \arrow[r] \arrow[d] & \displaystyle\bigoplus_{x\in\Sigma}\big(R^i\Phi_{\bar\eta}(K)\big)_x \arrow[r] \arrow[d, equal] & \cdots \\
\cdots \arrow[r] & \mathbb{H}^i(X_s,K) \arrow[r] & \mathbb{H}^i(X_{\bar\eta},K) \arrow[r] & \displaystyle\bigoplus_{x\in\Sigma}R^i\Phi_{\bar\eta}(K)_x \arrow[r] & \cdots
\end{tikzcd}$$

The printed reference (2.4.4.1) in the discussion following (2.4.6.2) is the variation morphism (2.4.5.1).

2.4.6. (second occurrence in the source)

Let $X_{s(x)}$ be the henselization of X_s at x, and put

$$X^*_{s(x)} = X_{s(x)} - \{x\}.$$

The long hypercohomology exact sequences of $(X_{s(x)},\{x\})$, with coefficients in K_s, $R\Psi_{\bar\eta}(K)$, and $R\Phi_{\bar\eta}(K)$, give the following commutative diagram of long exact sequences:

(2.4.6.1)

$$\begin{tikzcd}
 & 0 \arrow[r] \arrow[d] & R^{n-1}\Phi_{\bar\eta}(K)_x \arrow[r] \arrow[d] & R^{n-1}\Phi_{\bar\eta}(K)_x \arrow[r] \arrow[d] & 0 \arrow[d] & \\
{} \arrow[r] & \mathbb{H}^{n-1}(X^*_{s(x)},K) \arrow[r, "j"] \arrow[d, equal] & \mathbb{H}^n_{\{x\}}(X_s,K) \arrow[r] \arrow[d] & \underline{H}^n(K)_x \arrow[r] \arrow[d] & \mathbb{H}^n(X^*_{s(x)},K) \arrow[r] \arrow[d, equal] & {} \\
{} \arrow[r] & \mathbb{H}^{n-1}(X^*_{s(x)},K) \arrow[r, "j", "①"'] \arrow[d] & \mathbb{H}^n_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(K)\big) \arrow[r] \arrow[d] & R^n\Psi_{\bar\eta}(K)_x \arrow[r] \arrow[d] & \mathbb{H}^n(X_{s(x)},K) \arrow[r] \arrow[d] & {} \\
 & 0 \arrow[r] & R^n\Phi_{\bar\eta}(K)_x \arrow[r, equal] & R^n\Phi_{\bar\eta}(K)_x \arrow[r] & 0 &
\end{tikzcd}$$

The image of the arrow ① is contained in the inertia invariants.

The source prints both this paragraph and the preceding paragraph as 2.4.6, and prints both principal diagrams as (2.4.6.1). The repeated numbers are retained here; this second occurrence is the one cited in Exposé XV, 2.2.6.

Bibliography

[1] J. Giraud, "Classifying topos," pp. 43–56 in *Toposes, Algebraic Geometry and Logic*, Lecture Notes in Mathematics 274, Springer-Verlag, 1971.

[2] M. Nagata, "A generalization of the imbedding problem," *J. Math. Kyoto* **3**, no. 1 (1963), pp. 89–102.

SGA 7 — EXPOSÉ XIV

Comparison with Transcendental Theory

by P. Deligne

Source contents

Introduction

To compare the theory of vanishing cycles in Exposé XIII with its transcendental analogue over $\mathbb{C}$, we proceed in two stages.

(a) We give a transcendental formalism modeled on that of Exposé XIII and prove a comparison theorem by the usual methods (already used to prove the finiteness theorem in XIII, 2.3).

(b) In the case of isolated singularities, we explain in more concrete terms the meaning of the transcendental formalism introduced here.

Besides the comparison theorem, this exposé contains a proof of the Picard–Lefschetz formula in the transcendental case.

In §4 we give a result preliminary to the Hodge theory of vanishing cycles. That section will not be used later in the seminar.

1. Transcendental formalism of vanishing cycles

1.1. The fundamental group of the punctured disc: sign conventions

We shall have to consider punctured discs D^* and their fundamental group $\mathbb{Z}$. Since this group is abelian, sign ambiguities may arise. We eliminate them by forgetting that it is abelian.

1.1.1.

Let X be a nice connected topological space and let $x \in X$. The elements of the fundamental group are homotopy classes of loops based at x. If $\alpha, \beta \in \pi_1(X, x)$ are represented by loops a, b, then $\alpha\beta$ is represented by the loop obtained by traversing b and then a.

1.1.2.

Let F be a locally constant sheaf on X. For every path $a : [0,1] \longrightarrow X$, the inverse image a^*F on $[0,1]$ is locally constant, hence constant, and therefore gives a bijection

$$a : F_{a(0)} \xleftarrow{\sim} H^0([0,1], a^*F) \xrightarrow{\sim} F_{a(1)}.$$

This bijection depends only on the homotopy class of a. Through this construction, $\pi_1(X, x)$ acts on the left on F_x. The functor $F \longmapsto F_x$ is an equivalence from the category of locally constant sheaves on X to the category of sets equipped with an action of $\pi_1(X, x)$.

1.1.3.

As x varies, the groups $\pi_1(X, x)$ are the fibers of a locally constant sheaf of groups $\Pi_1(X)$, which acts on every locally constant sheaf F. If $\pi_1(X, x)$ is abelian, then $\Pi_1(X)$ is constant; its value is denoted by $\pi_1(X)$.

1.1.4.

The group $\pi_1(X, x)$ acts on the universal covering

$$p : (\widetilde{X}, \widetilde{x}) \longrightarrow (X, x).$$

Regarded as an étalé space over X, and hence as a sheaf (the sheaf of its local sections), this covering corresponds under 1.1.2 to the set $\pi_1(X, x)$ with distinguished element e, on which $\pi_1(X, x)$ acts by left translation. The group also acts on this π_1-set by automorphisms of π_1-sets: an element $\sigma \in \pi_1(X, x)$ acts by

$$g \longmapsto g\sigma^{-1}.$$

This construction identifies

$$\pi_1(X, x) = \mathrm{Aut}_X(\widetilde{X}).$$

If F is locally constant on X, the bijection

$$H^0(\widetilde{X}, p^*F) \longrightarrow (p^*F)_{\widetilde{x}} = F_x \tag{1.1.4.1}$$

commutes with the action of $\pi_1(X, x)$: on the left the action is transport of structure through its action on $\widetilde{X}$, while on the right it is the action of 1.1.2. The equivalence of 1.1.2 may therefore be reformulated by saying that

$$F \longmapsto \big(\widetilde{X} \longmapsto H^0(\widetilde{X}, p^*F)\big)$$

is an equivalence from the category of locally constant sheaves on X to the category of functors

$$(\text{universal coverings of } X) \longrightarrow (\text{Sets}).$$

1.1.5.

If $\pi_1(X, x)$ is abelian and hence independent of x, the action of 1.1.4 of $\pi_1(X)$ on $\widetilde{X}$ is independent of the base point: $\pi_1(X)$ acts on every universal covering $\widetilde{X}$ of X.

One must bear in mind that this action, which is the only one we shall use, corresponds to the inverse of the action in 1.1.3.

1.1.6.

Let D be a disc, that is, a topological space homeomorphic to the standard disc

$$\{z \in \mathbb{C} : |z| < 1\}.$$

Let $0 \in D$ and put $D^* = D - \{0\}$. Suppose that D is equipped with the structure of an analytic manifold, and hence is oriented. The fundamental group $\pi_1(D^*)$ is then canonically isomorphic to $\mathbb{Z}$. For the standard disc, with center 0 and $t \in D^*$, the positive generator of $\pi_1(D^*, t)$ is represented by the loop

$$x \longmapsto \exp(2\pi i x)t.$$

1.1.7.

Let $z : D \longrightarrow \mathbb{C}$ be a uniformizing parameter, so that $(dz)_0 \neq 0$, and let $\widetilde{D}^*$ be a universal covering of D^*. Write $\log z$ for any function on $\widetilde{D}^*$ such that

$$\exp(\log z) = z.$$

The pair $(\widetilde{D}^*, \log z)$ is unique up to unique isomorphism.

For $\alpha \in \mathbb{C}$, put

$$z^\alpha = \exp(\alpha \log z).$$

Let T be the positive generator of $\pi_1(D^*)$, acting on $\widetilde{D}^*$ as in 1.1.5 and on functions on $\widetilde{D}^*$ by transport of structure. Then

$$T \log z = \log z + 2\pi i, \tag{1.1.7.1}$$

and

$$T z^\alpha = \exp(2\pi i \alpha) z^\alpha. \tag{1.1.7.2}$$

1.1.8.

Let D be a disc, let $0 \in D$, and put $D^* = D - \{0\}$. If $D' \subset D$ is a smaller disc containing 0, restriction to D'^* is an equivalence from the category of locally constant sheaves on D^* to the category of locally constant sheaves on D'^*. Indeed, for $t \in D'^*$,

$$\pi_1(D'^*, t) \xrightarrow{\sim} \pi_1(D^*, t).$$

Suppose now that D is equipped with a differentiable-manifold structure. Let T be the tangent space to D at 0, and put $T^* = T - \{0\}$. For $D' \subset D$ as above and an embedding $f : D' \hookrightarrow T$ satisfying $f(0) = 0$ and $(df)_0 = \mathrm{id}$, the composite

$$(\text{restriction to } D')^{-1} \circ f^*$$

is an equivalence $[f]$ from the category of locally constant sheaves on T^* to the category of locally constant sheaves on D^*. Up to canonical isomorphism it depends only on the germ of f at 0, subject to $(df)_0 = \mathrm{id}$. A path from one such germ to another defines an isomorphism of equivalences, and homotopic paths define the same isomorphism. Since the space of such germs is contractible—by the retraction to a fixed germ f_0 given by $(1-\lambda)f + \lambda f_0$—we obtain an equivalence, well defined up to unique isomorphism, from locally constant sheaves on T^* to locally constant sheaves on D^*.

1.1.9.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering. We write

$$p : \widetilde{D} \longrightarrow D$$

for the topological space over D obtained from $\widetilde{D}^*$ by adjoining a point 0, in such a way that:

(a) $\widetilde{D}^*$ is open in $\widetilde{D}$;

(b) $p : \widetilde{D} \longrightarrow D$ extends $p : \widetilde{D}^* \longrightarrow D^*$ and $p(0) = 0$;

(c) the sets $p^{-1}(U)$, as U ranges over the neighborhoods of 0 in D, form a neighborhood basis of 0 in $\widetilde{D}$.

1.1.10.

Let X be a nonsingular complex algebraic curve, let $0 \in X$, and let D be a disc neighborhood of 0 in X. The ring of germs at 0 of holomorphic functions on D which are algebraic over the field $K(X)$ of rational functions on X is the henselization $\mathcal{O}_0^h$ of the local ring $\mathcal{O}_0$ of X at 0. Let $\widetilde{D}^*$ be a universal covering of $D^* = D - \{0\}$. The field of germs at $0 \in \widetilde{D}$ of meromorphic functions on $\widetilde{D}^*$ which are algebraic over $K(X)$ is an algebraic closure $\overline{K}$ of $K(X)$, and hence also of the fraction field K_0^h of $\mathcal{O}_0^h$.

The group $\pi_1(D^*)$ of D-automorphisms of $\widetilde{D}^*$ (1.1.5) acts by transport of structure on $\overline{K}$, giving

$$\begin{array}{ccc} \pi_1(D^*) & \longrightarrow & \mathrm{Gal}(\overline{K}/K_0^h) \\ \| & & \| \\ \mathbb{Z} & & \widehat{\mathbb{Z}}(1) \end{array} \qquad . \tag{1.1.10.1}$$

It follows from (1.1.7.2) that this diagram is compatible with the natural inclusion, over $\mathbb{C}$, of $\mathbb{Z}$ into $\widehat{\mathbb{Z}}(1)$. This inclusion is the composite of $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ and the isomorphism

$$\widehat{\mathbb{Z}} \xrightarrow{\sim} \widehat{\mathbb{Z}}(1),$$

which is the inverse limit of the isomorphisms

$$\mathbb{Z}/n \longrightarrow \mu_n, \qquad k \longmapsto \exp\left(\frac{2\pi i}{n} k\right).$$

1.1.11.

Every sheaf F on $\mathrm{Spec}(K_0^h)$ defines a locally constant sheaf on D^*. We use the equivalence of 1.1.4 and, to each universal covering $\widetilde{D}^*$ of D^*, associate $F_{\overline{K}}$ for the field $\overline{K}$ constructed in 1.1.10.

1.2. Sheaves of sets

1.2.1.

Let D be a disc (1.1.6), let $0 \in D$, and put $D^* = D - \{0\}$. We denote by $\mathcal{D}^*$ the topos of locally constant sheaves of sets on D^*. By 1.1.8 this topos does not "change" if D is replaced by a smaller disc D' containing 0, nor, when D is endowed with the structure of a differentiable manifold, if D is replaced by its tangent space at 0.

For $t \in D^*$, the functor $F \longmapsto F_t$ identifies this topos with the topos of sets equipped with an action of $\pi_1(D^*)$ (1.1.2). By 1.1.4, the functor

$$F \longmapsto \left(\widetilde{D}^* \longmapsto H^0(\widetilde{D}^*, p^*F)\right)$$

identifies it with the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\text{Sets}),$$

and these identifications are related by (1.1.4.1).

1.2.2.

Let F be a sheaf on D whose restriction to D^* is locally constant. There is a unique morphism of sheaves

$$\underline{F_0} \xrightarrow{\alpha} F$$

whose map on the stalk at 0 is the identity. As explained in XIII, 0.1 (with $X_0 = (P^t)$), the functor

$$F \longmapsto (F_0, F|_{D^*}, \alpha)$$

identifies these sheaves with triples consisting of a set F_0, a locally constant sheaf F_η on D^*, and a morphism

$$\alpha : (\text{constant sheaf with value } F_0 \text{ on } D^*) \longrightarrow F_\eta.$$

1.2.3.

We denote by $\mathcal{D}$ the topos of sheaves of sets on D whose restriction to D^* is locally constant. The inclusion of $\mathcal{D}$ into the topos of sheaves on D is the inverse-image functor of a morphism of topoi $D \longrightarrow \mathcal{D}$, which fits into the commutative diagram

$$\begin{tikzcd} \{0\} \arrow[r, hook] \arrow[d, equal] & D \arrow[d, "\beta"] & D^* \arrow[l, hook'] \arrow[d, "\beta"] \\ \{0\} \arrow[r, hook] & \mathcal{D} & \mathcal{D}^* \arrow[l, hook'] \end{tikzcd} .$$

As in 1.1.1, one checks that this topos does not change when D is replaced by a smaller disc or by the tangent space to D at 0.

1.2.4.

Let Y be a topological space. The product topos $Y \times \mathcal{D}$ has the following three descriptions (for their equivalence, see XIII, 0.1).

(a) $Y \times \mathcal{D}$ is the topos of sheaves F on $Y \times D$ satisfying

$(*)$ D^* is covered by open subsets U such that $F|_{Y \times U}$ is the inverse image of a sheaf on Y.

(b) Let $t \in D^*$. The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_t, α) in which

α) F_0 is a sheaf on Y;
β) F_t is a sheaf on Y equipped with an action of $\pi_1(D^*)$;
γ) $\alpha : F_0 \longrightarrow F_t$ is a morphism whose image is contained in the invariants.

(c) The topos $Y \times \mathcal{D}$ is the topos of triples (F_0, F_η, α) in which

α') F_0 is a sheaf on Y;
β') F_η is a functor, with values denoted by $F_{\bar{\eta}}$,

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y);$$

γ') α is a morphism from F_0 (regarded as the constant functor with value F_0 from the universal coverings of D^* to the sheaves on Y) to F_η.

It is immaterial here whether this topos is literally the product topos of Y and $\mathcal{D}$. Likewise, we define $Y \times \mathcal{D}^*$, at our choice, as

(a^*) the topos of sheaves on $Y \times D^*$ satisfying $(*)$;
(b^*) for $t \in D^*$, the topos of sheaves F_t on Y equipped with an action of $\pi_1(D^*)$;
(c^*) the topos of functors

$$(\text{universal coverings of } D^*) \longrightarrow (\text{sheaves on } Y).$$

1.2.5.

The topos $Y \times \mathcal{D}$ has the same functoriality properties in Y as the topoi $Y \times_S S$ of Exposé XIII, and we shall not list them. To fix notation, we merely record the diagram

$$\begin{tikzcd}
Y \arrow[d] & Y \arrow[r, hook, "i"] \arrow[d] \arrow[dl, "\mathrm{sp}"'] & Y \times \mathcal{D} \arrow[d] & Y \times \mathcal{D}^* \arrow[l, hook', "j"'] \arrow[d] \\
(P^t) & \{0\} \arrow[r, hook, "i"] & \mathcal{D} & \mathcal{D}^* \arrow[l, hook', "j"']
\end{tikzcd}$$

1.2.6.

Let K be a complex of sheaves of abelian groups on $Y \times \mathcal{D}$. As in XIII, 1.4.2, one obtains from K a distinguished triangle in $K(Y \times \mathcal{D}^*)$:

$$\begin{tikzcd}
& \Phi(K) \arrow[dl] & \\
\mathrm{sp}^* K_0 \arrow[rr] & & K_\eta \arrow[ul]
\end{tikzcd}$$

This construction passes to the derived category and defines a functor

$$D(Y \times \mathcal{D}) \longrightarrow (\text{distinguished triangles of } D(Y \times \mathcal{D}^*)).$$

Suppose that D is oriented (for example, complex analytic), and let T be the positive generator of $\pi_1(D^*)$ (1.1.6). For $K \in \mathrm{Ob}\, D(Y \times \mathcal{D})$, we define, as in XIII, 1.4.3, the variation $V = \mathrm{Var}(T)$,

$$V : \Phi(K) \longrightarrow K_\eta.$$

The endomorphism $T - \mathrm{Id}$ of K_η is the composite

$$T - \mathrm{Id} : K_\eta \longrightarrow \Phi(K) \longrightarrow K_\eta.$$

1.2.7.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. We shall define a functor

$$\Psi : (\text{sheaves on } X) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}).$$

Let $\widetilde{D}^*$ be a universal covering of D^*, let $\widetilde{D}$ be as in 1.1.9, and put

$$\overline{X} = X \times_D \widetilde{D}, \qquad X^* = X - X_0 = X \times_D D^*, \qquad \overline{X}^* = \overline{X} - X_0 = X \times_D \widetilde{D}^*.$$

There is a commutative diagram

(1.2.7.1)
$$\begin{tikzcd}
X_0 \arrow[r, hook, "\bar{i}"] \arrow[d, equal] & \overline{X} \arrow[d, "p"] & \overline{X}^* \arrow[l, hook', "\bar{j}"'] \arrow[d, "p'"] \\
X_0 \arrow[r, hook, "i"] & X & X^* \arrow[l, hook', "j"']
\end{tikzcd}$$

cartesian over the diagram

(1.2.7.2)
$$\begin{tikzcd}
\{0\} \arrow[r, hook, "\bar{i}"] \arrow[d, equal] & \widetilde{D} \arrow[d, "p"] & \widetilde{D}^* \arrow[l, hook', "\bar{j}"'] \arrow[d, "p'"] \\
\{0\} \arrow[r, hook, "i"] & D & D^* \arrow[l, hook', "j"']
\end{tikzcd}$$

Editorial note. In (1.2.7.1) the printed top-right node lacks the overline. The preceding definition and the map p' force that node to be $\overline{X}^*$, which is the notation used here.

For a sheaf F on X, we set (with the notation of 1.2.4(c))

$$\Psi(F) = \big(i^*F,\ \overline{i}^*\overline{j}_*((jp')^*F),\ \alpha\big),$$

where

$$\begin{aligned}\alpha : i^*F = \overline{i}^*p^*F &\longrightarrow \overline{i}^*\overline{j}_*((jp')^*F)\\ &= \overline{i}^*\overline{j}_*\overline{j}^*p^*F\end{aligned}$$

is obtained, after applying $\overline{i}^*$, from the adjunction map $\mathrm{Id} \longrightarrow \overline{j}_*\overline{j}^*$.

We also define a functor

$$\Psi_\eta : (\text{sheaves on } X^*) \longrightarrow (\text{sheaves on } X_0 \times \mathcal{D}^*)$$

by

$$\Psi_\eta(F) = \overline{i}^*\overline{j}_*(p'^*F).$$

1.2.8.

The functoriality properties of Ψ are the same as those of its analogue studied in XIII, 1.3.5–1.3.9.

1.3. The functor $R\Psi$

1.3.1.

Let D be a disc, let $0 \in D$, put $D^* = D - \{0\}$, let $f : X \longrightarrow D$ be a continuous map, and put $X_0 = f^{-1}(0)$. The functors $R\Psi$ and $R\Psi_\eta$ are the derived functors of Ψ and Ψ_η:

$$R\Psi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}),$$

$$R\Psi_\eta : D^+(X^*) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

We denote by $R\Phi$ the composite functor $\Phi \circ R\Psi$:

$$R\Phi : D^+(X) \longrightarrow D^+(X_0 \times \mathcal{D}^*).$$

The properties of these functors are exactly analogous to those of the functors with the same names in XIII, 2.1. Their proofs are much more elementary, since in the topological setting the base-change theorems for a proper morphism or for a smooth morphism (that is, one locally isomorphic to $\mathbb{R}^n \times S \longrightarrow S$) are easy to prove. The minor changes required are as follows.

1.3.2.

The functor $Rf_!$ is defined only for a separated morphism between locally compact spaces, and is the derived functor of the direct-image functor with proper supports $f_!$.

1.3.3.

For a proper (and separated) morphism $f : X \longrightarrow D$, the fundamental isomorphism (XIII, (2.1.8.3)) must be rewritten as follows. Let $p : \widetilde{D}^* \longrightarrow D^*$ be a universal covering of D^*. For every neighborhood U of 0 in D, put

$$\widetilde{U}^* = p^{-1}(U), \qquad X_{\widetilde{U}^*} = X \times_D \widetilde{U}^*.$$

For $K \in \operatorname{Ob} D^+(f^{-1}(D^*))$, one has

$$\varinjlim_{U} \mathbb{H}^i(X_{\widetilde{U}^*}, K) \xrightarrow{\sim} \mathbb{H}^i(X_0, R\Psi_{\bar\eta}(K)). \tag{1.3.3.1}$$

This isomorphism is especially useful when $X^* = f^{-1}(D^*)$ is a topological fiber bundle over D^*. As a disc $U \subset D$ shrinks to 0, the Leray spectral sequence for $X_{\widetilde{U}^*} \longrightarrow \widetilde{U}^*$ then shows that, for every locally constant sheaf Λ on X^* and every $t \in \widetilde{U}^*$,

$$H^i(X_{\widetilde{U}^*}, \Lambda) \xrightarrow{\sim} H^i(X_t, \Lambda).$$

Thus, for $t \in D^*$,

$$H^i(X_t, \Lambda) \xrightarrow{\sim} H^i(X_0, R\Psi_t(\Lambda)), \tag{1.3.3.2}$$

provided that X/D is proper, X^* is a fiber bundle, and Λ is locally constant.

In practice, to verify that X^* is a fiber bundle over D^* one uses Thom's isotopy theorem, one form of which is stated below. For the proof of this theorem, none of whose consequences will be used in the subsequent exposés, we refer to Whitney and Mather.

1.3.4. Preliminary.

Let $f : X \longrightarrow Y$ be a smooth morphism (by definition, one with no critical point) of differentiable manifolds. A connection ∇ on X/Y is a vector subbundle of the tangent bundle of X complementary to $\ker(df)$. For a vector field u on Y, there is a unique vector field $\nabla(u)$ on X which lifts u and lies in this subbundle. The connection is called integrable if

$$\nabla([u, v]) = [\nabla(u), \nabla(v)].$$

When f is proper, an integrable connection gives local trivializations over Y of X/Y,

$$\alpha : f^{-1}(U) \xrightarrow{\sim} X_0 \times U,$$

compatible with projection to U and satisfying $d\alpha(\nabla(u)) = (0, u)$. When f is not proper, the same result holds under suitable conditions at infinity; for example, it is enough that $\nabla(u)g$ vanish for a suitable exhaustion function g.

1.3.5. Thom's isotopy theorem.

Let

$$D = \{z \in \mathbb{C} : |z| < 1\}, \qquad D^* = D - \{0\},$$

and let $f : X \longrightarrow D$ be a proper morphism of analytic spaces. After replacing D by a smaller disc and X by its inverse image, there exists a stratification Σ of $X^* = f^{-1}(D^*)$ having the following properties:

(a) *Σ is a filtration*

$$X^* = F_0 \supset F_1 \supset \cdots \supset F_n \supset F_{n+1} = \varnothing$$

of X^* by closed subsets such that the closure $\overline{F}_i$ of F_i in X is an analytic subspace of X.

(b) *The open strata $F_i - F_{i+1}$ are smooth over D^*.*

(c) *Σ satisfies Whitney's conditions (a) and (b).*

For such a stratification Σ, there are integrable connections ∇_i on the $F_i - F_{i+1}$ such that

(i) *locally on D^*, ∇_i defines a trivialization of $F_i - F_{i+1}$;*

(ii) *these trivializations glue to give local trivializations of X^*/D^*.*

Finally, if (Y_α) is a finite family of closed analytic subspaces of X, then, after shrinking D further, one can choose Σ so that every Y_α is a union of connected components of open strata $(F_i - F_{i+1})$. A connection ∇ as above then locally trivializes $(X, (Y_\alpha))$.

1.3.6.

The isomorphism (1.3.3.2) remains valid for hypercohomology with values in a complex of sheaves K when X is proper over D and the following condition holds:

($*$) For every $i \in \mathbb{Z}$, the space D^* is covered by open subsets U for which there exist an isomorphism compatible with projection to U,

$$\varphi : f^{-1}(U) \xrightarrow{\sim} Y \times U,$$

and a sheaf G on Y such that, on $f^{-1}(U)$, $\underline{H}^i(K)$ is isomorphic to $\varphi^* \operatorname{pr}_1^* G$.

Under this hypothesis, the triangle of 1.2.6 gives a long exact sequence

$$\cdots \longrightarrow \mathbb{H}^i(X_0, K) \longrightarrow \mathbb{H}^i(X_t, K) \longrightarrow \mathbb{H}^i(X_0, R\Phi_t(K)) \longrightarrow \cdots . \tag{1.3.6.1}$$

Editorial note. The print cites the triangle "1.2.1" here; the distinguished triangle used is the one constructed in 1.2.6.

By Thom's theorem, condition ($*$) holds (after shrinking D) whenever X admits a complex analytic stratification such that all the $\underline{H}^i(K)$ are locally constant on every open stratum.

1.4. A special case

This section will not be used later in the seminar. It formalizes a construction sketched in the Introduction to Exposé XIII.

1.4.1.

Let D be an oriented disc, let $0 \in D$, and put $D^* = D - \{0\}$. A <u>topological space</u> $\mathcal{X}^*$ over $\mathcal{D}^*$ is a functor

$$\text{(universal coverings of } D^*) \longrightarrow \text{(topological spaces)}.$$

After a universal covering $\widetilde{D}^*$ has been chosen, this is a topological space X_η equipped with an action of $\pi_1(D^*)$; when $\widetilde{D}^*$ is the universal covering of (D^*, t), we denote it by $\mathcal{X}_t^*$.

A <u>sheaf</u> on $\mathcal{X}^*$ is a functor

$$\text{(universal coverings of } D^*) \longrightarrow \text{(topological spaces equipped with a sheaf)},$$

or, equivalently, a $\pi_1(D^*)$-equivariant sheaf on X_η.

To $\mathcal{X}^*$ we associate the topological space

$$\beta^* \mathcal{X}^* = X_\eta \times \widetilde{D}^* / \pi_1(D^*)$$

over D^*. Every sheaf F on $\mathcal{X}^*$ descends to a sheaf $\beta^* F$ on $\beta^* \mathcal{X}^*$, giving a commutative diagram of topoi

$$\begin{tikzcd} \beta^*\mathcal{X}^* \arrow[r] \arrow[d] & \mathcal{X}^* \arrow[d] \\ D^* \arrow[r, "\beta"] & \mathcal{D}^*. \end{tikzcd}$$

1.4.2.

A <u>topological space</u> $\mathcal{X}$ over $\mathcal{D}$ consists of

(a) a topological space $\mathcal{X}^*$ over $\mathcal{D}^*$;

(b) a topological space X_0;

(c) a $\pi_1(D^*)$-invariant morphism
$$\mathrm{sp} : X_\eta \longrightarrow X_0,$$
where $\pi_1(D^*)$ acts trivially on X_0.

A sheaf on $\mathcal{X}$ consists of a sheaf F_0 on X_0, a sheaf on $\mathcal{X}^*$ given by F_η, and an equivariant morphism
$$\mathrm{sp} : \mathrm{sp}^* F_0 \longrightarrow F_\eta \qquad (\text{or } F_0 \longrightarrow \mathrm{sp}_* F_\eta).$$
There is a commutative diagram
$$\begin{tikzcd} X_0 \arrow[r, hook] \arrow[d] & \mathcal{X} \arrow[d] & \mathcal{X}^* \arrow[l, hook'] \arrow[d] \\ \{0\} \arrow[r, hook] & \mathcal{D} & \mathcal{D}^* \arrow[l, hook'] \end{tikzcd}$$
and, in addition, a morphism of topoi
$$\Psi : \mathcal{X} \longrightarrow X_0 \times \mathcal{D}.$$
It is obtained from the functoriality of the preceding construction, applied to
$$(X_\eta, X_0, \mathrm{sp}) \longrightarrow (X_0, X_0, \mathrm{Id}),$$
and satisfies
$$\Psi(F_0, F_\eta, \mathrm{sp}) = (F_0, \mathrm{sp}_* F_\eta, \mathrm{sp}).$$
This morphism induces
$$\Psi_\eta : \mathcal{X}^* \longrightarrow X_0 \times \mathcal{D}^*.$$

1.4.3.

To a topological space $\mathcal{X}$ over $\mathcal{D}$ we associate a topological space $X = \beta^*\mathcal{X}$ over D as follows. To $X^* = \beta^*\mathcal{X}^*$ we adjoin the fiber X_0 over 0. A fundamental system of neighborhoods of $x_0 \in X_0$ is formed by the sets
$$V \cup \big(\mathrm{sp}^{-1}(V) \times \widetilde{U}^*/\pi_1(D^*)\big),$$
where V is a neighborhood of x_0 in X_0, U is a neighborhood of 0 in D, and $\widetilde{U}^* = U \times_D \widetilde{D}^*$. Let j (respectively i) denote the inclusion of X^* (respectively X_0) into X. For a sheaf F on $\mathcal{X}^*$, one has
$$i^* j_* \beta^* F = (\mathrm{sp}_* F_\eta)^{\pi_1(D^*)}.$$
It follows that there is a commutative diagram of topoi
$$\begin{tikzcd} X \arrow[r, "\beta"] \arrow[d] & \mathcal{X} \arrow[d] \\ D \arrow[r, "\beta"] & \mathcal{D} \end{tikzcd}$$
such that, for every sheaf F on $\mathcal{X}$, there is an isomorphism
$$\Psi(F) \xrightarrow{\sim} \Psi(\beta^* F). \tag{1.4.3.1}$$

Proposition 1.4.4.

The map (1.4.3.1) derives to an isomorphism

$$R\Psi \xrightarrow{\sim} R\Psi \circ \beta^*$$

between functors from $D^+(\mathcal{X})$ to $D^+(X_0 \times \mathcal{D})$.

Indeed, let $\overline{j}$ be the inclusion of

$$\overline{X}^* = X \times_D \widetilde{D}^* \simeq X_\eta \times \widetilde{D}^*$$

into $\overline{X} = X \times_D \widetilde{D}$. The proposition reduces to checking that, for every flabby sheaf of abelian groups F on X_η, the sheaf $\mathrm{pr}_1^* F$ is acyclic for $\overline{j}_*$. This in turn reduces (with $B = \widetilde{D}^*$) to the following familiar assertion, proved by a homotopy argument. If F is a sheaf of abelian groups on a space Y and B is a disc, then

$$H^i(Y, F) \xrightarrow{\sim} H^i(Y \times B, \mathrm{pr}_1^* F).$$

1.4.5.

Proposition 1.4.4 gives a method for computing $R\Psi$ for sheaves on X of the form $\beta^* F$, reducing the computation of $R\Psi$ essentially to that of $R\,\mathrm{sp}_*$.

1.4.6.

Later we shall consider the case in which X_η is locally compact and X_0 is obtained from X_η by contracting a compact subset $V \subset X_\eta$ to a point 0, with sp the contraction map and the action of $\pi_1(D^*)$ trivial outside V. Proper base change for sp then gives, for $K \in \mathrm{Ob}\, D^+(\mathcal{X}^*)$,

$$R\Psi_\eta(\beta^* K)_0 = R\Gamma(V, K) \tag{1.4.6.1}$$

and

$$R\Gamma_{\{0\}} R\Psi_\eta(\beta^* K) = R\Gamma_V(X_\eta, K), \tag{1.4.6.2}$$

the latter being cohomology with supports.

For K on $\mathcal{X}$, suppose that

$$\mathrm{sp} : \mathrm{sp}^* K_0 \longrightarrow K_\eta$$

is a quasi-isomorphism outside V. Then $R\Phi(K) = 0$ outside 0. In particular, for a constant sheaf Λ on $\mathcal{X}$,

$$R^i\Phi(\Lambda) = \widetilde{H}^i(V, \Lambda), \qquad \text{supported at } 0,$$

where the right-hand side is reduced cohomology.

1.4.7.

The case of interest to us is that in which X_η is a differentiable manifold and $V \subset X_\eta$ is a submanifold with boundary and interior V°. Then $X - \{0\}$ is a differentiable manifold obtained by gluing $\beta^* X_\eta$ to the constant bundle over D with fiber $X_\eta - V$. The map

$$H^i_c(V^\circ, \Lambda) \xrightarrow{\sim} H^i_V(X_\eta, \Lambda) \tag{1.4.7.1}$$

is an isomorphism. These groups are also denoted by $H^i(V \bmod \partial V, \Lambda)$. The isomorphism (1.4.7.1) defines an isomorphism

$$R\Gamma_c(V^\circ, \Lambda) \xrightarrow{\sim} R\Gamma_{\{0\}} R\Psi_\eta(\Lambda), \tag{1.4.7.2}$$

and the variation is defined, by the procedure of XIII, 2.4.5 and through (1.4.7.2), as a morphism whose source is the reduced cohomology of V:

$$\mathrm{Var} : R\widetilde{\Gamma}(V, \Lambda) \longrightarrow R\Gamma_c(V^\circ, \Lambda). \tag{1.4.7.3}$$

The automorphism induced by T on the pair $(V, \partial V)$ is again denoted by T, and its restriction ∂T to ∂V is the identity. The variation (1.4.7.3) should coincide with the classical variation morphism (3.1.9) attached to such an automorphism T. A statement of this kind will be proved in 3.1.10.

2. The comparison theorem

2.1.

For every scheme X of finite type over $\mathbb{C}$, we denote by X_{cl} the space $X(\mathbb{C})$ endowed with its usual topology.

Let S be a smooth curve over $\mathbb{C}$, that is, a smooth separated scheme of finite type over $\mathbb{C}$, pure of dimension 1. Let $0 \in S(\mathbb{C})$, let $f : X \longrightarrow S$ be a separated S-scheme of finite type, and let Λ be a noetherian torsion ring (for example $\mathbb{Z}/n\mathbb{Z}$). Put $X_0 = f^{-1}(0)$.

2.2.

We denote by S^h the henselized henselian trait of S at 0, and by η the generic point of S^h. There is a functor

$$\begin{aligned}\Psi : (\text{sheaves on } X) &\xrightarrow{\text{restriction}} (\text{sheaves on } X \times_S S^h)\\ &\xrightarrow{\Psi} (\text{sheaves on } X_0 \times S^h).\end{aligned}$$

Its derived functor is the composite

$$R\Psi : D(X, \Lambda) \longrightarrow D(X \times_S S^h, \Lambda) \xrightarrow{R\Psi} D(X_0 \times S^h, \Lambda).$$

2.3.

Let $D \subset S_{\mathrm{cl}}$ be a disc containing 0. The composite functor Ψ_{cl} (or simply Ψ)

$$\begin{aligned}\Psi_{\mathrm{cl}} : (\text{sheaves on } X_{\mathrm{cl}}) &\xrightarrow{\text{restriction}} (\text{sheaves on } X_{\mathrm{cl}} \times_{S_{\mathrm{cl}}} D)\\ &\xrightarrow{\Psi} (\text{sheaves on } (X_0)_{\mathrm{cl}} \times \mathcal{D})\end{aligned}$$

is independent, up to unique isomorphism, of the choice of D; recall that $\mathcal{D}$ does not change when D is replaced by a smaller disc. Its derived functor is the composite

$$R\Psi_{\mathrm{cl}} : D(X_{\mathrm{cl}}, \Lambda) \longrightarrow D(f_{\mathrm{cl}}^{-1}(D), \Lambda) \xrightarrow{R\Psi} D((X_0)_{\mathrm{cl}} \times \mathcal{D}, \Lambda).$$

2.4.

Likewise, we define

$$\begin{aligned}R\Psi_\eta &: D(X - X_0, \Lambda) \longrightarrow D(X_0 \times \eta, \Lambda),\\ R\Psi_{\eta\,\mathrm{cl}} &: D((X - X_0)_{\mathrm{cl}}, \Lambda) \longrightarrow D((X_0)_{\mathrm{cl}} \times \mathcal{D}^*, \Lambda).\end{aligned}$$

2.5.

The construction of 1.1.11 defines a morphism of topoi from $\mathcal{D}^*$ to η, which extends to

$$\varepsilon : \mathcal{D} \longrightarrow S^h.$$

Together with $\varepsilon : (X_0)_{\mathrm{cl}} \longrightarrow X_0$ (SGA 4, XI, 4), this morphism defines

$$\varepsilon : (X_0)_{\mathrm{cl}} \times \mathcal{D} \longrightarrow X_0 \times S^h.$$

If $\widetilde{D}^*$ is a universal covering of D^* and $k(\bar{\eta})$ is the corresponding algebraic closure (1.1.10) of $k(\eta)$, then

$$\varepsilon^*(F_0, F_\eta, \alpha) = \big(\varepsilon^* F_0, \varepsilon^* F_{\bar{\eta}}, \varepsilon^* \alpha\big).$$

2.6.

Let $\widetilde{D}^*$ be a universal covering of D^*, let $k(\bar{\eta})$ be the corresponding algebraic closure of $k(\eta)$, and let $\overline{S}$ be the normalization of S^h in $k(\bar{\eta})$. Write $\overline{S}$ as an inverse limit of affine S-schemes S_i. For every i, after replacing D by a smaller disc D_i, there is a natural morphism of S_{cl}-spaces

$$a_i : \widetilde{D}_i \longrightarrow (S_i)_{\mathrm{cl}}.$$

Let $j_i : U_i \hookrightarrow S_i$ be the open complement in S_i of the image 0 of the closed point of $\overline{S}$, and let $i : 0 \hookrightarrow S_i$.

Let F be a sheaf on X, and put $F_{\mathrm{cl}} = \varepsilon^* F$. We again denote by j_i the inclusion of $X \times_S U_i$ into $X \times_S S_i$, by i the inclusion of X_0 into $X \times_S S_i$, and by F the inverse image of F on $X \times_S S_i$. Then

$$\Psi(F) = \left(i^* F,\ \varinjlim_i i^* j_{i*} j_i^* F,\ \varinjlim_i \varphi_i \right).$$

The functors ε^* and inverse image commute. The morphism

$$\varepsilon^* j_{i*} \longrightarrow (j_{i,\mathrm{cl}})_* \varepsilon^*$$

therefore gives

$$c_i : (i^* j_{i*} j_i^* F)_{\mathrm{cl}} \longrightarrow i^* j_{i*} j_i^* (F_{\mathrm{cl}}).$$

Let $\bar{j}$ (respectively $\bar{j}_i$) be the inclusion of $\widetilde{D}^*$ (respectively $\widetilde{D}_i^*$) into $\widetilde{D}$ (respectively $\widetilde{D}_i$), as well as the induced inclusions

$$X \times_S \widetilde{D}^* \hookrightarrow X \times_S \widetilde{D}, \qquad X \times_S \widetilde{D}_i^* \hookrightarrow X \times_S \widetilde{D}_i.$$

The adjunction map $\mathrm{Id} \longrightarrow a_{i*} a_i^*$ defines

$$i^* j_{i*} j_i^* F_{\mathrm{cl}} \longrightarrow i^* \bar{j}_{i*} \bar{j}_i^* \overline{F}_{\mathrm{cl}} = i^* \bar{j}_* \bar{j}^* \overline{F}_{\mathrm{cl}}.$$

Passing to the limit gives

$$c_\eta : \Psi_{\bar{\eta}}(F)_{\mathrm{cl}} \longrightarrow \Psi_{\mathrm{cl}}(F_{\mathrm{cl}}).$$

This morphism defines the <u>comparison morphism</u>

(2.6.1) $$c : \varepsilon^* \Psi(F) \longrightarrow \Psi_{\mathrm{cl}}(\varepsilon^* F).$$

Similarly, one defines

(2.6.2) $$c_\eta : \varepsilon^* \Psi_\eta(F) \longrightarrow \Psi_\eta(\varepsilon^* F).$$

2.7.

The morphisms (2.6.1) and (2.6.2) derive to

$$c : \varepsilon^* R\Psi \longrightarrow R\Psi_{\mathrm{cl}} \circ \varepsilon^*$$

and

$$c_\eta : \varepsilon^* R\Psi_\eta \longrightarrow R\Psi_{\eta\,\mathrm{cl}} \circ \varepsilon^*.$$

Theorem 2.8.

Let $K \in \mathrm{Ob}\, D(X, \Lambda)$ (respectively, $K \in \mathrm{Ob}\, D(X - X_0, \Lambda)$) be a complex of sheaves of Λ-modules whose cohomology sheaves $\underline{H}^i(K)$ are constructible. Then the comparison morphism

$$c : \varepsilon^* R\Psi(K) \longrightarrow R\Psi_{\mathrm{cl}}(\varepsilon^* K)$$

(respectively,

$$c : \varepsilon^* R\Psi_\eta(K) \longrightarrow R\Psi_{\eta\,\mathrm{cl}}(\varepsilon^* K))$$

is an isomorphism.

It suffices to prove the assertion for $R\Psi_\eta$. Moreover, both sides of 2.7 are triangulated functors of finite cohomological dimension, so it is enough to treat the case in which K is a single sheaf F placed in degree 0. The proof is exactly analogous to that of the finiteness theorem XIII, 2.3.1. Since $R\Psi$ and $R\Psi_{\mathrm{cl}}$ have the same variance properties, the dévissages used there (with permission to shrink S) reduce the problem to the case in which X is regular, $f^{-1}(0)$ is a normal crossings divisor in X, and $F = \mathbb{Z}/\ell\mathbb{Z}$. One then compares Exposé I, 3.3 with its transcendental analogue.

3. Isolated singularities

3.1. The manifold of vanishing cycles

3.1.1.

Let

$$D = \{z \in \mathbb{C} \mid |z| < 1\},$$

let X be an analytic space with $0 \in X$, and let $f : X \to D$ be a morphism such that $f(0) = 0$. We suppose that f is smooth away from 0 (in particular, X is smooth away from 0). We intend to study the vanishing cycles near 0; hence there is no loss of generality in supposing that X is embedded in $\mathbb{C}^n \times D \subset \mathbb{C}^{n+1}$ compatibly with the projection to D, and in such a way that 0 maps to 0.

3.1.2.

Let $\delta(x)$ be a real-analytic function on X (the restriction to X of a real-analytic function on $\mathbb{C}^{n+1}$) such that, for $x \in X_0 = f^{-1}(0)$,

(a) $\delta(x) \geq 0$;

(b) $\delta(x) = 0$ if and only if $x = 0$.

For example, we may take δ to be the square of the Euclidean distance from x to 0 in $\mathbb{C}^{n+1}$. The critical locus of the restriction of δ to X_0 is real analytic and therefore, in a suitable neighborhood of 0, has only finitely many connected components. Replacing X by an open neighborhood of 0 and applying Sard's theorem, we may consequently obtain the following situation:

(c) for a suitable $r > 0$, the space $X_0 \cap \delta^{-1}([0, r])$ is compact;

(d) on X_0, δ has no critical value in $(0, r]$.

After shrinking X further, we then have:

(e) there exists a C^∞ function $\tau(s)$, $0 \leq s \leq r$, with $\tau(0) = 0$ and $\tau'(s) > 0$ for $s > 0$, such that, for $0 < s \leq r$ and $0 \leq |t|^2 \leq \tau(s)$, the fiber $X_t = f^{-1}(t)$ is transverse to the real hypersurface $\delta(x) = s$ in X, and the box

$$B_{t,d} = \{x \in X \mid \delta(x) \leq d \text{ and } |f(x)|^2 < t\}$$

is proper over the open disc D_t of radius $t^{1/2}$.

Let $L = X_0 \cap \delta^{-1}(r)$. By (c) and (d), there is a homeomorphism from $X_0 \cap \delta^{-1}([0, r])$ to the cone

$$L \times [0, r] / \big(L \times \{0\} \text{ contracted to a point}\big)$$

which transforms δ into the second coordinate and is C^∞ away from the origin.

3.1.3.

Let $d \leq r$ and $t \leq \tau(d)$. Away from 0, $B_{t,d}$ is a manifold with corners, and the projection $f : B_{t,d} \to D_t$ is a fibration. In particular:

(a) over $D_t^* = D_t - \{0\}$, $B_{t,d}$ induces a C^∞ bundle of compact manifolds with boundary;

(b) the boundary $\delta(x) = d$ of this bundle extends to a C^∞ bundle $\partial B_{t,d}$ of compact manifolds over D_t.

Let $y \in D_t^*$. We call the manifold with boundary $V = f^{-1}(y)$ the manifold of vanishing cycles. Since the disc is contractible, the bundle $\partial B_{t,d}$ over D_t is isomorphic to the constant bundle with fiber ∂V, and the space of bundle isomorphisms

$$\alpha : \partial B_{t,d} \xrightarrow{\sim} \partial V \times D_t$$

that restrict to the identity on the fiber over y is contractible. Choose one, denoted by α. Lift the vector field $\partial/\partial\theta$ on $S_{|y|}$, the circle of radius $|y|$, to a vector field u on $f^{-1}(S_{|y|})$ which is compatible with α on the boundary, so that

$$d\alpha(u) = (0, \partial/\partial\theta).$$

The space of such lifts is contractible. Integrating u once around the circle in the positive direction defines a diffeomorphism T of V onto itself which is the identity on ∂V; this is the monodromy transformation. It is well defined up to isotopy relative to ∂V.

In a sense whose precise formulation is left to the reader, (V, T) is independent of the choices of t and d. I do not know whether it is independent of the choice of δ. This is nevertheless the case if we restrict δ to be the restriction to X of a real-analytic function g on $\mathbb{C}^{n+1}$ such that, on the Zariski tangent space of X_0 at 0, one has $dg = 0$ and $d^2g > 0$ (positive definite). Indeed, for δ and δ' of this kind, r can be chosen as in 3.1.2(c),(d), uniformly for every $\lambda\delta + (1 - \lambda)\delta'$, $0 \leq \lambda \leq 1$.

3.1.4.

Let (V_1, T_1) be obtained from (V, T) by adjoining a collar:

$$V_1 - V^\circ \simeq \partial V \times [0,1], \qquad T_1|_{V_1 - V^\circ} = \mathrm{id}\,.$$

Define the following space over D_t (cf. 1.4.3):

$$f' : W \longrightarrow D_t.$$

(a) Let $\widetilde{D}_t^*$ be a universal covering of D_t^* with base point y, and let W^* be the bundle over D_t^* obtained as the quotient of $V_1 \times \widetilde{D}_t^*$ by the equivalence relation generated by

$$(v, y) \sim (T_1 v, e^{2\pi i} y).$$

(b) Since $T_1|_{V_1 - V^\circ}$ is trivial, W^* can be glued to the trivial bundle over D_t with fiber $V_1 - V$, giving a space W_1 over D_t.

(c) Put $W = W_1 \cup \{0\}$ and $f'(0) = 0$. Give W the unique topology inducing that of W_1, for which f' is continuous and, for $t' < t$, $f'^{-1}(\overline{D}_{t'})$ is the one-point compactification of $W_1 \cap f'^{-1}(\overline{D}_{t'})$.

Equip $W - \{0\}$ with the integrable connection of 1.3.4 for which the trivialization, supplied by (a), of the inverse image of W on $\widetilde{D}^*$ is horizontal.

Proposition 3.1.5.

There exists a homeomorphism

$$\Psi : B_{t,d} \xrightarrow{\sim} W$$

compatible with the projections to D_t and carrying 0 to 0; this homeomorphism may be chosen to be a diffeomorphism away from 0.

This result will not be used in the remainder of the seminar.

Let $d_0 < d_1 \leq r$, let $t_0 \leq \tau(d_0)$, and let $y \in D_{t_0}^*$. We have

$$B_{t_0,d_0} \subset B_{t_0,d_1},$$

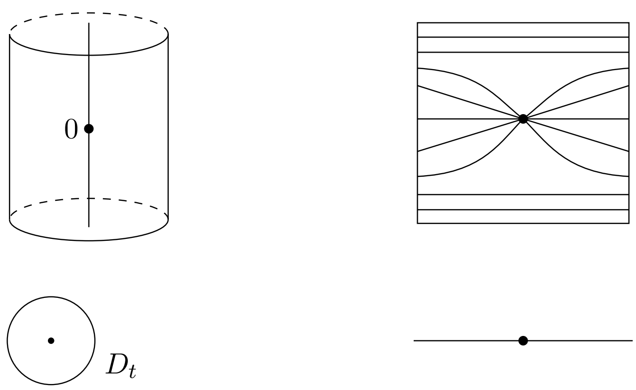
and $B_{t_0,d_1} - B_{t_0,d_0}^\circ$ is a bundle over D_{t_0}. Trivialize it and, over the circle $S_{|y|}$, lift $\partial/\partial\theta$ compatibly with this trivialization. Let T_1 be the corresponding endomorphism of

$$V_1 = f^{-1}(y) \cap B_{t_0,d_1}, \qquad V_0 = f^{-1}(y) \cap B_{t_0,d_0}.$$

Then $(V_0 \subset V_1, T_1)$ is isomorphic to $(V \subset V_1, T_1)$ of 3.1.4. Let W be constructed from $(V_0 \subset V_1, T_1)$. We have $f'^{-1}(y) = V_1$ canonically. The proof will show that Ψ can be chosen to induce the identity

$$f^{-1}(y) \xrightarrow{\sim} f'^{-1}(y).$$

The following drawings of f and of its restriction to $f^{-1}(\mathbb{R})$ summarize the proof.



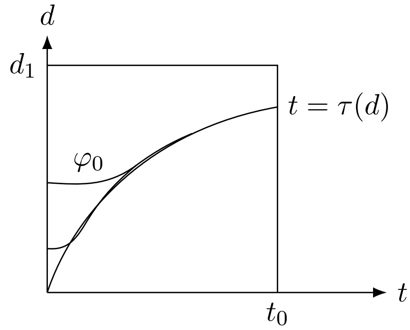
Let $B^+ \subset B_{t_0,d_1}$ be the set of points x such that

$$|f(x)|^2 \leq \tau(\delta(x)).$$

Let φ_n be a sequence of C^∞ functions from $[0, t_0]$ to $(0, d_1]$ such that, for a sequence $t_0 > r_1 > r_2 > \cdots$ tending to 0,

(a) for $|t|^2 \geq r_i$, φ_i is the inverse function of τ;

(b) for $|t|^2 < r_i$ and $j > i$, $\varphi_j(t) < \varphi_i(t)$.



The space $B^+ - \{0\}$ is the union of the bundles B_i^+ over D_t defined by

$$\delta(x) \geq \varphi_i(|f(x)|^2).$$

A standard argument, using the fact that δ has no critical points on the fibers of $B^+ - \{0\} \to D_t$, shows that there is a diffeomorphism $\alpha_i : B_i^+ \to B_{i+1}^+$, compatible with the projection to D_t and equal to the identity outside a small neighborhood of the subspace

$$\delta(x) = \varphi_i(|f(x)|^2), \qquad |f(x)|^2 \leq r_i,$$

of B_i^+. Composing the α_i and passing to the limit gives a diffeomorphism

$$\alpha : B_0^+ - \{x \mid f(x) = 0,\ \delta(x) = \varphi_0(0)\} \xrightarrow{\sim} B^+ - \{0\}.$$

Transport a trivialization of the bundle B_0^+ over D_t by α to obtain an integrable connection ∇' on $B^+ - \{0\}$. We may then:

(a) extend the restriction of this connection to $f^{-1}(S_{|y|}) \cap B^+$ to a connection on $f^{-1}(S_{|y|})/S_{|y|}$; this extension gives T_1;

(b) extend to B_{t_0,d_1} the lift, supplied by ∇' on $B^+ - \{0\}$, of the radial vector field on D_t.

There is then a unique integrable connection ∇ on B_{t_0,d_1} extending ∇', (a), and (b), and a unique horizontal diffeomorphism from $B_{t_0,d_1} - \{0\}$ to $W - \{0\}$ that is the identity on $f^{-1}(y)$. This is the required diffeomorphism.

3.1.6. A generalization.

The preceding discussion generalizes to the case in which $f : X \to D$ is a morphism such that $0 \in X$ is an isolated non-smooth point $\underline{\text{in}}$ X_0. After shrinking X and D, the non-smooth locus of f is then finite over D. The manifold of vanishing cycles V has a finite set Σ of critical points; $V - \Sigma$ is a differentiable manifold with boundary (its boundary is compact, although the manifold is not). The monodromy transformation T is the identity on the boundary and stabilizes Σ. Finally, Proposition 3.1.5 remains valid: the fibered homeomorphism Ψ is a diffeomorphism away from the critical locus, which it carries to the part of W arising from Σ.

3.1.7.

Proposition 3.1.5 allows us to use the calculations of 1.4. We obtain:

(A) The vanishing-cycle groups $(R^i\Phi(\Lambda))_0$ are the reduced cohomology groups, with coefficients in Λ, of the manifold of vanishing cycles V.

(B) The groups $H^i_{\{0\}}(X_0, R\Psi(\Lambda))$ are the cohomology groups with proper support of V°, with coefficients in Λ.

(C) The variation

$$\mathrm{Var} : \widetilde{H}^i(V, \Lambda) \longrightarrow H^i_c(V^\circ, \Lambda) = H^i(V \bmod \partial V, \Lambda)$$

is the classical variation defined by the automorphism T of V, which is the identity on ∂V.

3.1.8.

Assertions (A), (B), and (C) are in fact more elementary than Proposition 3.1.5. Here is a direct proof and construction which spares us from checking that the isomorphisms in (A) and (B) are independent of the choices made in the construction of 3.1.5.

Let $d \le r$ and $t \le \tau(d)$, and let B be the open box

$$B = \{x \mid \delta(x) < d \text{ and } |f(x)|^2 < t\}.$$

Let $\widetilde{D}^*$ be a universal covering of D^*, and consider the diagram

(3.1.8.1)
$$\begin{tikzcd} \{0\} \arrow[r, hook, "\bar\imath"] \arrow[d, equal] & \widetilde{D}_t \arrow[d, "p"] & \widetilde{D}^*_t \arrow[l, hook', "\bar\jmath"'] \arrow[d, "p'"] \\ \{0\} \arrow[r, hook, "i"] & D_t & D^*_t \arrow[l, hook', "j"'] \end{tikzcd}$$

obtained from (1.2.7.2) by restriction to D_t. Let

(3.1.8.2)
$$\begin{tikzcd} B_0 \arrow[r, hook, "\bar\imath"] \arrow[d, equal] & \overline{B} \arrow[d, "p"] & \overline{B}^* \arrow[l, hook', "\bar\jmath"'] \arrow[d, "p'"] \\ B_0 \arrow[r, hook, "i"] & B & B^* \arrow[l, hook', "j"'] \end{tikzcd}$$

be the inverse image of $f : B \to D$ over (3.1.8.1). We again write f for the projection from (3.1.8.2) to (3.1.8.1).

(A) By definition, $R^i\Psi_\eta(\Lambda)_0$ is the direct limit, as U runs through the neighborhoods of 0 in X, of the cohomology groups

$$H^i(U \times_D \widetilde{D}^*, \Lambda).$$

Let U run through the fundamental system of neighborhoods of 0 formed by boxes B as above. The fiber product $U \times_D \widetilde{D}^* = \overline{B}^*$ is a bundle with fiber V° over the contractible space $\widetilde{D}^*_t$. Hence the direct system of the groups $H^i(\overline{B}^*, \Lambda)$ is constant, with value

$$H^i(V^\circ, \Lambda) = H^i(V, \Lambda).$$

The resulting isomorphisms

(3.1.8.3)
$$R^i\Psi_\eta(\Lambda)_0 \simeq H^i(V, \Lambda)$$

may be refined to the following composite isomorphism (cf. (B) below):

$$
\begin{tikzcd}
R\Psi_\eta(\Lambda)_0 & R\Gamma(B_0, R\Psi_\eta(\Lambda)) \arrow[l, "\sim"'] & (Rf_* R\bar{\jmath}_*\Lambda)_0 \arrow[l, "\sim"'] & R\Gamma(\widetilde{D}_t, Rf_* R\bar{\jmath}_*\Lambda) \arrow[l, "\sim"'] \arrow[d, equal] \\
 & R\Gamma(V, \Lambda) \arrow[r, "\sim"] & R\Gamma(V^\circ, \Lambda) & R\Gamma(\overline{B}^*, \Lambda) \arrow[l, "\sim"']
\end{tikzcd}
\tag{3.1.8.4}
$$

The complex $R\Phi(\Lambda)_0$ is defined by the distinguished triangle

$$\Lambda \longrightarrow (R\Psi_\eta\Lambda)_0 \longrightarrow R\Phi(\Lambda)_0 \longrightarrow .$$

Consequently, the groups $R^i\Phi(\Lambda)_0$ are the reduced cohomology groups of V.

(B) Let $\mathfrak{F}$ be the family of closed subsets of $\overline{B}^*$ whose closure in $\overline{B}$ is proper over $\widetilde{D}$. The arrows in the following diagram are isomorphisms:

$$
\begin{tikzcd}
R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) \arrow[r, "\sim", "①"'] & R\Gamma_c(B_0, R\Psi_\eta(\Lambda)) & (Rf_! R\bar{\jmath}_*\Lambda)_0 \arrow[l, "\sim", "②"'] \\
R\Gamma_c(V^\circ, \Lambda) & R\Gamma_{\mathfrak{F}}(\overline{B}^*, \Lambda) \arrow[l, "\sim", "⑤"'] \arrow[r, equal, "④"'] & R\Gamma(\widetilde{D}_t, Rf_! R\bar{\jmath}_*\Lambda) \arrow[u, "③", "\wr"']
\end{tikzcd}
\tag{3.1.8.5}
$$

Indeed, ① is an isomorphism because B_0 is a cone (3.1.2) and $R^i\Psi_\eta(\Lambda)$ is constant away from 0; ② by the proper base-change theorem; ③ because $Rf_! R\bar{\jmath}_*\Lambda$ is constant away from 0; ④ is the derived-category version of the spectral sequence of a composite functor; and ⑤ because $\widetilde{D}^*$ is contractible.

The isomorphism (3.1.8.5) supplies the desired isomorphisms

$$R^i\Gamma_{\{0\}}(R\Psi_\eta(\Lambda)) \simeq H^i_c(V^\circ, \Lambda). \tag{3.1.8.6}$$

We shall verify (C) in 3.1.10.

3.1.9.

Cf. [3]. Let X be a topological space, let $i : Y \hookrightarrow X$ be a closed subset, let T be an automorphism of X inducing the identity on Y, and let Λ be an abelian group. To define the variation Var, choose a T-equivariant resolution Λ^*_X of Λ on X, a resolution Λ^*_Y of the constant sheaf Λ on Y on which, by convention, T acts trivially, and an equivariant morphism

$$\varphi : \Lambda^*_X \longrightarrow i_*\Lambda^*_Y$$

resolving the evident morphism $\varphi_0 : \Lambda_X \to i_*\Lambda_Y$. Suppose that the sheaves Λ^i_X and Λ^i_Y are acyclic for the functors to be considered. Passing to global sections gives

$$\Gamma(\varphi) : \Gamma(\Lambda^*_X) \longrightarrow \Gamma(\Lambda^*_Y).$$

For simplicity, suppose that every $\Gamma(\varphi)^i$ is surjective; otherwise, make it surjective by adding the cone of $\Gamma(\Lambda^*_Y)$ to $\Gamma(\Lambda^*_X)$. Put

$$R\Gamma(X \bmod Y, \Lambda) = \operatorname{Ker}(\Gamma(\varphi)).$$

Under reasonable hypotheses, if $\mathfrak{F}$ is the family of closed subsets of X disjoint from Y, there is a quasi-isomorphism

$$\Gamma_{\mathfrak{F}}(X - Y, \Lambda^*_X) \xrightarrow{\sim} R\Gamma(X \bmod Y, \Lambda).$$

The endomorphism $T-1$ of $\Gamma(\Lambda_X^*)$, where T acts by transport of structure, is zero on Λ (the constants) and factors through $\mathrm{Ker}(\Gamma(\varphi))$. Hence, in the derived category and under the same hypotheses, it gives a variation morphism

$$\mathrm{Var} : R\widetilde{\Gamma}(X, \Lambda) \longrightarrow R\Gamma_{\mathfrak{F}}(X - Y, \Lambda).$$

Without these hypotheses, one obtains instead

$$\mathrm{Var} : R\widetilde{\Gamma}(X, \Lambda) \longrightarrow R\Gamma(X \bmod Y, \Lambda).$$

This description simplifies when T is the identity on a neighborhood of Y and acts as the identity on Λ_X^* in that neighborhood. The endomorphism $T-1$ of $\Gamma(\Lambda_X^*)$ then factors through $\Gamma_{\mathfrak{F}}(\Lambda_X^*)$ and supplies the variation.

Proposition 3.1.10.

Under the isomorphisms of 3.1.8(A),(B), the variation

$$(R\Phi_\eta(\Lambda))_0 \longrightarrow R\Gamma_{\{0\}}(R\Psi_\eta(\Lambda))$$

is equal to the variation of 3.1.9 defined by the automorphism T of V, which is the identity on ∂V.

The reader is invited to draw a small picture rather than read the proof. Let ${}_fB$ be the closed box of 3.1.2 and let ${}_fC \subset {}_fB$ be a collar along the boundary; thus ${}_fC \simeq \partial_f B \times [0,1]$ as a bundle over D_t, through an isomorphism carrying $\partial_f B \times \{1\}$ identically onto $\partial_f B$. Let

$${}_f\alpha : {}_fC \xrightarrow{\sim} (\partial V \times [0,1]) \times D_t$$

be a trivialization of the bundle ${}_fC$ over D_t whose fiber at $y \in D_t$ induces the identity

$$\partial V = \partial(f^{-1}(y)) \xrightarrow{\sim} \partial V \times \{1\}.$$

Put $C = {}_fC \cap B$ and let

$$\alpha : C \xrightarrow{\sim} (\partial V \times [0,1)) \times D_t$$

be induced by ${}_f\alpha$. We retain the notation of (3.1.8.2).

Let R be the equivalence relation on $\overline{B}$ defined by xRy if either $x = y$, or $p(x), p(y) \in C$ and $\alpha(p(x))$, $\alpha(p(y))$ have the same image in $\partial V \times [0,1)$. Put

$${}_1\overline{B} = \overline{B}/R, \qquad {}_1\overline{B}^* = \overline{B}^*/R, \qquad {}_1\overline{C}^* = p^{-1}(C)/R.$$

There is a commutative diagram

$$\begin{tikzcd}
B_0 \arrow[r, hook, "\bar\imath"] \arrow[d, equal] & \overline{B} \arrow[d, "q"] & \overline{B}^* \arrow[l, hook', "\bar\jmath"'] \arrow[d, "q'"] \\
B_0 \arrow[r, hook, "\bar\imath"] & {}_1\overline{B} & {}_1\overline{B}^* \arrow[l, hook', "{}_1\bar\jmath"']
\end{tikzcd}$$

Let Λ^* be a $\pi_1(D^*)$-equivariant resolution of Λ by acyclic sheaves on $\overline{B}^*$. By definition, the variation is computed from the morphism of complexes

$$\Lambda[0] \longrightarrow \bar\imath^* \bar\jmath_*(\Lambda^*).$$

Since $\Lambda \xrightarrow{\sim} Rq_*\Lambda$ and $\Lambda \xrightarrow{\sim} Rq'_*\Lambda$, it may equally be computed using an equivariant resolution of Λ on ${}_1\overline{B}^*$ and the morphism

$$\Lambda[0] \longrightarrow \bar\imath^* {}_1\bar\jmath_*(\Lambda).$$

Under (A) and (B), the variation is thereby identified with the composite morphism

$$\begin{tikzcd} R\Gamma(V,\Lambda) \arrow[r, "\sim"] & R\Gamma(V^\circ,\Lambda) & R\Gamma({}_1\overline{B}^*,\Lambda) \arrow[l, "\sim"', "①"] \arrow[d, "\mathrm{Var}"] \\ & R\Gamma_c(V^\circ,\Lambda) & R\Gamma({}_1\overline{B}^* \bmod {}_1\overline{C}^*,\Lambda) \arrow[l, "\sim"', "②"] \end{tikzcd} \tag{3.1.10.1}$$

where ① and ② are induced by the inclusion of V° in ${}_1\overline{B}^*$, and where Var is the variation of 3.1.9 for ${}_1\overline{B}^*$ and the action of the positive generator of $\pi_1(D^*)$. This group acts on ${}_1\overline{B}^*$ through the action described in 1.1.5 and acts trivially on ${}_1\overline{C}^*$.

Choose a connection ∇ on B^*/D^* compatible with the trivialization of C. This connection trivializes $\overline{B}^*$ and, on passing to the quotient, defines a map $u : {}_1\overline{B}^* \to V$. For T defined as in 3.1.3, the diagram

$$\begin{tikzcd} {}_1\overline{B}^* \arrow[r, "u"] \arrow[d, "T"'] & V \arrow[d, "T"] \\ {}_1\overline{B}^* \arrow[r, "u"] & V \end{tikzcd}$$

commutes. The map u induces isomorphisms in cohomology, so Proposition 3.1.10 follows from (3.1.10.1) and the functoriality of variation.

Proposition 3.1.10 is meant to justify an explicit calculation to be made later in which, as here, T is the identity on a neighborhood of ∂V. We shall take a cohomology class c represented by a current C and use the fact that $\mathrm{Var}(c)$ is represented by the current $TC - C$, compactly supported in V°.

3.1.11.

In practice, T is often calculated in two steps.

(a) Fix t and d with $t \le \tau(d)$, choose $y \in D_t^*$, and choose a connection ∇ on $f^{-1}(S_{|y|}) \cap B_{t,d}$. This connection defines a diffeomorphism T_0 of

$$V_0 = f^{-1}(y) \cap B_{t,d}$$

onto itself, which need not be the identity on the boundary.

(b) Choose a trivialization of $\partial B_{t,d}$ over D_t. Integrating ∇ gives a diffeomorphism $\varphi(\theta)$ from ∂V to

$$f^{-1}(e^{2\pi i\theta}y) \cap \partial B_{t,d},$$

which is identified with ∂V by the trivialization. For $0 \le \theta \le 1$, $\varphi(\theta)$ is a path in $\mathrm{Diff}(\partial V)$ joining the identity to ∂T_0.

The resulting pair (T_0, φ) depends on the arbitrary choices made. Since the space of these choices is contractible, it is independent of them up to homotopy. For a transformation T of the kind considered in 3.1.3, $(T, \text{constant path})$ is a pair of the kind considered here.

3.1.12.

The map

$$\mathrm{Diff}(V) \longrightarrow \mathrm{Diff}(\partial V)$$

is a fibration. Consequently, the assignment $T \mapsto (T, \text{constant path})$ induces a bijection between:

a) diffeomorphisms $T : V \to V$ satisfying $\partial T = \mathrm{id}$, up to isotopy; and

b) pairs (T_0, φ), where $T_0 : V \xrightarrow{\sim} V$ and φ is a homotopy class of paths from id to ∂T_0, up to isotopy.

A pair (T_0, φ) calculated by 3.1.11(a),(b) therefore determines T. Concretely, calculate T from (T_0, φ) as follows.

(α) Choose a collar embedding

$$[0,1] \times \partial V \xhookrightarrow{\alpha} V, \qquad \alpha(1, x) = x,$$

and a representative $\varphi : [0,1] \to \mathrm{Diff}(\partial V)$ such that its extension by id for negative parameters is C^∞. On the collar put

$$\beta(x) = \mathrm{pr}_1\, \alpha^{-1}(x).$$

Thus β has no critical point, takes values in $[0,1]$, and equals 1 on ∂V.

(β) Take $T = T_0$ away from the collar and, on the collar, set

$$T(x) = T_0\big(\varphi(\beta(x))^{-1}(x)\big).$$

3.1.13.

Here is a variant of 3.1.11(b) for calculating the homotopy class of the path φ. On every ray $\arg(z) = \theta$, choose a trivialization of the bundle $\partial B_{t,d}$ which varies continuously with θ. Use the trivialization over the ray through y to identify

$$(\partial V)_0 = f^{-1}(0) \cap \partial B_{t,d}$$

with ∂V. Let

$$\Psi_\theta : V \longrightarrow f^{-1}(e^{i\theta} y) \cap B_{t,d}$$

be obtained by integrating ∇. Transporting $\partial\Psi_\theta$ along the rays gives a diffeomorphism of $(\partial V)_0$, hence of ∂V, onto itself. This is $\varphi(\theta)$.

3.2. The Picard–Lefschetz formula (transcendental case)

3.2.1.

We shall make the results of 3.1 explicit for the map

$$f = \sum_{i=1}^{n+1} z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

For δ take the function $\sum |z_i|^2$, and take the box $B = B_{a,b}$ defined by

$$\left|\sum z_i^2\right| \leq a^2, \qquad \sum |z_i|^2 \leq b^2,$$

where $0 < a < b$. It is only to simplify the notation that we use a closed box rather than a box as in 3.1.2.

3.2.2.

The manifold of vanishing cycles V has equations

(3.2.2.1)
$$\begin{aligned} \sum z_i^2 &= a^2, \\ \sum |z_i|^2 &\le b^2. \end{aligned}$$

Write $z = (z_i)$ in terms of its real and imaginary parts, $z = x + iy$. The preceding equations become

(3.2.2.2)
$$\begin{aligned} x^2 - y^2 &= a^2, \\ x \cdot y &= 0 \qquad \big(\text{i.e. } x \perp y \text{ in } \mathbb{R}^{n+1}\big), \\ x^2 + y^2 &\le b^2. \end{aligned}$$

[16] Introduce on V the new coordinate system

$$X = \frac{x}{|x|}, \qquad Y = y\left(\frac{b^2 - a^2}{2}\right)^{-1/2}.$$

Then $x = X(a^2 + y^2)^{1/2}$, and (3.2.2.2) becomes

(3.2.2.3)
$$\begin{aligned} X^2 &= 1, \\ X \cdot Y &= 0, \\ Y^2 &\le 1. \end{aligned}$$

This identifies V with the tangent ball bundle of the sphere S^n.

3.2.3.

The map
$$\theta \longmapsto \exp(2\pi i\theta)a$$
identifies $\mathbb{R}$ with the universal covering of the circle S_a in $\mathbb{C}$ having radius a and center 0. The pullback to $\mathbb{R}$ of the bundle B is isomorphic to $\mathbb{R} \times V$. We trivialize it by means of the isomorphism

$$v \longmapsto \exp(\pi i\theta)v$$

from V to the fiber of B over $\exp(2\pi i\theta)a$. Let T_0 be the map furnished by this trivialization as in 3.1.11. In the coordinates (3.2.2.3),

$$T_0(X, Y) = (-X, -Y).$$

3.2.4.

The bundle over the disc D_a of radius a which forms the boundary of the "bundle" B has equations

$$\sum |z_i|^2 = b^2, \qquad f(z) \in D_a,$$

the projection to the disc being f. We shall describe a trivialization of the inverse image of this bundle over each ray
$$e^{2\pi i\theta}\mathbb{R}^+ \cap D_a.$$

Over the ray $\mathbb{R}^+ \cap D_a$, take the isomorphism with $(\mathbb{R}^+ \cap D_a) \times \partial V$ which, in the coordinates (3.2.2.3) on ∂V, is given by

$$z = x + iy \longmapsto (f(z), X, Y) = \left(f(z), \frac{x}{|x|}, \frac{y}{|y|}\right).$$

[16] The printed source has $\mathbb{R}^n$ in (3.2.2.2), while z has $n+1$ coordinates. The English text corrects this evident dimension typo to $\mathbb{R}^{n+1}$.

Over $(e^{2\pi i\theta}\mathbb{R}^+) \cap D_a$, use the image of this trivialization under multiplication by $e^{\pi i\theta}$. Apply 3.1.13 and use on ∂V the coordinates (X, Y) satisfying

$$X^2 = Y^2 = 1, \qquad X \cdot Y = 0. \tag{3.2.4.1}$$

If $\varphi(\theta)$ takes (X, Y) to (X_θ, Y_θ), one finds

$$X_\theta + iY_\theta = e^{\pi i\theta}(X + iY) \qquad (0 \leq \theta \leq 1). \tag{3.2.4.2}$$

3.2.5.

In summary, the manifold of vanishing cycles V is the tangent ball bundle of S^n. The monodromy transformation is represented by (T_0, φ), where T_0 is the differential of the antipodal map and φ is the following path from Id to ∂T_0: under $\varphi(\theta)$, each unit tangent vector is parallel-transported along the geodesic which it determines through a distance $+\pi\theta$ (the plus sign means "in the direction in which the vector points"). At $\theta = 1$ one reaches the antipodal vector.

By 3.1.13 this gives the following description of T. Let $\varphi : [0, 1] \to [0, 1]$ be a C^∞ map which is 0 near 0 and 1 near 1. The map T moves u, along the geodesic which u determines, through the distance

$$-\pi\bigl(1 - \varphi(|u|)\bigr).$$

Near $|u| = 1$ this is the identity; near $|u| = 0$ it is the antipodal map.

3.2.7.

Let W be a differentiable manifold of dimension d, and let

$$p : T(W) \longrightarrow W$$

be its tangent bundle, regarded as a manifold. If $(x_1, \ldots, x_d)$ is a local coordinate system on W, then

$$(x_1, \ldots, x_d, dx_1, \ldots, dx_d)$$

is a local coordinate system on $T(W)$. Let ε' be the orientation of $T(W)$ defined by this coordinate system. It is independent of the choice of the x_i, and is therefore globally defined.

Suppose that W is orientable. The submanifold $W \subset T(W)$ (the zero section), endowed with an orientation (respectively, with an orientation of its normal bundle), defines an element w of $H_d(T(W))$ (respectively, of $H^d_c(T(W))$). The self-intersection of $W \subset T(W)$ is the Euler–Poincare characteristic:

$$\int w \wedge w = \chi(W)$$

for the orientation ε'.[17]

For the manifold of vanishing cycles V, its complex structure defines an orientation ε. It is related to the orientation ε' of the tangent bundle of S^n by

$$\varepsilon = (-1)^{\frac{n(n-1)}{2}}\varepsilon'.$$

3.2.8.

The vanishing cycle δ is well defined up to sign. For $n \neq 0$, it is the element of $H^n_c(V^\circ)$ defined by the submanifold $S^n \subset V^\circ$. For $n = 0$, the space V consists of two points, and δ is their difference. With the complex orientation of V, 3.2.7 gives

$$(\delta, \delta) = (-1)^{\frac{n(n-1)}{2}}\bigl(1 + (-1)^n\bigr). \tag{3.2.8.1}$$

[17] The printed sentence mixes three dimension symbols: it calls the dimension x, indexes the coordinates by n, and places the classes in degree d. The English text consistently uses d in the general statement; the specialization below uses $d = n$.

3.2.9.

Put

$$\widetilde{H}_c^i(V^\circ) = \mathrm{Ker}\Big(H_c^i(V^\circ) \xrightarrow{\mathrm{Tr}} \mathbb{Z}\Big)\,;$$

this is $H_c^i(V^\circ)$ for $i \neq 2n$. Then

$$\begin{aligned} \widetilde{H}_c^i(V^\circ) &= 0 \quad\ \text{for } i \neq n, \\ \widetilde{H}_c^n(V^\circ) &= \mathbb{Z}, \ \ \text{generated by } \delta. \end{aligned}$$

Dually,

$$\begin{aligned} &\widetilde{H}^i(V) = 0 \qquad\qquad \text{for } i \neq n, \\ &\widetilde{H}^n(V) \xrightarrow[\sim]{(\,\cdot\,,\delta)} \mathbb{Z}. \end{aligned}$$

For $n \neq 0$, the image of $\delta \in H_c^n(V^\circ)$ in $H^n(V)$ is zero when n is odd, and is divisible by two (and no further) when n is even.

3.2.10.

Let us calculate the variation

$$\mathrm{Var} : \widetilde{H}^n(V) \longrightarrow H_c^n(V^\circ).$$

We first use the orientation ε' and suppose that $n \neq 0$. Let α be an orientation of the tangent bundle of S^n, and let $\delta \in H_c^n(V^\circ)$ be defined by $S^n \subset V^\circ$ together with the orientation α of its normal bundle. The cocycle $\delta' \in H^n(V)$ such that $(\delta', \delta) = 1$ is represented by a fiber of the fibration of V° over S^n. The normal bundle of this fiber is the constant bundle with value the tangent space at one point of S^n, and it is endowed with the orientation α.

This assertion is independent of the sign convention chosen to define the map

$$(\text{submanifold} + \text{orientation of its normal bundle}) \longrightarrow \text{cohomology}.$$

Indeed, changing the convention changes δ to $-\delta$ and δ' to $-\delta'$. The assertion is valid when

$$\mathrm{Tr} : H_c^{2n}(V, \mathbb{R}) \longrightarrow \mathbb{R}$$

is defined as follows:

(α) since Ω_V^* is a soft resolution of $\underline{\mathbb{R}}$, $\Gamma_c(\Omega_V^*)$ represents $R\Gamma_c(V, \mathbb{R})$, whence an isomorphism

$$H^i\big(\Gamma_c(V, \Omega_V^*)\big) \simeq H_c^i(V, \mathbb{R})$$

(the iterated coboundary is not used);

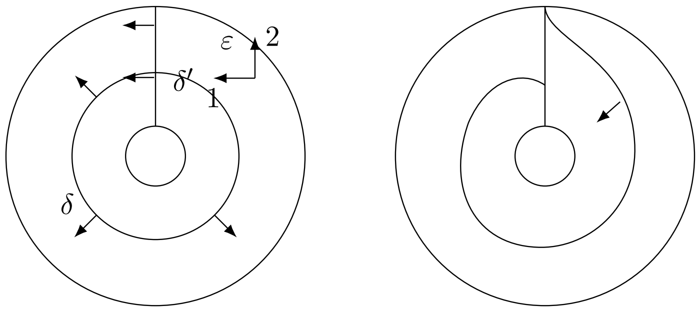
(β) Tr is integration.

The class $T(\delta')$ coincides with δ' in a neighborhood of ∂V, so that $\mathrm{Var}(\delta') = T(\delta') - \delta'$ has compact support in V°. We have

$$T(\delta') - \delta' \sim -\delta, \qquad \text{and hence}$$

$$\mathrm{Var}(\delta') = -(\delta', \delta)\, \delta.$$

Here is the picture for $n = 1$. We draw the tangent bundle of S^1 so that vectors pointing in the chosen counterclockwise orientation of S^1 point outward.



$\delta,\ \delta'$ and ε $\qquad\qquad$ $\mathrm{Var}(\delta')$

For the orientation ε, we therefore have

$$\mathrm{Var}(x) = -\frac{\varepsilon}{\varepsilon'}(x,\delta)\,\delta.$$

Theorem 3.2.11.

The group $\widetilde{H}^n_c(V^\circ)$ is isomorphic to $\mathbb{Z}$, generated by the vanishing cycle δ. The group $\widetilde{H}^n(V)$ is also isomorphic to $\mathbb{Z}$ and is dual to the preceding group. All the other groups $\widetilde{H}^i_c(V^\circ)$ and $\widetilde{H}^i(V)$ vanish.

The variation is given by

$$\mathrm{Var}(x) = -(-1)^{\frac{n(n-1)}{2}}(x,\delta)\,\delta, \qquad \text{and}$$

$$(\delta,\delta) = (-1)^{\frac{n(n-1)}{2}}\big(1+(-1)^n\big).$$

According to the congruence class of n modulo 4, we have the following table:

n mod 4	0	1	2	3
(δ,δ)	2	0	-2	0
$\mathrm{Var}(x) = \pm(x,\delta)\delta$	$-$	$-$	$+$	$+$
$T(\delta) = (-1)^{n+1}\delta$	$-\delta$	δ	$-\delta$	δ

As a check, we indeed have

$$T(\delta) = \delta + \mathrm{Var}(\delta).$$

4. De Rham cohomology

4.1.

Let D be a complex analytic disc, let $0 \in D$, put $D^* = D - \{0\}$, and let $\widetilde{D}^*$ be a universal covering of D^*. Let X be an analytic space, let $f : X \longrightarrow D$ be a morphism, and consider the diagram of (1.2.7.1):

(4.1.1)
$$\begin{tikzcd}
X_0 \arrow[r, hook, "\bar{i}"] \arrow[d, equal] & \bar{X} \arrow[d, "p"] & \bar{X}^* \arrow[l, hook', "\bar{j}"'] \arrow[d, "p'"] \\
X_0 \arrow[r, hook, "i"] & X & X^* \arrow[l, hook', "j"']
\end{tikzcd}$$

Suppose that $X^* = X - X_0$ is nonsingular (that is, smooth). Its covering $\bar{X}^*$ is therefore smooth. Denote by $\Omega^*_{\bar{X}^*}$ the De Rham complex of sheaves of holomorphic differential forms on $\bar{X}^*$. It is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincare lemma), and hence gives a morphism

(4.1.2)
$$\bar{j}_*\Omega^*_{\bar{X}^*} \longrightarrow R\bar{j}_*\mathbb{C}$$

in $D^+(\bar{X},\mathbb{C})$.

Proposition 4.2.

The morphism (4.1.2) is an isomorphism.

It is clear that (4.1.2) is a quasi-isomorphism at every point of $\bar{X}^* \subset \bar{X}$.[18] It is therefore enough to show that every point $x \in X_0$ has a fundamental system of neighborhoods U in $\bar{X}$ such that

$$H^q\big(\bar{j}^{-1}(U), \Omega^p\big) = 0 \qquad \text{for } q > 0.$$

As V runs through a fundamental system of neighborhoods of x in X, the sets $p^{-1}(V)$ form a fundamental system of neighborhoods of x in $\bar{X}$. In particular, one may take the $p^{-1}(V)$ for V a Stein neighborhood of x in X. If V is Stein, then $V - X_0$ is still Stein (since $X_0 \cap V$ is defined by one equation in V), as is its covering

$$p'^{-1}(V - X_0) = \bar{j}^{-1}p^{-1}(V).$$

Consequently, for every coherent sheaf $\mathcal{F}$ on $\bar{X}^*$,

$$H^q\big(\bar{j}^{-1}p^{-1}(V), \mathcal{F}\big) = 0 \qquad \text{for } q > 0,$$

which proves the proposition.

Corollary 4.3.

The isomorphism (4.1.2) induces an isomorphism

$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*} \xrightarrow{\sim} R\Psi_\eta(\mathbb{C}). \tag{4.3.1}$$

Variant 4.4.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a complex local system). The quasi-isomorphism

$$V \longrightarrow \Omega^*_{X^*}(V)$$

of the Poincare lemma induces a quasi-isomorphism

$$\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}(p'^*V) \xrightarrow{\sim} R\Psi_\eta(V). \tag{4.4.1}$$

We shall show that (4.3.1) remains true when $\bar{i}^*\bar{j}_*\Omega^*_{\bar{X}^*}$ is replaced by a subcomplex defined by growth and monodromy conditions.

4.5.

Let $f : X^* \longrightarrow D^*$ be a topological space over D^*, let F be a sheaf of $f^\circ\mathcal{O}_{D^*}$-modules on X^*, and let $\bar{F}$ be its inverse image on

$$\bar{X}^* = X^* \times_{D^*} \widetilde{D}^*.$$

Let $z : D \hookrightarrow \mathbb{C}$ be a holomorphic embedding with $z(0) = 0$. We shall always be free to shrink D below; for a sufficiently small disc the existence of z is trivial. Choose on $\widetilde{D}^*$ a function $\log z$ such that $\exp(\log z) = z$, and put

$$z^\alpha = \exp(\alpha \log z) \qquad (\alpha \in \mathbb{C}).$$

The group $\pi_1(D^*)$ acts, by transport of structure, on $H^0(\bar{X}^*, \bar{F})$. An element $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if its transforms under $\pi_1(D^*)$ span a finite-dimensional complex vector

[18] The printed proof says (4.1.1), which is the preceding diagram rather than the morphism under discussion; the English restores the evident reference to (4.1.2).

space $V(f)$. Let T be the positive generator of $\pi_1(D^*)$. The section f is of finite determination if and only if there is a nonzero polynomial $P \in \mathbb{C}[X]$ such that $P(T)(f) = 0$. The polynomial of least degree with this property is then both the minimal polynomial and the characteristic polynomial of T acting on $V(f)$.

We say that f is of unipotent finite determination if there is an A such that

$$(T-1)^A f = 0.$$

On the space of these sections define

$$N = \frac{\log T}{2\pi i}$$

by the finite sum

$$N(f) = -\frac{1}{2\pi i} \sum_{n>0} \frac{(1-T)^n}{n} f.$$

If f (respectively, g) is a section of unipotent finite determination in $H^0(D^*, \mathcal{O})$ (respectively, in $H^0(\bar{X}^*, \bar{F})$), then

$$N(fg) = N(f)g + fN(g),$$

since the logarithm of an automorphism is a derivation.

We say that f is of quasi-unipotent finite determination if there are A and $B > 0$ such that

$$(T^B - 1)^A f = 0.$$

Examples 4.6.

For $X = D^*$ and $F = \mathcal{O}$, one has

$$\big(T - \exp(2\pi i\alpha)\big)^{k+1} \big(z^\alpha (\log z)^k\big) = 0,$$

$$N(\log z) = 1,$$

and

$$N\left(\frac{(\log z)^k}{k!}\right) = \frac{(\log z)^{k-1}}{(k-1)!}, \qquad (0 \text{ for } k = 0).$$

Proposition 4.7.

A section $f \in H^0(\bar{X}^*, \bar{F})$ is of finite determination if and only if it can be written

$$f = \sum_{\alpha,k} f_{\alpha,k}\, z^\alpha \frac{(\log z)^k}{k!}, \tag{4.7.1}$$

where $0 \leq \mathrm{Re}(\alpha) < 1$, $k \in \mathbb{N}$, and the $f_{\alpha,k} \in H^0(X^*, F)$ are almost all zero. For f of finite determination, the decomposition (4.7.1) is unique.

By 4.6, every section of the form (4.7.1) is of finite determination. Conversely, let f be of finite determination, with $P(T)(f) = 0$. Write

$$P(X) = \prod_{\beta \in \mathbb{C}} (X - \beta)^{n'(\beta)} = \prod_{0 \leq \mathrm{Re}(\alpha) < 1} \big(X - \exp(2\pi i\alpha)\big)^{n(\alpha)},$$

where $n(\alpha) = n'(\exp(2\pi i\alpha))$ is almost always zero. Decomposing the $\mathbb{C}[X]$-module $V(f)$ into its isotypic components, one finds a unique decomposition

$$f = \sum_\alpha f'_\alpha$$

such that

$$\big(T - \exp(2\pi i\alpha)\big)^{n(\alpha)} f'_\alpha = 0.$$

Choose a polynomial $Q_\alpha \in \mathbb{C}[X]$ congruent to 1 modulo $\big(X - \exp(2\pi i\alpha)\big)^{n(\alpha)}$ and to 0 modulo $\big(X - \exp(2\pi i\alpha')\big)^{n(\alpha')}$ for every $\alpha' \neq \alpha$. Then $f'_\alpha = Q_\alpha(T)(f)$. Put

$$f_\alpha = z^{-\alpha} f'_\alpha = z^{-\alpha} Q_\alpha(T)(f). \tag{4.7.2}$$

Since

$$T(z^{-\alpha} f'_\alpha) = \exp(-2\pi i\alpha) z^{-\alpha} T(f'_\alpha),$$

we have

$$(T-1)^{n(\alpha)} f_\alpha = 0.$$

If

$$f_\alpha = \sum_k f_{\alpha,k} \frac{(\log z)^k}{k!}, \qquad f_{\alpha,k} \in H^0(X^*, F),$$

then 4.6 and the identity $\sum_j (-1)^j \binom{k}{j} = \delta_{0,k}$ give

$$f_{\alpha,k} = \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+k} f_\alpha, \tag{4.7.3}$$

which proves uniqueness.

Conversely, if f_α is a section of unipotent finite determination, the $f_{\alpha,k}$ defined by (4.7.3) vanish for large k (because $N^k f_\alpha = 0$ for large k) and are invariant under T; hence they identify with sections of F on X^*. Equivalently, we must show that $N f_{\alpha,k} = 0$. Apply to $N^k f_\alpha$ the identity

$$\begin{aligned} N \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^j &= \sum_{j>0} (-1)^j \frac{(\log z)^{j-1}}{(j-1)!} N^j \\ &\quad + \sum_{j\geq 0} (-1)^j \frac{(\log z)^j}{j!} N^{j+1} = 0. \end{aligned}$$

Finally,

$$\begin{aligned} \sum_{k\geq 0} f_{\alpha,k} \frac{(\log z)^k}{k!} &= \sum_{j,k\geq 0} (-1)^j \frac{(\log z)^{j+k}}{j!k!} N^{j+k} f_\alpha \\ &= \sum_{n\geq 0} \left(\sum_j (-1)^j \binom{n}{j} \right) \frac{(\log z)^n}{n!} N^n f_\alpha = f_\alpha, \end{aligned}$$

which proves 4.7.

Remark 4.8.1.

The condition $\mathrm{Re}(\alpha) \in [0,1)$ in 4.7 may be replaced by $\alpha \in C$, where C is any fundamental domain for $\mathbb{C} \longrightarrow \mathbb{C}/\mathbb{Z}$.

Remark 4.8.2.

Let f be of finite determination, with decomposition (4.7.1). For $0 \leq \mathrm{Re}(\alpha) < 1$ and for α such that some $f_{\alpha,k} \neq 0$, put

$$m(\alpha) = 1 + \sup\{k \mid f_{\alpha,k} \neq 0\}.$$

If all the $f_{\alpha,k}$ vanish, put $m(\alpha) = 0$. The characteristic polynomial of T acting on $V(f)$ is

$$\mathrm{Pc}(T, V(f)) = \prod_{\alpha} \big(X - \exp(2\pi i\alpha)\big)^{m(\alpha)}.$$

In particular, f is of quasi-unipotent finite determination if and only if the α in (4.7.1) may all be taken in $\mathbb{Q}$.

4.9.

Return to the more restrictive hypotheses of 4.1, and let F be a coherent sheaf on X whose restriction to X^* is locally free.

Proposition 4.10.

Let f be a section of finite determination of the inverse image $\bar{F}$ of F on $\bar{X}^*$. Then f has moderate growth along X_0 if and only if the $f_{\alpha,k}$ in the decomposition (4.7.1) are meromorphic along X_0.

For the definition of "moderate growth," we refer to [1], 2.10. It is clear from (4.7.1), (4.7.2), and (4.7.3) that f has moderate growth if and only if the $f_{\alpha,k}$ do. For sections of F on X^*, moderate growth along X_0 is known to be equivalent to being meromorphic along X_0.

For $X = D$ and $F = \mathcal{O}$, 4.10 has the following elementary corollary.

Corollary 4.11.

Let K be the field of germs at 0 of meromorphic functions on D, and let K^m (respectively, K^{mu} and K^{mqu}) be the K-algebra of germs at 0 of holomorphic functions on $\widetilde{D}^*$ which have moderate growth at 0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Then the following are bases over K:

$$\begin{array}{lll} K^m: & z^\alpha(\log z)^k, & 0 \le \mathrm{Re}(\alpha) < 1,\ k \in \mathbb{N}, \\ K^{mu}: & (\log z)^k, & k \in \mathbb{N}, \\ K^{mqu}: & z^\alpha(\log z)^k, & \alpha \in \mathbb{Q},\ 0 \le \alpha < 1,\ k \in \mathbb{N}. \end{array}$$

Under the hypotheses of 4.9, denote by $\Psi_\eta^m(F)$ (respectively, $\Psi_\eta^{mu}(F)$ and $\Psi_\eta^{mqu}(F)$) the subsheaf of $\bar{i}^*\bar{j}_*\bar{F}$ consisting of sections of $\bar{F}$, near points of X_0, which have moderate growth along X_0 and finite determination (respectively, unipotent finite determination and quasi-unipotent finite determination). Let

$$F^m \subset j_*j^*F$$

be the sheaf on X of sections of j_*j^*F which are meromorphic along X_0. Proposition 4.10 has the following corollary, independent of the arbitrary choices of z and $\log z$.

Corollary 4.12.

The natural morphisms

$$\begin{array}{ll} \text{(i)} & K^m \otimes_K i^*F^m \longrightarrow \Psi_\eta^m(F), \\ \text{(ii)} & K^{mu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mu}(F), \\ \text{(iii)} & K^{mqu} \otimes_K i^*F^m \longrightarrow \Psi_\eta^{mqu}(F) \end{array}$$

are isomorphisms.

The main result of this section is the following.

Theorem 4.13.

The natural morphism induced by (4.3.1),

$$\Psi_\eta^{mqu}(\Omega_X^*) \longrightarrow R\Psi_\eta(\mathbb{C}), \tag{4.13.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

We shall in fact prove the more general results 4.15 and 4.17. Their statements require a few preliminaries which a reader interested only in 4.13 may omit.

4.14.

Let V be a locally constant sheaf of finite-dimensional complex vector spaces on X^* (a local system). Let

$$\mathcal{V} = V \otimes_{\mathbb{C}} \mathcal{O}$$

be the corresponding locally free $\mathcal{O}$-module, and let ∇ be the integrable connection on $\mathcal{V}$ for which the local sections of V are horizontal. It is known that $\mathcal{V}$ admits an extension $\widetilde{\mathcal{V}}$ to a coherent sheaf on X such that ∇ is regular along X_0 relative to this extension. It is also known that any two such extensions are meromorphically equivalent ([1], II, 5.7). Thus the sheaf $\Psi_\eta^m(\widetilde{\mathcal{V}})$, and its variants Ψ^{mu} and Ψ^{mqu}, does not depend on the chosen extension; we denote it by $\Psi_\eta^m(\mathcal{V})$. Likewise define

$$\Psi_\eta^m\big(\Omega_{X^*}^p(V)\big) = \Psi_\eta^m(\Omega_X^p \otimes \mathcal{V}),$$

and its variants Ψ^{mu} and Ψ^{mqu}.

Proposition 4.15.

The natural morphism induced by (4.4.1),

$$\Psi_\eta^m\big(\Omega_{X^*}^*(V)\big) \longrightarrow R\Psi_\eta(V), \tag{4.15.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

4.16.

A local system V as in 4.14 is said to be unipotent (respectively, quasi-unipotent) along X_0 if, for every morphism u from the standard disc D_0 to X such that $u^{-1}(X_0) = \{0\}$, the monodromy transformation of the local system u^*V on D_0^* is unipotent (respectively, quasi-unipotent).

Proposition 4.17.

If V is quasi-unipotent, the natural morphism induced by (4.4.1),

$$\Psi_\eta^{mqu}\big(\Omega_{X^*}^*(V)\big) \longrightarrow R\Psi_\eta(V), \tag{4.17.1}$$

is an isomorphism in $D^+(X_0, \mathbb{C})$.

Theorem 4.13 is the special case of 4.17 in which $V = \underline{\mathbb{C}}$.

4.18.

We shall first treat the case in which X is smooth and $(X_0)_{\text{red}}$ is a normal-crossings divisor in X. Since the theorems are local, we may suppose that f is the map

$$f(\underline{z}) = \prod_{i=1}^{d} z_i^{n_i}, \qquad 0 < d \le e, \quad n_i > 0,$$

from $X = D^e$ to D, where this time D denotes the standard disc $D = \{z \in \mathbb{C} \mid |z| < 1\}$. It is enough to prove that the fiber at $0 \in D^e$ of (4.15.1) (respectively, (4.17.1)) is a quasi-isomorphism.

Let U be a neighborhood of 0 in D^e of the form $|z_i| < a$, where $0 < a \leq 1$. Define U_0, $\bar{U}$, U^*, and $\bar{U}^*$ by taking the inverse image of U in (4.1.1). The cohomology of U^* or $\bar{U}^*$ is independent of a, and hence

$$H^*(\bar{U}^*, V) \xrightarrow{\sim} \big(R^*\Psi_\eta(V)\big)_0.$$

The fundamental group $\mathbb{Z}^d$ of $X^* = (D^*)^d \times D^{e-d}$ (which is also that of U^*) is abelian. Therefore V has a finite filtration whose successive quotients are rank-one local systems. By devissage, we may thus suppose that V has rank one.

For $\widetilde{\mathcal{V}}$ as in 4.14, write $j_*^m\Omega^*_{X^*}(V)$ for the subcomplex of $j_*\Omega^*_{X^*}(V)$ whose degree-p term is $(\Omega^p_X \otimes \widetilde{\mathcal{V}})^m$. By [1], II 6.4.1 and 6.8, the natural map induced by the quasi-isomorphism $V \xrightarrow{\approx} \Omega^*_{X^*}(V)$ (the holomorphic Poincare lemma),

(4.18.1) $$j_*^m\Omega^*_{X^*}(V) \longrightarrow Rj_*V,$$

is a quasi-isomorphism. Thus

(4.18.2) $$H^*\big((j_*^m\Omega^*_{X^*}(V))_0\big) \xrightarrow{\sim} (R^*j_*V)_0 \xleftarrow{\sim} H^*(U^*, V).$$

[19] This will be the only analytic ingredient in this part 4.18 of the proof.

Lemma 4.18.3.

For a rank-one V, one has
(i) if V is nontrivial: $H^*(U^*, V) = 0$,
(ii) if $V = \underline{\mathbb{C}}$: $H^*(U^*, V) = \bigwedge^* \mathbb{C}^d$.

Indeed, U^* has the homotopy type of a torus.

Let η be the generator of $H^1(D^*, \mathbb{Z})$, and also its inverse image in $H^1(U^*, \mathbb{Z})$.

Lemma 4.18.4.

For a rank-one V, the natural map

$$H^*(U^*, V) \xrightarrow[(4.18.1)]{\sim} H^*\big((j_*^m\Omega^*_{X^*}(V))_0\big) \longrightarrow H^*\big(\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0\big)$$

identifies $H^*(\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0)$ with $H^*(U^*, V)/\eta H^*(U^*, V)$. [20]

Use 4.12(ii) and the identity $d\log f = \sum_i n_i\, dz_i/z_i$. The complex $\Psi_\eta^{mu}(\Omega^*_{X^*}(V))_0$ identifies with the simple complex sK associated (using direct sums) with the double complex K whose terms are

$$K^{-p,n+p} = \frac{(\log f)^p}{p!} \cdot (j_*^m\Omega^n_{X^*}(V))_0.$$

This complex lies in a bad quadrant $(p \leq 0, q \geq 0, p+q \geq 0)$. Nevertheless, s$K$ is the inductive limit of the subcomplexes $\mathrm{s}\,\sigma_{p\geq p_0}(K)$ whose terms are

$$(\mathrm{s}\,\sigma_{p\geq p_0}K)^n = \bigoplus_{\substack{p+q=n\\ p\geq p_0}} K^{pq},$$

and this suffices to justify the argument below.

[19] The printed right-hand term in (4.18.2) is $H^*(U, V)$, although V is defined on U^*; the English restores the evident U^*.

[20] The printed left-hand term is $H^*(U^*, \mathbb{C})$, while both the hypothesis and the conclusion concern the general rank-one system V; the English restores V.

The spectral sequence for the filtration by the first degree is

$$E_1^{-p,n+p} = H^n(U^*, V) \qquad (p \geq 0),$$

with

$$d_1 = 2\pi i\, \eta\wedge : H^n(U^*, V) \longrightarrow H^{n+1}(U^*, V).$$

If V is nontrivial, then $E_1 = 0$ by 4.18.3, and 4.18.4 reduces to $0 = 0$. Suppose that $V = \underline{\mathbb{C}}$. The complexes (E_1, d_1) are then Koszul complexes, possibly truncated, so that, by acyclicity of the Koszul complex,

$$E_2^{pq} = 0 \qquad \text{for } p \neq 0,$$

and

$$E_2^{0n} = H^n(U^*, V)/\eta H^{n-1}(U^*, V),$$

which proves 4.18.4.

For $\beta \in \mathbb{C}^*$, let V_β denote the rank-one local system on D^* whose inverse image on $\widetilde{D}^*$ is trivialized and whose monodromy is β. We use the same notation for its inverse image on U^*. On $\bar{U}^*$, the inverse image of $V \otimes V_\beta$ is canonically isomorphic to that of V.

Lemma 4.18.5.

For a rank-one V, the map

$$\bigoplus_{\beta \in \mathbb{C}^*} H^*(U^*, V \otimes V_\beta)/\eta H^*(U^*, V \otimes V_\beta) \longrightarrow H^*(\bar{U}^*, V)$$

is bijective.

Let k be the greatest common divisor of the n_i. The space $\bar{U}^*$ has k connected components, indexed by the kth roots of unity. On the component $\bar{U}^*_\xi$ indexed by ξ, the inverse image of the function $z^{1/k}$ on $\widetilde{D}^*$ coincides with the inverse image of the function

$$\xi \prod_{i=1}^{d} z_i^{n_i/k}$$

on U^*. The component $\bar{U}^*_\xi$ has the homotopy type of a torus, and the sequence

$$0 \longrightarrow \pi_1(\bar{U}^*_\xi) \longrightarrow \pi_1(U^*) \longrightarrow \pi_1(D^*)$$

is exact. Write $\varphi : \mathbb{Z}^d \to \mathbb{Z}$ for the map $\pi_1(U^*) \to \pi_1(D^*)$. If $\ker(\varphi)$ acts nontrivially on V, both sides of 4.18.5 vanish.[21] Otherwise, after replacing V by a suitable $V \otimes V_\beta$, we may suppose that V is trivial, that is, $V = \mathbb{C}$. In complex cohomology we then have

$$H^0(\bar{U}^*) \otimes H^*(U^*)/\eta H^*(U^*) \xrightarrow{\sim} H^*(\bar{U}^*).$$

The result follows on observing that V_β is trivial on U^* when $\beta^k = 1$, that

$$\bigoplus_{\beta^k=1} H^0(U^*, V_\beta) = \bigoplus_{\beta^k=1} \mathbb{C} \xrightarrow{\sim} H^0(\bar{U}^*, \mathbb{C}),$$

and that $H^*(U^*, V_\beta) = 0$ when $\beta^k \neq 1$.

[21] The print refers here to 4.18.4; the two sides under discussion are those of 4.18.5.

Remark 4.18.6.

If V is quasi-unipotent, the sum over $\mathbb{C}^*$ in 4.18.5 may be replaced by a sum over the roots of unity; the proof is the same.

Applying 4.12 gives a commutative diagram

$$\begin{tikzcd}
H^*\Psi^m_\eta(\Omega^*_{X^*}(V))_0 \arrow[d] & \bigoplus\limits_\beta H^*\Psi^{mu}_\eta(\Omega^*_{X^*}(V\otimes V_\beta)) \arrow[l, "\sim"', "4.12"] & \bigoplus\limits_\beta H^*(U^*, V\otimes V_\beta)/\eta H^*(U^*, V\otimes V_\beta) \arrow[l, "\sim"', "4.18.4"] \arrow[d, "\wr", "4.18.5"'] \\
H^*R\Psi(V)_0 & & H^*(\bar{U}^*, V) \arrow[ll, "\sim"]
\end{tikzcd}$$

from which 4.15 follows. Proposition 4.17 is obtained by replacing Ψ^m with Ψ^{mqu}, the sum over β with a sum over the roots of unity, and 4.18.5 with 4.18.6.

4.19.

Let us deduce 4.15 and 4.17 from the special case 4.18. Let $u : Y \longrightarrow X$ be a resolution of (X, X_0), that is, a proper morphism inducing an isomorphism $Y^* \xrightarrow{\sim} X^*$, such that Y is smooth and $(Y_0)_{\text{red}}$ is a normal-crossings divisor in Y. If V is quasi-unipotent along X_0, its inverse image u^*V is quasi-unipotent along Y_0.

For every coherent sheaf F on X, denote by F^m, or by $j^m_* j^* F$, the subsheaf of j_*j^*F formed by the meromorphic sections. If f is an equation for X_0, then $j^m_*j^*F$ is the inductive limit of the system

$$F \xrightarrow{f} F \xrightarrow{f} F \longrightarrow \cdots . \tag{4.19.1}$$

The following lemma is proved in [2].

Lemma 4.19.2.

For every coherent sheaf F on X, one has

$$F^m \xrightarrow{\sim} u_*(u^*F)^m$$

and $R^i u_*(u^*F)^m = 0$ for $i > 0$.

By Grauert's theorem, the $R^i u_* u^* F$ are coherent sheaves. The kernels and cokernels of $F \longrightarrow u_*u^*F$, as well as the $R^iu_*u^*F$ for $i > 0$, vanish outside X_0 and are therefore locally annihilated by a power of f. Since u is proper, R^iu_* commutes with filtered inductive limits, in particular with the inductive limit (4.19.1), and (4.19.2) follows.

From 4.19.2 and 4.12 one deduces that, under the hypotheses just stated and still denoting by u the map $Y_0 \to X_0$ induced by u, one has the following.

Lemma 4.19.3.

The morphisms

$$\Psi^m_\eta(F) \longrightarrow u_*\Psi^m_\eta(u^*F) \longrightarrow Ru_*\Psi^m_\eta(u^*F)$$

are isomorphisms.

That the second arrow is an isomorphism means that

$$R^iu_*\Psi^m_\eta(u^*F) = 0 \qquad (i > 0).$$

This follows from 4.12 and from $R^iu_*(i^*u^*F^m) = 0$ for $i > 0$, which in turn follows from $R^iu_*(u^*F^m) = 0$ for $i > 0$ and the proper base-change theorem for u.

To prove 4.15, it remains to consider the commutative diagram

$$
\begin{tikzcd}
\Psi^m(\Omega^*_{X^*}(V)) \arrow[r, equal, "(4.19.3)"'] \arrow[d] & u_*\Psi^m(\Omega^*_{Y^*}(u^*V)) \arrow[r, "\sim", "(4.19.3)"'] & Ru_*\Psi^m(\Omega^*_{Y^*}(u^*V)) \arrow[d, "\wr"', "(4.18)"] \\
R\Psi(V) \arrow[rr, "(1.3.1;\ \mathrm{XIII}\ 2.1.7.1)"] & & Ru_*R\Psi(u^*V)
\end{tikzcd}
$$

The proof of 4.17 is the same, with Ψ^m replaced everywhere by Ψ^{mqu}.

Remark 4.20.

Theorem 4.13 gives an incarnation of the monodromy theorem at the level of complexes.

Bibliography

[1] P. Deligne, *Equations différentielles à points singuliers réguliers*, Lecture Notes in Mathematics 163, Springer-Verlag.

[2] A. Grothendieck, *On the De Rham cohomology of algebraic varieties*, Publ. Math. IHES $\underline{29}$, 95–103 (1966).

[3] K. Lamotke, *Isolated critical points and monodromy.*

SGA 7 — EXPOSÉ XV

The Picard–Lefschetz Formula

by P. Deligne

Source contents

1. Ordinary quadratic singularities (normal forms)

In this section we take up the results of VI 6 again, adopting a "henselian" rather than a "formal" point of view. We give normal forms for ordinary quadratic singularities and their deformations, into which they may be put locally for the etale topology.

These results follow easily from VI 6, the implicit-function theorem (in Tougeron's form), and R. Elkik's theorem on versal henselian moduli of isolated singularities. At the time of the oral seminar, the latter was not yet available; a weaker theorem, covering only the case of complete intersections (which was enough for us), had been proved and will not be published.

1.1. Reminders

1.1.1.

Let $f : X \longrightarrow S$ be a morphism, with X locally describable by p equations in relative affine space $\mathbb{A}_S^{n+p}$ of dimension $n+p$ over S. The Jacobian ideal $J^n(X/S)$ was defined in VI 5. If X is the closed subscheme of $\mathbb{A}_S^{n+p}$ defined by the equations $f_i = 0$ $(1 \leq i \leq p)$, it is the ideal sheaf generated by the $p \times p$ minors of the Jacobian matrix $(\partial f_i / \partial x_j)$. The following theorem is well known. A special case of its formal analogue is proved in Bourbaki, *Algèbre commutative*, III, §4, no. 6, Corollary 3 to Theorem 2; the C^∞ analogue is proved in Tougeron, *Idéaux de fonctions différentiables* (thesis, Rennes, May 1967); the case considered here is treated in Artin, *Algebraic approximation of structures over complete local rings*, Publ. Math. IHES 36 (Zariski's volume), Lemma 5.10.

Theorem 1.1.2 (Tougeron–Artin).

Let S be the spectrum of a henselian local ring A, let $f : X \to S$ be a morphism as in 1.1.1, let a and I be two proper ideals of A, and let δ' be an ideal sheaf of X contained in $J^n(X/S)$. Suppose that I is of finite type. Let s be a section of f over S/I, that is, $s \in X(A/I)$. Denote by $s^*\delta'^2$ the ideal of A inverse to the ideal $s^*\delta'^2$ of A/I.

If $(s^*\delta'^2)\, a \supset I$, then there is a section $s_1 \in X(A)$ of f congruent to s modulo $(s^*\delta')\, a$.

1.1.3.

Let S be the spectrum of a henselian local ring, with closed point s. Let X'_0 be a scheme of finite type over $k(s)$, smooth away from a closed point x_0, and let X_0 be the henselization of X'_0 at x_0. We are interested in systems

$$\begin{tikzcd} X_0 \arrow[r, hook] \arrow[d] & X \arrow[d] & \\ s \arrow[r, hook] & T \arrow[r] & S \end{tikzcd}$$

of the following kind:

(a) T is henselian local and essentially of finite presentation over S (I 0.0.5); the morphism $T \to S$ is local and induces an isomorphism of residue fields;

(b) X is henselian local, flat, and essentially of finite presentation over T;

(c) the special fiber of X/T is equipped with a $k(s)$-isomorphism to X_0.

These are the henselian S-deformations of X_0. A morphism from (X, T) to (X', T') is a commutative diagram of S-schemes

$$\begin{tikzcd} X \arrow[r] \arrow[d] & X' \arrow[d] \\ T \arrow[r] & T' \end{tikzcd}$$

which induces the identity of X_0 after passage to the special fibers. A deformation X/T is called versal (minimally versal) if:

(a) for every morphism $X''/T'' \longrightarrow X'/T'$ for which $T'' \to T'$ is a closed immersion, the map

$$\operatorname{Hom}(X'/T', X/T) \longrightarrow \operatorname{Hom}(X''/T'', X/T)$$

is surjective. In particular, on taking $X''/T'' = X_0/s$, one has $\operatorname{Hom}(X'/T', X/T) \neq \varnothing$;

(b) for $T' = \operatorname{Spec}(k(s)[\varepsilon]/(\varepsilon^2))$, the "inverse-image" map

$$\operatorname{Hom}(T', T) \longrightarrow \{\text{isomorphism classes of deformations of } X_0 \text{ over } T'\}$$

is bijective.

Theorem 1.1.4 (R. Elkik).

There exists a versal henselian S-deformation of X_0 (and it is unique up to a nonunique isomorphism). If S is noetherian, a deformation X/T is such a deformation if and only if X induces over the completion $\widehat{T}$ of T a versal formal S-deformation of X_0 (a minimally formal versal deformation; VI 1.9 and VI 4).

For the proof of this delicate theorem, we refer to R. Elkik (ENS seminar, 1971/72).

1.2. Ordinary quadratic singularities

Let y be a closed point of a scheme Y of finite type over a field k of characteristic p, and let n be the dimension of Y at y.

Definition 1.2.1 (= VI 6.6).

(i) If k is algebraically closed, y is called an ordinary quadratic point of Y if the completion $\widehat{\mathcal{O}}_{Y,y}$ of the local ring of Y at y is isomorphic to the quotient of $k[[x_1, \ldots, x_{n+1}]]$ by the ideal generated by a single formal power series f which can be written

$$f(\underline{X}) = Q(\underline{X}) + \text{terms of order } > 2, \tag{1.2.1.1}$$

where $Q(\underline{X})$ is an ordinary quadratic form. [22]

(ii) If $\bar{k}$ is an algebraic closure of k, y is called an ordinary quadratic point of Y if the points of $\bar{Y} = Y \otimes_k \bar{k}$ lying over y are ordinary quadratic points of $\bar{Y}$.

1.2.2.

Replacing "ordinary quadratic form" by "nondegenerate quadratic form" in this definition gives the notion of a nondegenerate quadratic point (VI 6.0). A point y is a nondegenerate quadratic point of Y if and only if it is an ordinary quadratic point and either $p \neq 2$ or n is odd.

Example 1.2.3.

Let $Q(\underline{X})$ be a nondegenerate quadratic form in $n+1$ variables over k. Let Y_0 be the quadratic cone in $\mathbb{A}_k^{n+1}$ with equation $Q = 0$. Then the origin, denoted by y_0, is a nondegenerate quadratic point of Y_0.

Example 1.2.4.

Suppose that $p = 2$ and $n = 2m$. Let $a \in k$, and let Q be a polynomial

$$Q(\underline{X}) = (X_0^2 - a) + \sum_{0 < i \leq j \leq 2m} a_{ij} x_i x_j.$$

Suppose that the quadratic form in $2m$ variables $\sum_{0<i\leq j} a_{ij} x_i x_j$ is nondegenerate, and denote by y_0 the closed point of $\mathbb{A}_k^{n+1}$ with coordinates $(\sqrt{a}, 0, \ldots, 0)$. Then y_0 is an ordinary quadratic point of the closed subscheme $Y_0 \subset \mathbb{A}_k^{n+1}$ with equation $Q = 0$.

Example 1.2.5.

Let k' be a finite separable extension of k, and let Y_0 be a k'-scheme of the kind considered in 1.2.3 (respectively, 1.2.4). If Y_0 is regarded as a k-scheme, then y_0 is a nondegenerate (respectively, ordinary) quadratic point of Y_0. The extension k'/k identifies with the largest separable subextension of $k(y_0)/k$.

Theorem 1.2.6.

Suppose that y is an ordinary quadratic point of the k-scheme Y, and let k' be the largest separable subextension of $k(y)/k$. There then exist a k'-scheme $Y_0 \subset \mathbb{A}_{k'}^{n+1}$ of the kind in 1.2.3 or 1.2.4 and a k-isomorphism φ between the henselization $Y_{(y)}$ of Y at y and the henselization $Y_{0(y_0)}$ of Y_0 at y_0.

First observe that there is an etale neighborhood Y' of y in Y

[22] The printed display is numbered (4.2.1) and has an extra equals sign after $Q(\underline{X})$; the English restores the local numbering and the asserted expansion $f = Q+$ higher-order terms.

$$\begin{tikzcd} & Y' \arrow[d] \\ y \arrow[ur, hook] \arrow[r, hook] & Y \end{tikzcd}$$

which is a k'-scheme. Indeed, $Y_{(y)}$ is canonically a k'-scheme. Replacing (k, Y, y) by (k', Y', y) reduces us to the case $k = k'$, that is, to the case in which $k(y)/k$ is radicial. The $k(y)$-vector space $(\Omega^1_{Y/k})_y$ has dimension $n+1$. It is enough to check this after extending scalars from $k(y)$ to its algebraic closure and completing at y, where it is immediate. A neighborhood of y in Y is therefore a closed subscheme of a smooth k-scheme Z of dimension $n+1$. Moreover, after replacing Y and Z by suitable neighborhoods of y, the ideal in $\mathcal{O}_Z$ defining Y is principal (the same argument as above), say with equation f.

Lemma 1.2.7.

Suppose that y is a nondegenerate quadratic point of Y and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $J^n_{Y/k}$ (1.1.1) is generated by one element, supported at y, and of rank one over k. In particular, $k(y) = k$.

It is enough to prove the first assertion after passing to an algebraic closure of k and completing at y. It is then immediate by direct calculation: with the notation of 1.2.1, Nakayama's lemma shows that the $\partial f/\partial x_i$ (and hence, a fortiori, f and the $\partial f/\partial x_i$) generate the maximal ideal.

Let us prove 1.2.6 when y is nondegenerate. We may and shall suppose that $k(y) = k$, that Y is defined by an equation f in a smooth scheme Z, and that there is an etale morphism

$$x : Z \longrightarrow \mathbb{A}^{n+1}_k$$

(a "system of local coordinates x_i") taking y to 0. There is then a nondegenerate quadratic form Q in $n+1$ variables such that, at y,

$$f = Q(x_i) + \text{terms of order } > 2.$$

Let Y_0 be defined by Q as in 1.2.3.[23]

On Y, the system (x_i) is an "approximate solution" of the equation $Q(x_i) = 0$: if $\mathfrak{m}$ is the ideal defining y, then

$$Q(x_i) \in \mathfrak{m}^3 \qquad \text{on } Y.$$

Since Q is nondegenerate, the ideal of $\mathbb{A}^{n+1}_k$ generated by the $\partial Q/\partial x_i$ is the maximal ideal at 0, and its inverse image under $x = (x_i) : Y \to \mathbb{A}^{n+1}_k$ is $\mathfrak{m}$. By the implicit-function theorem 1.1.2 (applied with $a = \mathfrak{m}$ and $\delta' = J$), there are functions x'_i on the henselization $Y_{(y)}$, congruent to the x_i modulo $\mathfrak{m}^2$, which satisfy $Q(x'_i) = 0$. They give the required isomorphism between $Y_{(y)}$ and $Y_{0(y_0)}$.

Lemma 1.2.8.

Suppose that y is a degenerate ordinary quadratic point ($p = 2$ and n even), and that $k(y)/k$ is radicial. Then, in a neighborhood of y, $T^1_{Y/k}$ is generated by one element, supported at y, and of rank two over k. In particular, either $k(y) = k$, or $k(y)$ is isomorphic to $k(\sqrt{a})$ for some $a \in k$, $a \notin k^2$.

Proceed as in 1.2.7. With the notation of 1.2.1, it is enough to check that

$$\dim_k \big(k[[x_1, \ldots, x_{n+1}]]/(f, \partial f/\partial x_i)\big) = 2.$$

[23] The print refers to 4.3 here; the normal form just invoked is Example 1.2.3.

In suitable coordinates $x_0, x_1, \ldots, x_{2m}$, the formal power series f can be written

$$f = x_0^2 + \sum_{i=1}^{m} x_i x_{i+m} + R, \qquad \operatorname{ord}(R) \geq 3.$$

Let $\mathfrak{n}'$ be the ideal generated by f and the $\partial f/\partial x_i$, let $\mathfrak{n}$ be the ideal generated by x_0^2 and the x_i for $i \neq 0$, and let $\mathfrak{q}$ be the maximal ideal. We prove that $\mathfrak{n} = \mathfrak{n}'$. Plainly $\mathfrak{n}' \subset \mathfrak{n}$, and by Nakayama it is enough to check that

$$\mathfrak{n}' + \mathfrak{n}\mathfrak{q} = \mathfrak{n}.$$

Modulo $\mathfrak{n}\mathfrak{q}$, one has $f \equiv x_0^2$; modulo $\mathfrak{n}\mathfrak{q} + (x_0^2)$, one has

$$\frac{\partial f}{\partial x_i} \equiv x_{i+m}, \qquad \frac{\partial f}{\partial x_{i+m}} \equiv x_i \qquad (1 \leq i \leq m),$$

which proves the assertion.

Under the hypotheses of 1.2.8, if $k = k(y)$, we shall let a denote any square in k.

Lemma 1.2.9.

Let V be a vector bundle on a scheme S, and let T be a closed subscheme of $\mathbb{A}_S(V)$ locally free of rank two over S. Then T is contained in one and only one line in $\mathbb{A}_S(V)$.

The question is local for the fppf topology on S, so we may suppose that T/S has a section. By translation, we may suppose that T contains the zero section. Let $\mathfrak{n}$ be the ideal of T defining 0; it is locally free of rank one over S. Every homogeneous linear form $v \in V$ on $\mathbb{A}_S(V)$ defines an element of $\mathfrak{n}$, and hence an epimorphism $V \twoheadrightarrow \mathfrak{n}$. Its kernel defines the required line.

1.2.10.

In particular, if A is a radicial rank-two closed subscheme of affine space $\mathbb{A}_k^{n+1}$ over a field k of characteristic 2, there is an $a \in k$ such that, in suitable affine coordinates, A has equations

$$x_0^2 - a = 0, \qquad x_i = 0 \quad (1 \leq i \leq n).$$

It is enough to take the x_i which vanish on the line defined by A.

1.2.11.

Let us prove 1.2.6 for a degenerate y ($p = 2$ and $n = 2m$). We may and shall suppose that $k(y) = k(\sqrt{a})$, that Y is defined by an equation f in a smooth scheme Z, and that there is a system of local coordinates at y

$$x = (x_i)_{0 \leq i \leq 2m} : Z \longrightarrow \mathbb{A}_k^{2m+1}$$

taking y to the closed point $(\sqrt{a}, 0, \ldots, 0)$, such that, in a neighborhood of y,

$$J^n_{Y/k} \simeq \mathcal{O}_Y/(x_0^2 - a, x_1, \ldots, x_{2m}).$$

After replacing f by a suitable multiple, one has

$$f = (x_0^2 - a) + \sum_{1 \leq i \leq j \leq 2m} a_{ij} x_i x_j + R,$$

where

(a) the quadratic form in $2m$ variables $\sum a_{ij} x_i x_j$ is nondegenerate;

(b) after extending scalars to $\bar{k}$, R has order at least 3 in $x_0 - \sqrt{a}$ and the x_i.

Let Y_0 be defined, as in 1.2.4, by

$$Q = x_0^2 - a + \sum a_{ij} x_i x_j.$$

On Y, the system (x_i) may again be regarded as an approximate solution of $Q(X_i) = 0$. This time, the ideal δ, the inverse image under x of the ideal generated by the $\partial Q / \partial X_i$, satisfies $\mathcal{O}_Y/\delta \simeq J^n_{Y/k}$; at y it is

$$\delta = (x_0^2 - a, (x_i)_{1 \le i}).$$

Let $\mathfrak{q}$ be the maximal ideal defining y. To conclude as in the nondegenerate case by applying the implicit-function theorem, it remains to verify that

$$Q(x_i) \equiv 0 \pmod{\delta^2 \mathfrak{q}} \qquad \text{on } Y. \tag{1.2.11.1}$$

Let $\bar{\mathfrak{q}}$ be the ideal defining y after scalar extension to an algebraic closure of k. One checks that

$$\delta^2 \bar{\mathfrak{q}} \cap \mathcal{O}_{Y,y} = \delta^2 \mathfrak{q}.$$

It is therefore enough to verify (1.2.11.1) after extending scalars to $\bar{k}$; we may suppose that $a = 0$.[24] At y one then has

$$\delta^2 \mathfrak{q} = (x_0^5, (x_0^3 x_i)_{1 \le i}, (x_0 x_i x_j)_{1 \le i,j}, (x_i x_j x_k)_{1 \le i,j,k}).$$

It follows that $\delta^2 \mathfrak{q} \supset \mathfrak{q}^5$ and that

$$x_0^2 f \equiv x_0^4 \pmod{\delta^2 \mathfrak{q}}, \qquad \delta^2 \mathfrak{q} + (x_0^2 f) \supset \mathfrak{q}^4,$$

$$x_0 f \equiv x_0^3 \pmod{\delta^2 \mathfrak{q} + \mathfrak{q}^4}, \qquad \delta^2 \mathfrak{q} + (x_0 f) \supset \mathfrak{q}^3.$$

Finally,

$$Q(x_i) \equiv 0 \pmod{\delta^2 \mathfrak{q} + (f)} \qquad \text{on } Z,$$

which is (1.2.11.1).

Remark 1.2.12.

Let Q be a not necessarily homogeneous, nonzero quadratic form in $n+1$ variables, and suppose that the closed subscheme $Y \subset \mathbb{A}_k^{n+1}$ with equation Q has an ordinary quadratic singularity at a point y. A nonhomogeneous linear change of variables then puts Y into the form 1.2.3 or 1.2.4. In the nondegenerate case $k(y) = k$, and it is enough to put y at the origin. In the degenerate case it is enough that the support of $T^1_{Y/k}$ lie on the line $x_i = 0$ $(1 \le i \le 2m)$ (cf. 1.2.9).

1.3. Moduli of quadratic singularities

The following proposition follows from R. Elkik's theorem 1.1.4 and from explicit calculations left to the reader (and essentially contained in VI 6).

[24] The print refers at this point to (4.11.1); the displayed congruence under proof is (1.2.11.1).

Proposition 1.3.1.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $Y_0 \subset \mathbb{A}^{n+1}_{k(s)}$ be a scheme of the kind described in 1.2.3 or 1.2.4.

(i) Suppose that Y_0 is of the kind described in 1.2.3, defined by the equation

$$Q(\underline{X}) = \sum_{i \leq j} \bar{a}_{ij} x_i x_j = 0 \qquad (\bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ to elements a_{ij} of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}^{n+1}_T$ with equation

$$\sum a_{ij} x_i x_j - b = 0.$$

(ii) Suppose that Y_0 is of the kind described in 1.2.4, defined by the equation

$$Q(\underline{x}) = (x_0^2 - \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0 \qquad (\bar{a}, \bar{a}_{ij} \in k(s)),$$

and lift the $\bar{a}_{ij}$ and $\bar{a}$ to elements a_{ij} and a of A. Then the versal henselian S-scheme of deformations of the singularity of Y_0 at y_0 is

$$T = \mathrm{Spec}(A\{b, c\}),$$

the versal deformation being the henselization at y_0 of the T-scheme $Y \subset \mathbb{A}^{n+1}_T$ with equation

$$(x_0^2 - a) + \sum a_{ij} x_i x_j + b x_0 + c = 0.$$

The following corollary follows at once from 1.2.6 and 1.3.1.

Corollary 1.3.2.

Let $S = \mathrm{Spec}(A)$ be a henselian local scheme with closed point s, and let $f : X \to S$ be a flat morphism of finite presentation. Suppose that, at the closed point x of X_s, the scheme X_s has an ordinary quadratic singularity, that $k(x)$ is radicial over $k(s)$, and that X_s has dimension n.

(i) If x is nondegenerate, there exist a nondegenerate quadratic form Q in $n+1$ variables with coefficients in A, and an element b in the maximal ideal of A, such that the henselized S-scheme of X at x is isomorphic to the henselization at the origin of the closed subscheme of $\mathbb{A}^{n+1}_S$ with equation $Q - b = 0$, as in 1.3.1(i).

(ii) If x is degenerate ($n = 2m$, $\mathrm{char}(k(s)) = 2$), there exists a nonhomogeneous quadratic form in $n+1$ variables with coefficients in A

$$Q(\underline{X}) = x_0^2 + b x_0 + c + \sum_{1 \leq i \leq j \leq 2m} a_{ij} x_i x_j, \tag{1.3.2.1}$$

such that b lies in the maximal ideal, the quadratic form $\sum a_{ij} x_i x_j$ in $2m$ variables is nondegenerate, and the henselized S-scheme of X at x is isomorphic to the henselization at $(\sqrt{c}, 0, \ldots, 0)$ of the closed subscheme of $\mathbb{A}^{n+1}_S$ with equation $Q = 0$. [25]

[25] The print labels this display (4.14.2) and writes x_0 in its first term. The surrounding normal forms and Remark 1.3.3 require the local tag (1.3.2.1) and the quadratic term x_0^2; the English restores both.

Remark 1.3.3.

In case (i), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at O of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$\sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = \sum a_{ij} x_i x_j.$$

In case (ii), suppose that the henselization of X_s at x is isomorphic, by an isomorphism φ, to the henselization at $(\sqrt{a}, 0, \ldots, 0)$ of the closed subscheme $X_0 \subset \mathbb{A}_k^{n+1}$ with equation

$$(x_0^2 + \bar{a}) + \sum \bar{a}_{ij} x_i x_j = 0,$$

and suppose that the a_{ij} lift the $\bar{a}_{ij}$. One may then take

$$Q = x_0^2 + bx_0 + c + \sum a_{ij} x_i x_j, \qquad c \equiv \bar{a} \pmod{\mathfrak{m}_A}.$$

In both cases, the isomorphism promised in 1.3.2 may be chosen to extend φ.

Corollary 1.3.4.

Let $f : X \to S$ be as in 1.3.2. There is a neighborhood U of x in X such that the nonsmooth points of f contained in U are ordinary quadratic points of their respective fibers.

2. Computation of vanishing cycles in a standard ordinary quadratic case

2.1. Preliminary: the cohomology of a cone

2.1.1.

Let $Y \subset \mathbb{P}^r$ be a projective variety over an algebraically closed field k. We use the following notation.

The affine cone over Y is denoted by $X \subset \mathbb{A}^{r+1}$, and its vertex by O;
$X_{(O)}$ is the henselization of X at O;
$X^* = X - \{O\}$ and $X^*_{(O)} = X_{(O)} - \{O\}$;
$X_1 \subset \mathbb{P}^{r+1}$ is the projective cone over Y, so $X = X_1 - Y$;
$\widetilde{X}$, $\widetilde{X}_{(O)}$, and $\widetilde{X}_1$ are the blowups of X, $X_{(O)}$, and X_1 at O;
Y_0 is the inverse image of O in $\widetilde{X}$, $\widetilde{X}_{(O)}$, or $\widetilde{X}_1$;
$h : \widetilde{X}_1 \to Y$ is the natural projection;
$y_0, y_\infty : Y \to \widetilde{X}_1$ are the sections of h whose images are Y_0 and Y;
F is an abelian torsion group of order prime to the characteristic of k. All cohomology groups considered below have coefficients in F, which will not be displayed.

Our purpose in this subsection is to relate the cohomology of $X^*_{(O)}$ to that of Y.

Proposition 2.1.2.

There are isomorphisms

$$\text{(i)} \qquad H^i(X) \xrightarrow{\sim} H^i(\{O\}) = \begin{cases} F, & i = 0, \\ 0, & i > 0; \end{cases}$$

$$\text{(ii)} \qquad H^i_{\{O\}}(X) \xrightarrow{\sim} H^i_c(X).$$

This proposition is proved by a homotopy argument, which we now formalize.

Let k be an algebraically closed field, let U and V be two k-schemes, let Λ be a torsion ring of order prime to the characteristic of k, and let

$$K \in \operatorname{Ob} D^+(U,\Lambda), \qquad L \in \operatorname{Ob} D^+(V,\Lambda).$$

By a "morphism" f from (U,K) to (V,L) we mean a pair consisting of a morphism $f : U \to V$ and a morphism $\varphi : f^*L \to K$. Such a morphism induces

$$f^* : H^*(V,L) \longrightarrow H^*(U,K).$$

We say that two "morphisms" f_0 and f_1 are homotopic if there exist a connected k-scheme T of finite type, two points 0 and 1 of T, and a "morphism" H from $(U \times T, \operatorname{pr}_1^* K)$ to (V,L) whose fibers at 0 and 1 induce f_0 and f_1.

Lemma 2.1.3.

If f_0 is homotopic to f_1, then $f_0^* = f_1^*$.

There is a sequence of points $0 = x_0, x_1, \ldots, x_n = 1$ of T, smooth connected curves Γ_i, and morphisms $a_i : \Gamma_i \to T$ such that x_i and x_{i+1} belong to the image of a_i. To construct the Γ_i, normalize a connected one-dimensional closed subscheme of T containing 0 and 1. This reduces us to the case in which T is a smooth curve. Consider the commutative square

$$\begin{tikzcd}
U \times T \arrow[r, "\operatorname{pr}_1"] \arrow[d, "\operatorname{pr}_2"'] & U \arrow[d, "f"] \\
T \arrow[r, "t"'] & k.
\end{tikzcd}$$

By smooth base change, applied to t, one has

$$t^* Rf_* K \xrightarrow{\sim} R\operatorname{pr}_{2*}(\operatorname{pr}_1^* K).$$

The map f_i^* $(i = 0,1)$ therefore factors through

$$\begin{tikzcd}
H^n(V,L) \arrow[r, "H^*"] & H^n(U \times T, \operatorname{pr}_1^* K) \arrow[r] & H^0\big(T, R^n \operatorname{pr}_{2*}(\operatorname{pr}_1^* K)\big) \arrow[dd, "\text{fiber at } i"] \\
& H^0\big(T, t^* H^n(U,K)\big) \arrow[ur, "\sim"] & \\
& & H^n(U,K) \arrow[ul, "\sim"]
\end{tikzcd}$$

and is consequently independent of i.

2.1.3.

Let us prove 2.1.2. Scalar multiplication gives a homotopy between the identity map of X and the constant map from X to X with value O; this proves (i).

Using scalar multiplications whose factors tend to infinity, one sees that Y is a deformation retract of X_1^*. Assertion (ii) then follows from the five lemma and the diagram

$$\begin{tikzcd}
{} \arrow[r, "\partial"] & H^i_{\{O\}}(X) \arrow[r] \arrow[d] & H^i(X_1) \arrow[r] \arrow[d, equal] & H^i(X_1^*) \arrow[r, "\partial"] \arrow[d, "\wr"] & {} \\
{} \arrow[r, "\partial"] & H^i_c(X) \arrow[r] & H^i(X_1) \arrow[r] & H^i(Y) \arrow[r, "\partial"] & {} .
\end{tikzcd}$$

Corollary 2.1.4.

$$H^i(X^*) \xrightarrow{\sim} H^i(X^*_{(O)}).$$

Indeed, apply the five lemma to

$$\begin{tikzcd}
{} \arrow[r] & H^i_{\{O\}}(X) \arrow[r] \arrow[d, "\wr"] & H^i(X) \arrow[r] \arrow[d, "\wr"', "(2.1.2(i))"] & H^i(X^*) \arrow[r] \arrow[d] & {} \\
{} \arrow[r] & H^i_{\{O\}}(X_{(O)}) \arrow[r] & H^i(X_{(O)}) \arrow[r] & H^i(X^*_{(O)}) \arrow[r] & {} .
\end{tikzcd}$$

2.1.5.

The Leray spectral sequences for the restrictions of h to $\widetilde{X}_1 - Y_0$ and $\widetilde{X}_1 - Y$ show that the restriction maps

$$H^i(\widetilde{X}_1 - Y_0) \xrightarrow{\sim} H^i(Y), \qquad H^i(\widetilde{X}) \xrightarrow{\sim} H^i(Y_0) = H^i(Y)$$

are isomorphisms. Under these isomorphisms, the long exact cohomology sequences of the pairs $(\widetilde{X}_1 - Y_0, Y)$ and $(\widetilde{X}_1 - Y, Y_0)$ form the rows of the diagram

$$\begin{tikzcd}
H^{i-1}(X^*) \arrow[r] \arrow[d, equal] & H^{i-2}(Y)(-1) \arrow[r, "\eta"'] \arrow[d, equal] & H^i(Y) \arrow[r] \arrow[d, dash, "-1"] & H^i(X^*) \arrow[r] \arrow[d, equal] & {} \\
H^{i-1}(X^*) \arrow[r] & H^{i-2}(Y)(-1) \arrow[r, "-\eta"'] & H^i(Y) \arrow[r] & H^i(X^*) \arrow[r] & {} .
\end{tikzcd} \tag{2.1.5.1}$$

Lemma 2.1.6.

The arrows denoted by η and $-\eta$ in (2.1.5.1) are cup products with the cohomology class of a hyperplane section. Diagram (2.1.5.1) is commutative.

The first assertion follows because η (respectively $-\eta$) is the restriction to Y (respectively to $Y_0 \simeq Y$) of the cohomology class of the invertible sheaf $\mathcal{O}(Y)$ (respectively $\mathcal{O}(Y_0)$) on $\widetilde{X}_1 - Y_0$ (respectively $\widetilde{X}_1 - Y$).

Use the Leray spectral sequence for h. The two rows of (2.1.5.1) are obtained by applying the functor $H(Y, -)$ to the distinguished triangles

$$y_{\infty *} R y^!_\infty F \longrightarrow R(h|\widetilde{X}_1 - Y_0)_* F \xrightarrow{①} R(h|X^*)_* F \xrightarrow{②}$$

and

$$y_{0*} R y^!_0 F \longrightarrow R(h|\widetilde{X}_1 - Y)_* F \longrightarrow R(h|X^*)_* F \longrightarrow .$$

The nonzero cohomology sheaves of the complexes at the vertices of these triangles occur in degrees (2), (0), and (0 and 1). The arrows ① and ② are therefore determined by their effect on cohomology sheaves. This reduces us to the case in which Y is a point, which is left to the reader.

2.1.7.

The long exact sequence of the pair $(\widetilde{X}_{(O)}, Y_0)$ is a local analogue of the second row of (2.1.5.1). One has

$$H^i(\widetilde{X}_{(O)}) \xrightarrow{\sim} H^i(Y_0)$$

by proper base change for the morphism $\widetilde{X}_{(O)} \to X_{(O)}$. The long exact sequence in question may therefore be written

(2.1.7.1)
$$\begin{tikzcd}
H^{i-1}(X^*_{(O)}) \arrow[r] & H^{i-2}(Y_0)(-1) \arrow[r, "-\eta"] & H^i(Y_0) \arrow[r] & H^i(X^*_{(O)}) \\
H^{i-1}(X^*) \arrow[r] \arrow[u, "\wr"] & H^{i-2}(Y)(-1) \arrow[r, "-\eta"] \arrow[u, equal] & H^i(Y) \arrow[r] \arrow[u, equal] & H^i(X^*) \arrow[u, leftarrow, "\wr"']
\end{tikzcd},$$

the second row of this commutative diagram being the second row of (2.1.5.1). [26]

The following technical lemma will be used in 2.2.7.

Lemma 2.1.8.

$$\begin{tikzcd}
H^{n-1}(Y_0) \arrow[d, equal] & H^{n-1}(\widetilde{X}) \arrow[l, "\sim"'] \arrow[r] & H^{n-1}(X^*) \arrow[r] & H^n_{\{x\}}(X) \arrow[r] & H^n_c(X_s) \arrow[d, equal] \\
H^{n-1}(Y) \arrow[rrrr, "\partial"] & & & & H^n_c(X_s)
\end{tikzcd}$$

is anticommutative. [27]

By 2.1.7, this is equivalent to proving the commutativity of

$$\begin{tikzcd}
H^{n-1}(X_1 - \{O\}) \arrow[r, "\partial"] \arrow[d, "\wr"] & H^n_{\{O\}}(X_1) = H^n_{\{O\}}(X) \arrow[d] \\
H^{n-1}(Y) \arrow[r, "\partial"] & H^n_c(X).
\end{tikzcd}$$

This is general nonsense.

2.2. Vanishing cycles

2.2.1.

Let S be a henselian trait, and let $s, \eta, \bar{s}, \bar{\eta}$ have the meaning of XIII, 0.2.5. Put $\Lambda = \mathbb{Z}/k$, where k is prime to the residual characteristic of S. Let $\mathbb{A}^{n+1}_S$ be standard affine $(n+1)$-space over S, and let X be the closed subscheme of $\mathbb{A}^{n+1}_S$ defined by a quadratic equation $Q = 0$ which is nonzero modulo a uniformizer:

$$Q(\underline{x}) = \sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i + c.$$

Regard $\mathbb{A}^{n+1}_S$ as the complement in $\mathbb{P}^{n+1}_S$ of the hyperplane at infinity H. Then X is obtained from the quadric $X_1 \subset \mathbb{P}^{n+1}_S$ with homogeneous equation

(2.2.1.1)
$$\sum_{i \le j} a_{ij} x_i x_j + \sum_i b_i x_i z + c z^2 = 0$$

by removing $Y = X_1 \cap H$.

We make the following assumptions:

(a) Y is a smooth quadric over S, that is, the quadratic form $\sum_{i \le j} a_{ij} x_i x_j$ is ordinary;

(b) the quadric X_s is singular, that is, $X_{\bar{s}}$ is a quadratic cone.

[26] The printed tag omits the dot between 1 and 7; the English uses the local number (2.1.7.1).

[27] The print numbers this lemma 2.7.8. Its position and the forward reference in 2.2.7 require 2.1.8.

[28]

Let x_0 denote the singular point of X_s, and let $f : X \to S$ be the projection.

Remark 2.2.2.

If x_0 is a rational point, a translation puts it at the origin of the coordinates. If $x_0 = 0$, the b_i and c are divisible by a uniformizer.

The subscheme A of X_s defined by the equations $\partial Q/\partial x_i$ is concentrated at x_0. Except in the exceptional case where $k(s)$ has characteristic 2 and $n+1$ is even, this scheme has degree one over $k(s)$, and hence x_0 is rational. In the exceptional case it is ramified over $k(s)$ and has rank 2, so that $k(x_0)$ is either $k(s)$ or a purely inseparable quadratic extension of $k(s)$. The assertions about A are checked most easily after passing to the algebraic closure $k(\bar{s})$ of $k(s)$.

Proposition 2.2.3.

Under the assumptions of 2.2.1, the morphisms

$$H^i(X_{\bar{\eta}}, \Lambda) \xrightarrow{\sim} \mathbb{H}^i\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xrightarrow{\sim} R^i\psi_{\bar{\eta}}(\Lambda)_{x_0}$$

$$H^i_c(X_{\bar{\eta}}, \Lambda) \xleftarrow{\sim} \mathbb{H}^i_c\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big) \xleftarrow{\sim} \mathbb{H}^i_{\{x_0\}}\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big)$$

are isomorphisms.

That the left-hand arrows are isomorphisms follows from 2.1.8.6 and 2.1.10.5; the assertion for the right-hand arrows is a corollary of 2.1.2. Indeed, there is a distinguished triangle

$$(\Lambda \text{ on } X_{\bar{s}})[0] \longrightarrow R\psi_{\bar{\eta}}(\Lambda) \longrightarrow R\Phi_{\bar{\eta}}(\Lambda),$$

and 2.1.2 applies to (Λ on $X_{\bar{s}}$), whereas $R\Phi_{\bar{\eta}}(\Lambda)$ is supported at x_0.

Corollary 2.2.4.

If the geometric generic fiber $X_{\bar{\eta}}$ is singular, that is, if it is still a quadratic cone, then all the sheaves $R^i\Phi(\Lambda)$ vanish.

By 2.1.2,

$$\Lambda = H^0(X_{\bar{s}}, \Lambda) \xrightarrow{\sim} H^0(X_{\bar{\eta}}, \Lambda), \qquad 0 = H^i(X_{\bar{s}}, \Lambda) = H^i(X_{\bar{\eta}}, \Lambda) \quad (i > 0).$$

We then apply the long exact sequence XIII, 2.1.8.9 (valid by XIII, 2.1.9). [29]

2.2.5.

Suppose now that

$$(*) \qquad \text{the generic fiber } X_\eta \text{ is smooth.}$$

For simplicity we shall also assume that S is strictly henselian. The sheaves of vanishing cycles are easily computed with the aid of 2.2.3 and XII, 3.7. The cohomology with coefficients in Λ considered here follows from the ℓ-adic cohomology of the cited passage by the universal-coefficient formula (cf. 2.1.13). The results are as follows.

A. Cohomology without supports.

One has $R^i\psi_{\bar{\eta}}(\Lambda) = 0$ for $i \neq 0, n$.

1) If $n \neq 0$: $\psi_{\bar{\eta}}(\Lambda) = \Lambda$. Moreover, $R^n\psi_{\bar{\eta}}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

[28] The print has X'_s in (b). The equivalent clause and the immediately following definition of the singular point show that X_s is intended.

[29] The print uses the letter o in the two occurrences of H^o; the cohomological degree is 0.

2) <u>For arbitrary n:</u> even when $n = 0$, one checks that $R^i\Phi_{\bar\eta}(\Lambda) = 0$ for $i \neq n$, and that $R^n\Phi_{\bar\eta}(\Lambda)$ is (noncanonically) isomorphic to the sheaf Λ on x_0, extended by zero to X_s.

<u>B. Cohomology with support at x_0.</u>

One has

$$H^i_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda)\big) = 0 \qquad (i \neq n, 2n).$$

There is, moreover, a trace morphism

$$\mathrm{tr} : H^{2n}_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big) \longrightarrow \Lambda. \tag{2.2.5.1}$$

<u>If $n \neq 0$:</u> the morphism (2.2.5.1) is an isomorphism. Moreover,

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big)$$

is (noncanonically) isomorphic to Λ.

<u>C. Duality.</u> For

$$a \in R^n\psi_{\bar\eta}(\Lambda)_{x_0}, \qquad b \in H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big),$$

put $(a, b) = \mathrm{Tr}(a \wedge b)$, where Tr is the morphism XIII, 2.2.6

$$\mathrm{Tr} : H^{2n}_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big) \longrightarrow \Lambda.$$

It follows from Poincaré duality on $X_{\bar\eta}$ and the isomorphisms 2.2.3 that (a, b) gives a perfect duality between the free Λ-modules

$$R^n\psi_{\bar\eta}(\Lambda)_{x_0} \quad \text{and} \quad H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(n))\big).$$

[30] <u>D. $n = 2m > 0$.</u>

By XII, 3.7, the groups

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(m))\big) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. Let ϕ be the natural map from the first of these groups to the second. The signs of δ and δ' may be normalized so that

$$(\delta', \delta) = 1; \tag{2.2.5.2}$$

then

$$\phi(\delta) = (-1)^m\, 2\, \delta'. \tag{2.2.5.3}$$

Let Z be the separable quadratic extension of $k(\eta)$ which is the center of the even part of the Clifford algebra of the quadratic form (2.2.1.1), and let ϵ be the corresponding character, of order 1 or 2, of the inertia group I. For $\sigma \in I$, one has

$$\sigma\delta = \epsilon(\sigma)\delta \tag{2.2.5.4}$$

and

$$\sigma\delta' = \epsilon(\sigma)\delta'. \tag{2.2.5.5}$$

[30] The print drops the bar from $\bar\eta$ in the displayed trace map; the surrounding formulas determine the indicated term.

Formulas (2.2.5.2), (2.2.5.3), and (2.2.5.4), in cohomology with coefficients in $\mathbb{Z}/2k$, imply that the variation is

(2.2.5.6) $$\mathrm{Var}(\sigma)(a) = \frac{\epsilon(\sigma) - 1}{2}\,(-1)^m\,(a, \delta)\,\delta.$$

E. $n = 2m + 1$—Vanishing cycles.

By XII, 3.7, the groups

$$H^n_{\{x_0\}}\big(X_s, R\psi_{\bar\eta}(\Lambda(m))\big) \quad \text{and} \quad R^n\psi_{\bar\eta}(\Lambda(m+1))_{x_0}$$

have natural generators, defined up to sign; denote them by δ and δ', respectively. With the notation of D, their signs may be normalized so that

(2.2.5.7) $$(\delta', \delta) = 1;$$

then

(2.2.5.8) $$\phi(\delta) = 0.$$

F. $n = 2m + 1$—The variation.

The inertia group I acts trivially on the cohomology of X_η. This follows from XII, 3.7 and from the fact that I acts trivially on the cohomology of Y_η, since Y is proper and smooth over S.

For σ in the inertia group one necessarily has

(2.2.5.9) $$D(\sigma)(\delta') = \lambda(\sigma)\delta, \qquad \lambda(\sigma) \in \Lambda(1).$$

The additivity formula (XIII, 1.4.3.4) becomes

$$\lambda(\sigma\tau) = \lambda(\sigma) + \lambda(\tau).$$

Let

$$\epsilon : I \longrightarrow \Lambda(1) = \mu_k$$

be the homomorphism such that

$$\sigma\big(\sqrt[k]{t}\big) = \epsilon(\sigma)\,\sqrt[k]{t}$$

for a uniformizer t. In view of the structure of the tame inertia group, λ is necessarily a multiple of ϵ. Formula (2.2.5.9) may therefore be rewritten

(2.2.5.10) $$\mathrm{Var}(\sigma)(a) = \lambda_X \epsilon(\sigma)(a, \delta)\delta$$

for a suitable $\lambda_X \in \Lambda$ (depending on X/S). We shall determine λ_X later. [31]

2.2.6.

We shall give the cycles δ and δ' of D and E a description which is local for the ètale topology.

The case of even n is easy. If $2 \nmid k$ ($\Lambda = \mathbb{Z}/k$), then $\pm\delta$ is characterized by (2.2.5.3) and (2.2.5.4), that is, by

(2.2.6.1) $$(\delta, \delta) = (-1)^m\,2.$$

In general, $\pm\delta$ is the image of any cycle $\widetilde{\delta}$, in cohomology with coefficients in $\mathbb{Z}/2^a k$ for a sufficiently large, which satisfies (2.2.6.1).

[31] In (2.2.5.6) and (2.2.5.10), the print suppresses the comma in (a, δ); this is the pairing defined in C.

Suppose that n is odd. Then x_0 is a rational point (2.2.2). Let $X_{s(0)}$ be the henselization of X_s at x_0, let $\widetilde{X}_{s(0)}$ be its blowup at x_0, let Y_0 be the inverse image of x_0 in $\widetilde{X}_{s(0)}$ (a quadric), and put

$$X^*_{s(0)} = X_{s(0)} - \{x_0\} = \widetilde{X}_{s(0)} - Y_0.$$

Consider the composite map (cf. XIII, 2.4.6.1)
(2.2.6.2)

$$\begin{tikzcd}
H^{n-1}(Y_0, \Lambda(m)) & H^{n-1}(\widetilde{X}_{s(0)}, \Lambda(m)) \arrow[l, "\sim"'] \arrow[r] & H^{n-1}(X^*_{s(0)}, \Lambda(m)) \arrow[d] \\
 & H^n_{\{x_0\}}\big(X_s, R\Psi_{\bar\eta}(\Lambda(m))\big) & H^n_{\{x_0\}}(X_s, \Lambda(m)) \arrow[l]
\end{tikzcd}$$

Lemma 2.2.7.

The map (2.2.6.2) identifies $H^n_{\{x_0\}}(X_s, R\Psi_{\bar\eta}(\Lambda(m)))$ with the primitive quotient of the middle-dimensional cohomology of the $2m$-dimensional quadric Y_0; the two classes $\pm\delta$ are the images of the natural generators of this primitive quotient.

The cycles $\pm\delta$ are characterized by their images in $H^n_c(X_{\bar\eta}, \Lambda(m))$. Using (2.1.7.1), it is therefore enough to prove that the composite map

$$\begin{tikzcd}
H^{n-1}(Y_0, \Lambda(m)) & H^{n-1}(\widetilde{X}_s, \Lambda(m)) \arrow[l, "\sim"'] \arrow[r] & H^{n-1}(X^*_s, \Lambda(m)) \arrow[d] \\
H^n_c(X_{\bar\eta}, \Lambda(m)) & H^n_c(X_s, \Lambda(m)) \arrow[l, "\mathrm{sp}"'] & H^n_{\{x_0\}}(X_s, \Lambda(m)) \arrow[l]
\end{tikzcd}$$

sends the distinguished generators of the primitive quotient of the first group to the distinguished generators of the last group.

Apply 2.1.8. Up to sign, the map may be rewritten

$$\begin{tikzcd}
H^{n-1}(Y_s, \Lambda(m)) \arrow[r, "\partial"] \arrow[d, equal] & H^n_c(X_s, \Lambda(m)) \arrow[d] \\
H^{n-1}(Y_{\bar\eta}, \Lambda(m)) \arrow[r] & H^n_c(X_{\bar\eta}, \Lambda(m)),
\end{tikzcd}$$

and 2.2.7 follows. [32]

3. The Picard–Lefschetz formula

3.1. Preliminary results

3.1.1.

Let S be a henselian trait and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n. Suppose that the special fiber X_s is smooth except at a finite set of points Σ', at which X_s has an ordinary quadratic singularity.

Let Σ be the set of points $x \in \Sigma'$ such that the generic fiber X_η is smooth in a neighborhood of x. Finally, let Λ be a torsion ring whose order is prime to the residual characteristic of S.

[32] In this proof the print has H^i_c for the group in which the degree-n cycle δ is characterized; the displayed maps and all surrounding terms require H^n_c. The print also abbreviates the support point in (2.2.6.2) as x; the English uses the fixed singular point x_0.

Proposition 3.1.2.

(i) The sheaves $R^i\Phi(\Lambda)$ vanish for $i \neq n$.

(ii) The sheaf $R^n\Phi(\Lambda)$ is zero outside Σ. Its restriction to Σ is a sheaf of Λ-modules of rank one.

(iii) For every geometric point $\bar{x}$ of Σ, the pairing (a, b) of 2.2.5(C) puts the Λ-modules

$$\big(R^n\psi(\Lambda)\big)_{\bar{x}} \quad \text{and} \quad H^n_{\{\bar{x}\}}\big(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda)\big)$$

in duality. They are free of rank one if $n \neq 0$, and of rank two if $n = 0$.

33

The assertions of 3.1.2 are local for the ètale topology. We may therefore treat only the case where S is strictly henselian and X is the hypersurface in relative affine space $\mathbb{A}^{n+1}_S$ defined by a quadratic equation satisfying 2.2.1(a) and (b) (apply 1.3.2).

The morphism f is smooth at every point of $X_s - \Sigma'$, hence $R\Phi = 0$ outside Σ'. By 2.2.4, $R\Phi$ even vanishes outside Σ. We may now assume 2.2.5(*); assertions (ii) and (iii) then follow from 2.2.5(A), (B), and (C).

3.2. Even relative dimension

We retain the notation of 3.1 and take $n = 2m$ to be even. For simplicity we suppose that S is strictly henselian (the general case would follow by descent).

Proposition 3.2.1.

(i) For $x \in \Sigma$, the group $H^n_{\{x\}}(X_s, R\psi_{\bar{\eta}}(\Lambda(m)))$ has a natural generator δ, well defined up to sign, characterized by being natural in Λ and by satisfying

$$(\delta, \delta) = (-1)^m\, 2.$$

For $n = 0$, add the condition $\mathrm{Tr}(\delta) = 0$.

(ii) For a suitable character, independent of Λ,

$$\varepsilon_x : I \longrightarrow \{\pm 1\}$$

of the inertia group I, the variation is

$$\mathrm{Var}(\sigma)(a) = (-1)^m \left(\frac{\varepsilon_x(\sigma) - 1}{2}\right)(a, \delta)\delta$$

for $a \in R^n\Phi_{\bar{\eta}}(\Lambda(m))$. Consequently,

$$\sigma(\delta) = \varepsilon_x(\sigma)\delta.$$

34

Passing to the inverse limit, it is enough to treat the universal case $\Lambda = \mathbb{Z}_\ell$. Up to sign, only one δ can then satisfy the conditions in (i). As in 3.1.2, we reduce to the case 2.2.5(*) and apply 2.2.5(D).

33 The print says that $R^n\Phi(\Lambda)$ is "alone" outside Σ (*seul*); the proof and the asserted support on Σ require "zero" (*nul*). In (iii) it prints $H^n_{\{\bar{x}\}}(R^n\psi_{\bar{\eta}}(\Lambda))$; the pairing cited from 2.2.5(C) has the second term $H^n_{\{\bar{x}\}}(X_{\bar{s}}, R\psi_{\bar{\eta}}(\Lambda))$.

34 The print has $H^n_{\{x\}}(R^n\psi_{\bar{\eta}}(\Lambda(m)))$ in (i). The proof explicitly invokes 2.2.5(D), where the natural generator belongs to $H^n_{\{x\}}(X_s, R\psi_{\bar{\eta}}(\Lambda(m)))$.

Complement 3.2.2.

Suppose that, at $x \in \Sigma$, the henselization $X_{(x)}$ of X at x is isomorphic to the henselization at x_0 of the projective scheme Q with equation

$$\sum_{i \leq j} a_{ij} x_i x_j = 0.$$

Then the character ε_x is defined by the separable quadratic extension $Z(C^+(Q))$ of the fraction field. This follows from 2.2.5(D).

3.2.3.

In residual characteristic different from 2, Complement 3.2.2 can be made more explicit. In that case I has a unique nontrivial character ε of order 2. For a uniformizer t, one has

$$\sigma(\sqrt{t}) = \varepsilon(\sigma)\sqrt{t}.$$

On the other hand, $X_{(x)}$ is then isomorphic to the henselization at 0 of an affine scheme

$$\sum a_{ij} x_i x_j = b,$$

where b lies in the maximal ideal and $Q(x) = \sum a_{ij} x_i x_j$ is nondegenerate. The ideal (b) defines the image of the subscheme of $X_{(x)}$ which is the support of $J^n_{X_{(x)}/S}$. The center of the Clifford algebra is isomorphic to

$$k(\eta)\left(\left((-1)^{m+1} 2b \det(a_{ij})\right)^{1/2}\right)$$

(Bourbaki, *Algèbre*, Ch. 9, §9, no. 4, final remark), whence

$$\varepsilon_x = \varepsilon^{v(b)}.$$

The variation is zero if b has even valuation; otherwise $\varepsilon_x = \varepsilon$.

3.3. Odd relative dimension

To prove the Picard–Lefschetz formula in odd relative dimension, we shall have to reduce to the transcendental case treated in XIV, 3.2.1.1. The specialization argument required for this forces us to work over a base which is no longer one-dimensional. We shall use ad hoc arguments adapted to the precise case we need: I do not have a usable theory over a base of dimension greater than one, and I have preferred not to cover my ignorance with generalities.

3.3.1.

Let $S = \mathrm{Spec}(A)$ be a strictly local noetherian scheme, and let $f : X \to S$ be a flat morphism of finite type, purely of relative dimension n, whose special fiber X_s has no nonsmooth points other than nondegenerate quadratic points. Let s be the closed point of S, let X_s be the special fiber, and let $x_0 \in X_s$ be a singular point.

By 1.3.2(i), in a neighborhood of x_0 the scheme X/S is locally isomorphic to an S-scheme defined in $\mathbb{A}^{n+1}_S$ by an equation $Q - b = 0$, where b belongs to the maximal ideal and Q is a nondegenerate quadratic form. The ideal (b) of A depends only on $(X/S, x_0)$: in a neighborhood of x_0 there is a section

$$s \in X(A/(b))$$

of X over $\mathrm{Spec}(A/(b))$ such that

$$J^n_{X/S} \simeq \mathcal{O}_{s(\mathrm{Spec}(A/(b)))}.$$

We denote this section by s_x, and denote by $b(x)$ any generator of (b).

3.3.2.

Suppose that S is normal, and let $b \in A$. If n is invertible in A and $b \neq 0$, write $\varepsilon_b^{(n)}$ for the following homomorphism from the Galois group of the fraction field of A to $\mathbb{Z}/(n)(1)$:

$$\varepsilon_b^{(n)}(\sigma) = (\sigma \sqrt[n]{b})/\sqrt[n]{b}.$$

One has

$$\bigl(\varepsilon_b^{(nm)}\bigr)^m = \varepsilon_b^{(n)}.$$

If Λ is a $\mathbb{Z}/n$-algebra, we again denote by ε_b the composite

$$\mathrm{Gal} \longrightarrow \mathbb{Z}/(n)(1) \longrightarrow \Lambda(1).$$

It is independent of n.

The character ε_b factors through

$$\pi_1\bigl(S - (\text{the subscheme } b = 0)\bigr).$$

3.3.3.

Suppose that S is a strictly henselian trait and that n is odd, $n = 2m+1$. Let x be a singular point of X_s in a neighborhood of which X_η is smooth, and let X_1/S be a subscheme of $\mathbb{A}_S^{n+1}$ with equation $Q - b = 0$, as in 1.3.2(i). The assumption on x means that b, although zero modulo the maximal ideal, is not zero.

Let $X_{s(x)}$ be the henselization of X_s at x, let $\widetilde{X}_{s(x)}$ be the blowup of X_s at x, let Y be the quadric which is the inverse image of x, and put

$$X^*_{s(x)} = X_{s(x)} - \{x\} = \widetilde{X}_{s(x)} - Y.$$

The composite morphism

$$\begin{tikzcd}
H^{n-1}(Y,\Lambda(m)) & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) \arrow[l, "\sim"'] \arrow[r] & H^{n-1}(X^*_{s(x)},\Lambda(m)) \arrow[d] \\
& H^n_{\{x\}}\bigl(X_s, R\Psi_{\bar\eta}(\Lambda(m))\bigr) & H^n_{\{x\}}(X_s,\Lambda(m)) \arrow[l]
\end{tikzcd}$$

identifies the latter group with the primitive quotient (XII, 3.5) of the cohomology of the quadric Y. Indeed, this follows from 2.2.7 in a locally isomorphic situation. The images of the two natural generators of this quotient (the cohomology classes of the rulings of Y) are the two basis vectors

$$\pm\delta \in H^n_{\{x\}}\bigl(X_s, R\Psi_{\bar\eta}(\Lambda(m))\bigr).$$

[35]

3.3.4.

Return to the more general assumptions of 3.3.1, with $n = 2m+1$. For every singular point x of X_s, we again denote by $\pm\delta_x$ the sections of $R^n f_! \Lambda(m)$ over S which are the images of the generators of the primitive quotient of $H^{n-1}(Y,\Lambda(m))$ (with the same notation as above) under the composite morphism

[35] At the definition of the punctured henselization, the print places a tilde on the first $X^*_{s(x)}$. The equality with $X_{s(x)} - \{x\}$ and the identical construction in 2.2.6 determine the untilded term used here.

$$\begin{array}{ccccc}
H^{n-1}(Y,\Lambda(m)) & \xleftarrow{\sim} & H^{n-1}(\widetilde{X}_{s(x)},\Lambda(m)) & \longrightarrow & H^{n-1}(X^*_{s(x)},\Lambda(m)) \\
 & & & & \downarrow \\
H^0\big(S,R^nf_!\Lambda(m)\big) & \xleftarrow{\sim} & H^n_c(X_s,\Lambda(m)) & \longleftarrow & H^n_{\{x\}}(X_s,\Lambda(m))
\end{array}$$

The class $\pm\delta_x$ is the vanishing cycle at x.

When S is a trait, the class

$$\pm\delta \in H^n_c(X_{\bar\eta},\Lambda(m))$$

is the image under XIII, 2.1.8.4 of the class of 3.3.3

$$\pm\delta \in H^n_{\{x\}}\big(X_s,R\Psi_{\bar\eta}(\Lambda(m))\big).$$

We shall prove the following two results simultaneously.

Proposition 3.3.5.

Let S be a henselian trait and let X/S be as in 3.3.3. The variation is given by

$$\mathrm{Var}(\sigma)(a) = (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta)\delta.$$

Proposition 3.3.6.

Let S be strictly local and regular, and let $f : X \to S$ be a proper flat morphism of relative dimension $n = 2m+1$ such that X_s has exactly one nonsmooth point x, assumed to be a nondegenerate quadratic point. Suppose also that $b(x)$ is a parameter. Then Galois acts on the cohomology of the geometric generic fiber by

$$\sigma(a) = a + (-1)^{m+1}\varepsilon_{b(x)}(\sigma)(a,\delta_x)\delta_x.$$

(A) Let k be an algebraically closed field of characteristic 0, and let S be the henselization of $\mathrm{Spec}(k[T])$ at 0. We prove 3.3.5 when $b(x)$ is a uniformizer.

By the Lefschetz principle, we reduce to the case $k=\mathbb{C}$. Since the question is local, we may suppose that $f : X \to S$ is obtained by base change from the map

$$\sum_i z_i^2 : \mathbb{C}^{n+1} \longrightarrow \mathbb{C}.$$

The construction of $\pm\delta$ given above has a transcendental analogue in integral cohomology. Via XIV, 2.1, the class $\pm\delta$ therefore corresponds to a transcendental vanishing cycle, the image of

$$\pm\delta_{\mathrm{cl}} \in H^n_{\{0\}}\big(X_0,R\Psi_{\mathrm{cl}\eta}(\mathbb{Z})\big).$$

By XIV, 2.1 and the corresponding property of the algebraic cycles $\pm\delta$, the image of $\pm\delta_{\mathrm{cl}}$ in

$$H^n_{\{0\}}\big(X_0,R\Psi_{\mathrm{cl}\eta}(\mathbb{Z})\big)\otimes\mathbb{Z}_\ell$$

is a generator of this group. Thus $\pm\delta_{\mathrm{cl}}$ are the two generators of $H^n_{\{0\}}(X_0,R\Psi_{\mathrm{cl}\eta}(\mathbb{Z}))$, and coincide with the vanishing cycles of XIV, 3.2. Formula 3.3.5 consequently follows from XIV, 3.2.11. Strictly speaking, one should have checked compatibility between the isomorphism of XIV, 2.1 and cup products and traces.

(B) We prove 3.3.6 for X/S as in (A), with X/S proper in addition. This follows from XIII, 2.4.6.2.

(C) Let p be a prime number, and let S_0 be the strict henselization of $\mathrm{Spec}(\mathbb{Z}[T])$ at (p, T). We prove 3.3.6 for X/S_0 such that $b(x) = T$.

The restriction of the sheaf $R^i f_* \Lambda$ to the strict henselization of $\mathrm{Spec}(\mathbb{Z}[T])$ at (T) is governed by (B). This sheaf is locally constant on the complement of the smooth divisor $T = 0$, which has mixed characteristic. Assertion (C) therefore follows from (B) and Abhyankar's lemma.

(D) In at least one case to which (C) applies, δ_x is not a torsion element in ℓ-adic cohomology.

Let S_1 be the strict henselization of $\mathrm{Spec}(\mathbb{Z})$ at (p), and let $V \subset \mathbb{P}^r/S_1$ be a smooth projective variety over S_1, purely of relative dimension $n+1$. Suppose that there is a Lefschetz pencil of hyperplane sections for the special fiber V_p of V/S_1 (XVII, 2.5). This pencil is defined by a line d' in the special fiber of the projective space $\check{\mathbb{P}}^{,r}/S_1$ dual to $\mathbb{P}^r/S_1$. Lift d' to a line d in $\check{\mathbb{P}}^{,r}/S_1$, and let $X \to d$ be the pencil of hyperplane sections of V parametrized by d (XVIII, 3.1). The exceptional locus Δ of d is ètale over S_1 (XVII, 6). After ètale localization at a point of $\Delta_p \subset d$, we therefore obtain a situation of type (C).

To check that δ_x is not torsion, it is enough to do so after extending scalars from S_1 to $\mathbb{C}$. Take $V = \mathbb{P}^{n+1}$ with the projective embedding defined by $\mathcal{O}(N)$. For $N \geq 2$, XVII, 2.5 applies (it is valid for every N in this particular case). We then use the following lemma.

Lemma 3.3.7.

For a Lefschetz pencil of hypersurfaces of sufficiently large degree N in $\mathbb{P}^{n+1}(\mathbb{C})$ ($N \geq 3$ suffices), a vanishing cycle in $\mathbb{Q}_\ell$-cohomology is nonzero.

This lemma follows from the theory of Lefschetz pencils, for example from XVIII, 6.6.1. There is no circularity in using here this result of XVIII, based on the "Picard–Lefschetz formula," since (A) is already available. One may instead use the more direct, purely transcendental variant XIX, no. 4, which depends only on XIV, 3.2.

(E) We prove 3.3.5 in residual characteristic p for $(X/S, x \in X_s)$.

Let X_0/S_0 be as in (D). By 1.3.2(i), there is a morphism $g : S \to S_0$ such that the henselization at x_0 of

$$g^* X_0 = X_0 \times_{S_0} S$$

is S-isomorphic to the henselization of X at x. It is therefore enough to prove 3.3.5 for $g^* X_0$. By base change from (C) for X_0/S_0, one obtains

$$\sigma(a) = a - (-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$, for σ in the inertia group. Let π be a uniformizer and compare this formula with (2.2.5.9), where $\varepsilon = \varepsilon_\pi$. We find

$$\lambda_X \varepsilon_\pi(\sigma)(a, \delta_x)\delta_x = -(-1)^m \varepsilon_{b(x)}(\sigma)(a, \delta_x)\delta_x$$

in $H^n(X_{0\bar{\eta}})$. Since δ_x is not torsion, we may cancel $(a, \delta_x)\delta_x$ from this identity. Formula (2.2.5.9), with the resulting value of λ_X, is precisely 3.3.5.

(F) Proof of 3.3.6 in the general case.

The general case of 3.3.6 reduces to the general case of 3.3.5 just as (B) and (C) reduce to (A).

3.4. Summary

Propositions 3.1.2, 3.2.1, 3.2.2, 3.2.3, and 3.3.5 constitute the "Picard–Lefschetz formula." This formula is the algebraic analogue of XIV, 3.2.11, but takes very different forms according to the parity of the relative dimension.

In the most interesting case, where X/S is proper, the results may be summarized as follows.

Theorem 3.4 (Picard–Lefschetz formula).

Let S be a strictly henselian trait and let $f : X \to S$ be a proper flat morphism, purely of relative dimension $n = 2m$ or $2m+1$. Suppose that X_η is smooth and that X_s has only ordinary quadratic singularities. Let Σ be the singular locus of X_s.

(i) To every $x \in \Sigma$ is attached a vanishing cycle

$$\pm\delta_x \in H_n(X_{\bar\eta})(m).$$

The δ_x are pairwise orthogonal. When n is even,

$$(\delta_x, \delta_x) = (-1)^m, 2.$$

(ii) There are isomorphisms

$$H^i(X_s) \xrightarrow{\sim} H^i(X_{\bar\eta})$$

for $i \neq n, n+1$, and an exact sequence

$$0 \longrightarrow H^n(X_s) \longrightarrow H^n(X_{\bar\eta}) \xrightarrow{((\,\cdot\,,\delta_x))_{x\in\Sigma}} \bigoplus_{x\in\Sigma} \Lambda(n-m) \longrightarrow H^{n+1}(X_s) \longrightarrow H^{n+1}(X_{\bar\eta}) \longrightarrow 0.$$

(iii) There are defined local characters of the inertia group I:

$$\begin{aligned} n \text{ even} : \quad & \varepsilon_x : I \longrightarrow \{\pm 1\}, \\ n \text{ odd} : \quad & \varepsilon_x : I \longrightarrow \Lambda(1). \end{aligned}$$

When X is regular at x, these characters are surjective; when x is nondegenerate, they are respectively the unique character of order 2 and the canonical character.

The action of the inertia group I is given by

$$n = 2m \qquad \sigma(a) = a + (-1)^m \sum_{x\in\Sigma} \frac{\varepsilon_x(\sigma) - 1}{2} (a, \delta_x)\delta_x,$$

$$n = 2m+1 \qquad \sigma(a) = a - (-1)^m \sum_{x\in\Sigma} \varepsilon_x(\sigma)(a, \delta_x)\delta_x.$$

In particular,

$$\begin{aligned} n \text{ even} : \quad & \sigma(\delta_x) = \varepsilon_x(\sigma)\delta_x, \\ n \text{ odd} : \quad & \sigma(\delta_x) = \delta_x. \end{aligned}$$

SGA 7 — EXPOSÉ XVI

The Milnor Formula

by P. Deligne

In this exposé, Milnor's formula [3] (see also [2]) is transposed to characteristic p. In certain cases this formula makes it possible to compute the dimensions of spaces of vanishing cycles. The proof is global in nature and treats only the "geometric" case in which the base is a henselian trait of equal characteristic with perfect residue field.

1. Statement of the problem

1.1.

Let S be a regular scheme, purely of dimension one, let s be a closed point of S with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, let x_0 be a closed point of the fiber X_s, and let $\bar{x}_0$ be a geometric point of X localized at x_0. Suppose that

(a) x_0 is an isolated point at which f is not smooth: there are an integer n and a neighborhood U of x_0, purely of relative dimension n, such that $U - \{x_0\}$ is smooth over S;

(b) X is regular at x_0.

Under these assumptions, X is a relative complete intersection over S at x_0. We refer to VI, 3.10 for the definition of $D^1_{X/S}$. On the neighborhood U one has

$$D^1_{X/S} = \mathcal{H}^1\big(R\mathcal{H}om(L\Omega_{X/S}, \mathcal{O}_X)\big) = \mathcal{E}xt^1(\Omega^1_{X/S}, \mathcal{O}_X);$$

on this neighborhood, $D^1_{X/S}$ has finite length and support contained in $\{x_0\}$.

Definition 1.2.

The Milnor number $\mu = \mu(X/S, \bar{x}_0)$ of X/S at $\bar{x}_0$ is the length of the $\mathcal{O}_{X,x_0}$-module defined by $D^1_{X/S}$.

Example 1.3.

Take S to be a smooth curve over an algebraically closed field k. After replacing X by a neighborhood of x_0, we may suppose that X is smooth over k. The cotangent complex then reduces to the following resolution of $\Omega^1_{X/S}$:

$$L\Omega_{X/S} = \big[f^*\Omega^1_{S/k} \longrightarrow \Omega^1_{X/k}\big].$$

Choose a local coordinate $t : S \to \mathbb{E}^1_k$ on S at s (with t étale at s), again denote by f the composite $t \circ f$, and let $(x_i)_{1 \le i \le n+1}$ be a system of local coordinates on X at x_0. At x_0, the module $f^*\Omega^1_{S/k}$ then has basis df, while $\Omega^1_{X/k}$ has basis the dx_i. Thus $D^1_{X/S}$ is identified with the quotient of $\mathcal{O}_X$ by the ideal generated by the $\partial f/\partial x_i$, and

(1.3.1) $$\mu = \dim_k\left(\mathcal{O}_{X,x_0}\Big/\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_{n+1}}\right)\right).$$

This is the definition given by Milnor.

Example 1.4.

Suppose that X is the closed subscheme of $\mathbb{E}_S^{n+1}$ defined by an equation g, and let $i : X \hookrightarrow \mathbb{E}_S^{n+1}$ be the inclusion. The cotangent complex then reduces to

$$\big[(g)/(g)^2 \longrightarrow i^*\Omega^1_{\mathbb{E}^{n+1}/S}\big],$$

and $D^1_{X/S}$ is isomorphic to the quotient of $\mathcal{O}_X$ by the ideal generated by $(\partial g/\partial x_i)_{1\le i\le n+1}$. Equivalently, it is the quotient of $\mathcal{O}_{\mathbb{E}^{n+1}}$ by the ideal generated by g and the $\partial g/\partial x_i$.

Example 1.5.

Let S be the spectrum of a discrete valuation ring V with perfect residue field and fraction field K, and let X be the normalization of S in a separable extension K' of K. By 1.4 and [5], III, 7.14, p. 68, $D^1_{X/S}$ is then isomorphic to the quotient of $\mathcal{O}_X$ by the different $\mathfrak{d}_{X/S}$. If $\bar{x}_0$ is a geometric point of X_s, localized at $x_0 \in X_s$, its Milnor number is therefore

$$\mu = \operatorname{length}\big(\mathcal{O}_{X,x_0}/\mathfrak{d}_{X/S}\big). \tag{1.5.1}$$

Let $\bar{s}$ be a geometric point of S localized at the closed point s, and let $d_{K'/K} \in V$ generate the discriminant ideal, with valuation $v(d_{K'/K})$. It follows from (1.5.1) that

$$v(d_{K'/K}) = \sum_{x\in X(\bar{s})} \mu(X/S, x). \tag{1.5.2}$$

1.6.

Retain the assumptions of 1.1. Let $\bar{s}$ be the geometric point of S which is the image of $\bar{x}_0$, let $S_{(\bar{s})}$ be the spectrum of the strict henselization of $\mathcal{O}_{S,s}$ at $\bar{s}$, and let $X_{(\bar{s})}$ be the inverse image of X over $S_{(\bar{s})}$. Let $\bar{\eta}$ be the spectrum of an algebraic closure of the fraction field $k(\eta)$ of $S_{(\bar{s})}$, and put

$$I = \operatorname{Gal}(k(\bar{\eta})/k(\eta)).$$

Let ℓ be a prime number different from the residual characteristic of S. In a neighborhood of $\bar{x}_0$ in $X_{\bar{s}}$, the sheaves of vanishing cycles

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell) = R^i\Phi_{\bar{\eta}}(\mathbb{Z}/\ell)$$

relative to the constant sheaf $\mathbb{Z}/\ell$ on $X_{(\bar{s})}$ are concentrated at $\bar{x}_0$. Moreover, the inertia group I acts on the $\mathbb{Z}/\ell$-vector spaces

$$\Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0}.$$

These spaces are finite-dimensional (XIII, 2.4.2).

1.7.

Let I_1 be a finite quotient of the inertia group. Recall that the Swan character Sw of I_1 is the difference $a - u$ between the Artin character a and the character u of the augmentation representation. Recall also that, up to isomorphism, there is a unique projective $\mathbb{Z}_\ell[I_1]$-module $S(I_1)$ with character Sw (see [6]).

Let V be a finite-dimensional $\mathbb{Z}/\ell$-vector space with an action $\rho : I \to \mathrm{GL}(V)$, and suppose that ρ factors through I_1. The wild dimension of V is then

$$\dim \mathrm{Sw}(V) = \dim_{\mathbb{Z}/\ell} \operatorname{Hom}_{I_1}(S(I_1) \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, , V)\,. \tag{1.7.1}$$

This integer is independent of the choice of I_1, since for a normal subgroup H of I_1 one has $S(I_1/H) \sim S(I_1)_H$.

We further put

$$\dim \operatorname{tot}(V) = \dim \mathrm{Sw}(V) + \dim_{\mathbb{Z}/\ell}(V)$$

and call this number the total dimension of V.

Definition 1.8.

With the terminology of 1.6 and 1.7, the number of vanishing cycles of X/S at $\bar{x}_0$ (respectively the wild number of vanishing cycles, respectively the total number of vanishing cycles) is the integer

$$(-1)^n \dim \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$(-1)^n \dim \operatorname{Sw} \Phi_{\bar{x}_0} \underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \dim \operatorname{Sw} \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0},$$

$$\begin{aligned}(-1)^n \dim \operatorname{tot} \Phi_{\bar{x}_0} &\underset{\text{def}}{=} (-1)^n \sum_i (-1)^i \\ &\qquad \cdot \dim \operatorname{tot} \Phi^i_{\bar{\eta}}(\mathbb{Z}/\ell)_{\bar{x}_0} \\ &= (-1)^n \big(\dim \Phi_{\bar{x}_0} + \dim \operatorname{Sw} \Phi_{\bar{x}_0}\big).\end{aligned}$$

Conjecture 1.9.

Under the assumptions of 1.1, the Milnor number μ of X/S at $\bar{x}_0$ is equal to the total number of vanishing cycles of X/S at $\bar{x}_0$.

Remark 1.10.

Let S be a henselian trait with imperfect residue field and let $f : X \to S$ be a morphism which is smooth except at $x_0 \in X_s$. Suppose that X is regular at x_0. Definition 1.2 of the Milnor number μ still makes sense under these assumptions, but I no longer have a geometric interpretation of μ to propose.

1.11.

For the remainder of this section, suppose that $S = \operatorname{Spec}(V)$ is a henselian trait with algebraically closed residue field and closed point s, that $f : X \to S$ is a flat morphism of finite type, purely of relative dimension n, that X is regular, that x_0 is a closed point of X_s, and that $X - \{x_0\}$ is smooth over S.

This entails no loss of generality.

Proposition 1.12.

If $n = 0$ (that is, if X/S is finite), Conjecture 1.9 holds for $(X/S, x_0)$.

We may suppose that X is the normalization of S in a finite extension K' of the fraction field K. By 1.5, $\mu = v(d_{K'/K})$.

The extension K' is defined by an open subgroup H of the inertia group I. Let $H' \subset H$ be an open normal subgroup of I, and put $I' = I/H'$. Let $s_{I/H}$ be the character of the permutation representation of I' on I/H, let $a_{I'}$ be the Artin character, let $\operatorname{Sw}_{I'}$ be the Swan character, and let $u_{I'}$ be the character of the augmentation representation. By Artin's "Führerdiskriminantenproduktformel," the discriminant ideal is the Artin conductor of $s_{I/H}$ ([5], VI, 3, Corollary 1, p. 111):

$$\mu = (a_{I'}, s_{I/H}) = |I'|^{-1} \sum_{i \in I'} a_{I'}(i) s_{I/H}(i).$$

One has

$$a_{I'} = u_{I'} + \operatorname{Sw}_{I'}, \qquad (u_{I'}, s_{I/H}) = \#(I/H) - 1,$$

and

$$
\begin{aligned}
(\mathrm{Sw}_{I'}, s_{I/H}) &= \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, (\mathbb{Z}/\ell)^{I/H}\Big) \\
&= \dim_{\mathbb{Z}/\ell} \mathrm{Hom}_I\Big(S(I') \otimes_{\mathbb{Z}_\ell} \mathbb{Z}/\ell, \mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big)\Big).
\end{aligned}
$$

Consequently,

$$\mu = \#(I/H) - 1 + \dim \mathrm{Sw}\,\mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big).$$

One checks immediately that $\Phi^i_{x_0} = 0$ for $i \neq 0$ and that

$$\Phi^0_{x_0} = \mathrm{Coker}\big(\mathbb{Z}/\ell \longrightarrow (\mathbb{Z}/\ell)^{I/H}\big);$$

the formula follows.

Proposition 1.13.

If X_s has an ordinary quadratic singularity at x_0, Conjecture 1.9 holds for $(X/S, x_0)$.

Case 1: $n = \dim(X_s) = 2m - 1$.

The problem is local for the étale topology on X. We may therefore suppose that X is the closed subscheme of $\mathbb{E}^{n+1}_S$ defined by an equation

$$f = \sum_{i=1}^{m} x_i x_{i+m} + a = 0,$$

where a lies in the maximal ideal of V and x_0 is the origin of the coordinates. Moreover, X is regular at x_0 if and only if a is a uniformizer. By 1.4,

$$\mu = \mathrm{length}\left(\mathcal{O}_{\mathbb{E}^{2m}_S, x_0} / \left(f, \frac{\partial f}{\partial x_i}\right)\right) = 1.$$

By the ordinary-quadratic vanishing-cycle calculation of Exposé XV, $\Phi^i_{x_0} = 0$ for $i \neq n$, while $\Phi^n_{x_0}$ has dimension one and inertia acts trivially on it.[36] The wild number of vanishing cycles is therefore zero and the total number is 1, which verifies 1.9.

Case 2: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) \neq 2$.

Under these assumptions we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + x^2_{2m+1} + a = 0,$$

where a is a uniformizer. Again $\mu = 1$, $\Phi^i_{x_0} = 0$ for $i \neq n$, and $\Phi^n_{x_0}$ has dimension one. This time the inertia group acts by ± 1 on $\Phi^n_{x_0}$, hence tamely. The wild number of vanishing cycles is again zero, and 1.9 is verified.

Case 3: $n = \dim(X_s) = 2m$ and $\mathrm{char}(k(s)) = 2$.

This time we may take $X \subset \mathbb{E}^{n+1}_S$ to be defined by

$$f = \sum_{i=1}^{m} x_i x_{i+m} + (x^2_{2m+1} + a x_{2m+1} + b) = 0,$$

where a and b lie in the maximal ideal. The scheme X is regular if and only if b is a uniformizer, and its generic fiber is smooth if and only if $a^2 - 4b \neq 0$.

[36] After the printed word "By," the source leaves the expected reference blank. The stated concentration and inertia action are the ordinary-quadratic vanishing-cycle calculation established in Exposé XV.

The Milnor number of X/S is the same as that of $X_0 \subset \mathbb{E}^1_S$ defined by

$$x^2 + ax + b = 0. \tag{1}$$

Likewise, $\Phi^i_{x_0} = 0$ for $i \neq n$, $\Phi^n_{x_0}$ has dimension one, and I acts on it by ± 1; the index-two subgroup of I acting trivially corresponds to the extension of K defined by (1). The total number of vanishing cycles of X/S at x_0 therefore coincides with the one for X_0/S, and 1.9 in this case follows from 1.12. The Milnor number is given by the discriminant formula of 1.5:

$$\mu = v(a^2 - 4b). \tag{1.13.1}$$

2. The "geometric" case

We shall prove that Conjecture 1.9 holds in equal characteristic. The key geometric result is the following. Its possible use was suggested to me by conversations with B. Saint-Donat.

Proposition 2.1.

Let S be a smooth connected projective curve over an algebraically closed field k, let X be a smooth proper variety over k, purely of dimension $n+1$, and let $f : X \to S$ be a morphism which is smooth away from a finite set F of points of X. Then the Euler–Poincaré characteristics in étale cohomology of S, X, and the geometric generic fiber $X_{\bar\eta}$ of f satisfy

$$\chi(X) = \chi(S)\chi(X_{\bar\eta}) - (-1)^n \sum_{x \in F} \mu(X/S, x). \tag{2.1.1}$$

The Euler–Poincaré characteristic of X is the trace (or integral over X) of the top Chern class of the tangent bundle:

$$\chi(X) = \int_X c_{n+1}(\Theta_X) = (-1)^{n+1} \int_X c_{n+1}(\Omega^1_X).$$

Write

$$\Omega^1_X = \left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \otimes f^*\Omega^1_S.$$

The total Chern class of the invertible sheaf $f^*\Omega^1_S$ is the inverse image of the total Chern class of Ω^1_S on S:

$$c^*(f^*\Omega^1_S) = 1 + f^*c^1(S, \Omega^1_S),$$

and the cup powers of c^1 vanish:

$$\left(f^*c^1(S, \Omega^1_S)\right)^n = f^*\left(c^1(S, \Omega^1_S)^n\right) = 0 \qquad (n > 1).$$

It follows that

$$\begin{aligned} c^{n+1}(\Omega^1_X) = c^n&\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S, \Omega^1_S) \\ &+ c^{n+1}\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right). \end{aligned}$$

If S has genus g, then $c_1(S, \Omega^1_S)$ is $(2g-2)$ times the cohomology class of any closed point s of S. Consequently, when X_s is nonsingular,

$$\begin{aligned} \mathrm{I} &= \int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*c_1(S, \Omega^1_S) \\ &= (2g-2) \int_X c^n\left(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\right) \smile f^*\,\mathrm{cl}(s) \\ &= (2g-2) \int_{X_s} c^n\left(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\right). \end{aligned}$$

Moreover, on X_s the sheaf $f^*\Omega^1_S$ is isomorphic to $\mathcal{O}_{X_s}$, and $\Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{O}_{X_s}$ occurs in the exact sequence

$$0 \longrightarrow f^*\Omega^1_S \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_X \otimes \mathcal{O}_{X_s} \longrightarrow \Omega^1_{X_s} \longrightarrow 0.$$

Hence

$$c^n\big(X_s, \Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1} \otimes \mathcal{O}_{X_s}\big) = c^n(X_s, \Omega^1_{X_s}),$$

and

$$\mathrm{I} = (2g-2)\int_{X_s} c^n(X_s, \Omega^1_{X_s}) = (-1)^{n+1}\chi(S)\chi(X_s).$$

The canonical map

$$f^* : f^*\Omega^1_S \longrightarrow \Omega^1_X$$

is identified with a section a of $\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}$. After choosing systems of local coordinates on S and X, thereby identifying this bundle with $\mathcal{O}_X^{n+1}$, the coordinates of a are the partial derivatives $\partial f/\partial x_i$. Thus a vanishes only on F, and by 1.3 the multiplicity of its zero at $x \in F$ is the Milnor number $\mu(X/S, x)$.

Recall that if a section a of a vector bundle G of rank d on a smooth proper k-scheme X, purely of dimension d, vanishes only at isolated points, then $\int_X c^d(G)$ is the number of zeros of a, counted with multiplicity. Here

$$\mathrm{II} = \int_X c^{n+1}\big(\Omega^1_X \otimes (f^*\Omega^1_S)^{\otimes -1}\big) = \sum_{x\in F}\mu(X/S, x).$$

Collecting terms gives

$$\chi(X) = (-1)^{n+1}(\mathrm{I}+\mathrm{II}) = \chi(S)\chi(X_s) - (-1)^n\sum_{x\in F}\mu(X/S, x),$$

which proves 2.1, since $\chi(X_s) = \chi(X_{\bar\eta})$.

Corollary 2.2.

Under the preceding assumptions, if $n(X/S, x)$ is the total number of vanishing cycles of X/S at x (1.8), then

$$\sum_{x\in F}\mu(X/S, x) = \sum_{x\in F} n(X/S, x).$$

The Leray spectral sequence for f gives

$$\begin{aligned}\chi(X) &= \sum_{p,q}(-1)^{p+q}\dim_{\mathbb{Z}/\ell} H^p(S, R^q f_*\mathbb{Z}/\ell)\\ &= \sum_q (-1)^q \chi(S, R^q f_*\mathbb{Z}/\ell).\end{aligned}$$

For every constructible sheaf G of $\mathbb{Z}/\ell$-modules on S, one has ([4] or SGA 5, X, 7.2)

$$\chi(S, G) = \chi(S)\dim_{\mathbb{Z}/\ell}(G_{\bar\eta}) - \sum_{s\in S}\varepsilon_s(G),$$

where, for each $s \in S$, after identifying $\bar\eta$ with the algebraic closure of the fraction field of the henselization of S at s,

$$\varepsilon_s(G) = \dim(G_{\bar\eta}) - \dim(G_s) + \dim \mathrm{Sw}(G_{\bar\eta}).$$

[37]

[37] The print has a minus sign before the Swan term. The plus sign is forced both by the standard Artin-conductor formula cited in the preceding sentence and by the printed identity (c), which identifies the alternating sum with the *total* dimension $\dim + \dim \mathrm{Sw}$.

For $s \in S$, the exact sequences XIII, 2.4.6.3,

$$\cdots \longrightarrow (R^q f_* \mathbb{Z}/\ell)_s \longrightarrow (R^q f_* \mathbb{Z}/\ell)_{\bar\eta} \longrightarrow \bigoplus_{x \in F(s)} \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x \longrightarrow \cdots$$

give

$$\begin{aligned}\sum_q (-1)^q \left(\dim(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \dim(R^q f_* \mathbb{Z}/\ell)_s\right)\\ = \sum_{x \in F(s)} \sum_q (-1)^q \dim \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x,\end{aligned}$$

and

$$\sum_q (-1)^q \dim \mathrm{Sw}(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} = \sum_{x \in F(s)} \sum_q (-1)^q \dim \mathrm{Sw}\, \Phi^q_{\bar\eta}(\mathbb{Z}/\ell)_x.$$

Consequently,

(c) $$\sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell) = \sum_{x \in F(s)} \dim \operatorname{tot} \Phi_x.$$

The Euler–Poincaré formula therefore becomes

$$\begin{aligned}\chi(X) &= \sum_q (-1)^q \chi(S, R^q f_* \mathbb{Z}/\ell)\\ &= \sum_q (-1)^q \left(\chi(S) \dim(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \sum_s \varepsilon_s(R^q f_* \mathbb{Z}/\ell)\right)\\ &= \chi(S) \sum_q (-1)^q \dim(R^q f_* \mathbb{Z}/\ell)_{\bar\eta} - \sum_s \sum_q (-1)^q \varepsilon_s(R^q f_* \mathbb{Z}/\ell)\\ &= \chi(S)\chi(X_{\bar\eta}) - \sum_s \sum_{x \in F(s)} \dim \operatorname{tot} \Phi_x\\ &= \chi(S)\chi(X_{\bar\eta}) - (-1)^n \sum_{x \in F} n(X/S, x).\end{aligned}$$

Comparison with (2.1.1) proves 2.2.

Corollary 2.3.

Under the preceding assumptions and with the preceding notation, suppose that $x_0 \in F$ and that, for every $x \in F$ distinct from x_0, the fiber of f through x has an ordinary quadratic singularity at x. Then

$$\mu(X/S, x_0) = n(X/S, x_0).$$

For $x \neq x_0$, Proposition 1.13 gives

$$\mu(X/S, x) = n(X/S, x),$$

and it remains to apply 2.2.

Theorem 2.4.

Let S be a henselian trait of equal characteristic with perfect residue field, let $f : X \to S$ be a flat morphism of finite type, and let $\bar x$ be a geometric point of X whose image in X is a closed point x of X_s at which X/S is not smooth, and which is isolated as such. Suppose that X is regular at

x. Then the total number (1.8) of vanishing cycles of X/S at $\bar{x}$ coincides with the Milnor number $\mu(X/S, x)$.

Let S' be the henselian trait obtained from S by extending the residue field $k(s)$ to its separable closure, and put $X' = X \times_S S'$. The assertion of 2.4 for X/S is equivalent to the assertion for X'/S'. Since $k(s)$ is already assumed perfect, we may therefore suppose that the residue field of S is algebraically closed.

For such an S, let S' be its completion and put $X' = X \times_S S'$. The Milnor number at x is the same for X/S and X'/S', and by XIII, 2.4.2 the same is true of the total number of vanishing cycles. We may thus suppose in addition that S is complete, hence isomorphic to $k[[t]]$, with k algebraically closed. If X_s has dimension n at x, we then have:

Proposition 2.5.

There exist a smooth projective variety Y over k, purely of dimension $n+1$, a morphism $f : Y \to \mathbb{P}^1_k$ which is smooth outside a finite set F of points of Y, and a point $y_0 \in F$ such that:

(a) if $y \in F$ and $y \neq y_0$, then y is an ordinary quadratic singular point of the fiber of f through y;

(b) $f(y_0) = 0$, and if $a : S \to \mathbb{P}^1_k$ is the natural identification of S with the completion of $\mathbb{P}^1_k$ at 0, there is an S-isomorphism between the henselization of X at x and the henselization of a^*Y at y_0.

Let us show that 2.5 implies 2.4. By 2.3 and 2.5(a), the Milnor number of $Y/\mathbb{P}^1_k$ at y_0 coincides with the total number of vanishing cycles.[38] As above, XIII, 2.4.2 gives the same result for a^*Y/S, hence also for $(X/S, x)$, which by 2.5(b) is locally isomorphic to $(a^*Y/S, y_0)$ for the étale topology.
We prove 2.5. As a $k[[t]]$-algebra, the completion of the local ring $\mathcal{O}_{X,x}$ is isomorphic to a quotient

$$\widehat{\mathcal{O}}_{X,x} = k[[t, x_1, \ldots, x_{n+1}]]/(f).$$

The implicit-function theorem XV, 1.1.2 (that is, [1], Lemma 5.10) then shows that there is an integer N such that condition 2.5(b) holds whenever

$$\widehat{\mathcal{O}}_{Y,y_0} \simeq k[[t, x_1, \ldots, x_{n+1}]]/(f_1)$$

as $k[[t]]$-algebras, with $f_1 - f$ in the Nth power $\mathfrak{m}^N$ of the maximal ideal of $k[[t, x_1, \ldots, x_{n+1}]]$.

Let (t_0, t_1) and $(x_0, \ldots, x_{n+1})$ be the usual homogeneous coordinate systems on $\mathbb{P}^1$ and $\mathbb{P}^{n+1}$. We take $Y/\mathbb{P}^1$ to be the subvariety of $\mathbb{P}^1 \times \mathbb{P}^{n+1}$ defined by an equation

$$g(t_0, t_1; x_0, \ldots, x_{n+1}) = 0, \tag{2.5.1}$$

homogeneous of degree d' in (t_0, t_1) and of degree d'' in the x_i. More intrinsically, g is a section of

$$\mathcal{O}(d', d'') = \mathrm{pr}_1^* \mathcal{O}(d') \otimes \mathrm{pr}_2^* \mathcal{O}(d'').$$

Condition (b) holds if

$$g(1, t; 1, x_1, \ldots, x_{n+1}) \equiv f \pmod{\mathfrak{m}^N}. \tag{2.5.2}$$

Fix d' and d''. The sections g of type (2.5.1) are the k-rational points of an affine space A. For $u \in \mathbb{P}^1 \times \mathbb{P}^{n+1}$, let $B(u, r)$ denote the affine space whose points are the r-jets at u of sections of $\mathcal{O}(d', d'')$. Put $u_0 = (1, 0; 1, 0, \ldots, 0)$. If $u \neq u_0$, then for sufficiently large d' and d'' the always-linear restriction map

$$A \longrightarrow B(u_0, N) \times B(u, 2) \tag{2.5.3}$$

[38] The print says Y/T here, but no base T is introduced; Proposition 2.5 defines the morphism $Y \to \mathbb{P}^1_k$.

is surjective.

Let $C(u) \subset B(u,2)$ be the set of 2-jets which are the 2-jets either of the zero germ or of a germ $\widetilde{h}$ such that

(a) u lies on the variety V defined by $\widetilde{h} = 0$;

(b) u is neither a smooth point nor an ordinary quadratic point of the fiber through u of $V/\mathbb{P}^1$.

It is readily checked that $C(u)$ is closed of codimension $n+3$ in $B(u,2)$.

Let D be the affine subspace of A consisting of the g which satisfy (2.5.2). For $u \neq u_0$, let $D(u)$ be the subspace of D consisting of the zero g and of those g such that $g(u) = 0$ and u is neither a smooth point nor an ordinary quadratic point of the fiber through u of the variety V defined by $g = 0$. If d' and d'' are sufficiently large for (2.5.3) to be surjective, then $D(u)$ has codimension $n+3$ in D. The union of the $D(u)$ for $u \neq u_0$ therefore has codimension at least

$$1 = (n+3) - (n+2),$$

and a general element g of D lies in none of the $D(u)$. It solves the problem.

Bibliography

[1] M. Artin, "Algebraic approximation of structures over complete local rings," *Publ. Math. IHES* 36 (1969) (Zariski's volume).

[2] Lê Dũng Tráng, *Singularités isolées des hypersurfaces complexes*, Publ. Centre Math. École Polytechnique, May 1969.

[3] J. Milnor, *Singular Points of Complex Hypersurfaces*, Princeton University Press, 1968.

[4] M. Raynaud, "Caractéristique d'Euler–Poincaré d'un faisceau et cohomologie des variétés abéliennes," Séminaire Bourbaki 286, February 1965.

[5] J.-P. Serre, *Corps locaux*, Publ. Inst. Math. Univ. Nancago VIII, Hermann, 1962.

[6] J.-P. Serre, "SGA 5, IX. Introduction à la théorie de Brauer," reproduced in *Représentations linéaires des groupes*, 3rd part, Hermann, Paris, 1967.

SGA 7 — EXPOSÉ XVII

Lefschetz Pencils: Existence Theorem

by N. M. Katz

0. Introduction

In the present exposé (which, in the oral seminar, came before Exposés XIV and XV devoted to the Picard–Lefschetz formula), we introduce the "Lefschetz pencils" of hyperplane sections of a smooth projective scheme and give conditions for their existence. In the following exposé we make an elementary cohomological study of them, based on the known structure of the cohomology of blown-up varieties. The fact that certain hypotheses on Lefschetz pencils which we shall introduce are often satisfied will follow from the Picard–Lefschetz formula of Exposé XV, which is less elementary in nature than the results of the present exposé and the next.

Some developments of the present exposé and the next will be given elsewhere in a more general setting (general base schemes, quasi-projectivity hypotheses in place of projectivity, and so forth): in particular, the precohomological study of pencils of hyperplane sections, and especially of Lefschetz pencils, in EGA V, and the cohomological study of blown-up schemes in Jouanolou's Exposé VII of SGA 5. The reader will also find in EGA V proofs in which intrinsic diagram chasing replaces the use of coordinates favored by the writer of the present exposé.

I wish to thank P. Deligne and P. Griffiths for the help they gave me in conceiving this exposé.
Throughout it, k denotes an algebraically closed field unless expressly stated otherwise.

1. A reminder on quadratic singularities (cf. VI, 6)

1.1.

Let Y be a k-scheme pure of dimension $n-1$. [39] A closed point $y \in Y$ is called an ordinary quadratic singularity of Y if:

(a) the completion $\widehat{\mathcal{O}}_{Y,y}$ of the local ring of Y at y is isomorphic to

$$k[[T_1, \dots, T_n]]/(Q(T)),$$

the quotient of the formal power-series ring in $n = 1 + \dim Y$ variables by the ideal generated by an element $Q(T)$;

(b) Q begins in degree two, and the subvariety of $\mathbb{P}_k^{n-1}$ defined by the vanishing of the homogeneous degree-two part of $Q(T)$ is smooth over k when $n \geq 2$, while the degree-two part of $Q(T)$ is nonzero when $n = 1$.

If the characteristic of k is different from two, or if n is even, condition (b) is equivalent to:

(b′) Q begins in degree two and its Hessian matrix

$$\left(\frac{\partial^2 Q}{\partial T_i \partial T_j}(0) \right)_{1 \leq i,j \leq n}$$

is invertible.

[39] The print first says "of dimension n" and then, in the hypersurface presentation below, imposes $n = 1 + \dim Y$. The latter is the convention used by the displayed Hessian and normal form.

This follows from the Jacobian criterion for smoothness.

The standard form for the homogeneous degree-two part Q_2 of a series Q satisfying (b), obtainable after a linear change of variables, is

$$
\begin{cases}
Q_2(T) = \sum_{i=1}^{n/2} T_{2i-1}T_{2i} & \text{if } n \text{ is even,} \\[2ex]
Q_2(T) = T_n^2 + \sum_{i=1}^{(n-1)/2} T_{2i-1}T_{2i} & \text{if } n \text{ is odd.}
\end{cases}
\tag{1.1.1}
$$

This follows from the theory of quadratic forms (Bourbaki, *Algèbre*, Ch. X).

1.2.

A closed point $y \in Y$ satisfying (a) and (b) is called a nondegenerate quadratic singularity. Since the determinant of the Hessian matrix is identically zero in characteristic two when n is odd, nondegenerate quadratic singularities presented in n variables exist only when n is even or the characteristic of k is different from two; in those cases they are precisely the ordinary quadratic singularities.

2. Lefschetz pencils: statement of the results

2.1.

Let $\mathbb{P}^r$ denote projective r-space over k, $\check{\mathbb{P}}^r$ the dual projective space of hyperplanes in $\mathbb{P}^r$, and $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ the variety of lines in $\check{\mathbb{P}}^r$. A point $D \in \mathrm{Gr}(1, \check{\mathbb{P}}^r)$ is called a pencil of hyperplanes in $\mathbb{P}^r$. Regarding D as a line in $\check{\mathbb{P}}^r$, we write, by abuse of notation, $D = \{H_t\}_{t \in D}$, where the H_t are the hyperplanes of $\mathbb{P}^r$ which are the points of the line D.

The axis of D is the codimension-two linear subvariety of $\mathbb{P}^r$ which is the intersection $H_0 \cap H_\infty$ of any two distinct members of the pencil. Regarding this axis as a point of $\mathrm{Gr}(r-2, \mathbb{P}^r)$, the variety of $(r-2)$-dimensional linear subvarieties of $\mathbb{P}^r$, the map sending D to its axis is the familiar orthogonality isomorphism

$$\mathrm{Gr}(1, \check{\mathbb{P}}^r) \xrightarrow{\sim} \mathrm{Gr}(r-2, \mathbb{P}^r).$$

2.2.

Let X be a proper, smooth, irreducible k-scheme of dimension $n \geq 1$, equipped with an embedding $X \hookrightarrow \mathbb{P}^r$. A pencil $D = \{H_t\}_{t \in D}$ of hyperplanes in $\mathbb{P}^r$ is called a Lefschetz pencil (relative to this embedding) if:

(a) the axis of D meets X transversely (EGA IV, 17.13.7);

(b) there is a dense open subset $U \subset D$ such that, for every $t \in U$, the hyperplane H_t meets X transversely;

(c) for $t_0 \in D - U$, the hyperplane H_{t_0} meets X transversely except at one point, which is an ordinary quadratic singular point of $X \cap H_{t_0}$.

We shall see below in 3.2.1 that the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$.

2.3.

The embedding $X \hookrightarrow \mathbb{P}^r$ is called a Lefschetz embedding if the Lefschetz pencils form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. [40]

[40] More precisely, they form a dense open subset in the set of closed points of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$. We shall permit ourselves such abuses of language below.

2.4.

Recall the notion of a multiple of a given embedding $X \hookrightarrow \mathbb{P}^r$. It is the composite of that embedding with the d-th Veronese embedding [41]

$$S_d : \mathbb{P}^r \hookrightarrow \mathbb{P}^{\binom{r+d}{d}-1}, \qquad (X_0, \ldots, X_r) \longmapsto (\ldots, X^W, \ldots),$$

where

$$X^W = X_0^{W_0} \cdots X_r^{W_r} \quad \text{and} \quad W_0 + \cdots + W_r = d.$$

Thus a "section of X by a hypersurface of degree d" for the given embedding—in English, a hypersurface section of degree d—is simply a hyperplane section for the d-th multiple of that embedding.

Theorem 2.5.

Let X be irreducible, proper, and smooth over k, of dimension $n \geq 1$, and equipped with an embedding $i : X \hookrightarrow \mathbb{P}^r$. Then:

2.5.1. every d-th multiple of i, with $d \geq 2$, is a Lefschetz embedding;

2.5.2. if k has characteristic zero, the given embedding i is a Lefschetz embedding.

The proof is divided into several pieces, which follow.

3. The dual variety

3.1.

Let I be the ideal in $\mathbb{P}^r$ defining X. Write $N = I/I^2$ for the conormal sheaf, locally free of rank $r - n$ on X, and let $\mathbb{P}(N)$ be the projective bundle over X formed by the lines in $\check{N}$. (When $X = \mathbb{P}^r$, the bundle $\mathbb{P}(N)$ is empty.) The bundle $\mathbb{P}(N)$ identifies with the subvariety of $X \times \check{\mathbb{P}}^r$ formed by the pairs (x, H) for which H is tangent to X at x, by the map

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r.$$

More explicitly,

$$\mathbb{P}(I/I^2)_x \ni F \longmapsto \left(x,\ \text{the hyperplane with equation } \sum_i \frac{\partial F}{\partial X_i}(x) T_i = 0\right), \tag{3.1.1}$$

where F denotes a local section of I/I^2. Define

$$\varphi : \mathbb{P}(N) \longrightarrow \check{\mathbb{P}}^r \tag{3.1.2}$$

to be the composite

$$\mathbb{P}(N) \xrightarrow{\nu} X \times \check{\mathbb{P}}^r \xrightarrow{\mathrm{pr}_2} \check{\mathbb{P}}^r.$$

3.1.3.

Let $\check{X}$ denote the image of φ. It is the set of hyperplanes which do not meet X transversely, and is called the dual variety of X for the embedding $i : X \hookrightarrow \mathbb{P}^r$.

[41] The print calls this the d-th Segre embedding; the displayed monomial map is the Veronese embedding.

Proposition 3.1.4.

The dual variety $\check{X}$ is irreducible and proper, of dimension at most $r-1$.
Proof. It is the image under φ of $\mathbb{P}(N)$, which is itself irreducible of dimension $r-1$ if $X \neq \mathbb{P}^r$, and empty if $X = \mathbb{P}^r$.

Remark 3.1.5.

The interpretation of $\mathbb{P}(N)$ can be made more precise as follows. Let $\mathcal{Y} \subset X \times \check{\mathbb{P}}^r$ be the incidence subscheme whose fiber over every point $H \in \check{\mathbb{P}}^r(k)$ is $X \cap H$. The scheme $\mathcal{Y}$ is plainly smooth of dimension $n+r-1$, being a projective bundle of relative dimension $r-1$ over X. Thus $\mathcal{Y} \to \check{\mathbb{P}}^r$ makes $\mathcal{Y}$ a relative complete intersection of virtual relative dimension $(n+r-1)-r = n-1$.

The Jacobian subscheme

$$J_{n-1}(\mathcal{Y}/\check{\mathbb{P}}^r)$$

(VI, 5.2) is then precisely $\mathbb{P}(N)$. Formation of the Jacobian subscheme of a relative complete intersection is compatible with arbitrary base change. Consequently, for every line $D \subset \check{\mathbb{P}}^r$, the Jacobian subscheme $J_{n-1}(\widetilde{X}/D)$, where

$$\widetilde{X} = \mathcal{Y} \times_{\check{\mathbb{P}}^r} D,$$

is precisely $\mathbb{P}(N) \times_{\check{\mathbb{P}}^r} D$.

Proposition 3.2.

The subset $B \subset \check{X}$ formed by the hyperplanes H such that $X \cap H$ has exactly one singular point, and that point is an ordinary quadratic singularity, is open.

This is a special case of XV, 1.3.2, based on R. Elkik's theory of "henselian modules" (XV, 1.1.4). The case in which one restricts to nondegenerate quadratic points—that is, where $\mathrm{char}(k) \neq 2$ or $n = \dim X$ is even—is also immediate from 3.3 below, whose proof does not use 3.2.

Corollary 3.2.1.

Under the hypotheses of 3.2, let $F = \check{X} - B$, a closed subset of $\check{X}$. Then the Lefschetz pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$, and the following conditions are equivalent:

3.2.2. the given embedding $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding;

3.2.3. there exists a Lefschetz pencil of hyperplanes for the given embedding;

3.2.4. the codimension of F in $\check{\mathbb{P}}^r$ is at least two.

Proof. A line $D \subset \check{\mathbb{P}}^r$ is a Lefschetz pencil if and only if:

(3.2.5) the axis of D meets X transversely;

(3.2.6) D is not contained in $\check{X}$;

(3.2.7) D does not meet F.

Conditions (3.2.5) and (3.2.6) define open subsets of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$; since F is closed, so does (3.2.7). Hence the Lefschetz pencils form an open subset.

Because $\dim X \leq r-1$, the open subset in (3.2.6) is nonempty and therefore dense. The open subset in (3.2.5) is likewise nonempty and dense. Indeed, choose a hyperplane H meeting X transversely; such an H exists because $\dim X \leq r-1$. Since $\dim(X \cap H) \leq r-1$, there is a hyperplane G meeting $X \cap H$ transversely, and the line D through H and G then has an axis meeting X transversely. Thus the embedding is Lefschetz if and only if the open subset in (3.2.7) is nonempty, hence dense. Since F is closed, this occurs if and only if F has codimension at least two in $\check{\mathbb{P}}^r$ (cf. EGA V, or Mumford, *Introduction to Algebraic Geometry*, p. 88, Cor. 3).

Corollary 3.2.8.

Under the hypotheses of 3.2, suppose that $i : X \hookrightarrow \mathbb{P}^r$ is a Lefschetz embedding. Let H_0 be a hyperplane meeting X transversely, and let $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ be the subvariety of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)$ formed by the lines through H_0. Then the Lefschetz pencils through H_0 form a dense open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$.

Proof. By 3.2.1 these pencils form an open subset of $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$. The argument just given to prove that the open subsets (3.2.5) and (3.2.6) are nonempty also shows that their intersections with this subvariety are nonempty. It remains to prove that some lines through H_0 do not meet F.

Choose a hyperplane $T \subset \check{\mathbb{P}}^r$ not containing H_0. There is an isomorphism

$$\begin{array}{ccc} \mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0} & \xleftarrow{\ \sim\ } & T \\ & & \\ \text{the line through } H_0 \text{ and } \tau & \longleftarrow\!\shortmid & \tau \end{array}$$

so $\mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0}$ "is" a projective space of dimension $r-1$. The lines in it which pass through a point of F form a closed subset of dimension at most $r-2$, being the image of the morphism

$$\begin{array}{ccc} F & \longrightarrow & \mathrm{Gr}(1, \check{\mathbb{P}}^r)_{H_0} \\ & & \\ f & \longmapsto & \text{the line through } H_0 \text{ and through } f \end{array}.$$

Proposition 3.3.

The morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is unramified at a closed point (x_0, H) if and only if x_0 is a nondegenerate quadratic singularity (1.2) of $X \cap H$. In particular, the subset of $\mathbb{P}(N)$ formed by the points (x, H) for which x is a nondegenerate quadratic singularity of $X \cap H$ is open. Proof of Proposition 3.3. We may restrict to the case $X \neq \mathbb{P}^r$. The two properties asserted to be equivalent are independent of the particular choice of coordinates, so we may describe them using coordinates chosen in a very special way. Since X is a smooth subvariety of $\mathbb{P}^r$ of dimension $n \geq 1$, choose homogeneous coordinates $X_0, \ldots, X_r$ so that:

3.3.1. the point $x_0 \in \mathbb{P}^r$ is $(1, 0, \ldots, 0)$;

3.3.2. putting $x_i = X_i/X_0$ for $i = 1, \ldots, r$, so that $\widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ identifies with $k[[x_1, \ldots, x_r]]$, the ideal defining the trace of X in $\operatorname{Spec} \widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}$ is generated by elements g_i $(i = 1, \ldots, r-n)$ which, in the completion, have the form

$$g_i = x_{n+i} - h_i(x_1, \ldots, x_n), \qquad i = 1, \ldots, r-n,$$

where the h_i are power series in $x_1, \ldots, x_n$ alone and lie in the square of the maximal ideal of $k[[x_1, \ldots, x_n]]$;

3.3.3. under (3.1.1), the hyperplane H comes from g_{r-n}.

Put $V = \operatorname{Spec}(\widehat{\mathcal{O}}_{X, x_0})$. It follows from 3.3.2 that there is a trivialization

$$(I/I^2)|_V \simeq \mathcal{O}_V^{r-n}$$

given by

$$(\Lambda_1, \ldots, \Lambda_{r-n}) \longmapsto \sum_{i=1}^{r-n} \Lambda_i g_i. \tag{3.3.4}$$

Together with 3.3.3 this gives a neighborhood W of (x_0, H) in $\mathbb{P}(N)|_V$ and an isomorphism $V \times \mathbb{A}^{r-n-1} \simeq W$, given by

$$\begin{aligned} V \times \mathbb{A}^{r-n-1} &\longrightarrow \mathbb{P}(I/I^2)|_V, \\ (v, (\lambda_1, \dots, \lambda_{r-n-1})) &\longmapsto \left(v, \left[g_{r-n} + \sum_{i=1}^{r-n-1} \lambda_i g_i\right]\right), \end{aligned} \tag{3.3.5}$$

where $\lambda_i = \Lambda_i / \Lambda_{r-n}$.

In particular, the completion of the local ring of $\mathbb{P}(N)$ at (x_0, H) identifies with

$$k[[x_1, \dots, x_n, \lambda_1, \dots, \lambda_{r-n-1}]].$$

We now express φ in these coordinates at the level of completions:

$$\varphi(x, \lambda) = \text{the hyperplane with equation } \sum_{i=0}^{r} \varphi_i(x, \lambda) T_i = 0. \tag{3.3.6}$$

Since this image hyperplane contains the point x, whose homogeneous coordinates are

$$x = (1, x_1, \dots, x_n, h_1(x), \dots, h_{r-n}(x)), \tag{3.3.7}$$

we have

$$\varphi_0(x, \lambda) = -\sum_{i=1}^{n} x_i \varphi_i(x, \lambda) - \sum_{j=1}^{r-n} h_j(x) \varphi_{n+j}(x, \lambda) \tag{3.3.8}$$

in $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Moreover, by (3.1.1), if

$$D_i = \frac{\partial}{\partial x_i} \in \mathrm{Der}_k(\widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}, \widehat{\mathcal{O}}_{\mathbb{P}^r, x_0}),$$

then for $1 \le i \le r$,

$$\varphi_i(x, \lambda) = D_i\left(g_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j g_j\right)(x) \quad \text{in } \widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}. \tag{3.3.9}$$

Recalling that $g_i = x_{n+i} - h_i(x_1, \dots, x_n)$, one obtains

$$\varphi_i(x, \lambda) = -\frac{\partial}{\partial x_i}\left(h_{r-n} + \sum_{j=1}^{r-n-1} \lambda_j h_j\right), \qquad 1 \le i \le n, \tag{3.3.10}$$

$$\varphi_{n+j}(x, \lambda) = \lambda_j, \qquad 1 \le j \le r-n-1, \tag{3.3.11}$$

and

$$\varphi_r(x, \lambda) = 1. \tag{3.3.12}$$

[42]

Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{\mathbb{P}(N),(x_0,H)}$. Since $h_i \equiv 0 \pmod{\mathfrak{m}^2}$ for $i = 1, \dots, r-n$, we find

$$\varphi_0(x, \lambda) \equiv 0 \pmod{\mathfrak{m}^2} \tag{3.3.13}$$

[42] The print gives incompatible ranges in (3.3.11). The coordinates in (3.3.5), the coefficient formula (3.3.9), and the identity block of the printed Jacobian matrix force the indices shown here.

and

$$\varphi_i(x,\lambda) \equiv -\frac{\partial h_{r-n}}{\partial x_i} \pmod{\mathfrak{m}^2}, \qquad i = 1,\ldots,n. \tag{3.3.14}$$

Putting (3.3.11)–(3.3.14) together, the Jacobian matrix of φ at (x_0, H) has the block form

$$J_{(x_0,H)}(\varphi) = \left(\begin{array}{c|c} 0_{1\times n} & 0_{1\times(r-n-1)} \\ \hline -\operatorname{Hess}(h_{r-n})(0) & 0_{n\times(r-n-1)} \\ \hline 0_{(r-n-1)\times n} & I_{r-n-1} \\ \hline 0_{1\times n} & 0_{1\times(r-n-1)} \end{array}\right). \tag{3.3.15}$$

Thus φ is unramified at (x_0, H) if and only if the Hessian matrix

$$\left(\frac{\partial^2 h_{r-n}}{\partial x_i \partial x_j}(0)\right)_{1\le i,j\le n}$$

is invertible. On the other hand, 3.3.2 and 3.3.3 show that the completion of the local ring of $X \cap H$ at x_0 is

$$k[[x_1,\ldots,x_n]]/\big(h_{r-n}(x_1,\ldots,x_n)\big), \tag{3.3.16}$$

and the result follows from Definition 1.2.

Example 3.4.

Suppose that X is a hypersurface in $\mathbb{P}^r$, defined by a homogeneous equation $F(X_0,\ldots,X_r) = 0$. Then $\mathbb{P}(N) = X$, and in homogeneous coordinates $\varphi : X \to \check{\mathbb{P}}^r$ is given by

$$\varphi(X_0,\ldots,X_r) = \left(\frac{\partial F}{\partial X_0},\ldots,\frac{\partial F}{\partial X_r}\right). \tag{3.4.1}$$

This is the <u>Gauss morphism</u>.

Here is an example in which φ is everywhere ramified. Let p be the characteristic of k, let $r \ge 1$, and take the hypersurface X with equation

$$\sum_{i=0}^{r} X_i^{p^r+1} = 0. \tag{3.4.2}$$

Then

$$\varphi(X_0,\ldots,X_r) = (X_0^{p^r},\ldots,X_r^{p^r}). \tag{3.4.3}$$

Thus $\check{X}$ is reduced to a point if $p = 0$, while if $p > 0$ one has $\check{X} \simeq X$ and φ is the Frobenius morphism. It follows from 3.3 (or from 3.5.0 below) that if $p > 2$, this embedding is not a Lefschetz embedding.

We now return to the general case.

Proposition 3.5.

<u>Suppose that the morphism $\varphi : \mathbb{P}(N) \to \check{\mathbb{P}}^r$ is generically unramified (that is, is not everywhere ramified). Then φ induces a birational morphism from $\mathbb{P}(N)$ onto $\check{X}$. More precisely, the following three open subsets of $\check{X}$ are identical:</u>

(a) <u>the largest open subset on which $\check{X}$ is smooth;</u>

(b) <u>the largest open subset $V \subset \check{X}$ such that φ induces an isomorphism $\varphi^{-1}(V) \xrightarrow{\sim} V$;</u>

(c) <u>the open subset (cf. 3.2) formed by those hyperplanes H for which $X \cap H$ has exactly one singular point, that point being a nondegenerate quadratic singularity.</u>

Corollary 3.5.0.

Suppose that $\mathrm{char}(k) \neq 2$ or that $\dim X$ is even. The given embedding is a Lefschetz embedding if and only if one of the following two conditions holds:

(a) φ is generically unramified;

(b) $\dim \check{X} \leq r-2$.

Indeed, sufficiency follows from criterion (3.2.4). In case (a), Proposition 3.5 identifies F with the singular locus of $\check{X}$, so

$$\dim F < \dim \check{X} = r-1;$$

in case (b), the criterion is immediate. Conversely, suppose that D is a Lefschetz pencil and $\dim \check{X} = r-1$. There is a point $H \in D \cap \check{X}$, and 3.3 together with Definition 2.2(c) shows that φ is unramified at the points of $\varphi^{-1}(H)$. Hence it is generically unramified.

Proof of Proposition 3.5. The hypothesis implies

$$\dim \check{X} = \dim \mathbb{P}(N) = r-1,$$

and therefore 3.2 shows that $\check{X}$ is an irreducible hypersurface in $\check{\mathbb{P}}^r$.

Let R be the ramification locus of φ. By hypothesis it is a closed subset of $\mathbb{P}(N)$ of dimension at most $r-2$; hence $\varphi(R)$ is a closed subset of $\check{X}$ of dimension at most $r-2$. Let V' be the dense open subset

$$V' = (\check{X} - \varphi(R)) \cap (\check{X} - \mathrm{Sing}(\check{X})) \tag{3.5.1}$$

of $\check{X}$, and denote its complement by F:

$$F = \varphi(R) \cup \mathrm{Sing}(\check{X}). \tag{3.5.1.1}$$

We first prove that φ induces an isomorphism $\varphi^{-1}(V') \simeq V'$ by constructing an inverse $\psi : V' \to \varphi^{-1}(V')$. Regard $\varphi^{-1}(V')$ as embedded in $X \times V'$ by (3.1.1), so that φ is the second projection. Constructing ψ is then equivalent to constructing a morphism $\psi_1 : V' \to \mathbb{P}^r$ which makes the diagram

$$\begin{tikzcd} \mathbb{P}(N) \arrow[d] & \supset \varphi^{-1}(V') \arrow[r, "\varphi"] & V' \arrow[d, "\psi_1"] \\ X \arrow[rr, hookrightarrow] & & \mathbb{P}^r \end{tikzcd} \tag{3.5.2}$$

commute; one then puts $\psi(v) = (\psi_1(v), v) \in \mathbb{P}^r \times V'$.

For any hypersurface in projective space, the Gauss morphism (3.4.1) is defined at every smooth point. Apply this to the hypersurface $\check{X} \subset \check{\mathbb{P}}^r$. Since V' lies in the smooth locus V of $\check{X}$, the Gauss morphism gives

$$\psi_1 : V' \longrightarrow \check{\check{\mathbb{P}}}^r \simeq \mathbb{P}^r. \tag{3.5.3}$$

It remains to check that (3.5.2) commutes. Let (x_0, H_0) be a closed point of $\varphi^{-1}(V')$. We must show that x_0, viewed as a hyperplane in $\check{\mathbb{P}}^r$, is tangent to $\check{X}$ at H_0: in other words, tangency of H_0 to X at x_0 implies tangency of x_0 to $\check{X}$ at H_0. Since φ is unramified at (x_0, H_0) and $\check{X}$ is smooth at $H_0 = \varphi(x_0, H_0)$, its differential is an isomorphism

$$d\varphi : T_{\mathbb{P}(N),(x_0,H_0)} \xrightarrow{\sim} T_{\check{X},H_0}. \tag{3.5.4}$$

The first part of the proof is therefore completed by the following lemma.

Lemma 3.6 (Euler).

Let (x_0, H_0) be a closed point of $\mathbb{P}(N)$, and let T_L denote the tangent bundle of the hyperplane $L \subset \check{\mathbb{P}}^r$ defined by x_0. Then, inside $T_{\check{\mathbb{P}}^r, H_0}$,

$$d\varphi\big(T_{\mathbb{P}(N),(x_0,H_0)}\big) \subset T_{L,H_0}.$$

Proof. Choose homogeneous coordinates $X_0, \ldots, X_r$ on $\mathbb{P}^r$ in which x_0 is the point $\underline{A} = (A_0, \ldots, A_r)$. A point of T_{X,x_0} is a point of $\mathbb{P}^r$ with values in $k[\epsilon]/(\epsilon^2)$ of the form

$$\underline{A} + \epsilon\underline{B}, \tag{3.6.1}$$

such that, for every local section H of I,

$$\sum_{i=0}^{r} B_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0. \tag{3.6.2}$$

Choose also a local section H of I which gives rise to H_0 by (3.1.1). Since $\mathbb{P}(N)$ is locally trivial over X, a point of $T_{\mathbb{P}(N),(x_0,H_0)}$ can be represented as

$$(\underline{A} + \epsilon\underline{B},\ H + \epsilon F), \tag{3.6.3}$$

where F is a local section of I and $\underline{A}, \underline{B}$ are as above. Its image under φ is the hyperplane over $k[\epsilon]/(\epsilon^2)$ with equation

$$\sum_{i=0}^{r} \frac{\partial(H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon\underline{B})T_i = 0. \tag{3.6.4}$$

Thus the assertion is equivalent to

$$\sum_{i=0}^{r} A_i \frac{\partial(H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon\underline{B}) = 0. \tag{3.6.5}$$

Because $\epsilon^2 = 0$, Taylor's formula gives

$$\frac{\partial(H + \epsilon F)}{\partial X_i}(\underline{A} + \epsilon\underline{B}) = \frac{\partial H}{\partial X_i}(\underline{A}) + \epsilon \frac{\partial F}{\partial X_i}(\underline{A}) + \epsilon \sum_{j=0}^{r} B_j \frac{\partial^2 H}{\partial X_j \partial X_i}(\underline{A}). \tag{3.6.6}$$

It is therefore enough to verify

$$\sum_{i=0}^{r} A_i \frac{\partial H}{\partial X_i}(\underline{A}) = 0, \tag{3.6.7}$$

$$\sum_{i=0}^{r} A_i \frac{\partial F}{\partial X_i}(\underline{A}) = 0, \tag{3.6.8}$$

and

$$\sum_{i,j=0}^{r} A_i B_j \frac{\partial^2 H}{\partial X_i \partial X_j}(\underline{A}) = 0. \tag{3.6.9}$$

Here H and F are represented by homogeneous functions of degree zero, so Euler's formula gives

$$\sum_i X_i \frac{\partial H}{\partial X_i} = 0, \qquad \sum_i X_i \frac{\partial F}{\partial X_i} = 0. \tag{3.6.10}$$

Since $\partial H/\partial X_j$ is homogeneous of degree -1, the same formula gives

$$\sum_i X_i \frac{\partial^2 H}{\partial X_i \partial X_j} = -\frac{\partial H}{\partial X_j}.$$

Consequently, (3.6.9) becomes

$$-\sum_j B_j \frac{\partial H}{\partial X_j}(\underline{A}) = 0, \tag{3.6.11}$$

which is precisely (3.6.2).

This proves that φ induces an isomorphism

$$\varphi^{-1}(V') \xrightarrow{\sim} V', \tag{3.6.12}$$

whose inverse is ψ. Now observe that the morphism

$$\begin{cases} \psi : V' \longrightarrow \mathbb{P}^r \times V' \subset \mathbb{P}^r \times \check{X}, \\ \psi(v) = (\psi_1(v), v) \end{cases} \tag{3.6.13}$$

extends to a morphism

$$\begin{cases} \widetilde{\psi} : V \longrightarrow \mathbb{P}^r \times V, \\ \widetilde{\psi}(v) = (\widetilde{\psi}_1(v), v), \end{cases} \tag{3.6.14}$$

where

$$\widetilde{\psi}_1 : V \longrightarrow \mathbb{P}^r \tag{3.6.15}$$

is the Gauss morphism, now regarded as defined on all of V.

We have just proved

$$\psi(V') \subset \mathbb{P}(N). \tag{3.6.16}$$

Since V' is dense in V and $\mathbb{P}(N)$ is closed in $\mathbb{P}^r \times \check{X}$, this implies

$$\widetilde{\psi}(V) \subset \mathbb{P}(N), \tag{3.6.17}$$

and in fact plainly

$$\widetilde{\psi}(V) \subset \varphi^{-1}(V). \tag{3.6.18}$$

Again by continuity, the identities

$$\psi \circ \varphi = \mathrm{id}_{\varphi^{-1}(V')}, \qquad \varphi \circ \psi = \mathrm{id}_{V'}$$

extend to

$$\widetilde{\psi} \circ \varphi = \mathrm{id}_{\varphi^{-1}(V)}, \qquad \varphi \circ \widetilde{\psi} = \mathrm{id}_V .$$

Thus the open subsets in 3.5(a) and 3.5(b) are identical.

We next show that the open subsets in 3.5(a) and 3.5(c) are identical. By 3.3, the latter is the subset $U \subset \check{X}$ defined by

$$U = \left\{ H \in \check{X} \;\middle|\; \begin{array}{l} \varphi^{-1}(H) \text{ consists of exactly one point, and} \\ \varphi \text{ is unramified at that point} \end{array} \right\}. \tag{3.6.19}$$

The scheme $\varphi^{-1}(U)$ is normal, since it is smooth, and the induced morphism

$$\varphi : \varphi^{-1}(U) \longrightarrow U$$

is birational, as already proved, radicial by the definition of U, and proper. It is therefore the normalization of U. In particular, U is geometrically unibranch. Since this morphism is unramified, EGA IV, 18.10.1 shows that it is étale; EGA IV, 18.10.18 then shows that it is an open immersion. It is consequently an isomorphism, and U is smooth. This proves $3.5(c) \subset 3.5(a)$. Formula (3.6.19) makes the inclusion $3.5(b) \subset 3.5(c)$ clear, and the equality $3.5(a) = 3.5(b)$ completes the proof.

3.7.

Theorem 2.5 is true when $\mathrm{char}(k) \neq 2$, and also when X has even dimension.
Proof. In view of 3.5.0 and 3.3, it is enough to verify the following assertion:

(3.7.1) For every closed point $x_0 \in X$ and every integer $d \geq 2$, there is a hypersurface H of degree d, tangent to X at x_0, such that x_0 is an ordinary quadratic singularity of $X \cap H$.

Choose homogeneous coordinates $X_0, \dots, X_r$ so that $x_0 = (1, 0, \dots, 0)$ and the functions X_i/X_0, $i = 1, \dots, n$, are local coordinates on X near x_0. One may take H to be the hypersurface with equation

$$H = X_0^{d-2} \sum_{i=1}^{n} X_i^2 \qquad \text{if } \mathrm{char}(k) \neq 2, \tag{3.7.1.1}$$

or

$$H = X_0^{d-2} \sum_{i=1}^{n/2} X_i X_{n/2+i} \qquad \text{if } n \text{ is even.} \tag{3.7.1.2}$$

When n is odd (under the first hypothesis), the equivalent standard choice recorded in the source is

$$H = X_0^{d-2} \left(X_n^2 + \sum_{i=1}^{\lfloor n/2 \rfloor} X_i X_{\lfloor n/2 \rfloor + i} \right). \tag{3.7.1.3}$$

For assertion 2.5.2, note that either φ is generically unramified, in which case 3.5.0 applies, or it is everywhere ramified. In characteristic zero the latter case implies $\dim \check{X} \leq r - 2$, so criterion (3.2.4) is then automatically satisfied.[43]

4. The general case

We shall sketch another method of proving 2.5, valid in every characteristic, assuming the following statement, which is close to 3.1 and is a special case of XV, 1.1.4.

(4.1) The subset of $\mathbb{P}(N)$ formed by the points (x, H) such that x is an ordinary quadratic singularity of $X \cap H$ is open.

Observe that 4.1 follows from 3.3, except when $\mathrm{char}(k) = 2$ and $\dim X$ is odd.

4.1.1.

Let R_1 be the closed complement in $\mathbb{P}(N)$ of the open set 4.1, and let

$$F_1 = \varphi(R_1) \subset \check{X} \subset \check{\mathbb{P}}^r$$

be its image.

Lemma 4.1.2.

If the open set 4.1 is nonempty, then $\mathrm{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$.
Proof. By hypothesis,

$$\dim R_1 < \dim \mathbb{P}(N) = r - 1,$$

and $F_1 = \varphi(R_1)$.

[43] The print has $\dim X \leq r - 2$ here. The alternative in Corollary 3.5.0 and the criterion being invoked require $\dim \check{X} \leq r - 2$.

4.1.3.

Let F_2 be the subset of $\check{X}$ formed by the hyperplanes which are tangent to X at at least two points. Notice that F_2 is not closed, but that

$$\overline{F_2} \subset F_1 \cup F_2. \tag{4.1.4}$$

This follows from the fact that $F_1 \cup F_2 = F$ (cf. 3.2) is closed. (Inclusion (4.1.4) "says" that a confluence of two singularities is not an ordinary quadratic singularity.) In any case, the construction in 4.2.4 will show that F_2 is constructible.

To prove 2.5.1 we shall apply criterion 3.2.4. For this we need to know that

$$\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2 \quad \text{and} \quad \operatorname{codim}_{\check{\mathbb{P}}^r}(F_2) \geq 2.$$

On the one hand, 3.7.1 and 4.1.2 show that, after replacing the given embedding by its d-th multiple, $d \geq 2$, one has $\operatorname{codim}_{\check{\mathbb{P}}^r}(F_1) \geq 2$. [44] Thus 2.5.1 will follow once we know the following proposition.

Proposition 4.2.

Let X be a smooth irreducible subvariety of $\mathbb{P}^r$, of dimension $n \geq 1$. For every integer $d \geq 2$, the subset F_2 of

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

formed by the hypersurfaces of degree d in $\mathbb{P}^r$ which are tangent to X at at least two points has codimension at least 2 in $\check{\mathbb{P}}^N$. [45]

4.2.1.

Begin with the special case $d = 2$ and X a linear subvariety of $\mathbb{P}^r$. It is enough to find a line in $\check{\mathbb{P}}^N$ which does not meet F (cf. 3.2.4). Choose homogeneous coordinates $X_0, \ldots, X_r$ on $\mathbb{P}^r$ so that X is defined by

$$X_{n+1} = \cdots = X_r = 0,$$

and homogeneous coordinates (λ, μ) on $\mathbb{P}^1$.

For n even, $n = 2k$, take the pencil of quadrics

$$H_{(\lambda,\mu)} = \lambda X_0^2 + (\lambda X_k + \mu X_0)X_{2k} + \sum_{i=1}^{k-1}(\lambda X_i + \mu X_{i+1})X_{k+i}. \tag{4.2.2}$$

For n odd, $n = 2k+1$, take the pencil

$$H_{(\lambda,\mu)} = (\lambda X_1 + \mu X_0)X_0 + \sum_{i=1}^{k}(\lambda X_{2i} + \mu X_i)X_{2i+1}. \tag{4.2.3}$$

In both cases one checks that $X \cap H_{(\lambda,\mu)}$ is smooth for $(\lambda, \mu) \neq (0, 1)$, while $X \cap H_{(0,1)}$ has just one singular point, which is moreover an ordinary quadratic singularity. Thus the pencil does not meet F.

[44] The print reads "3.8.1" and F_2 in this sentence. The ordinary-quadratic construction is (3.7.1), and Lemma 4.1.2 applies to F_1; Proposition 4.2 supplies the required assertion for F_2.

[45] The print has $\binom{n+d}{d} - 1$ here and $\binom{r+d}{d} - d$ in Proposition 4.3. The degree-d Veronese embedding defined in 2.4 gives $N = \binom{r+d}{d} - 1$ in both places.

4.2.4.

Let $\operatorname{Btg}^d(X)$ be the subvariety of

$$\check{\mathbb{P}}^N \times \big((X \times X) - \Delta(X \times X)\big)$$

formed by triples (H, x_1, x_2) such that H is a hypersurface of degree d in $\mathbb{P}^r$, tangent to X at x_1 and at x_2, with $x_1 \neq x_2$. The image of $\operatorname{Btg}^d(X)$ under pr_1 is the set F_2 of Proposition 4.2, so it is enough to prove

$$\dim \operatorname{Btg}^d(X) \leq N - 2 \quad \text{if } d \geq 3\text{, or if } d = 2 \text{ and } X \text{ is not linear.} \tag{4.2.5}$$

For this purpose, let $\operatorname{Lin}(X \times X)$ be the subvariety of $(X \times X) - \Delta(X \times X)$ given by

$$\begin{aligned} \operatorname{Lin}(X \times X) &= \big\{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad\qquad \text{the line spanned by } x_1, x_2 \text{ is tangent to } X \text{ at } x_1\big\} \\ &= \big\{(x_1, x_2) \in X \times X \mid x_1 \neq x_2, \\ &\qquad\qquad x_2 \in X \cap \operatorname{Tg}_X(x_1)\big\}. \end{aligned} \tag{4.2.6}$$

Clearly:

(4.2.7) If X is not a linear subvariety of $\mathbb{P}^r$, then

$$\dim \operatorname{Lin}(X \times X) < 2 \dim X.$$

Consequently, to prove (4.2.5) it is enough to show that the fibers of the morphism

$$\operatorname{Btg}^d(X) \xrightarrow{\operatorname{pr}_{2,3}} (X \times X) - \Delta(X \times X), \tag{4.2.8}$$

regarded as subvarieties of $\check{\mathbb{P}}^N$, have

$$\begin{cases} \text{codimension at least } 2\dim X + 1 & \text{over } \operatorname{Lin}(X \times X), \\ \text{codimension at least } 2\dim X + 2 & \text{over } (X \times X) - \Delta(X \times X) - \operatorname{Lin}(X \times X). \end{cases} \tag{4.2.9}$$

Statement (4.2.9) follows from the next proposition by taking L_i to be the tangent space to X at x_i, for $i = 1, 2$.

Proposition 4.3.

Let x_1 and x_2 be two distinct closed points of $\mathbb{P}^r$, and let L_1 and L_2 be two linear subspaces of $\mathbb{P}^r$ such that $x_1 \in L_1$ and $x_2 \in L_2$. In the space

$$\check{\mathbb{P}}^N = \check{\mathbb{P}}^{\binom{r+d}{d}-1}$$

of hypersurfaces of degree d, the linear subspace formed by the hypersurfaces tangent to L_1 at x_1 and to L_2 at x_2 has codimension

$$\begin{cases} \dim L_1 + \dim L_2 + 2, & \text{if } d \geq 3\text{, or if } d = 2 \text{ and } L_1 \cap L_2 \text{ does not} \\ & \text{contain the line } (x_1, x_2) \text{ through } x_1, x_2, \\ \dim L_1 + \dim L_2 + 1, & \text{if } d = 2 \text{ and } L_1 \cap L_2 \text{ contains the line } (x_1, x_2). \end{cases}$$

Proof. For $d \geq 3$, one must prove that $\dim L_1 + \dim L_2 + 2$ linear conditions are independent. It is therefore enough to treat the case $L_1 = L_2 = \mathbb{P}^r$, that is, to determine how many independent conditions make a hypersurface singular at both x_1 and x_2. Choose homogeneous coordinates so that

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0).$$

The hypersurface with equation $\sum_W A_W X^W = 0$ is singular at $(1, 0, \ldots, 0)$ if and only if all the coefficients of the monomials $X_0^{d-1} X_i$, $i = 0, \ldots, r$, vanish. It is singular at $(0, 1, 0, \ldots, 0)$ if and only if all the coefficients of the monomials $X_1^{d-1} X_i$, $i = 0, \ldots, r$, vanish. For $d \geq 3$ all these monomials are distinct, which proves the assertion in this case.

For $d = 2$, the same calculation gives

$$\operatorname{codim} \geq \dim L_1 + \dim L_2 + 1,$$

because the monomial $X_0 X_1$ is counted twice.

Suppose first that $L_1 \cap L_2$ does not contain the line (x_1, x_2). We reduce to the case $L_1 = \mathbb{P}^r$ and L_2 is a hyperplane in $\mathbb{P}^r$ which does not contain (x_1, x_2). Choose homogeneous coordinates $X_0, \ldots, X_r$ so that

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0), \qquad L_2 = (X_0 = 0).$$

A quadric $\sum_{i \leq j} A_{ij} X_i X_j$ is singular at $(1, 0, \ldots, 0)$ if and only if $A_{0j} = 0$ for $j = 0, \ldots, r$, and is tangent to $X_0 = 0$ at $(0, 1, \ldots, 0)$ if and only if $A_{1j} = 0$ for $j = 1, \ldots, r$. Together these are

$$2r + 1 = \dim L_1 + \dim L_2 + 2$$

independent conditions.

Finally, suppose that $L_1 \cap L_2$ contains the line (x_1, x_2). We reduce to the case $L_1 = L_2 = (x_1, x_2)$. In homogeneous coordinates with

$$x_1 = (1, 0, \ldots, 0), \qquad x_2 = (0, 1, 0, \ldots, 0),$$

the quadric $\sum_{i \leq j} A_{ij} X_i X_j$ passes through $(1, 0, \ldots, 0)$ and is tangent there to L_1 if and only if $A_{00} = A_{01} = 0$. It passes through $(0, 1, 0, \ldots, 0)$ and is tangent there to L_2 if and only if $A_{11} = A_{01} = 0$. Thus there are altogether only

$$3 = \dim L_1 + \dim L_2 + 1$$

conditions.

5. The degree of the dual variety by an "elementary" method

(Cf. XVIII, 3.2 for another method when $\dim X$ is even or $\operatorname{char}(k) \neq 2$.)

We continue to use the notation of 2.2: thus $X \hookrightarrow \mathbb{P}^r$ is a smooth irreducible subvariety of dimension n.

5.1.

We first consider the morphism (3.1.1)

$$\nu : \mathbb{P}(N) \longrightarrow X \times \check{\mathbb{P}}^r. \tag{5.1.0}$$

By definition, the conormal sheaf $\check{N}$ occurs in the exact sequence of coherent sheaves on X

$$0 \longrightarrow \check{N} \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \Omega^1_X \longrightarrow 0. \tag{5.1.1}$$

On the other hand, writing differential forms on $\mathbb{P}^r$ in homogeneous coordinates gives an exact sequence of sheaves on $\mathbb{P}^r$

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k} \longrightarrow \mathcal{O}_{\mathbb{P}^r}(-1)^{r+1} \longrightarrow \mathcal{O}_{\mathbb{P}^r} \longrightarrow 0. \tag{5.1.2}$$

Its restriction to X is therefore the exact sequence

$$0 \longrightarrow \Omega^1_{\mathbb{P}^r/k}|_X \longrightarrow \mathcal{O}_X(-1)^{r+1} \longrightarrow \mathcal{O}_X \longrightarrow 0. \tag{5.1.3}$$

[46] Combining (5.1.1) and (5.1.3) gives an inclusion

$$\check{N} \hookrightarrow \mathcal{O}_X(-1)^{r+1}, \tag{5.1.4}$$

which, by transposition, gives a surjection

$$\mathcal{O}_X(1)^{r+1} \longrightarrow N. \tag{5.1.5}$$

Hence there is a closed immersion

$$\alpha : \mathbb{P}(N) \longrightarrow \mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r, \tag{5.1.6}$$

which is precisely (5.1.0).

5.2. The idea of the calculation.

From now on suppose that $\check{X} = \varphi(\mathbb{P}(N))$ has dimension $r-1$, so that it is a hypersurface in $\check{\mathbb{P}}^r$. Its degree is naturally calculated by means of Chern classes [1].

For every smooth projective variety T, let $\mathrm{CH}(T)$ denote its Chow ring, graded by codimension. For $z \in \mathrm{CH}(T)$, write

$$\int_T z$$

for the degree of its zero-dimensional component. The class in $\mathrm{CH}^1(T)$ of an invertible sheaf $\mathcal{L}$ on T will also be denoted by $\mathcal{L}$ whenever no confusion can arise (for example, under the integral sign).

Let L be the class of a line in $\mathrm{CH}(\check{\mathbb{P}}^r)$. Then

$$\deg(\check{X}) = \int_{\check{\mathbb{P}}^r} L \cdot \check{X}. \tag{5.2.1}$$

Set-theoretically, $\check{X}$ is the image of $\mathbb{P}(N)$ under φ (3.1.2). From the viewpoint of the proper pushforward

$$\varphi_* : \mathrm{CH}(\mathbb{P}(N)) \longrightarrow \mathrm{CH}(\check{\mathbb{P}}^r),$$

however, one has

$$\varphi_*[\mathbb{P}(N)] = \deg(\varphi)[\check{X}].$$

Consequently, (5.2.1) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\check{\mathbb{P}}^r} L \cdot \varphi_*[\mathbb{P}(N)]. \tag{5.2.2}$$

Here, by abuse of notation, φ in $\deg(\varphi)$ denotes the morphism $\mathbb{P}(N) \to \check{X}$ induced by (3.1.2); its degree is a nonnegative integer, taken to be zero precisely when $\dim \check{X} < r-1 = \dim \mathbb{P}(N)$. [47] The projection formula rewrites (5.2.2) as

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \varphi^*(L). \tag{5.2.3}$$

[46] The print leaves the last term as $\mathcal{O}_{\mathbb{P}^r}$ after restricting the sequence to X; the restricted term is $\mathcal{O}_X$.

[47] The print has $\dim X < r-1$. The condition for the induced map to the dual variety to be generically finite is $\dim \check{X} = r-1$.

Let $\eta \in \mathrm{CH}(\check{\mathbb{P}}^r)$ be the class of a hyperplane. Since $L = \eta^{r-1}$, equation (5.2.3) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\varphi^*(\eta)\big)^{r-1}. \tag{5.2.4}$$

We now calculate $\varphi^*(\eta)$ concretely.

Let ξ (respectively τ) denote the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(N)}(1)$ on $\mathbb{P}(N)$ (respectively $\mathcal{O}_{\mathbb{P}(\mathcal{O}_X(1)^{r+1})}(1)$ on $\mathbb{P}(\mathcal{O}_X(1)^{r+1})$). Functoriality gives, by (5.1.6),

$$\alpha^*(\tau) = \xi. \tag{5.2.5}$$

Under the identification

$$\mathbb{P}\big(\mathcal{O}_X(1)^{r+1}\big) \simeq X \times \check{\mathbb{P}}^r,$$

and writing $H = \mathcal{O}_X(1)$, one plainly has

$$\tau = \mathrm{pr}_1^*(H) \otimes \mathrm{pr}_2^*(\eta). \tag{5.2.6}$$

Let $\pi : \mathbb{P}(N) \to X$ be the structural morphism. Then

$$\pi = \mathrm{pr}_1 \circ \alpha, \qquad \varphi = \mathrm{pr}_2 \circ \alpha.$$

Applying α^* to (5.2.6) and using (5.2.5), we obtain

$$\xi = \alpha^*(\tau) = \pi^*(H) \otimes \varphi^*(\eta). \tag{5.2.6$'$}$$

[48] Thus, in $\mathrm{CH}(\mathbb{P}(N))$,

$$\varphi^*(\eta) = \xi - \pi^*(H), \tag{5.2.7}$$

and (5.2.3) becomes

$$\deg(\check{X}) = \frac{1}{\deg(\varphi)} \int_{\mathbb{P}(N)} \big(\xi - \pi^*(H)\big)^{r-1}. \tag{5.2.8}$$

The right-hand side of (5.2.8) is calculated by means of the following lemma (cf. 5.3.3).

Lemma 5.3.

Let X be a smooth connected quasi-projective variety of dimension n, and let M be a locally free sheaf of finite rank a on X. Let ξ denote the invertible sheaf $\mathcal{O}_{\mathbb{P}(M)}(1)$ on $\mathbb{P}(M)$, and let $\pi : \mathbb{P}(M) \to X$ be the structural morphism. Recall that $\mathrm{CH}(\mathbb{P}(M))$ is a free $\mathrm{CH}(X)$-module (via π^*), with basis $1, \xi, \dots, \xi^{a-1}$, where ξ satisfies

$$\xi^a - C_1\xi^{a-1} + C_2\xi^{a-2} - C_3\xi^{a-3} + \cdots = 0, \tag{5.3.1}$$

the C_i being the Chern classes of M in $\mathrm{CH}(X)$. Put $C(M) = 1 + C_1 + C_2 + \cdots$, the total Chern class of M; it is an invertible element of $\mathrm{CH}(X)$. Then:

(i) one has

$$\pi_*\left(\frac{\xi^{a-1}}{1+\xi}\right) = \frac{1}{C(M)}; \tag{5.3.2}$$

[48] The print numbers both this display and the preceding one (5.2.6); the prime distinguishes the second occurrence.

(ii) let $F(T) \in \mathrm{CH}(X)[T]$ be a polynomial such that the coefficient of T^i is homogeneous of degree $n+a-1-i$. Thus $F(\xi)$ has maximal degree $n+a-1 = \dim \mathbb{P}(M)$ in $\mathrm{CH}(\mathbb{P}(M))$. Then

$$\int_{\mathbb{P}(M)} F(\xi) = (-1)^{a-1} \int_X \frac{F(-1)}{C(M)}. \tag{5.3.3}$$

Proof. We first deduce (5.3.3) from (5.3.2). Write

$$\frac{1}{C(M)} = \sum_{j \geq 0} b_j, \qquad b_0 = 1, \qquad b_j \in \mathrm{CH}^j(X), \qquad b_j = 0 \text{ for } j < 0. \tag{5.3.4}$$

Then (5.3.2) is equivalent to

$$\pi_*(\xi^{a-1+j}) = (-1)^j b_j \qquad \text{whenever } a-1+j \geq 0, \tag{5.3.5}$$

since $\pi_*(\xi^q) = 0$ for $q < a-1$ for dimensional reasons.

By linearity it suffices to prove (5.3.3) when $F(T) = uT^{a-1+j}$, with $u \in \mathrm{CH}^{n-j}(X)$ homogeneous. By (5.3.5) and the projection formula,

$$\begin{aligned} \int_{\mathbb{P}(M)} F(\xi) &= \int_{\mathbb{P}(M)} \pi^*(u)\xi^{a-1+j} \\ &= \int_X u\, \pi_*(\xi^{a-1+j}) = (-1)^j \int_X u\, b_j. \end{aligned} \tag{5.3.6}$$

On the other hand,

$$\begin{aligned} (-1)^{a-1} \int_X \frac{F(-1)}{C(M)} &= (-1)^{a-1} \int_X \frac{(-1)^{a-1+j} u}{C(M)} \\ &= (-1)^j \int_X \sum_i u\, b_i. \end{aligned} \tag{5.3.7}$$

Because $\deg u = n-j$ and $\deg b_i = i$, one has

$$\int_X u\, b_i = 0 \qquad \text{for } i \neq j.$$

Thus (5.3.7) reduces to

$$(-1)^{a-1} \int_X \frac{F(-1)}{C(M)} = (-1)^j \int_X u\, b_j, \tag{5.3.8}$$

which proves the asserted implication.

It remains to prove (5.3.2), or equivalently (5.3.5).

For dimensional reasons,

$$\pi_*(\xi^j) = 0 \qquad \text{if } j < a-1,$$

whereas

$$\pi_*(\xi^{a-1}) = 1_X$$

for the geometric reason that ξ^{a-1} meets each fiber in one point. The projection formula therefore gives

$$\pi_*\left(\sum_{i=0}^{a-1} \pi^*(U_i)\xi^i\right) = U_{a-1}. \tag{5.3.9}$$

[49] By (5.3.9), assertion (5.3.5) follows from the next lemma, in which ξ is replaced by T and C_i by S_i, in view of (5.3.1).

[49] The printed display mixes the indices i, j and gives U_{j-i} on the right. The two immediately preceding push-forward identities force the coefficient U_{a-1}.

Lemma 5.4.

Let $a > 0$ be an integer and let $A = \mathbb{Z}[[X_1, \ldots, X_a]]$ be the ring of formal power series in a indeterminates. Let B be the A-algebra

$$B = A[[T]] \Big/ \left(\prod_{i=1}^{a} (T - X_i) \right).$$

Thus B is a free A-module with basis $1, T, \ldots, T^{a-1}$:

$$B = AT^{a-1} \oplus AT^{a-2} \oplus \cdots \oplus A. \tag{5.4.1}$$

Let $S_i \in A$ be the i-th elementary symmetric function

$$S_i = \sum_{1 \le \alpha_1 < \cdots < \alpha_i \le a} X_{\alpha_1} \cdots X_{\alpha_i},$$

so that

$$\sum_{i=0}^{a} (-1)^i S_i T^{a-i} = 0 \qquad \text{in } B. \tag{5.4.2}$$

The element $1 + S_1 + \cdots + S_a$ is invertible in A; write its inverse as

$$\frac{1}{\sum_{i=0}^{a} S_i} = \sum_{j \ge 0} b_j, \qquad b_j \text{ homogeneous of degree } j \text{ in } A, \qquad b_0 = 1. \tag{5.4.3}$$

For every integer $j \ge 0$, define a homogeneous element $a_j \in A$ of degree j by writing, using (5.4.1),

$$T^{a-1+j} = a_j T^{a-1} + ?\,T^{a-2} + ?\,T^{a-3} + \cdots + ?, \qquad ? \in A. \tag{5.4.4}$$

Then, for every $j \ge 0$,

$$a_j = (-1)^j b_j. \tag{5.4.5}$$

Proof. In view of the definition (5.4.3) of the b_j, it is enough to prove

$$\begin{cases} a_0 = 1, \\ \displaystyle\sum_{i=0}^{a} (-1)^i S_i \, a_{j-i} = 0 \quad \text{if } j > 0. \end{cases} \tag{5.4.6}$$

Clearly $a_0 = 1$. For a fixed $j > 0$, the second relation is obtained by taking the coefficient of T^{a-1}, using (5.4.1), in the relation

$$\sum_{i=0}^{a} (-1)^i S_i T^{a-1+j-i} = 0 \qquad \text{in } B. \tag{5.4.7}$$

[50] This relation is obtained by multiplying (5.4.2) by T^{j-1}, which proves the lemma.

[50] The print omits the upper summation bound a in (5.4.7); the relation is (5.4.2) multiplied by T^{j-1}.

Corollary 5.5.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{(1+H)^{r-1}}{C(N)}. \tag{5.5.1}$$

Proof. Apply (5.3.3), with $M = N$ the normal sheaf of $X \hookrightarrow \mathbb{P}^r$, to (5.2.8).

For a smooth quasi-projective variety S, write

$$C(S) \overset{\text{def}}{=} C(T_S) \tag{5.5.2}$$

for its total Chern class, where T_S denotes the tangent bundle of S.

Dualizing the exact sequence (5.1.2) gives

$$C(\mathbb{P}^r) = (1+H)^{r+1} \qquad \text{in } \mathrm{CH}(\mathbb{P}^r). \tag{5.5.3}$$

Dualizing the exact sequence (5.1.1) gives

$$C(N)\, C(X) = C(\mathbb{P}^r)|_X \qquad \text{in } \mathrm{CH}(X). \tag{5.5.4}$$

Combining (5.5.3) and (5.5.4), formula (5.5.1) becomes:

Corollary 5.6.

One has

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \int_X \frac{C(X)}{(1+H)^2}. \tag{5.6.1}$$

[51]

5.7.

For a smooth projective variety S, define its Euler–Poincaré characteristic by

$$\chi(S) = \int_S C(S). \tag{5.7.1}$$

Proposition 5.7.2.

Let Y be a smooth hyperplane section of X, and let Δ be a smooth section of X by a codimension-two linear subspace of $\mathbb{P}^r$. Then

$$\deg(\check{X}) = \frac{(-1)^n}{\deg(\varphi)} \big[\chi(X) + \chi(\Delta) - 2\chi(Y)\big]. \tag{5.7.3}$$

Proof. Let $i : Y \hookrightarrow X$ be the inclusion. One has

$$N_{Y/X} = H|_Y = i^*H.$$

The exact sequence

$$0 \longrightarrow T_Y \longrightarrow i^*T_X \longrightarrow i^*H \longrightarrow 0$$

[51] The print has $(-1)^r$. Substitution of (5.5.3) and (5.5.4) into (5.5.1) does not alter its sign $(-1)^n$; formula (5.7.3) immediately below confirms the same exponent.

therefore gives

$$C(Y) = i^*\left(\frac{C(X)}{1+H}\right). \tag{5.7.4}$$

Hence

$$\chi(Y) = \int_Y i^*\left(\frac{C(X)}{1+H}\right),$$

and the projection formula gives

$$\chi(Y) = \int_X i_* i^*\left(\frac{C(X)}{1+H}\right) = \int_X \frac{H\,C(X)}{1+H}. \tag{5.7.5}$$

Iterating this calculation for Δ, one obtains

$$\begin{aligned}\chi(\Delta) &= \int_Y \frac{H\,C(Y)}{1+H} \\ &= \int_Y i^*\left(\frac{H}{1+H}\cdot\frac{C(X)}{1+H}\right) = \int_X \frac{H^2 C(X)}{(1+H)^2}.\end{aligned} \tag{5.7.6}$$

It follows that

$$\begin{aligned}\chi(X) - 2\chi(Y) + \chi(\Delta) &= \int_X C(X)\left[1 - \frac{2H}{1+H} + \frac{H^2}{(1+H)^2}\right] \\ &= \int_X \frac{C(X)}{(1+H)^2}.\end{aligned}$$

Together with (5.6.1), this proves the proposition.

6. The case of a general base

6.1.

Let S be a scheme and let $\bar{s}$ be a geometric point of S, defined by an algebraic closure of the residue field $k(s)$ for a point $s \in S$. Let $\mathbb{P}^r_S$ be projective r-space over S, let $\check{\mathbb{P}}^r_S$ be its dual projective space, and let

$$I \subset \mathbb{P}^r_S \times_S \check{\mathbb{P}}^r_S$$

be the incidence variety. Let $D \subset \check{\mathbb{P}}^r_S$ be a line, defined by an S-valued point of the appropriate Grassmannian, and let Δ be the axis of the corresponding pencil of hyperplanes (cf. 2.1).

Let X be a proper smooth S-scheme, pure of dimension $n \geq 1$, equipped with a projective embedding $X \hookrightarrow \mathbb{P}^r_S$. Put

$$\widetilde{X} = (X \times_S D) \cap I.$$

Thus $\widetilde{X}$ is the scheme over D whose fibers are the hyperplane sections of X parametrized by D; let $u : \widetilde{X} \to D$ be the projection.

We use a subscript $\bar{s}$ to denote base change by $\bar{s} \to S$. Suppose that $\Delta_{\bar{s}}$ is transverse to $X_{\bar{s}}$. Then, over a neighborhood U of s in S, Δ is transverse to X. In what follows we replace S by this neighborhood and write $U = S$. The scheme $\widetilde{X}$ is then smooth over S, and a neighborhood in $\widetilde{X}$ of the inverse image of Δ is smooth over D.

Suppose moreover that one of the hyperplane sections of $X_{\bar{s}}$ belonging to the pencil $D_{\bar{s}}$ is smooth. After again replacing S by a suitable neighborhood of s, the closed subscheme $F \subset \widetilde{X}$ defined by the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ is finite and flat over S.

6.2.

The exceptional locus of the pencil D is the relative Cartier divisor E of D/S defined by the finite flat $\mathcal{O}_D$-module $u_*\mathcal{O}_F$. Locally on D, if

$$\mathcal{O}_D^m \xrightarrow{v} \mathcal{O}_D^m \longrightarrow u_*\mathcal{O}_F \longrightarrow 0$$

is a presentation of $u_*\mathcal{O}_F$, then E has equation $\det(v) = 0$. Formation of E commutes with arbitrary base change.

The divisor $E_{\bar{s}}$ on $D_{\bar{s}}$ can be written

$$E_{\bar{s}} = \sum_P n(P)[P],$$

where $n(P)$ is the sum of the colengths of the Jacobian ideal $J^{n-1}(\widetilde{X}/D)$ at the critical points of u mapping to P.

Proposition 6.3.

Suppose that $D_{\bar{s}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{s}}$ and that char $k(s) \neq 2$ or n is even. There is a neighborhood U of s in S such that:

(a) for every geometric point $\bar{t}$ of U, $D_{\bar{t}}$ is a Lefschetz pencil of hyperplane sections of $X_{\bar{t}}$;

(b) over U, the exceptional locus E is étale over S.

To prove (a) and (b), it is enough to observe that, under these hypotheses, $D_{\bar{s}}$ satisfies condition 2.2(c) if and only if the divisor $E_{\bar{s}}$ is reduced. Invoking 3.3, one could use only the easier implication from left to right.

6.3 (exceptional case).

[52] The proposition is false in the exceptional case just mentioned. Nevertheless, if S has characteristic two and n is odd, there is a neighborhood U of s over which (a) holds, and one can define a divisor E' étale over U such that $E = 2E'$. We shall not use this result.

References

[1] A. Grothendieck, *La théorie des classes de Chern*, Bull. Soc. Math. France **86** (1958), 137–154.

[52] The print repeats the number 6.3 after Proposition 6.3.

SGA 7 — EXPOSÉ XVIII

Cohomological Study of Lefschetz Pencils

by N. Katz[53]

Contents

Introduction

In the present exposé we give the elementary cohomological structure of Lefschetz pencils. Sections 1–4 give the "well-known" facts concerning the cohomological structure of projective bundles and blow-ups. In Section 5 we carry out the study named in the title, assuming that X satisfies the "Lefschetz vache theorem" (5.2) and that the pencil D in question satisfies a certain condition (A), local in nature on D (5.3). The principal results are 5.6, 5.7, and 5.8.7; the last plays an essential role in the proof of Griffiths's theorem in Exposé XX. In the final section, which is distinctly less elementary, we use Mme Raynaud's Exposé XIII of SGA 1 on the tame fundamental group, and the Picard–Lefschetz formula (XVI), to prove:

(1) condition (A) is often satisfied (6.3, 6.4);

(2) except when $\dim(X)$ is odd and the characteristic is 2, the monodromy group of a Lefschetz pencil acts irreducibly on the "vanishing cohomology," and transitively (up to sign) on the set of "local vanishing cycles" associated, via the Picard–Lefschetz formula, with the various "critical points" of the pencil (6.6 and 6.7).

1. The cohomology of a projective bundle

Theorem 1.1 (Deligne).

Let $f : X \longrightarrow S$ be a proper morphism, let ℓ be a prime number invertible on S, let $u \in H^2(X, \mathbb{Z}_\ell(1))$ (respectively, $u \in H^2(X, \mathbb{Q}_\ell(1))$) be a cohomology class on X, and let $\mathfrak{F}$ be a $\mathbb{Z}_\ell$-sheaf (respectively, a $\mathbb{Q}_\ell$-sheaf) on X. Suppose that there is an integer n such that, for every integer j and every integer $i > 0$, the map

(1.1.1) $$R^{n-i} f_*\big(\mathfrak{F}(j)\big) \xrightarrow{\wedge u^i} R^{n+i} f_*\big(\mathfrak{F}(j+i)\big)$$

[53] Based on succinct notes of Grothendieck.

is an isomorphism of $\mathbb{Z}_\ell$-sheaves (respectively, $\mathbb{Q}_\ell$-sheaves) on S. Then, for every morphism $g : S \longrightarrow T$, the Leray spectral sequence

$$E_2^{p,q} = R^p g_* R^q f_*(\mathfrak{F}) \Longrightarrow R^{p+q}(gf)_*(\mathfrak{F})$$

degenerates.

The proof is found in [1, Prop. 2.4]. Notice that, by the proper base-change theorem (SGA 4 XII, 5.2), the hypothesis can be checked fibre by fibre.

Theorem 1.2 (cf. SGA 5 VII, 2.2.1).

Let $\mathcal{E} \longrightarrow X$ be a vector bundle, locally free of rank $1 + r$. Let $f : \mathbb{P}(\mathcal{E}) \longrightarrow X$ denote the projective bundle defined by $\mathcal{E}$, and let $u \in H^2(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell(1))$ be the first Chern class of the canonical invertible sheaf $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$. For ℓ a prime number invertible on X, one has:

a) the spectral sequence

$$E_2^{p,q} = H^p\big(X, R^q f_* \mathbb{Z}_\ell\big) \Longrightarrow H^{p+q}\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big) \tag{1.2.1}$$

degenerates;

b) the map

$$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big) \longrightarrow H^q\big(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell\big), \qquad (\tau_{q-2j})_j \longmapsto \sum_j f^*(\tau_{q-2j}) \wedge u^j \tag{1.2.2}$$

is an isomorphism.

[54]

Proof. The fibres of f are projective spaces of dimension r, and one knows that

$$R^i f_* \mathbb{Z}_\ell \xleftarrow[\sim]{\wedge^{i/2} u} \begin{cases} \mathbb{Z}_\ell\left(-\dfrac{i}{2}\right), & \text{if } i \text{ is even and } 0 \le i \le 2r, \\ 0, & \text{otherwise.} \end{cases} \tag{1.2.3}$$

Theorem 1.1 applies to $(f, \ell, u, \mathbb{Z}_\ell, r)$, which proves (a). By (1.2.3),

$$E_2^{a,b} = \begin{cases} H^a\left(X, \mathbb{Z}_\ell\left(-\dfrac{b}{2}\right)\right), & \text{if } b \text{ is even and } 0 \le b \le 2r, \\ 0, & \text{otherwise.} \end{cases} \tag{1.2.4}$$

Thus the graded object associated with $H^q(\mathbb{P}(\mathcal{E}), \mathbb{Z}_\ell)$ for the filtration coming from the spectral sequence is

$$\bigoplus_{0 \le j \le r} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big).$$

If this direct sum is equipped with the filtration

$$F^i = \bigoplus_{\substack{0 \le j \le r \\ q-2j \le i}} H^{q-2j}\big(X, \mathbb{Z}_\ell(-j)\big),$$

the map (1.2.2) is compatible with the filtrations and induces the identity on the associated graded objects. This proves (b).

[54] In (1.2.2) the print has the upper bound n, the degree $q - 2 - j$, and one fewer closing parenthesis. The rank $1 + r$, the defining map, and the proof immediately below require $0 \le j \le r$ and $q - 2j$; the corrected formula is printed here.

2. The cohomology of blown-up varieties (cf. SGA 5 VII, 8)

2.1.

Let X be a smooth scheme over a field k, and let $\Delta \subset X$ be an everywhere smooth subscheme of codimension $1 + r$. It is known (EGA IV, 19.4.9) that there is a <u>smooth</u> scheme $\widetilde{X}$, equipped with a projective morphism

$$f : \widetilde{X} \longrightarrow X, \tag{2.1.1}$$

such that, on setting

$$\widetilde{\Delta} = f^{-1}(\Delta) = \Delta \times_X \widetilde{X}, \tag{2.1.2}$$

f induces an isomorphism

$$f|_{\widetilde{X}-\widetilde{\Delta}} : \widetilde{X} - \widetilde{\Delta} \xrightarrow{\sim} X - \Delta, \tag{2.1.3}$$

and there is a Δ-isomorphism between $\widetilde{\Delta}$ and $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$, the projective bundle of lines in the normal bundle $\underline{N}_{\Delta/X}$ of Δ in X. Thus one has a commutative diagram

$$\begin{tikzcd} \widetilde{\Delta} \arrow[r, phantom, "\simeq"] \arrow[dr, "f|_{\widetilde{\Delta}}"'] & \check{\mathbb{P}}(\underline{N}_{\Delta|X}) \arrow[d, "\text{canonical projection}"] \\ & \Delta \end{tikzcd} . \tag{2.1.4}$$

2.1.5.

In the special case

$$X = \operatorname{Spec}(A), \qquad \Delta = \operatorname{Spec}(A/I), \qquad I = (g_0, \ldots, g_r),$$

$\widetilde{X}$ is obtained by the Hopf construction (cf. EGA IV, 19.4.11). One defines a morphism from $X - \Delta$ to $\mathbb{P}^r$ by

$$x \in X - \Delta \longmapsto \big(g_0(x), \ldots, g_r(x)\big) \in \mathbb{P}^r$$

and regards its graph Γ as embedded in $X \times \mathbb{P}^r$. Then $\widetilde{X}$ is defined to be the schematic closure of Γ in $X \times \mathbb{P}^r$, and the morphism $f : \widetilde{X} \longrightarrow X$ is the composite

$$\widetilde{X} \hookrightarrow X \times \mathbb{P}^r \xrightarrow{\mathrm{pr}_1} X.$$

2.1.6.

Similarly (cf. again EGA IV, 19.4.11), if X is projective and Δ is the intersection of X with $1+r$ hypersurfaces, say with equations $G_i = 0$, $i = 0, \ldots, r$, one again considers the morphism

$$X - \Delta \longrightarrow \mathbb{P}^r, \qquad x \longmapsto \big(G_0(x), \ldots, G_r(x)\big),$$

and obtains $\widetilde{X}$ as the closure of its graph in $X \times \mathbb{P}^r$. Concretely, in terms of homogeneous coordinates $\lambda_0, \ldots, \lambda_r$ on $\mathbb{P}^r$, $\widetilde{X}$ is the subscheme of $X \times \mathbb{P}^r$ defined by the equations

$$\lambda_i G_j(X) - \lambda_j G_i(X), \qquad 0 \leq i < j \leq r. \tag{2.1.6.1}$$

These equations show that the point (x, λ) lies in $\widetilde{\Delta}$ if and only if $x \in \Delta$; in other words, $\check{\mathbb{P}}(\underline{N}_{\Delta/X})$ is the trivial projective bundle[55]

$$\begin{tikzcd}
\check{\mathbb{P}}(\underline{N}_{\Delta/X}) \arrow[dr, "\text{canonical projection}"'] & \simeq & \Delta \times \mathbb{P}^r \arrow[dl, "\mathrm{pr}_1"] \\
& \Delta &
\end{tikzcd} \quad . \tag{2.1.6.2}$$

One has the following commutative diagram:

$$\begin{tikzcd}
\Delta \times \mathbb{P}^r \arrow[r, "\sim"] \arrow[dr, "\mathrm{pr}_1"'] & \widetilde{\Delta} \arrow[r, hook] \arrow[d, "f|_{\widetilde{\Delta}}"] & \widetilde{X} \arrow[r, hook] \arrow[d, "f"] & X \times \mathbb{P}^r \arrow[dl, "\mathrm{pr}_1"] \\
& \Delta \arrow[r, hook] & X &
\end{tikzcd} \quad , \tag{2.1.6.3}$$

and the composite inclusion $\Delta \times \mathbb{P}^r \hookrightarrow X \times \mathbb{P}^r$ is obtained from $\Delta \hookrightarrow X$ by base change.

Theorem 2.2.

Let X be smooth over a field k, let Δ be an everywhere smooth subscheme of codimension $1 + r$, and let $f : \widetilde{X} \longrightarrow X$ be the blow-up of X along Δ. Then:

(2.2.1) The Leray spectral sequence

$$E_2^{p,q} = H^p\big(X, R^q f_* \mathbb{Z}_\ell\big) \Longrightarrow H^{p+q}\big(\widetilde{X}, \mathbb{Z}_\ell\big)$$

degenerates.

(2.2.2) There is an additive isomorphism

$$\begin{aligned}
H^q(\widetilde{X}, \mathbb{Z}_\ell) \xleftarrow{\ \sim\ } H^q(X, \mathbb{Z}_\ell) \oplus \frac{H^q(\widetilde{\Delta}, \mathbb{Z}_\ell)}{\operatorname{Im} H^q(\Delta, \mathbb{Z}_\ell)} \\
\simeq \left(\bigoplus_{j=1}^{r} H^{q-2j}(\Delta, \mathbb{Z}_\ell(-j)) \right) \oplus H^q(X, \mathbb{Z}_\ell).
\end{aligned}$$

Proof. Let

$$g : \widetilde{\Delta} \longrightarrow \Delta \tag{2.2.3}$$

be the restriction of f to $\widetilde{\Delta}$, and let

$$\begin{cases} j : \Delta \hookrightarrow X, \\ i : \widetilde{\Delta} \hookrightarrow \widetilde{X} \end{cases} \tag{2.2.4}$$

be the inclusions, so that one has the commutative diagram

$$\begin{tikzcd}
\widetilde{X} \arrow[d, "f"'] & \widetilde{\Delta} \arrow[l, "i"'] \arrow[d, "g"] \\
X & \Delta \arrow[l, "j"']
\end{tikzcd} \quad . \tag{2.2.5}$$

[55]The print labels the right-hand projection in (2.1.6.2) pr_i. Its target is Δ, and the same projection is labelled pr_1 in (2.1.6.3); the corrected label is used here.

It induces a homomorphism between the Leray spectral sequences for the morphisms f and g, with the sheaf $\mathbb{Z}_\ell$ on $\widetilde{X}$ and $\widetilde{\Delta}$:

$$E_s^{p,q}(f) \xrightarrow{(i^*,j^*)} E_s^{p,q}(g). \tag{2.2.6}$$

On the other hand, since f induces an isomorphism from $\widetilde{X}-\widetilde{\Delta}$ to $X-\Delta$, the proper base-change theorem gives

$$R^q f_* \mathbb{Z}_\ell \simeq \begin{cases} \mathbb{Z}_\ell, & q=0, \\ j_* R^q g_* \mathbb{Z}_\ell, & q>0. \end{cases} \tag{2.2.7}$$

Hence

$$E_2^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_2^{p,q}(g) \qquad (q \neq 0). \tag{2.2.8}$$

We shall prove by induction on s that

$$E_s^{p,q}(f) \xrightarrow[\sim]{(i^*,j^*)} E_s^{p,q}(g) \qquad (q \neq 0,\ s \geq 2) \tag{2.2.9}$$

and

$$d_s^{p,q}(f) = 0 \qquad (s \geq 2). \tag{2.2.10}$$

For the moment, admit the formula

$$d_s^{p,s-1}(f) = 0 \qquad (s \geq 2,\ \text{for every } p). \tag{2.2.11}$$

Begin with $s=2$. We have a commutative diagram

$$\begin{tikzcd} E_2^{p,q}(f) \arrow[r, "(i^*,j^*)"] \arrow[d, "d_2^{p,q}(f)"] & E_2^{p,q}(g) \arrow[d, "0=d_2^{p,q}(g)"] \\ E_2^{p+2,q-1}(f) \arrow[r, "(i^*,j^*)"] & E_2^{p+2,q-1}(g) \end{tikzcd} . \tag{2.2.12}$$

For $q \leq 0$, one has $E_2^{p+2,q-1}(f) = 0$, hence $d_2^{p,q}(f) = 0$. For $q = 1$, formula (2.2.11) gives $d_2^{p,1}(f) = 0$. For $q > 1$, the horizontal arrows in (2.2.12) are isomorphisms, while $d_2^{p,q}(g) = 0$ by 1.2 and 2.1.4. This proves (2.2.10) for $s = 2$, and (2.2.9) for $s = 2$ is exactly (2.2.8).

Suppose (2.2.9) and (2.2.10) are known for some $s \geq 2$. Hypothesis (2.2.10) implies $E_s^{p,q}(f) \simeq E_{s+1}^{p,q}(f)$, and by 1.2 one has $E_s^{p,q}(g) \simeq E_{s+1}^{p,q}(g)$, so there is a commutative diagram

$$\begin{tikzcd} E_s^{p,q}(f) \arrow[r, "(i^*,j^*)"] \arrow[d, "\wr"'] & E_s^{p,q}(g) \arrow[d, "\wr"] \\ E_{s+1}^{p,q}(f) \arrow[r, "(i^*,j^*)"] & E_{s+1}^{p,q}(g) \end{tikzcd} , \tag{2.2.13}$$

which proves (2.2.9) for $s+1$, in view of (2.2.9) for s. To deduce (2.2.10) for $s+1$, consider the commutative diagram

$$\begin{tikzcd} E_{s+1}^{p,q}(f) \arrow[r, "(i^*,j^*)"] \arrow[d, "d_{s+1}^{p,q}(f)"] & E_{s+1}^{p,q}(g) \arrow[d, "0=d_{s+1}^{p,q}(g)"] \\ E_{s+1}^{p+s+1,q-s}(f) \arrow[r, "(i^*,j^*)"] & E_{s+1}^{p+s+1,q-s}(g) \end{tikzcd} . \tag{2.2.14}$$

For $q \leq s-1$, one has $E_{s+1}^{p+s+1,q-s}(f) = 0$, hence $d_{s+1}^{p,q}(f) = 0$. For $q = s$, one has (2.2.11), and for $q > s$ the horizontal arrows in (2.2.14) are isomorphisms and $d_{s+1}^{p,q}(g) = 0$ by 1.2. This gives the desired conclusion. [56]

It remains to verify (2.2.11), which says that the composite edge homomorphism

$$E_2^{p,0}(f) \longrightarrow E_3^{p,0}(f) \longrightarrow E_4^{p,0}(f) \longrightarrow \cdots \longrightarrow E_\infty^{p,0}(f) \tag{2.2.15}$$

is an isomorphism; that is, that

$$f^* : H^p(X, \mathbb{Z}_\ell) \longrightarrow H^p(\widetilde{X}, \mathbb{Z}_\ell) \tag{2.2.16}$$

is injective. But it has a left inverse, namely the Gysin homomorphism (SGA 5 IV)

$$f_* : H^p(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^p(X, \mathbb{Z}_\ell), \tag{2.2.17}$$

which indeed satisfies

$$f_* f^* = \mathrm{id}. \tag{2.2.18}$$

This follows from the projection formula and the fact that $f_*(1) = 1$, since f is birational and therefore has degree one. This proves (2.2.1).

As for (2.2.2), passing to cohomology in diagram (2.2.5) gives a commutative diagram

$$\begin{tikzcd} H^p(\widetilde{X}, \mathbb{Z}_\ell) \arrow[r, "i^*"] & H^p(\widetilde{\Delta}, \mathbb{Z}_\ell) \\ H^p(X, \mathbb{Z}_\ell) \arrow[u, "f^*"] \arrow[r, "j^*"] & H^p(\Delta, \mathbb{Z}_\ell) \arrow[u, "g^*"'] \end{tikzcd} , \tag{2.2.19}$$

hence a morphism

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{f^* H^p(X, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^p(\Delta, \mathbb{Z}_\ell)}. \tag{2.2.20}$$

Let F^i denote the filtrations on $H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$ and $H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)$ coming from the Leray spectral sequences. Then (2.2.20) can be rewritten as[57]

$$\frac{H^p(\widetilde{X}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{X}, \mathbb{Z}_\ell)} \xrightarrow{(i^*, j^*)} \frac{H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}{F^p H^p(\widetilde{\Delta}, \mathbb{Z}_\ell)}. \tag{2.2.21}$$

By (2.2.9), the map induced by (2.2.21) on associated graded objects is an isomorphism, so (2.2.20) is an isomorphism. On the other hand, (2.2.18) gives a direct-sum decomposition

$$H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \underset{f_* \oplus (1 - f^* f_*)}{\overset{f^* + \text{inclusion}}{\rightleftarrows}} H^\bullet(X, \mathbb{Z}_\ell) \oplus \operatorname{Ker} f_* \ , \tag{2.2.22}$$

and (1.2.2) gives the direct-sum decomposition

$$\frac{H^\bullet(\widetilde{\Delta}, \mathbb{Z}_\ell)}{g^* H^\bullet(\Delta, \mathbb{Z}_\ell)} \xleftarrow{\sim} \bigoplus_{1 \leq j \leq r} H^{\bullet - 2j}(\Delta, \mathbb{Z}_\ell(-j)). \tag{2.2.23}$$

Combining the isomorphism (2.2.20) and decompositions (2.2.22) and (2.2.23) gives (2.2.2), as required.

[56] At this point the print writes $d_s^{p,q}(f) = 0$, although the inductive step and diagram (2.2.14) concern $d_{s+1}^{p,q}(f)$.

[57] In the right-hand denominator of (2.2.21), the print reverses the notation to $H^p F^p$. The filtration used in the preceding sentence and on the left is $F^p H^p$; that notation is restored here.

Remark 2.2.24.

Let us sketch a second, distinctly simpler proof of (2.2.2), valid when X is proper over k. More generally, without any hypothesis on X, one obtains the corresponding result (2.2.26) in cohomology with proper supports. The exact sequences of cohomology with proper supports, denoted $H_c^\bullet$, for the pairs $(\widetilde{X}, \widetilde{\Delta})$ and (X, Δ) form the commutative diagram

$$\begin{array}{lccccccccc}
(2.2.25) & \longrightarrow & H_c^i(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{X},\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(\widetilde{X}-\widetilde{\Delta},\mathbb{Z}_\ell) \\
 & & \wr\uparrow & & \uparrow f^* & & \uparrow g^* & & \wr\uparrow \\
(2.2.26) & \longrightarrow & H_c^i(X-\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(X,\mathbb{Z}_\ell) & \longrightarrow & H_c^i(\Delta,\mathbb{Z}_\ell) & \longrightarrow & H_c^{i+1}(X-\Delta,\mathbb{Z}_\ell)
\end{array}$$

The quotient of the acyclic complex (2.2.25) by the acyclic subcomplex (2.2.26) gives the acyclic complex

$$\longrightarrow 0 \longrightarrow \frac{H_c^i(\widetilde{X},\mathbb{Z}_\ell)}{f^*H_c^i(X,\mathbb{Z}_\ell)} \longrightarrow \frac{H_c^i(\widetilde{\Delta},\mathbb{Z}_\ell)}{g^*H_c^i(\Delta,\mathbb{Z}_\ell)} \longrightarrow 0 \longrightarrow . \tag{2.2.27}$$

Thus, without considering spectral sequences, one finds that (2.2.20) is an isomorphism. The arguments just given then apply and yield (2.2.2).

Corollary 2.3.

Set

$$b_q(X) = \dim_{\mathbb{Q}_\ell} H^q(X,\mathbb{Q}_\ell), \qquad \chi(X) = \sum_q (-1)^q b_q(X).$$

Then[58]

$$b_q(\widetilde{X}) = b_q(X) + \sum_{j=1}^{r} b_{q-2j}(\Delta), \tag{2.3.1}$$

$$\chi(\widetilde{X}) = \chi(X) + r\,\chi(\Delta). \tag{2.3.2}$$

3. The blow-up associated with a pencil

3.1.

Let X be projective, smooth, and irreducible over an algebraically closed field k, of dimension n, and let $D = \{H_\tau\}_{\tau\in\mathbb{P}^1}$ be a pencil of hypersurfaces whose axis meets X transversely, in Δ. We consider the blow-up of X along Δ, which we shall now describe explicitly.[59]

In terms of equations, let (λ,μ) be homogeneous coordinates on $\mathbb{P}^1$, and let F and G be two linear forms such that $F = G = 0$ defines the axis of the pencil. The pencil is written

$$\lambda F - \mu G. \tag{3.1.1}$$

By definition, the subscheme Δ is the intersection of X with the linear subvariety $G = F = 0$. By 2.1.6,

$$\widetilde{X} = \{(x,(\lambda,\mu)) \in X \times \mathbb{P}^1 \mid \lambda F - \mu G = 0\}; \tag{3.1.2}$$

[58]The print has $b_{q-j}(\Delta)$ in (2.3.1). The decomposition (2.2.2) shifts cohomological degree by $2j$, so the correct summand is $b_{q-2j}(\Delta)$.

[59]The print writes u for the dimension in this sentence. Section 3.2 immediately denotes the same dimension by n, which also governs the signs in (3.2.3)–(3.2.4); that notation is used here.

it is the scheme-theoretic closure, in $X \times \mathbb{P}^1$, of the graph of the map

$$\begin{array}{ccc} X - \Delta & \longrightarrow & \mathbb{P}^1 \\ x & \longmapsto & (G(x), F(x)). \end{array} \tag{3.1.3}$$

The great advantage of $\widetilde{X}$ is that it is projective (being embedded in $X \times \mathbb{P}^1$) and smooth (because Δ is smooth) over k, and admits a morphism ρ to $\mathbb{P}^1$, namely

$$\rho : \widetilde{X} \longrightarrow X \times \mathbb{P}^1 \xrightarrow{\mathrm{pr}_2} \mathbb{P}^1, \tag{3.1.4}$$

whose fibers are the hyperplane sections of X by the members of our pencil:

$$\rho^{-1}(t) = X_t \overset{\mathrm{def}}{=} X \cdot H_t.$$

3.2. The degree of the dual variety: topological method

Suppose in addition that the pencil of 3.1 is such that

$$X_t \text{ is smooth for every } t \text{ in a nonempty open subset of } \mathbb{P}^1, \tag{3.2.1}$$

and
(3.2.2)
if X_t is singular, its only singularities are nondegenerate quadratic singularities (XVII 1.1).

These conditions hold in particular for a Lefschetz pencil (XVII 2.21) if $\operatorname{char} k \neq 2$ or if X has even dimension.

Let η be the generic point of $\mathbb{P}^1$, let $\bar{\eta}$ be a geometric point above it, let $n = \dim X$, and let χ be the Euler–Poincaré characteristic with coefficients in $\mathbb{Q}_\ell$. The Euler–Poincaré formula XVI 2.1 applied to ρ gives

$$\chi(\widetilde{X}) = \chi(\mathbb{P}^1)\,\chi(X_{\bar{\eta}}) + (-1)^n \sum_{\substack{\text{singular points} \\ \text{of the fibers}}} 1, \tag{3.2.3}$$

which, in view of (2.2.2), can be rewritten

$$(-1)^n \begin{bmatrix} \text{total number of} \\ \text{singular points} \end{bmatrix} = \chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}}). \tag{3.2.4}$$

Notice that the quantity $\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})$ is independent of the particular choice of pencil. It depends only on the embedding of X in some $\mathbb{P}^r$, and indeed only on the Chern class of $\underline{\mathcal{O}}_X(1)$; see XVII 5.7.5 and 5.7.6.

3.2.5.

Let us make this explicit in a concrete case. Take $X = \mathbb{P}^2$ and let $\{H_\tau\}$ be a pencil of plane curves of degree d satisfying 3.1, (3.2.1), and (3.2.2). Then

$$\begin{array}{ll} \chi(X) = 3 & \text{(by 1.2)}, \\ \chi(X_{\bar{\eta}}) = -d(d-3) & \text{(see SGA 5 VII 7.3)}, \\ \chi(\Delta) = d^2, & \end{array} \tag{3.2.6}$$

whence

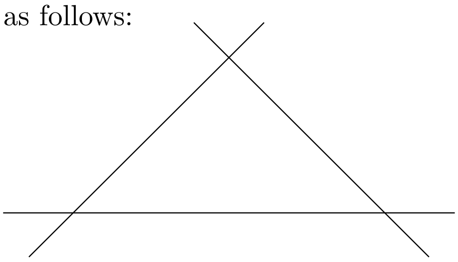
(3.2.7) the total number of singular points is $3(d-1)^2$.

Take the special case $d = 3$ and consider the family of elliptic curves (in characteristic different from 3)

$$X^3 + Y^3 + Z^3 + 3\mu XYZ. \tag{3.2.8}$$

The four singular fibers lie over $\mu^3 = -1$ and $\mu = \infty$, respectively,[60] and each singular fiber is the union of three lines as follows:

(3.2.9)



,

It has exactly three singular points, all nondegenerate quadratic singularities. Altogether this gives twelve singular points and verifies (3.2.7) in this example.

Proposition 3.2.10.

Let $X \hookrightarrow \mathbb{P}^r$ be a Lefschetz embedding (XVII 2.3), and suppose that $n = \dim X$ is even, or that $\operatorname{char} k \neq 2$. Then every Lefschetz pencil, regarded as a line in $\check{\mathbb{P}}^r$, meets $\check{X}$ in

$$(-1)^n \left[\chi(X) + \chi(\Delta) - 2\chi(X_{\bar{\eta}})\right] \tag{3.2.11}$$

points, and meets it transversely (EGA IV 17.13.7).

Proof. By hypothesis, every singularity of a fiber of the Lefschetz pencil is a nondegenerate quadratic singularity. Formula (3.2.4) therefore shows that the number of singular fibers in the pencil D—each containing exactly one singular point—is the number (3.2.11). Now the fiber X_τ is singular if and only if $\tau \in \check{X} \cap D$.

To see that the intersection is transverse, observe that (3.2.11) is independent of the particular Lefschetz pencil. Since the generic pencil is a Lefschetz pencil, (3.2.11) is the degree of $\check{X}$; that is, its degree as a hypersurface in $\check{\mathbb{P}}^r$, taken by convention to be 0 if $\dim \check{X} \leq r-2$. If a line D meets a hypersurface of degree $d > 0$ in d distinct points x_i $(1 \leq i \leq d)$, the intersection must be transverse. Indeed, the local intersection multiplicity

$$\mu_i = \operatorname{length}_{x_i}(D \cap \check{X})$$

is at least 1, while $\sum_{i=1}^{d} \mu_i = d$. Hence $\mu_i = 1$ for every i; equivalently, $D \cap \check{X}$ is smooth, which is precisely transversality.[61]

Remark 3.2.12.

For a Lefschetz pencil D already assumed to meet $\check{X}$ transversely (a condition satisfied in any case by sufficiently general D), the result 3.2.10 may also be regarded as the conjunction of XVII 5.7.2 and the fact that $\deg \varphi = 1$ contained in XVII 3.5. Of course, the fact proved here that every Lefschetz pencil D meets $\check{X}$ transversely—by means of the Euler–Poincaré formula XVI 2.1—may also be verified directly and elementarily, using the fact that the axis of D meets X transversely.

[60] The print says that the finite parameters are the cube roots of unity. With the printed plus sign in (3.2.8), the Hesse factorization gives $\mu^3 = -1$; cube roots of $+1$ correspond to the polynomial with coefficient -3μ.

[61] The print denotes the local length by rg; the standard local-length notation and the stated intersection-multiplicity argument require lg (or, equivalently, length).

4. Cohomology of the blow-up associated with a pencil

4.1.

We return to the situation of 3.1, summarized by the commutative diagram (see 2.1.6.3):
(4.1.1)

$$\begin{tikzcd}
X \times \mathbb{P}^1 & \widetilde{X} \arrow[l, hook', "f\times\rho"'] \arrow[dl, "\rho"'] \arrow[d, "f"] & \widetilde{\Delta} \arrow[l, hook', "i"'] \arrow[r, "(g{,}\rho_i)", "\sim"'] \arrow[d, "g"] & \Delta \times \mathbb{P}^1 \\
\mathbb{P}^1 & X & \Delta \arrow[l, hook', "j"] &
\end{tikzcd}$$

We begin by making (2.2.2) more explicit in this case.

Proposition 4.2.

Under the hypotheses of 3.1:
(4.2.1) The homomorphisms of $\mathbb{Z}_\ell$-modules

$$H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \mathrel{\underset{f^*+\mathrm{incl}}{\overset{f_*\oplus(1-f^*f_*)}{\rightleftarrows}}} H^\bullet(X,\mathbb{Z}_\ell) \oplus \operatorname{Ker}\Big(f_* : H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \to H^\bullet(X,\mathbb{Z}_\ell)\Big)$$

are mutually inverse isomorphisms.
(4.2.2) The homomorphisms of $\mathbb{Z}_\ell$-modules

$$H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \mathrel{\underset{f^*+i_*g^*}{\overset{f_*\oplus(-g_*i^*)}{\rightleftarrows}}} H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$$

are mutually inverse isomorphisms.[62]
(4.2.3) The algebra structure on

$$H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)),$$

transported by (4.2.2) from the cup product on $H^\bullet(\widetilde{X},\mathbb{Z}_\ell)$, is expressed, for

$$a,b \in H^\bullet(X,\mathbb{Z}_\ell), \qquad x,y \in H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)),$$

by the formulas

(4.2.3.1) $$(0\oplus x)\wedge(0\oplus y) = -j_*(xy) \oplus 2xy\, c_1(\underline{\mathcal{O}}_\Delta(1)),$$

(4.2.3.2) $$(a\oplus 0)\wedge(b\oplus 0) = ab \oplus 0,$$

(4.2.3.3) $$(a\oplus 0)\wedge(0\oplus y) = 0 \oplus j^*(a)y,$$

(4.2.3.4) $$(0\oplus x)\wedge(b\oplus 0) = 0 \oplus xj^*(b).$$

[62] The print omits the minus sign before g_*i^* in (4.2.2), (4.2.6), (4.2.9), and (4.2.10). But Lemma 4.3 gives $g_*i^*i_*g^* = -\mathrm{id}$, and the printed product calculation itself uses $-g_*i^*$. The sign is therefore restored consistently.

In particular, if

$$a \oplus x \in H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$$

and

$$b \oplus y \in H^{2n-\bullet}(X,\mathbb{Z}_\ell) \oplus H^{2n-2-\bullet}(\Delta,\mathbb{Z}_\ell(-1)),$$

then

(4.2.3.5) $$(a \oplus x) \wedge (b \oplus y) = ab - j_*(xy).$$

In other words, the total intersection matrix of $H^\bullet(\widetilde{X},\mathbb{Z}_\ell)$ is

$$\left(\begin{array}{c|c} \begin{array}{c}\text{total intersection}\\ \text{matrix on } X\end{array} & 0 \\ \hline 0 & -\begin{array}{c}\text{total intersection}\\ \text{matrix on } \Delta\end{array}\end{array}\right).$$

Proof. We have already proved (4.2.1); see (2.2.22). For (4.2.2), first observe that, since $g : \widetilde{\Delta} \to \Delta$ is a projective-line bundle, the sequence

(4.2.4) $$0 \longrightarrow H^\bullet(\Delta,\mathbb{Z}_\ell) \xrightarrow{g^*} H^\bullet(\widetilde{\Delta},\mathbb{Z}_\ell) \xrightarrow{g_*} H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \longrightarrow 0$$

is exact, by (1.2.2). Together with (2.2.26), this gives the commutative diagram of exact sequences
(4.2.5)

$$\begin{tikzcd} 0 \arrow[r] & H^\bullet(X,\mathbb{Z}_\ell) \arrow[r, "f^*"] \arrow[d, "j^*"] & H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \arrow[r, "g_*i^*"] \arrow[d, "i^*"] & H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & H^\bullet(\Delta,\mathbb{Z}_\ell) \arrow[r, "g^*"] & H^\bullet(\widetilde{\Delta},\mathbb{Z}_\ell) \arrow[r, "g_*"] & H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \arrow[r] & 0. \end{tikzcd}$$

In view of the splitting (4.2.1), this yields the isomorphism

(4.2.6) $$f_* \oplus (-g_*i^*) : H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \xrightarrow{\sim} H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)).$$

The transpose of (4.2.5), after the degree reversal $\bullet \mapsto 2n-\bullet$, gives the commutative diagram of exact sequences
(4.2.7)

$$\begin{tikzcd} 0 & H^\bullet(X,\mathbb{Z}_\ell) \arrow[l] & H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \arrow[l, "f_*"'] & H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \arrow[l, "i_*g^*"'] & 0 \arrow[l] \\ 0 & H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-2)) \arrow[l] \arrow[u, "j_*"] & H^{\bullet-2}(\widetilde{\Delta},\mathbb{Z}_\ell(-1)) \arrow[l, "g_*"'] \arrow[u, "i_*"] & H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \arrow[l, "g_*"'] \arrow[u, equal] & 0 \arrow[l] \end{tikzcd}$$

and hence the isomorphism

(4.2.8) $$H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \xrightarrow[\sim]{f^*+i_*g^*} H^\bullet(\widetilde{X},\mathbb{Z}_\ell).$$

It remains to prove that the composite

(4.2.9) $$\begin{aligned} H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) &\xrightarrow{f^*+i_*g^*} H^\bullet(\widetilde{X},\mathbb{Z}_\ell) \\ &\xrightarrow{f_*\oplus(-g_*i^*)} H^\bullet(X,\mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1)) \end{aligned}$$

is the identity. Now

$$f_*f^* = \mathrm{id} \quad \text{on } H^\bullet(X,\mathbb{Z}_\ell) \quad \text{by (2.2.18)},$$

$$g_*i^*f^* = 0 \quad \text{by (4.2.5)}, \qquad f_*i_*g^* = 0 \quad \text{by (4.2.7)}.$$

It therefore suffices to prove

$$-g_*i^*i_*g^*(x) = x \quad \text{for } x \in H^\bullet(\Delta, \mathbb{Z}_\ell(-1)). \tag{4.2.10}$$

To verify (4.2.10), first observe that $i : \widetilde{\Delta} \hookrightarrow \widetilde{X}$ is the inclusion of a smooth subvariety of codimension one. Thus (SGA 5 VII)

$$i^*i_*(x) = c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\, x, \tag{4.2.11}$$

and

$$i_*(x)i_*(y) = i_*\Big(xy\, c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\,. \tag{4.2.12}$$

Write $\xi_X = \underline{\mathcal{O}}_X(1)$, $\xi_\Delta = \underline{\mathcal{O}}_\Delta(1)$, and $\xi_{\mathbb{P}^1} = \underline{\mathcal{O}}_{\mathbb{P}^1}(1)$. By EGA II 8.1.7, if $\widetilde{\mathfrak{J}}$ is the ideal of $\widetilde{\Delta}$ in $\widetilde{X}$, then

$$\begin{aligned} \check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} &\simeq \widetilde{\mathfrak{J}}/\widetilde{\mathfrak{J}}^{,2} \simeq \widetilde{\mathfrak{J}} \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} \\ &\simeq \underline{\mathcal{O}}_{\widetilde{X}}(1) \otimes_{\underline{\mathcal{O}}_{\widetilde{X}}} \underline{\mathcal{O}}_{\widetilde{\Delta}} = \underline{\mathcal{O}}_{\widetilde{\Delta}}(1). \end{aligned} \tag{4.2.13}$$

Here $\widetilde{\Delta}$ is viewed as $\mathbb{P}(\mathfrak{J}/\mathfrak{J}^2)$, where $\mathfrak{J}$ is the ideal of Δ in X. Since Δ is the intersection of X with two given hyperplanes, there is a canonical isomorphism

$$\mathfrak{J}/\mathfrak{J}^2 \simeq \underline{\mathcal{O}}_\Delta^{,2}(-1),$$

which induces

$$\widetilde{\Delta} = \mathbb{P}(\mathfrak{J}/\mathfrak{J}^2) \simeq \mathbb{P}(\underline{\mathcal{O}}_\Delta^{,2}(-1)) \xrightarrow[\sim]{\alpha} \mathbb{P}(\underline{\mathcal{O}}_\Delta^{,2}) \simeq \Delta \times \mathbb{P}^1.$$

The canonical isomorphism α of EGA II 4.1.4 carries the tautological line bundle to

$$(\rho i)^*\underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^*\underline{\mathcal{O}}_\Delta(-1).$$

Consequently, (4.2.13) becomes

$$\check{\underline{N}}_{\widetilde{\Delta}/\widetilde{X}} \simeq (\rho i)^*\underline{\mathcal{O}}_{\mathbb{P}^1}(1) \otimes g^*\underline{\mathcal{O}}_\Delta(-1). \tag{4.2.14}$$

Lemma 4.3.

<u>One has</u>

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}}) = g^*c_1(\xi_\Delta) - (\rho i)^*c_1(\xi_{\mathbb{P}^1}).$$

Return now to the proof of (4.2.10). By (4.2.11) and Lemma 4.3, it is enough to prove

$$g_*(g^*(x)\,[(\rho i)^*c_1(\xi_{\mathbb{P}^1}) - g^*c_1(\xi_\Delta)]) = x.$$

By (4.2.4), $g_*g^* = 0$, while

$$\widetilde{\Delta} \xrightarrow[\sim]{g\times\rho_i} \Delta \times \mathbb{P}^1,$$

so

$$g_*(g^*(x)(\rho i)^*c_1(\xi_{\mathbb{P}^1})) = x, \tag{4.3.1}$$

and hence

$$g_*\Big(g^*(x)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big) = -x. \tag{4.3.2}$$

We now compute the algebra structure on

$$H^\bullet(X,\mathbb{Z}_\ell)\oplus H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$$

transported from the cup product by (4.2.2). Let a,b belong to $H^\bullet(X,\mathbb{Z}_\ell)$ and x,y to $H^{\bullet-2}(\Delta,\mathbb{Z}_\ell(-1))$. Begin with (4.2.3.1), and write

$$(0\oplus x)\wedge(0\oplus y)=A\oplus B.$$

By (4.2.12) and (4.3.2),

$$\begin{aligned}A&=f_*((i_*g^*x)(i_*g^*y))\\&=f_*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=j_*g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)=-j_*(xy),\end{aligned}$$

and

$$\begin{aligned}B&=-g_*i^*((i_*g^*x)(i_*g^*y))\\&=-g_*i^*i_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=-g_*\Big(g^*(xy)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2\Big).\end{aligned}$$

Lemma 4.3 gives

$$c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})^2=g^*\big(c_1(\xi_\Delta)^2\big)-2g^*c_1(\xi_\Delta)(\rho i)^*c_1(\xi_{\mathbb{P}^1}),$$

because $c_1(\xi_{\mathbb{P}^1})^2=0$. Therefore, by (4.2.4) and (4.3.1),

$$B=0+2xy\,c_1(\xi_\Delta),$$

which proves (4.2.3.1). Formula (4.2.3.2) expresses the fact that f^* is an algebra homomorphism.

To prove (4.2.3.3), the projection formula shows that $\operatorname{Ker} f_*$ is stable under multiplication by $\operatorname{im} f^*$. Hence

$$(a\oplus 0)\wedge(0\oplus y)=0\oplus C,$$

where

(4.3.3) $$C=-g_*i^*(f^*(a)i_*g^*(y))\,.$$

Using (4.3.2),

$$\begin{aligned}C&=-g_*\Big(i^*f^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=-g_*\Big(g^*j^*(a)g^*(y)c_1(\underline{N}_{\widetilde{\Delta}/\widetilde{X}})\Big)\\&=j^*(a)y.\end{aligned}$$

This proves (4.2.3.3). Finally, (4.2.3.4) follows from (4.2.3.3) by graded commutativity of the cup product.

5. Cohomological study of the morphism $\rho:\widetilde{X}\longrightarrow\mathbb{P}^1$ associated with a pencil

5.1. The restriction homomorphism $H^*(\widetilde{X})\longrightarrow H^*(Y)$

Let Y denote one of the smooth fibers of $\rho:\widetilde{X}\to\mathbb{P}^1$, say

$$Y=\rho^{-1}(t_0)=X\cdot H_{t_0}.$$

We have the commutative diagram

$$
\begin{tikzcd}
\mathbb{P}^1 & & \widetilde{X} \arrow[ll, "\rho"'] \arrow[d, "f"] & \widetilde{\Delta} \arrow[l, hook', "i"] \arrow[d, "g"] & \simeq & \Delta \times \mathbb{P}^1 \\
\mathrm{Sp}(k) = \{t_0\} \arrow[u, hook] & & X & \Delta \arrow[l, hook', "j"] \arrow[dll, hook', "h"] \arrow[urr, hook, "\ell=\mathrm{id}\times\{t_0\}"'] & & \\
& Y \arrow[ul] \arrow[uur, "k"] \arrow[ur, "m"] & & & &
\end{tikzcd}
$$

Proposition 5.1.1.

Under the hypotheses of (3.1), with Y denoting a smooth fiber:

(5.1.2) The restriction homomorphism

$$k^* : H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^\bullet(Y, \mathbb{Z}_\ell)$$

is expressed (via (4.2.2)) by

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{m^*+h_*} H^\bullet(Y, \mathbb{Z}_\ell).$$

(5.1.3) The Gysin map

$$k_* : H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \longrightarrow H^\bullet(\widetilde{X}, \mathbb{Z}_\ell)$$

is expressed by

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{m_* \oplus (-h^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)).$$

Proof. The smooth subschemes $\widetilde{\Delta}$ and Y of $\widetilde{X}$ intersect transversely in Δ; in other words, the square of smooth schemes

(5.1.4)
$$
\begin{tikzcd}
\Delta \arrow[r, "\ell"] \arrow[d, hook, "h"] & \widetilde{\Delta} \arrow[d, hook, "i"] \\
Y \arrow[r, "k"'] & \widetilde{X}
\end{tikzcd}
$$

is Cartesian. It therefore implies the formula [SGA 5, IV]

(5.1.5)
$$h_* \ell^* = k^* i_*, \qquad \ell_* h^* = i^* k_*.$$

The composite

$$H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{f^*+i_*g^*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{k^*} H^\bullet(Y, \mathbb{Z}_\ell)$$

is the sum of $k^*f^* = m^*$ and $k^*i_*g^* = h_*\ell^*g^*$ by (5.1.5). Since $g\ell = \mathrm{id}_\Delta$, this proves (5.1.2).

The composite

$$H^{\bullet-2}(Y, \mathbb{Z}_\ell(-1)) \xrightarrow{k_*} H^\bullet(\widetilde{X}, \mathbb{Z}_\ell) \xrightarrow{f_* \oplus (-g_* i^*)} H^\bullet(X, \mathbb{Z}_\ell) \oplus H^{\bullet-2}(\Delta, \mathbb{Z}_\ell(-1))$$

is the sum of

$$f_* k_* = m_* \qquad \text{and} \qquad -g_* i^* k_* = -g_* \ell_* h^* = -h^*,$$

which proves (5.1.3).

Corollary 5.1.6.

Under the hypotheses of 5.1.1, for $i \leq n-1$, the images of

$$m^* : H^i(X, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

and of

$$k^* : H^i(\widetilde{X}, \mathbb{Z}_\ell) \longrightarrow H^i(Y, \mathbb{Z}_\ell)$$

are equal.

Proof. By the weak Lefschetz theorem for hyperplane sections (SGA 5, VII 7.1), for $i \leq n-1$ the map

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \tag{5.1.7}$$

is an isomorphism. Thus, by (5.1.2), the image of k^* is the sum of the image of m^* and the image of the composite

$$H^{i-2}(X, \mathbb{Z}_\ell(-1)) \xrightarrow{j^*} H^{i-2}(\Delta, \mathbb{Z}_\ell(-1)) \xrightarrow{h_*} H^i(Y, \mathbb{Z}_\ell).$$

But the projection formula gives

$$\begin{aligned} h_* j^*(x) &= h_*\big(h^* m^*(x)\big) = m^*(x) h_*(1) = \xi_Y \, m^*(x) \\ &= m^*(x \xi_X), \qquad \text{since } m^*(\xi_X) = \xi_Y. \end{aligned}$$

Consequently $\operatorname{Im} k^* \subset \operatorname{Im} m^*$, and the reverse inclusion follows directly from (5.1.2). This proves the corollary.

5.2. The hard Lefschetz theorem (LV)

5.2.1.

Let S be projective of dimension n over the algebraically closed field k, and let $S \hookrightarrow \mathbb{P}$ be a projective embedding. Denote by

$$u \in H^2(S, \mathbb{Z}_\ell(1))$$

the cohomology class of a hyperplane section. Thus u is the pullback along $S \hookrightarrow \mathbb{P}$ of the first Chern class of $\underline{\mathcal{O}}_{\mathbb{P}}(1)$ in $H^2(\mathbb{P}, \mathbb{Z}_\ell(1))$. Write

$$L_S : H^\bullet(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{\bullet+2}(S, \mathbb{Q}_\ell(j+1))$$

for multiplication by u.

5.2.2.

We say that S (equipped with u) satisfies (LV), the hard Lefschetz theorem, if, for every (or for one) integer j and every integer $i > 0$, the maps

$$L_S^i : H^{n-i}(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+i}(S, \mathbb{Q}_\ell(i+j))$$

are isomorphisms.

5.2.2.1.

It is conjectured that this condition holds when S is smooth. It is true when k has characteristic zero [6]. The specialization theorem [SGA 4, XVI 2.2] therefore gives the same conclusion when S, together with its projective embedding, lifts to characteristic zero. In particular, it holds for a smooth complete intersection. It is also known for a smooth S of dimension at most two [2], and for an abelian variety, either directly (Lieberman [4] and [2]) or by Mumford's theorem [5] that a polarized abelian variety lifts, together with its polarization, to characteristic zero.

Recall that, for a projective variety S and $i \leq n$, the primitive part of cohomology is defined by

$$(5.2.2.2) \qquad \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) = \mathrm{Ker}\big(L^{n-i+1} : H^i(S, \mathbb{Q}_\ell(j)) \longrightarrow H^{2n-i+2}(S, \mathbb{Q}_\ell(j+n-i+1))\big)\,.$$

If S satisfies (LV), then for $i \leq n$ one has the decomposition

$$(5.2.2.3) \qquad H^i(S, \mathbb{Q}_\ell(j)) \xleftarrow[\sim]{1+L} \mathrm{Prim}^i(S, \mathbb{Q}_\ell(j)) \oplus H^{i-2}(S, \mathbb{Q}_\ell(j-1)).$$

Proposition 5.2.3.

Let X be smooth and projective and suppose that it satisfies (LV). Let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then

$$(5.2.3.1) \qquad \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) = \mathrm{Ker}(\tau^* : H^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^n(Y, \mathbb{Q}_\ell(j)))\,.$$

Proof. The projection formula for $\tau : Y \to X$ gives the commutative diagram

$$(5.2.3.2) \qquad \begin{tikzcd} H^n(X, \mathbb{Q}_\ell(j)) \arrow[rr, "L"] \arrow[dr, "\tau^*"'] & & H^{n+2}(X, \mathbb{Q}_\ell(j+1)) \arrow[dl, "\tau_*"] \\ & H^n(Y, \mathbb{Q}_\ell(j)) & \end{tikzcd}$$

By duality and weak Lefschetz,

$$\tau_* : H^n(Y, \mathbb{Q}_\ell(j)) \longrightarrow H^{n+2}(X, \mathbb{Q}_\ell(j+1))$$

is the transpose of an isomorphism and hence is itself an isomorphism. The assertion follows.

Lemma 5.2.4.

Let X be smooth and projective, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Then X satisfies (LV) if and only if the following two conditions hold:

5.2.4.1. Y satisfies (LV).

5.2.4.2. The restriction of the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell)$ to the subspace $\mathrm{Im}\, H^{n-1}(X, \mathbb{Q}_\ell)$ is nondegenerate.

Proof. We have

$$(5.2.4.3) \qquad \begin{cases} L_X = \tau_* \tau^*, \\ L_Y = \tau^* \tau_* \qquad \text{(by 4.2.11)}, \end{cases}$$

and consequently

$$(5.2.4.4) \qquad L_X^{i+1} = \tau_* L_Y^i \tau^*.$$

In the commutative diagram

$$\begin{tikzcd}
H^{n-i-1}(X,\mathbb{Q}_\ell(j)) \arrow[r, "L_X^{i+1}"] \arrow[d, "\tau^*"] & H^{n+i+1}(X,\mathbb{Q}_\ell(j+i+1)) \\
H^{n-1-i}(Y,\mathbb{Q}_\ell(j)) \arrow[r, "L_Y^i"] & H^{n-1+i}(Y,\mathbb{Q}_\ell(j+i)) \arrow[u, "\tau_*"]
\end{tikzcd} \tag{5.2.4.5}$$

τ^* is an isomorphism for $i \geq 1$ by weak Lefschetz, and τ_* is an isomorphism by duality. Hence X satisfies (LV) if and only if Y satisfies (LV) and

$$L_X : H^{n-1}(X,\mathbb{Q}_\ell(j)) \longrightarrow H^{n+1}(X,\mathbb{Q}_\ell(j+1)) \tag{5.2.4.6}$$

is an isomorphism.

It remains to identify (5.2.4.6) with 5.2.4.2. Let $a, b \in H^{n-1}(X,\mathbb{Q}_\ell(j))$. By Poincaré duality on X, (5.2.4.6) is equivalent to nondegeneracy of the bilinear form

$$\begin{aligned} H^{n-1}(X,\mathbb{Q}_\ell(j)) \times H^{n-1}(X,\mathbb{Q}_\ell(j)) &\longrightarrow H^{2n}(X,\mathbb{Q}_\ell(2j+1)), \\ (a,b) &\longmapsto a\,L_X(b). \end{aligned} \tag{5.2.4.7}$$

But by (5.2.4.3) and the projection formula,

$$aL_X(b) = a\tau_*\tau^*(b) = \tau_*\big(\tau^*(a)\tau^*(b)\big).$$

This proves the equivalence of (5.2.4.7) and 5.2.4.2.

Corollary 5.2.5.

Let X be smooth and projective of dimension n, and let $\tau : Y \hookrightarrow X$ be the inclusion of a smooth hyperplane section. Suppose that X satisfies (LV). Then, for $q \geq n$, the map

$$\tau^* : H^q(X,\mathbb{Q}_\ell(j)) \longrightarrow H^q(Y,\mathbb{Q}_\ell(j))$$

is surjective.

Proof. By (5.2.4.3), for $i \geq 0$ one has

$$L_Y^{i+1} = \tau^* L_X^i \tau_*,$$

and hence the commutative diagram

$$\begin{tikzcd}
H^{n-i}(X,\mathbb{Q}_\ell(j+1)) \arrow[r, "L_X^i"] & H^{n+i}(X,\mathbb{Q}_\ell(j+i+1)) \arrow[d, "\tau^*"] \\
H^{n-i-2}(Y,\mathbb{Q}_\ell(j)) \arrow[r, "L_Y^{i+1}"] \arrow[u, "\tau_*"] & H^{n+i}(Y,\mathbb{Q}_\ell(j+i+1)).
\end{tikzcd}$$

This gives the required result.

Lemma 5.2.6.

Let $f : F \to B$ be a smooth proper morphism with B connected. Let $\mathcal{O}_F(1)$ be a relatively ample invertible sheaf on F, and equip each geometric fiber $F_{\bar{b}}$ with the ample invertible sheaf induced by $\mathcal{O}_F(1)$. If one geometric fiber satisfies (LV), then so do all the others.

Proof. Let $\xi_F \in H^2(F, \mathbb{Z}_\ell(1))$ be the first Chern class of $\mathcal{O}_F(1)$. This class and its powers $\xi_F^i \in H^{2i}(F, \mathbb{Z}_\ell(i))$ induce, by cup product, the homomorphisms

$$\xi_F^i : R^{n-i} f_* \mathbb{Q}_\ell(j) \longrightarrow R^{n+i} f_* \mathbb{Q}_\ell(j+i), \tag{5.2.6.1}$$

where n is the relative dimension of the fibers. By proper base change [SGA 4, XII 5.2], all geometric fibers satisfy (LV) if and only if all the maps (5.2.6.1) are isomorphisms. By the specialization theorem [SGA 4, XVI 2.2], the sheaves $R^i f_* \mathbb{Q}_\ell(j)$ are twisted constant sheaves. Thus the maps (5.2.6.1) are isomorphisms if and only if they are so over one given geometric point.

Corollary 5.2.7.

Let X be smooth and projective. If one smooth hyperplane section satisfies (LV), then every smooth hyperplane section does.

Proof. The universal family of smooth hyperplane sections of X satisfies the hypotheses of Lemma 5.2.6.

5.3. Condition (A)

5.3.1.

Let

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

be the "fibration" associated with a pencil whose axis meets X transversely and whose generic fiber is assumed to be smooth.

Let U be a nonempty open subset of $\mathbb{P}^1$ over which ρ is smooth, and let

$$\nu : U \hookrightarrow \mathbb{P}^1 \tag{5.3.1.1}$$

be the inclusion. Since the sheaves $R^i \rho_* \mathbb{Q}_\ell(j)$ are twisted constant over U, the sheaves

$$\nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j)$$

are seen to be essentially independent of the choice of U; the canonical "adjunction" homomorphisms

$$R^i \rho_* \mathbb{Q}_\ell(j) \longrightarrow \nu_* \nu^* R^i \rho_* \mathbb{Q}_\ell(j) \tag{5.3.2}$$

are likewise independent of the choice of U.

5.3.3.

Observe that, for an arbitrary pencil of hyperplane sections, the weak Lefschetz theorem implies that, for $i \leq n-2$, the immersion

$$\widetilde{X} \overset{(f \times \rho)}{\hookrightarrow} X \times \mathbb{P}^1$$

gives an isomorphism of sheaves on $\mathbb{P}^1$

$$R^i \rho_* \mathbb{Q}_\ell(j) \xleftarrow{\sim} H^i(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1}, \qquad i \leq n-2.$$

In other words, $R^i \rho_* \mathbb{Q}_\ell(j)$ is constant for $i \leq n-2$ and, a fortiori, the maps (5.3.2) are isomorphisms for $i \leq n-2$.

5.3.4.

The local theory of vanishing cycles, and in particular the Picard–Lefschetz formula [SGA 7, XV 3.4], shows that, for a Lefschetz pencil, the homomorphisms (5.3.2) are isomorphisms for $i \neq n$, and are surjective for $i = n$.

5.3.5.

We say that ρ, or the pencil from which it arises, satisfies condition (A) if all the homomorphisms (5.3.2) are isomorphisms (equivalently, if this holds for every i with $j = 0$).

We shall see below (6.3, 6.4) that condition (A) has a strong tendency to hold.

Lemma 5.4.

Suppose that one smooth hyperplane section of X (and hence every such section, by Corollary 5.2.7) satisfies (LV)—for example, suppose that X satisfies (LV), cf. Lemma 5.2.4—and suppose that ρ satisfies condition (A). Then every fiber (including every singular fiber) satisfies (LV), and the sheaves $R^i\rho_*\mathbb{Q}_\ell(j)$ are constant for $i \neq n-1$.

Proof. Let

$$L : R^i\rho_*\mathbb{Q}_\ell(j) \longrightarrow R^{i+2}\rho_*\mathbb{Q}_\ell(j+1)$$

denote multiplication by $f^*c_1(\mathcal{O}_X(1)) \in H^2(\widetilde{X}, \mathbb{Z}_\ell(1))$; fiber by fiber, this is the operation considered in (5.2.1). By hypothesis, over U we have an isomorphism

$$\nu^*(L^i) : \nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i) \tag{5.4.1}$$

for every $i > 0$ and every j. Applying the functor ν_* gives the isomorphisms

$$(\nu_*\nu^*)L^i : \nu_*\nu^*R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} \nu_*\nu^*R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.2}$$

In view of (A), these identify with the morphisms

$$L^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell(j) \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell(j+i). \tag{5.4.3}$$

The conclusion follows from 5.3.3.

5.4.4.

Let Y be a smooth fiber of $\rho : \widetilde{X} \to \mathbb{P}^1$. By weak Lefschetz, the restriction map

$$H^{n-1}(X, \mathbb{Q}_\ell(j)) \overset{m^*}{\hookrightarrow} H^{n-1}(Y, \mathbb{Q}_\ell(j))$$

is injective. We saw in 5.2.4 that, if X satisfies (LV), then the cup product on $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ remains nondegenerate when restricted to $m^*H^{n-1}(X, \mathbb{Q}_\ell(j))$. The orthogonal complement of $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$ is therefore a complementary subspace to $H^{n-1}(X, \mathbb{Q}_\ell(j))$ in $H^{n-1}(Y, \mathbb{Q}_\ell(j))$; following the classical terminology, it is called the vanishing-cohomology space of Y.

Let $\nu : U \hookrightarrow \mathbb{P}^1$ be any nonempty open subset over which ρ is smooth. The corresponding subsheaf of $\nu_*\nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)$, the "vanishing-cohomology sheaf" $E^{n-1}\rho_*\mathbb{Q}_\ell(j)$ on $\mathbb{P}^1$, is defined by

$$E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \nu_* \left\{ \begin{array}{l} \text{the orthogonal complement in } \nu^*R^{n-1}\rho_*\mathbb{Q}_\ell(j) \text{ of the} \\ \text{constant subsheaf } H^{n-1}(X, \mathbb{Q}_\ell(j))_U \end{array} \right\}. \tag{5.4.4.1}$$

It is clear that this sheaf is independent of the choice of U.

Lemma 5.5.

Suppose that X satisfies (LV) and that the maps (5.3.2) are isomorphisms for $i = n-1$—for example, suppose that ρ arises from a Lefschetz pencil, cf. 5.3.4. Then there is a direct-sum decomposition

$$R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_*\mathbb{Q}_\ell(j). \tag{5.5.1}$$

Proof. By (LV), over U there is a decomposition

$$\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j), \tag{5.5.2}$$

to which we apply ν_*. This proves the assertion.

Theorem 5.6.

Let X be a smooth, irreducible, projective variety satisfying (LV) (5.2.2). Let $\{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely, in X_η and Δ respectively. Suppose that the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ satisfies condition (A). Then: [63]

(5.6.1) the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \Longrightarrow H^{p+q}(\widetilde{X}, \mathbb{Q}_\ell)$$

[64] degenerates.

(Note.) This remains valid if, instead of assuming that X satisfies (LV), one assumes only that a smooth hyperplane section Y does.

5.6.2.

Let η denote the generic point of $\mathbb{P}^1$, $\bar\eta$ the geometric generic point, $X_{\bar\eta}$ the geometric generic fiber of ρ, and π the Galois group of $k(\bar\eta)/k(\eta)$.

For $q \neq n-1$ there are canonical isomorphisms

$$E_2^{0,q} = H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar\eta}, \mathbb{Q}_\ell), \tag{5.6.2.1}$$
$$E_2^{2,q} = H^2(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \simeq H^q(X_{\bar\eta}, \mathbb{Q}_\ell(-1)), \tag{5.6.2.2}$$
$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) = 0, \qquad p \neq 0, 2. \tag{5.6.2.3}$$

For $q = n-1$ one has

$$E_2^{0,n-1} = H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell), \tag{5.6.2.4}$$
$$E_2^{2,n-1} = H^2(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(-1))^\pi \simeq H^{n-1}(X, \mathbb{Q}_\ell(-1)). \tag{5.6.2.5}$$

(The missing value $E_2^{1,n-1}$ will be found in 5.7 below.)

Proof. By (5.4.3), Theorem 1.1 applies and gives (5.6.1). The first three formulas (5.6.2.1)–(5.6.2.3) follow from the constancy of $R^q\rho_*\mathbb{Q}_\ell$ for $q \neq n-1$ (Lemma 5.4); for (5.6.2.3) one also uses, of course, the simple connectedness of $\mathbb{P}^1$.

The first isomorphism in (5.6.2.4) follows from

$$R^{n-1}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell.$$

Thus

$$H^0(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \simeq H^0(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell),$$

[63] The print says "condition (5.3.2)", but (5.3.2) is the displayed family of adjunction maps; condition (A) is defined from those maps in 5.3.5.

[64] The printed abutment has H^{m+q}; the bidegree and the parallel statement in Theorem 5.6.8 require H^{p+q}.

and $\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell$ is a "twisted constant" sheaf on U, hence corresponds to the π-module $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$. The second isomorphism in (5.6.2.4) follows from degeneration together with Corollary 5.1.6.

We turn to (5.6.2.5). Let $F^\bullet$ denote the filtration on $H^\bullet(\widetilde{X}, \mathbb{Q}_\ell)$ defined by the spectral sequence. By degeneration, $F^2 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) = E_2^{2,n-1}$ and $\operatorname{gr}^1 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) = E_2^{1,n} = 0$ by (5.6.2.3). Consequently (cf. 5.1.0),

$$\begin{aligned} E_2^{2,n-1} &= F^1 H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \\ &= \operatorname{Ker}\Big(H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^{n+1}\rho_*\mathbb{Q}_\ell)\Big) \\ &\overset{(A)}{=} \operatorname{Ker}\Big(k^* : H^{n+1}(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)\Big) . \end{aligned} \tag{5.6.3}$$

By (5.1.2), k^* may be written as

$$H^{n+1}(X, \mathbb{Q}_\ell) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \xrightarrow{m^* + h_*} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell). \tag{5.6.4}$$

Using (5.2.2.3), this becomes

$$\begin{aligned} &\operatorname{Prim}^{n-1}(X, \mathbb{Q}_\ell(-1)) \oplus H^{n-3}(X, \mathbb{Q}_\ell(-2)) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \\ &\qquad\qquad \xrightarrow{m^* L + m^* L^2 + h_*} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell). \end{aligned} \tag{5.6.5}$$

[65] Now the map

$$H^{n-3}(X, \mathbb{Q}_\ell(-2)) \xrightarrow{m^* L^2} H^{n+1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$

is an isomorphism by (LV), since $m^* L_X^2 = L_{X_{\bar{\eta}}}^2 m^*$ (cf. 5.2.4 and 5.2.5). Hence the kernel of (5.6.5) identifies, by $\mathrm{pr}_1 \oplus \mathrm{pr}_3$, with

$$\operatorname{Prim}^{n-1}(X, \mathbb{Q}_\ell(-1)) \oplus H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)).$$

By (LV) and weak Lefschetz,

$$H^{n-1}(\Delta, \mathbb{Q}_\ell(-1)) \underset{\sim}{\xleftarrow{L}} H^{n-3}(\Delta, \mathbb{Q}_\ell(-2)) \underset{\sim}{\longleftarrow} H^{n-3}(X, \mathbb{Q}_\ell(-2)).$$

Together with (5.2.2.3), this gives

$$E_2^{2,n-1} \simeq H^{n-1}(X, \mathbb{Q}_\ell(-1)). \tag{5.6.6}$$

The other isomorphism in (5.6.2.5) follows by twisting (5.6.2.4).

Remark 5.6.7.

The assertion (5.6.1) can be proved under substantially weaker hypotheses, as Deligne explained to me:

Theorem 5.6.8.

Let X be smooth of dimension n over an algebraically closed field k, and let

$$\rho : X \longrightarrow \mathbb{P}_k^1$$

be a projective morphism whose generic fiber is smooth. Suppose that:

[65] The printed (5.6.5) omits several closing parentheses and the plus sign before the third-summand map h_*. The three displayed source summands and (5.6.4) force the corrected three-term map shown here.

(5.6.8.1) the geometric generic fiber, equipped with the restriction of the relatively ample invertible sheaf $\underline{\mathcal{O}}_X(1)$, satisfies (LV);

(5.6.8.2) the maps (5.3.2) are isomorphisms for $i \neq n$ and are surjective for $i = n$ (a condition satisfied in particular when ρ arises from a Lefschetz pencil, cf. 5.3.4). Then the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell)$$

degenerates.

Remark. The same theorem, with the same proof, holds with $\mathbb{P}^1_k$ replaced by any smooth, irreducible, projective curve.

Proof. For the proof we identify the sheaf $\mathbb{Q}_\ell(1)$ with $\mathbb{Q}_\ell$; this is possible because k is algebraically closed. We may plainly suppose that X is connected and that ρ is surjective, hence flat with fibers of dimension $n-1$. Let

$$\xi_X \in H^2(X, \mathbb{Q}_\ell)$$

be the first Chern class of $\underline{\mathcal{O}}_X(1)$, and let

$$\nu : U \hookrightarrow \mathbb{P}^1$$

be the inclusion of an open subset over which ρ is smooth. By (5.6.8.1) and Lemma 5.2.6, the cup-product maps

$$\xi_X^i : \nu^* R^{n-1-i}\rho_*\mathbb{Q}_\ell \longrightarrow \nu^* R^{n-1+i}\rho_*\mathbb{Q}_\ell$$

are isomorphisms for $i \geq 1$. Applying ν_* and using (5.6.8.2) gives, for $i \geq 2$, the isomorphisms

$$\xi_X^i : R^{n-1-i}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} R^{n-1+i}\rho_*\mathbb{Q}_\ell, \tag{5.6.8.3}$$

and, for $i = 1$, the isomorphism

$$\xi_X : R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\sim} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.4}$$

The latter factors as

$$R^{n-2}\rho_*\mathbb{Q}_\ell \xrightarrow{\xi_X} R^n\rho_*\mathbb{Q}_\ell \xrightarrow{\text{adjunction}} \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell. \tag{5.6.8.5}$$

Consequently there is a canonical decomposition

$$R^n\rho_*\mathbb{Q}_\ell \xleftarrow[\sim]{\xi_X + \text{incl}} R^{n-2}\rho_*\mathbb{Q}_\ell \oplus \operatorname{Local}^n \mathbb{Q}_\ell, \tag{5.6.8.6}$$

where

$$\operatorname{Local}^n \mathbb{Q}_\ell = \operatorname{Ker}(R^n\rho_*\mathbb{Q}_\ell \longrightarrow \nu_*\nu^* R^n\rho_*\mathbb{Q}_\ell)\,. \tag{5.6.8.7}$$

The sheaf $\operatorname{Local}^n \mathbb{Q}_\ell$ is concentrated at the points of $\mathbb{P}^1$ where the fiber of ρ is not smooth. Combining this fact with (5.6.8.3), we obtain

$$\xi_X^{i+1} : H^j(\mathbb{P}^1, R^{n-2-i}\rho_*\mathbb{Q}_\ell) \xrightarrow{\sim} H^j(\mathbb{P}^1, R^{n+i}\rho_*\mathbb{Q}_\ell), \qquad i \geq 0, \quad j \geq 1. \tag{5.6.8.8}$$

[66]

We now prove degeneration. Since $E_2^{p,q} = 0$ unless $p = 0, 1, 2$, it is enough to prove that all the differentials

$$d_2 : H^0(\mathbb{P}^1, R^i\rho_*\mathbb{Q}_\ell) \longrightarrow H^2(\mathbb{P}^1, R^{i-1}\rho_*\mathbb{Q}_\ell)$$

[66] The printed (5.6.8.8) has an odd degree shift, labels the map only ξ_X, and starts at $i = 1$. The corrected form is forced by (5.6.8.3), (5.6.8.6), the lower arrow of the next commutative square, and the later specialization $E_2^{1,n-2} \simeq E_2^{1,n}$.

vanish. For $q \leq n-1$, set

$$\operatorname{Prim}(E_2^{p,q}) = \operatorname{Ker}\left(\xi_X^{n-q} : E_2^{p,q} \longrightarrow E_2^{p,2n-q}\right). \tag{5.6.8.9}$$

Equations (5.6.8.3) and (5.6.8.6), in particular the injectivity of

$$\xi_X : H^p(\mathbb{P}^1, R^{n-2}\rho_*\mathbb{Q}_\ell) \longrightarrow H^p(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell),$$

give, for $q \leq n-1$, the decomposition

$$E_2^{p,q} \xleftarrow[\sim]{1+\xi_X+\xi_X^2+\cdots} \operatorname{Prim}(E_2^{p,q}) \oplus \operatorname{Prim}(E_2^{p,q-2}) \oplus \operatorname{Prim}(E_2^{p,q-4}) \oplus \cdots . \tag{5.6.8.10}$$

[67] To prove that

$$d_2^{p,q} = 0 \qquad (q \leq n-1),$$

consider the commutative diagram

$$\begin{tikzcd}
\operatorname{Prim}(E_2^{p,q}) \arrow[r, "\xi_X^{n-q}=0"] \arrow[d, "d_2^{p,q}"] & E_2^{p,2n-q} \arrow[d, "d_2^{p,2n-q}"] \\
E_2^{p+2,q-1} \arrow[r, "\xi_X^{n-1-(q-1)}", "\sim"'] & E_2^{p+2,2n-q-1}.
\end{tikzcd}$$

The lower arrow is an isomorphism by (5.6.8.8). The diagram therefore shows that $d_2^{p,q}$ kills $\operatorname{Prim}(E_2^{p,q})$ for $q \leq n-1$, and (5.6.8.10) then shows that d_2 kills $E_2^{p,q}$ for $q \leq n-1$. By (5.6.8.3), it also kills $E_2^{p,q}$ for $q \geq n+1$. It remains only to prove that

$$d_2^{0,n} : E_2^{0,n} \longrightarrow E_2^{2,n-1}$$

is zero, or equivalently that

$$\dim E_3^{2,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.12}$$

We have already proved

$$\dim E_2^{p,q} = \dim E_3^{p,q} \qquad \text{for } (p,q) \neq (0,n), (2,n-1), \tag{5.6.8.13}$$

and for degree reasons

$$E_3^{p,q} = E_\infty^{p,q} \qquad \text{for every pair } (p,q). \tag{5.6.8.14}$$

It follows that

$$b_i(X) = \dim H^i(X, \mathbb{Q}_\ell) = \sum_{p+q=i} \dim E_3^{p,q}.$$

In view of (5.6.8.13), we therefore have

$$b_{n-1}(X) = \dim E_2^{0,n-1} + \dim E_2^{1,n-2} + \dim E_2^{2,n-3} \tag{5.6.8.15}$$

and

$$b_{n+1}(X) = \dim E_2^{0,n+1} + \dim E_2^{1,n} + \dim E_3^{2,n-1}. \tag{5.6.8.16}$$

[67] The print displays the first three primitive summands while its arrow label already contains an ellipsis; the final $\oplus\cdots$ makes the same finite decomposition explicit.

Poincaré duality gives

$$b_{n-1}(X) = b_{n+1}(X). \tag{5.6.8.17}$$

First, by (5.6.8.8),

$$\dim E_2^{1,n} = \dim E_2^{1,n-2}.$$

Thus (5.6.8.17) shows that the proof of (5.6.8.12) is complete once we know

$$\dim E_2^{2,n-3} = \dim E_2^{0,n+1}, \tag{5.6.8.18}$$

$$\dim E_2^{0,n-1} = \dim E_2^{2,n-1}. \tag{5.6.8.19}$$

These two formulas follow from (5.6.8.2), Poincaré duality on the smooth fibers of ρ, and the following lemma applied to $\mathbb{P}^1$ and the sheaves $R^{n-3}\rho_*\mathbb{Q}_\ell$ and $R^{n-1}\rho_*\mathbb{Q}_\ell$.

Lemma 5.6.9.

Let D be a smooth, connected, proper curve over an algebraically closed field k. Let $i : \eta \hookrightarrow D$ be the inclusion of the generic point, and let F be a constructible $\mathbb{Q}_\ell$-sheaf on D such that $F \xrightarrow{\sim} i_*i^*F$. Then

$$H^2(D, F) \simeq H^0(D, \check{F}(-1))^\vee,$$

where $\check{F} \simeq i_*((i^*F)^\vee)$.

Corollary 5.6.10.

(Uses Picard–Lefschetz.) Let $\rho : \widetilde{X} \to \mathbb{P}^1$ be the "fibration" arising from a Lefschetz pencil, and suppose that the smooth fibers satisfy (LV). Then

$$H^i(X_{\bar\eta}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\big(H^i(X, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar\eta}, \mathbb{Q}_\ell)\big)\,. \tag{5.6.10.1}$$

Proof. By 5.3.4, which uses the Picard–Lefschetz formula, Theorem 5.6.8 applies and gives

$$H^i(X_{\bar\eta}, \mathbb{Q}_\ell)^\pi = \mathrm{Im}\Big(H^i(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^i(X_{\bar\eta}, \mathbb{Q}_\ell)\Big)\,.$$

It remains to check that $H^i(X, \mathbb{Q}_\ell)$ and $H^i(\widetilde{X}, \mathbb{Q}_\ell)$ have the same image in $H^i(X_{\bar\eta}, \mathbb{Q}_\ell)$. This holds for $i \leq n-1$ by Corollary 5.1.6, and for $i \geq n$ by Corollary 5.2.5.

We now turn to the most important computation:

Theorem 5.7.

Under the hypotheses of 5.6, set

$$\begin{aligned} E^{n-2}(\Delta, \mathbb{Q}_\ell(j)) = {} & \text{the orthogonal complement in } H^{n-2}(\Delta, \mathbb{Q}_\ell(j)) \\ & \text{of the image of } H^{n-2}(X, \mathbb{Q}_\ell(j)). \end{aligned}$$

(This is the "vanishing" part of the cohomology of Δ; cf. 5.4.4.) Then there is a direct-sum decomposition (cf. 4.2)

$$E_2^{1,n-1} = H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell) \xleftarrow[f^*+i_*g^*]{\sim} \mathrm{Prim}^n(X, \mathbb{Q}_\ell) \oplus E^{n-2}(\Delta, \mathbb{Q}_\ell(-1)). \tag{5.7.1}$$

Proof. By degeneration,

$$E_2^{1,n-1} \simeq F^1H^n(\widetilde{X}, \mathbb{Q}_\ell)/F^2H^n(\widetilde{X}, \mathbb{Q}_\ell).$$

By (A),

$$F^1H^n(\widetilde{X},\mathbb{Q}_\ell)=\ker\Big(k^*:H^n(\widetilde{X},\mathbb{Q}_\ell)\longrightarrow H^n(X_{\bar\eta},\mathbb{Q}_\ell)\Big)\,, \tag{5.7.2}$$

which, by (5.1.2), identifies with

$$\ker\Big(H^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1))\xrightarrow{m^*+h_*}H^n(X_{\bar\eta},\mathbb{Q}_\ell)\Big)\,.$$

Under (LV), k^* can also be written as

$$\mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(X,\mathbb{Q}_\ell(-1))\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1))\xrightarrow{m^*+m^*L+h_*}H^n(X_{\bar\eta},\mathbb{Q}_\ell), \tag{5.7.3}$$

and the middle arrow

$$H^{n-2}(X,\mathbb{Q}_\ell(-1))\xrightarrow{m^*L}H^n(X_{\bar\eta},\mathbb{Q}_\ell) \tag{5.7.4}$$

is an isomorphism (cf. 5.2.3 and (5.2.2.3)). Thus, denoting by

$$\wp:H^n(X,\mathbb{Q}_\ell)\longrightarrow\mathrm{Prim}^n(X,\mathbb{Q}_\ell) \tag{5.7.5}$$

the projection onto the primitive part, we obtain [68]

$$F^1H^n(\widetilde{X},\mathbb{Q}_\ell)\xrightarrow[\wp f_*\oplus(-g_*i^*)]{\sim}\mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \tag{5.7.6}$$

By (5.6.2.2) and degeneration,

$$F^2H^n(\widetilde{X},\mathbb{Q}_\ell)\simeq E_2^{2,n-2}\simeq H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1)). \tag{5.7.7}$$

On the other hand, the inclusion of $F^2H^n(\widetilde{X},\mathbb{Q}_\ell)=H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))$ in $H^n(\widetilde{X},\mathbb{Q}_\ell)$ is the Gysin map k_*. [69] Strictly speaking, the Gysin map is defined only as a map

$$H^{i-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))\longrightarrow H^i(X\times_k k(\bar\eta),\mathbb{Q}_\ell),$$

but, in view of the canonical isomorphism (SGA 4 XII 5.4)

$$H^i(X,\mathbb{Q}_\ell)\xrightarrow{\sim}H^i(X\otimes_k K,\mathbb{Q}_\ell)$$

for every algebraically closed field K/k, we hope the reader will allow us such abuses. By (5.1.3), the Gysin map is expressed as

$$H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))\xrightarrow{m_*\oplus(-h^*)}H^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \tag{5.7.8}$$

By weak Lefschetz,

$$m_*H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1))=m_*m^*H^{n-2}(X,\mathbb{Q}_\ell(-1))=L_XH^{n-2}(X,\mathbb{Q}_\ell(-1))$$

is the orthogonal complement of the primitive part. Hence we have the commutative diagram

$$\begin{tikzcd} F^2H^n(\widetilde{X},\mathbb{Q}_\ell) \arrow[r, hook] & F^1H^n(\widetilde{X},\mathbb{Q}_\ell) \arrow[d, "\wr", "\wp f_*\oplus(-g_*i^*)"'] \\ H^{n-2}(X_{\bar\eta},\mathbb{Q}_\ell(-1)) \arrow[u, "\wr"', "k_*"] \arrow[r, "0\oplus h^*"] & \mathrm{Prim}^n(X,\mathbb{Q}_\ell)\oplus H^{n-2}(\Delta,\mathbb{Q}_\ell(-1)). \end{tikzcd} \tag{5.7.9}$$

This proves (5.7.1), as required.

[68] The printed label in (5.7.6) has 2^* after g_*. The inverse blow-up decomposition (4.2.2), used here verbatim, requires i^*.

[69] The print has F^2H^N in the first sentence and omits the Tate twist (-1) from the displayed source of the Gysin map. The surrounding identifications and the degree-two Gysin shift force the readings shown here.

Remark 5.7.10.

In 5.7, hypothesis (A) may be replaced by the weaker assumption that, for $q = n$, the maps

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^0(\mathbb{P}^1, R^q\rho_*\mathbb{Q}_\ell) \tag{5.7.11}$$

and

$$H^q(\widetilde{X}, \mathbb{Q}_\ell) \longrightarrow H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) \tag{5.7.12}$$

have the same kernel, as our proof shows. Without this condition, the conclusion can fail; cf. 5.9.

5.8. The Griffiths homomorphism

5.8.1.

Let $f : T \to S$ be a morphism of schemes. The Leray spectral sequence defines a filtration (F^i) on the groups $H^\bullet(T, \mathbb{Q}_\ell(j))$. Define

$$\operatorname{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = F^1 H^\bullet(T, \mathbb{Q}_\ell(j)). \tag{5.8.1.1}$$

Because the Leray spectral sequence lies in the first quadrant, we have

$$E_\infty^{0,\bullet} \hookrightarrow E_2^{0,\bullet}, \tag{5.8.1.2}$$

and hence

$$\operatorname{Prim}^\bullet(T/S, \mathbb{Q}_\ell(j)) = \ker\big(H^\bullet(T, \mathbb{Q}_\ell(j)) \longrightarrow H^0(S, R^\bullet f_*\mathbb{Q}_\ell(j))\big)\,. \tag{5.8.1.3}$$

For the same reason,

$$E_\infty^{1,\bullet} \hookrightarrow E_2^{1,\bullet}. \tag{5.8.1.4}$$

The composite

$$\begin{array}{ccc} \operatorname{Prim}^i(T/S, \mathbb{Q}_\ell(j)) = F^1 H^i(T, \mathbb{Q}_\ell(j)) & \longrightarrow & \operatorname{gr}^1 H^i(T, \mathbb{Q}_\ell(j)) = E_\infty^{1,i-1} \\ & & \big\downarrow \\ & & H^1(S, R^{i-1} f_*\mathbb{Q}_\ell(j)) = E_2^{1,i-1} \end{array} \tag{5.8.1.5}$$

will be denoted [70]

$$U^i : \operatorname{Prim}^i(T/S, \mathbb{Q}_\ell(j)) \longrightarrow H^1(S, R^{i-1} f_*\mathbb{Q}_\ell(j)). \tag{5.8.1.6}$$

5.8.2.

Now let X be projective, smooth, and irreducible, and let $D = \{H_\tau\}$ be a pencil of hyperplanes whose generic member and axis meet X transversely. Let $U \xrightarrow{\nu} \mathbb{P}^1$ be a nonempty open subset over which the projection $\rho : \widetilde{X} \to \mathbb{P}^1$ is smooth, and put $\widetilde{X}|_U = \widetilde{X} \times_{\mathbb{P}^1} U$. We have the commutative diagram

$$\begin{tikzcd} & \widetilde{X} \arrow[dl, "f"'] \arrow[d, "\rho"] & \widetilde{X}|_U \arrow[l, "\alpha"'] \arrow[d, "\rho|_U"] \\ X & \mathbb{P}^1 & U. \arrow[l, "\nu"'] \end{tikzcd} \tag{5.8.2.1}$$

[70] The print has a baseline dot in place of the degree index in (5.8.1.6). The source and target of (5.8.1.5) force $R^{i-1} f_*$.

Since $\rho|_U : \widetilde{X}|_U \to U$ is proper and smooth, the specialization theorem for cohomology (SGA 4 XVI 2.2) gives

$$\operatorname{Prim}^\bullet(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) = \ker\Big(H^\bullet(\widetilde{X}|_U, \mathbb{Q}_\ell(j)) \longrightarrow H^\bullet(X_{\bar{\eta}}, \mathbb{Q}_\ell(j))\Big)\,. \tag{5.8.2.2}$$

Consequently, by (5.2.3.1), we have:

Proposition 5.8.2.3.

Under the hypotheses of 5.8.2, for $x \in H^n(X, \mathbb{Q}_\ell(j))$, the following conditions are equivalent:

$$x \in \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff \alpha^* f^*(x) \in \operatorname{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)). \tag{5.8.2.4}$$

If, in addition, ρ satisfies hypothesis (A) of 5.3, then

$$x \in \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \iff f^*(x) \in \operatorname{Prim}^n(\widetilde{X}/\mathbb{P}^1, \mathbb{Q}_\ell(j)). \tag{5.8.2.5}$$

Proof. Formula (5.8.2.4) is proved. For (5.8.2.5), hypothesis (A) implies that

$$H^0(\mathbb{P}^1, R^n\rho_*\mathbb{Q}_\ell(j)) \xrightarrow[\nu^*]{\sim} H^0(U, \nu^*R^n\rho_*\mathbb{Q}_\ell(j)).$$

Thus $f^*(x)$ is primitive relative to $\mathbb{P}^1$ if and only if $\alpha^*f^*(x)$ is primitive relative to U. [71]

Definition 5.8.3.

In the situation of 5.8.2, the composite homomorphism [72]

$$\begin{tikzcd} \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \arrow[r, "\alpha^* f^*"] \arrow[dr, "\mathrm{Grif}"'] & \operatorname{Prim}^n(\widetilde{X}|_U/U, \mathbb{Q}_\ell(j)) \arrow[d, "U^n"] \\ & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \end{tikzcd} \tag{5.8.3.1}$$

is called the Griffiths homomorphism (for X, relative to the given pencil D).

Adjoin to diagram (5.8.2.1) the constant family over U with fiber X:

$$\begin{tikzcd} & \widetilde{X} \arrow[dl, "f"'] \arrow[dd] & \widetilde{X}|_U \arrow[l, "\alpha"'] \arrow[r, "f\times\rho|_U"] \arrow[dd, "\rho|_U"] & X \times U \arrow[ddl, "\mathrm{pr}_2"] \\ X & & & \\ & \mathbb{P}^1 & U \arrow[l, "\nu"'] & \end{tikzcd} \tag{5.8.3.2}$$

[71] The last printed implication has a stray α after $\mathbb{P}^1$ and substitutes α for the class x in $\alpha^*f^*(x)$. The displayed equivalence and the preceding restriction isomorphism determine the corrected sentence.

[72] The ink beside the vertical arrow is not independently legible in the scan. The map is nevertheless determined uniquely: (5.8.1.6) has just defined U^i with exactly this source and target, so the degree-n label is U^n.

Theorem 5.8.4.

<u>Suppose that X satisfies (LV) (5.2.2), and that ρ satisfies hypothesis (A) (5.3). Then, for every nonempty smoothness open $U \xrightarrow{\nu} \mathbb{P}^1$ of ρ, the Griffiths homomorphism "modulo the fixed part", which is by definition the composite</u>

(5.8.4.1)

$$\begin{tikzcd} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \arrow[r, "\mathrm{Grif}"] & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[d] \\ & H^1(U, \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq \dfrac{H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j))}{(f \times \rho|_U)^* H^1(U, H^{n-1}(X, \mathbb{Q}_\ell(j))_U)} \end{tikzcd}$$

<u>is injective.</u> [73]

<u>Proof.</u> By (5.5.1), over $\mathbb{P}^1$ we have

$$R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_{\mathbb{P}^1} \oplus E^{n-1}\rho_*\mathbb{Q}_\ell(j), \tag{5.8.4.2}$$

and over U we have

$$\nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq H^{n-1}(X, \mathbb{Q}_\ell(j))_U \oplus \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j). \tag{5.8.4.3}$$

This permits us to rewrite (5.8.4.1) as indicated:

$$\begin{tikzcd} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \arrow[r, "\mathrm{Grif}"] & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[d] \\ & H^1(U, \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{tikzcd} \tag{5.8.4.4}$$

By (5.7.1), the map (defined only by virtue of hypothesis (A))

$$u : \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \longrightarrow H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \tag{5.8.4.5}$$

is injective. Since $\mathbb{P}^1$ is simply connected, decomposition (5.8.4.2) gives [74]

$$H^1(\mathbb{P}^1, R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq H^1(\mathbb{P}^1, E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \tag{5.8.4.6}$$

We can therefore complete (5.8.4.4) to a commutative square:

$$\begin{tikzcd} \mathrm{Prim}^n(X, \mathbb{Q}_\ell(j)) \arrow[r, "\mathrm{Grif}"] \arrow[d, hook, "u"] & H^1(U, \nu^* R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[d] \\ H^1(\mathbb{P}^1, E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[r, "\nu^*"] & H^1(U, \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{tikzcd} \tag{5.8.4.7}$$

It remains only to see that the lower arrow is injective. This follows from

$$E^{n-1}\rho_*\mathbb{Q}_\ell(j) \simeq \nu_*\nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j) \tag{5.8.4.8}$$

and the following well-known lemma.

[73] The print omits ν^* from the lower-left sheaf in (5.8.4.1) and leaves its coefficient parenthesis unclosed. Formula (5.8.4.3) and the same lower node in (5.8.4.4) give the corrected reading.

[74] The print omits the twist (j) from the targets in (5.8.4.5) and from the left-hand side of (5.8.4.6), although the domain and the decomposition (5.8.4.2) retain it.

Lemma 5.8.5.

Let $\nu : A \to B$ be a morphism of topoi, and let F be a sheaf on A. Then the canonical map

(5.8.5.1) $$H^1(B, \nu_* F) \longrightarrow H^1(A, F)$$

is injective.

Proof. In a first-quadrant spectral sequence,

(5.8.5.2) $$E_2^{1,0} \simeq E_\infty^{1,0}.$$

Thus, for the Leray spectral sequence

$$E_2^{p,q} = H^p(B, R^q \nu_* F) \Longrightarrow H^{p+q}(A, F),$$

we have

(5.8.5.3) $$E_2^{1,0} = H^1(B, \nu_* F) \simeq E_\infty^{1,0} = F^1 H^1(A, F).$$

Taking $F = \nu^* E$, where E is a sheaf on B, gives

(5.8.5.4) $$H^1(B, \nu_* \nu^* E) \hookrightarrow H^1(A, \nu^* E).$$

5.8.6.

Let $L/k(\eta)$ be a finite extension of the rational function field of $\mathbb{P}^1$, and put

(5.8.6.1) $$S = \text{the normalization of } \mathbb{P}^1 \text{ in } L.$$

Make the base change $S \to \mathbb{P}^1$ in situation (5.8.3.2):

(5.8.6.1 bis) $$\begin{tikzcd} & \widetilde{X} \arrow[dl, "f"'] \arrow[d, "\rho"] & \widetilde{X}_S \arrow[l, "\alpha_S"'] \arrow[d, "\rho_S"] \\ X & \mathbb{P}^1 & S. \arrow[l, "\pi"'] \end{tikzcd}$$

For every open immersion $V \overset{\mu}{\hookrightarrow} S$ over which ρ_S is smooth, base change by $V \xrightarrow{\pi\mu} \mathbb{P}^1$ in (5.8.3.2) gives

(5.8.6.2) $$\begin{tikzcd} & \widetilde{X} \arrow[dl, "f"'] \arrow[d, "\rho"] & \widetilde{X}_V \arrow[l, "\alpha_V"'] \arrow[d, "\rho_V"] \arrow[dr, "f \times \rho_V"] & \\ X & \mathbb{P}^1 & V \arrow[l, "\pi\mu"'] & X \times V \arrow[l, "\mathrm{pr}_2"] \end{tikzcd}.$$

We still have (cf. (5.8.2.3))

(5.8.6.3) $$\alpha_V^* f^* \operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \subset \operatorname{Prim}^n(\widetilde{X}_V/V, \mathbb{Q}_\ell(j)),$$

so that we may again speak of the Griffiths homomorphism

(5.8.6.4) $$\operatorname{Prim}^n(X, \mathbb{Q}_\ell(j)) \xrightarrow{\text{Grif}} H^1(V, R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j)).$$

Corollary 5.8.7.

With the preceding notation and under the hypotheses of 5.8.4, the Griffiths homomorphism "modulo the fixed part," that is, the composite

(5.8.7.1)

$$\begin{tikzcd} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \arrow[r, "\mathrm{Grif}"] & H^1(V,R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j)) \arrow[d] \\ H^1(V,(\pi\mu)^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \simeq & \dfrac{H^1(V,R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j))}{(f\times\rho_V)^*H^1(V,H^{n-1}(X,\mathbb{Q}_\ell(j))_V)} \end{tikzcd}$$

is injective.

Proof. By the specialization theorem for cohomology,

$$R^{n-1}\rho_{V*}\mathbb{Q}_\ell(j) = (\pi\mu)^*R^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Thus decomposition (5.8.4.2) rewrites (5.8.7.1) as indicated:

$$\begin{tikzcd} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \arrow[r, "\mathrm{Grif}"] & H^1(V,(\pi\mu)^*R^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[d] \\ & H^1(V,(\pi\mu)^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{tikzcd}$$

It is therefore enough to prove the injectivity of

$$\mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \tag{5.8.7.3}$$

for arbitrarily small open immersions $V \overset{\mu}{\hookrightarrow} S$. We are reduced to the case

$$\begin{cases} U \overset{\nu}{\hookrightarrow} \mathbb{P}^1 & \text{is an open subset over which } \rho \text{ is smooth,} \\ V \xrightarrow{\pi_U} U & \text{is the normalization of } U \text{ in } L. \end{cases}$$

From the cartesian square

$$\begin{tikzcd} V \arrow[r, hook, "\mu"] \arrow[d, "\pi_U"'] & S \arrow[d, "\pi"] \\ U \arrow[r, hook, "\nu"] & \mathbb{P}^1 \end{tikzcd}$$

we obtain

$$\mu^*\pi^*E^{n-1}\rho_*\mathbb{Q}_\ell(j) = \pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j).$$

Consequently, (5.8.7.3) factors as

$$\begin{tikzcd} \mathrm{Prim}^n(X,\mathbb{Q}_\ell(j)) \arrow[r] & H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \arrow[d, "\pi_U^*"] \\ & H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)). \end{tikzcd}$$

By (5.8.4), it is enough to verify that

$$\pi_U^* : H^1(U,\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j)) \longrightarrow H^1(V,\pi_U^*\nu^*E^{n-1}\rho_*\mathbb{Q}_\ell(j))$$

is injective. Since $\pi_U : V \to U$ is finite and flat, trace theory [SGA 4 XVII 6.2] supplies, for every $\mathbb{Q}_\ell$-sheaf F on U (for example $F = \nu^* E^{n-1}\rho_*\mathbb{Q}_\ell(j)$), a map

$$\mathrm{tr}_{\pi_U} : H^1(V, \pi_U^* F) \longrightarrow H^1(U, F)$$

such that

$$\mathrm{tr}_{\pi_U} \circ \pi_U^* = \deg(\pi_U)\ \mathrm{id}\,.$$

It follows that $H^*(U, F) \to H^*(V, \pi_U^* F)$ is injective, completing the proof. [75]

5.9. An example in which hypothesis (A) is not satisfied.

Let X be a smooth quadric surface in $\mathbb{P}^3$. Then X satisfies (LV) (5.2.4). Take a pencil (a Lefschetz pencil, if desired) of hyperplanes whose generic member and axis meet X transversely. Then (as Deligne pointed out to me) the projection

$$\rho : \widetilde{X} \longrightarrow \mathbb{P}^1$$

does not satisfy (A). Indeed, $R^2\rho_*\mathbb{Q}_\ell$ is not constant: the singular fibers are singular conics, hence reducible, and in a Lefschetz pencil each is the union of two lines meeting at one point (this is the definition of a Lefschetz pencil in this simple case). Thus

$$\dim_{\mathbb{Q}_\ell} H^2(\text{singular fiber}, \mathbb{Q}_\ell) = 2,$$

whereas

$$\dim_{\mathbb{Q}_\ell} H^2(\text{smooth fiber}, \mathbb{Q}_\ell) = 1.$$

Moreover, the conclusion of Theorem 5.8.4 is false here. Let $U \overset{\nu}{\hookrightarrow} \mathbb{P}^1$ be a nonempty open subset over which ρ is smooth. Since the fibers of ρ over U are smooth curves of genus zero, we have

$$\nu^* R^1\rho_*\mathbb{Q}_\ell = 0$$

and hence

$$H^1(U, \nu^* R^1\rho_*\mathbb{Q}_\ell) = 0.$$

Consequently, the Griffiths homomorphism

$$\mathrm{Prim}^2(X, \mathbb{Q}_\ell) \xrightarrow{\mathrm{Grif}} H^1(U, \nu^* R^1\rho_*\mathbb{Q}_\ell)$$

is zero. But on a quadric surface,

$$\dim_{\mathbb{Q}_\ell} \mathrm{Prim}^2(X, \mathbb{Q}_\ell) = 1.$$

Likewise, the conclusion of 5.7 fails here.

6. Application of the local theory of vanishing cycles (Picard–Lefschetz)

Proposition 6.1.

Let Z be a hypersurface in projective space $\mathbb{P}^r_k$, $r \geq 2$, where k is an algebraically closed field. Let

$$i_o : D_o \longhookrightarrow \mathbb{P}^r$$

[75] The print drops the twist (j) from one base-change display and the frozen transcription corrupts both the source and target of the trace map. The corrected formulas are the standard finite-flat pullback–trace identities used by the concluding argument.

be the inclusion of a line which meets Z only in its smooth locus and meets it transversely. Choose a geometric point $\bar{a}$ of $D_o - D_o \cap Z$, and denote by

$$\pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

the quotient of $\pi_1(D_o - D_o \cap Z, \bar{a})$ classifying the étale coverings of $D_o - D_o \cap Z$ which are tamely ramified at every point of $D_o \cap Z$. Then:

6.1.1.

The canonical homomorphism

$$i_{o*} : \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}) \tag{6.1.1.1}$$

is surjective.

6.1.2.

Every choice of a geometric point $\bar{x}_o \in D_o \cap Z$ and of a "path" in D_o from $\bar{x}_o$ to $\bar{a}$ determines a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.2.1}$$

whose image is "the" inertia group at $\bar{x}_o$. Composing this morphism with i_{o*} gives a homomorphism

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}). \tag{6.1.2.2}$$

If Z is irreducible, all these homomorphisms are conjugate to one another.

Proof. Let T be the subscheme of the Grassmannian formed by the lines in $\mathbb{P}^r$ which

(i) pass through $\bar{a}$;

(ii) meet Z only in its smooth locus, and meet it there transversely.

Let η be its generic point, and let

$$i_\eta : D_\eta \longrightarrow \mathbb{P}^r \otimes_k k(\eta)$$

be the corresponding line (thus a line "joining $\bar{a}$ to the generic point of $\mathbb{P}^r_k$"). Choose also a geometric point $\bar{\eta}$ over η, and write $D_{\bar{\eta}}$ for the corresponding "geometric" line. Let $o \in T$ be the point corresponding to the given line D_o. By [SGA 1, XIII 2.8], there is a specialization homomorphism (depending on a path in T from $\bar{\eta}$ to o, and hence defined up to conjugacy)

$$\pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \tag{6.1.3}$$

such that the following assertions hold.

6.1.4.

The local-monodromy homomorphisms (6.1.2.1)

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a})$$

are obtained by composing the specialization arrow (6.1.3) with the local monodromy homomorphisms for $D_{\bar{\eta}}$:

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar{\eta}} - D_{\bar{\eta}} \cap Z, \bar{a}) \xrightarrow{\bar{\eta} \longrightarrow o} \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}). \tag{6.1.4.1}$$

6.1.5.

The following diagram is commutative:
(6.1.5)

$$\begin{tikzcd}
& \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a}) & \\
\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a}) \arrow[ur, "i_{\eta*}"] & & \pi_1^{\mathrm{mod}}(D_o - D_o \cap Z, \bar{a}) \arrow[ul, "i_{o*}"'] \\
& \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar{a}) \arrow[uu, "i_{\bar\eta*}"] \arrow[ul] \arrow[ur, "\bar\eta \longrightarrow o"'] &
\end{tikzcd}\quad .$$

Assume this for the moment, and first prove 6.1.1. By the commutativity of 6.1.5, it is enough to prove that

$$i_{\bar\eta*} : \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(\mathbb{P}^r - Z, \bar{a})$$

is surjective. For this, it is enough to prove that the analogous homomorphism of the ordinary fundamental groups

$$\text{(6.1.6)} \qquad i_{\bar\eta*} : \pi_1(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar{a}) \longrightarrow \pi_1(\mathbb{P}^r - Z, \bar{a})$$

is surjective. In other words, for every étale covering $E \to \mathbb{P}^r - Z$ with E connected (and hence irreducible, since $\mathbb{P}^r - Z$ is smooth and therefore unibranch [EGA IV, 18.10.3]), its inverse image

$$E_{\bar\eta} = E \times_{\mathbb{P}^r - Z} (D_{\bar\eta} - D_{\bar\eta} \cap Z)$$

over $D_{\bar\eta} - D_{\bar\eta} \cap Z$ must remain geometrically irreducible. It is enough to use the following variant of Bertini's theorem ([EGA V, citation left blank in the source]) for the composite $E \to \mathbb{P}^r - Z \hookrightarrow \mathbb{P}^r$ and $s = 1$.

6.1.6.1. Bertini's theorem.

Let k be a field, let $\mathbb{P}^r = \mathbb{P}^r_k$, and let $f : X \to \mathbb{P}^r$ be a nonconstant morphism, where X is a geometrically irreducible algebraic scheme. For every integer $s < \dim \overline{f(X)}$, let L_t denote the generic linear subspace of codimension s in $\mathbb{P}^r$. Then $X_t = f^{-1}(L_t)$ is geometrically irreducible. We now prove 6.1.2. By 6.1.4 and the commutativity of (6.1.5), it is enough to prove that all the local-monodromy homomorphisms (6.1.3) for $D_{\bar\eta}$,

$$\text{(6.1.7.1)} \qquad \prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar{a}),$$

become conjugate after composition with the homomorphism

$$\text{(6.1.7.2)} \qquad \pi_1^{\mathrm{mod}}(D_{\bar\eta} - D_{\bar\eta} \cap Z, \bar{a}) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a}).$$

To prove this, observe that for every finite Galois extension K of $k(\eta)$, with

$$k(\eta) \subset K \subset k(\bar\eta),$$

the scheme

$$D_K \overset{\mathrm{def}}{=} D_\eta \times_{k(\eta)} K$$

is an étale covering of D_η. Thus $D_K - D_K \cap Z$ is a tame covering of $D_\eta - D_\eta \cap Z$, and hence $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$ acts on the scheme $D_K \cap Z$. We take K sufficiently large that every point of $D_K \cap Z$ is K-rational; this is possible because, by hypothesis, $D_\eta \cap Z$ is étale over $k(\eta)$.

It remains to prove that $\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$ acts transitively on the points of $D_K \cap Z$. This will prove that the composite local-inertia homomorphisms

$$\prod_{\ell \neq \mathrm{char}(k)} \mathbb{Z}_\ell(1) \longrightarrow \pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a})$$

attached to the various points of $D_K \cap Z$ are conjugate. Now the action on $D_K \cap Z$ factors through the canonical quotient [76]

$$\pi_1^{\mathrm{mod}}(D_\eta - D_\eta \cap Z, \bar{a}) \longrightarrow \pi_1(\eta, \bar{\eta}) = \mathrm{Gal}\big(k(\bar{\eta})/k(\eta)\big).$$

The action of $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ on $D_K \cap Z$ is the one compatible with the inclusion

$$D_K \cap Z \hookrightarrow \mathbb{P}^r_k \times_k K,$$

where the Galois group acts on $\mathbb{P}^r_k \times_k K$ only through the second factor. It follows that the desired assertion is equivalent to saying that $D_\eta \cap Z$ has only one point, that is, that it is the spectrum of a field. In elementary terms, this is precisely the following Bertini-type fact (take an affine equation $F = 0$ for Z and suppose that $\bar{a}$ is the origin).

Fact 6.1.8.

Let k be an algebraically closed field, let $r \geq 2$, and let $F(X_1, \ldots, X_r) \in k[X_1, \ldots, X_r]$ be irreducible. Then

$$G(T) = F(X_1T, \ldots, X_rT) \in k(X_1, \ldots, X_r)[T]$$

is irreducible as a polynomial in T with coefficients in $k(X_1, \ldots, X_r)$.

This completes the proof of 6.1.2.

Remark 6.1.9.

Another method of proving 6.1.2, valid under considerably more general conditions, would be simply to establish a general compatibility for "monodromy homomorphisms": the composite homomorphisms in 6.1.2 are conjugate to the monodromy homomorphism at the generic point of Z, and therefore are conjugate to one another.

Corollary 6.1.10.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$.[77] Let D be a Lefschetz pencil for X, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the natural representation of $\pi_1(D - D \cap \check{X}, \bar{\eta})$ on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ factors through $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$. If this representation is nontrivial, then $D \cap \check{X} \neq \varnothing$, and the inertia group at every point of $D \cap \check{X}$ acts nontrivially.

[76] The numerator of the printed Galois group is blank. The field tower immediately preceding the display determines it as $k(\bar{\eta})$.

[77] The print has $\mathrm{card}(k) \neq 2$; the parallel hypothesis in (6.3.2) and the tame-monodromy argument require the characteristic.

Proof.

If $\dim(\check{X}) \leq r-2$, then $D \cap \check{X}$ is empty and there is nothing to prove, since $\mathbb{P}^1$ is simply connected. Otherwise $\check{X}$ is a hypersurface in $\mathbb{P}^r$ [XVII, 3.1.4], and 6.1.1 applies by 3.2.10 and XVII 3.5, with $Z = \check{X}$ and $D_o = D$. The fact that the action of π_1 on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is tame was established in the local theory by the Picard–Lefschetz formula [XV, 3.4], recalled below. Finally, the nontrivial fact that $\pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$ is generated by the inertia groups at the various points of $D \cap \check{X}$ [SGA 1, XIII 2.10] gives the required conclusion.

We shall now use the preceding result to study condition (A) of (5.3.5). First recall the following consequence of the Picard–Lefschetz formula [XV, 3.4].

Proposition 6.2.

Let X be a smooth connected projective variety of dimension n over the algebraically closed field k. Let D be a Lefschetz pencil, let η be its generic point, and let $\bar{\eta}$ be a geometric point over η. Then the pencil D satisfies condition (A) if and only if the inertia group at every point of $D \cap \check{X}$ acts nontrivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

Theorem 6.3.

Under the hypotheses of 6.2, the following assertions hold.

(6.3.1) If $n = \dim X$ is odd, condition (A) is satisfied by every Lefschetz pencil for X.

(6.3.2) Suppose that $n = \dim X$ is even, or that $\mathrm{char}(k) \neq 2$. Then the following conditions are equivalent:

(i) D does not satisfy condition (A) of (5.3.5);

(ii) $\pi_1(D - D \cap \check{X})$ acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$, and $D \cap \check{X} \neq \varnothing$;

(iii) there is a point of $D \cap \check{X}$ whose inertia group acts trivially on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$;

(iv) the sheaf $R^{n-1}\rho_*\mathbb{Q}_\ell$ is constant on D, and $D \cap \check{X}$ is nonempty;

(v) the restriction $R^{n-1}\rho_*\mathbb{Q}_\ell|_U$ is constant on U, where U is a nonempty open subset of D over which ρ is smooth, and ρ is not smooth.

(6.3.3) If, in addition, X satisfies condition (LV) of (5.2.2), conditions (i)–(v) are also equivalent to:

(vi) the restriction map
$$H^{n-1}(X, \mathbb{Q}_\ell) \longrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$$
is surjective (and hence is an isomorphism by the weak Lefschetz theorem).

Proof.

Assertion 6.3.1 follows from XV 3.4. For the equivalences in 6.3.2, (i) is equivalent to (iii) by 6.2, (ii) to (iii) by 6.1.6, (ii) to (v) tautologically, and (iv) to (v) by 5.3.4. Finally, if X satisfies condition (LV), 5.6.10 gives the equivalence of (ii) and (vi).

Remark 6.3.4.

Condition (A) does not depend on the choice of the Lefschetz pencil D, but only on the chosen (Lefschetz) immersion $i : X \longrightarrow \mathbb{P}^r$. This is clear from (6.3.1) when $\dim X$ is odd. When $\dim X$ is even, condition (A) also says that the local system on $\mathbb{P}^r - \check{X}$ formed by the groups $H^{n-1}(X \cap H_s, \mathbb{Q}_\ell)$ is constant. This follows from criterion (ii), from the fact that the local system in question is tamely ramified at the generic point of $\check{X}$, and from the surjectivity in (6.1.2).

Corollary 6.4.

Under the hypotheses of 6.2, suppose that $n = \dim X$ is even or $\mathrm{char}(k) \neq 2$, and that X satisfies condition (LV).[78] Then there is an integer d_o such that, for $d \geq d_o$, every Lefschetz pencil of sections of X by hypersurfaces of degree d satisfies condition (A).

Proof.

By the equivalence (i)$\Longleftrightarrow$(vi) of (6.3.2), it is enough to prove that, if $Y(d)$ denotes a smooth hypersurface section of X of degree d, then

$$\dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \longrightarrow \infty \qquad \text{as } d \longrightarrow \infty. \tag{6.4.1}$$

Indeed, we have the following precise result.

Lemma 6.4.2.

Let X be a smooth connected projective variety of dimension $n \geq 1$ over an algebraically closed field k, and let $Y(d)$ denote a hypersurface section of degree d. Then there is a polynomial $f(T) \in \mathbb{Z}[T]$ such that

$$f(T) = \deg(X)T^n + \text{a polynomial of degree at most } n-1, \tag{6.4.2.1}$$

$$f(d) = \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) \qquad (d \geq 1). \tag{6.4.2.2}$$

Proof.

By XVII 5.7.5, if $C(X)$ denotes the total Chern class of X and H the first Chern class of a hyperplane, the Euler–Poincaré characteristic of $Y(d)$ is

$$\chi(Y(d)) = \int_X \frac{dH\, C(X)}{1 + dH} = \sum_{i=1}^{n} (-1)^{i-1} d^i \int_X H^i C_{n-i}. \tag{6.4.2.3}$$

On the other hand, Poincaré duality and weak Lefschetz give

$$\chi(Y(d)) = (-1)^{n-1} \dim H^{n-1}(Y(d), \mathbb{Q}_\ell) + 2 \sum_{i \geq 1} (-1)^{n-1-i} \dim H^{n-1-i}(X, \mathbb{Q}_\ell). \tag{6.4.2.4}$$

Combining (6.4.2.3) and (6.4.2.4) gives the result.

6.5. A reminder on the Picard–Lefschetz formula (XV 3.4)

Let S be the spectrum of a strictly henselian valuation ring, with generic point η, geometric generic point $\bar{\eta}$, and closed point s. Let

$$\rho : X \longrightarrow S$$

be a proper morphism, where X is a connected regular scheme of dimension n. Suppose that ρ is smooth away from a single point x in the special fiber and that x is an ordinary quadratic singularity of that fiber.

Let I denote the inertia group of S at s. By XV 3.4, I acts trivially on $H^i(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ for $i \neq n-1$, and its action on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is described as follows.

[78] In the print, the placement of the conjunction attaches condition (LV) only to the second alternative. The proof invokes the equivalence (i)$\Longleftrightarrow$(vi) of (6.3.2), so condition (LV) is required in either case.

6.5.1.

If $n = 2m+1$, then I acts through a quotient isomorphic to $\mathbb{Z}/2\mathbb{Z}$ (if $\mathrm{char}(k) \neq 2$, there is only one such quotient). There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

determined uniquely up to sign and called the vanishing cycle at s, such that

$$\delta \cdot \delta = (-1)^m 2 \quad \text{in} \quad H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(n-1)) \simeq \mathbb{Q}_\ell. \tag{6.5.1.1}$$

The nontrivial element σ of $\mathbb{Z}/2\mathbb{Z}$ acts on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the reflection

$$\sigma x = x - (-1)^m (x \cdot \delta)\delta. \tag{6.5.1.2}$$

[79] Here $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m)$, so $(x \cdot \delta)\delta$ defines an element of $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

6.5.2.

If $n = 2m+2$, then I acts through its canonical quotient $\mathbb{Z}_\ell(1)$. There is an element

$$\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)),$$

canonically determined up to sign and called the vanishing cycle at s, such that $\sigma \in \mathbb{Z}_\ell(1)$ acts on $x \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ by the transvection

$$\sigma(x) = x - (-1)^m \sigma\,(x \cdot \delta)\delta. \tag{6.5.2.1}$$

Indeed, $x \cdot \delta$ belongs to $H^{2n-2}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \simeq \mathbb{Q}_\ell(-m-1)$. Multiplication by $\sigma \in \mathbb{Z}_\ell(1)$ therefore gives an element $\sigma(x \cdot \delta) \in \mathbb{Q}_\ell(-m)$, whose product with $\delta \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$ lies in $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

We return to the global situation.

Theorem 6.6.

Let X be a smooth connected projective variety of dimension n over an algebraically closed field k. Suppose that n is even or that $\mathrm{char}(k) \neq 2$. Let D be a Lefschetz pencil for X, let η be its generic point, let $\bar{\eta}$ be a geometric point over η, and let $s_1, \ldots, s_N$ be the points of $D \cap \check{X}$.

For each i, choose an isomorphism ℓ_i between $\bar{\eta}$ and the geometric generic point of S_i, the strict henselization of D at s_i. Via ℓ_i, let I_i be the inertia group at s_i acting on $H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(j))$, and let δ_i be the vanishing cycle at s_i defined in 6.5.[80] Thus

$$\delta_i \in H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)), \qquad m = \begin{cases} (n-1)/2, & n \text{ odd}, \\ n/2 - 1, & n \text{ even}. \end{cases}$$

[81] Choose the ℓ_i so that the subgroups

$$I_i \subset \pi = \pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta})$$

topologically generate π, as is possible by SGA 1, XIII 2.10. Then:

[79] The print numbers the reflection formula (6.5.2.2), although it is the second formula of 6.5.1 and is followed by (6.5.2.1).

[80] The print calls I_i the inertia group "of δ_i" before δ_i is defined. The standard construction is the inertia group at s_i, embedded by the chosen path ℓ_i.

[81] The print omits the generic-fiber subscript in this display; the preceding action and the definition in 6.5 place every δ_i in the cohomology of $X_{\bar{\eta}}$.

6.6.1. If X satisfies condition (LV), the vanishing cohomology space

$$E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)) \stackrel{\text{def}}{=} H^{n-1}(X, \mathbb{Q}_\ell(m))^\perp \subset H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$$

(cf. 5.4.4) is generated by the vanishing cycles δ_i, $i = 1, \ldots, N$.

6.6.2. The vanishing cycles δ_i are conjugate, up to sign, under the global monodromy group $\pi_1(D - D \cap \check{X})$.

Proof.

For 6.6.1, the Picard–Lefschetz formula 6.5 and the fact that the I_i generate $\pi = \pi_1(D - D \cap \check{X})$ give

$$H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))^\pi = \big\langle \delta_1, \ldots, \delta_N \big\rangle^\perp \quad \text{in } H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)). \tag{6.6.3}$$

By 5.6.10,

$$H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))^\pi = \operatorname{Im}\big(H^{n-1}(X, \mathbb{Q}_\ell(m)) \longrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))\big). \tag{6.6.4}$$

Taking orthogonals in (6.6.3) proves 6.6.1.

For 6.6.2, note simply that the vanishing cycle δ_i is determined up to sign by the representation of the inertia group at s_i. Apply 6.1.2, which gives conjugacy in $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta})$ of the composite inertia homomorphisms

$$\begin{array}{ccc} \mathbb{Z}_\ell(1) & \longrightarrow & \pi_1^{\mathrm{mod}}(D - D \cap \check{X}, \bar{\eta}) \\ & & \big\downarrow \\ & & \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta}) \end{array} .$$

Consequently, the vanishing cycles determined by these homomorphisms are conjugate up to sign by elements of $\pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}, \bar{\eta})$. The proof is completed by 6.1.1, which gives the surjection

$$\pi_1^{\mathrm{mod}}(D - D \cap \check{X}) \longrightarrow \pi_1^{\mathrm{mod}}(\check{\mathbb{P}}^r - \check{X}).$$

Corollary 6.7.

Under the hypotheses of 6.6, suppose that X satisfies condition (LV). Then:

6.7.1. The representation of $\pi_1(D - D \cap \check{X})$ on $E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is absolutely irreducible.

6.7.2. If n is even, every finite-index subgroup of $\pi_1(D - D \cap \check{X})$ acts absolutely irreducibly on $E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell)$.

Proof.

We may replace the module in question by its twist

$$E = E^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m)).$$

Let $\bar{E} = E \otimes_{\mathbb{Q}_\ell} \bar{\mathbb{Q}}_\ell$, and choose $0 \neq \chi \in \bar{E}$. To prove 6.7.1, we must show that the $\bar{\mathbb{Q}}_\ell$-linear subspace of $\bar{E}$ generated by χ and its conjugates under π is all of $\bar{E}$. By 6.6.1 and 6.6.2, it is enough to prove that this subspace contains one of the vanishing cycles δ_i.

Since X satisfies (LV), the restriction of the cup-product form to

$$H^{n-1}(X, \mathbb{Q}_\ell(m)) \hookrightarrow H^{n-1}(X_{\bar{\eta}}, \mathbb{Q}_\ell(m))$$

is nondegenerate; hence so is its restriction to the orthogonal $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(m))$. By 6.6.1, there is therefore at least one i such that

$$\chi \cdot \delta_i \neq 0.$$

For this i, 6.5 supplies an element σ_i of the inertia group at s_i such that

$$\sigma_i(\chi) - \chi = \text{a nonzero multiple of } \delta_i. \tag{6.7.3}$$

This proves 6.7.1.

For 6.7.2, $n-1$ is odd, so $\delta_i \cdot \delta_i = 0$. Iterating 6.5.2 gives, for every k,

$$\sigma_i^k(\chi) = \chi + k\,\sigma_i(\chi \cdot \delta_i)\delta_i. \tag{6.7.4}$$

Thus a $\mathbb{Q}_\ell$-subspace M of E is stable under the action of π if and only if it is stable under a finite-index subgroup: both conditions are equivalent to

$$\chi \in M \quad \Longrightarrow \quad \delta_i \in M \quad \text{for every } i \text{ with } \chi \cdot \delta_i \neq 0.$$

The assertion therefore reduces to 6.7.1.

Bibliography

[1] P. Deligne, *Théorème de Lefschetz et critères de dégénérescence de suites spectrales*, Publ. Math. IHÉS **35** (1968), 259–278.

[2] S. Kleiman, *Algebraic cycles and the Weil conjectures*, in *Dix exposés sur la cohomologie des schémas*, North-Holland, 1968.

[3] S. Lefschetz, *L'analyse situs et la géométrie algébrique*, Gauthier-Villars, Paris, 1924.

[4] D. Lieberman, *Higher Picard Varieties*, Amer. J. Math. **90** (4) (1968), 1165–1199.[82]

[5] D. Mumford, unpublished.

[6] A. Weil, *Variétés kählériennes*, Hermann, Paris, 1958.

[82]The print gives the journal volume as 15; the article appeared in volume 90.

SGA 7 — EXPOSÉ XIX

The Noether Theorem

by P. Deligne

Contents

1. Statement of the theorems

The present exposé updates Lefschetz's proof [3, p. 108] of the Noether theorem (1.2) stated below.

1.1.

Let k be an algebraically closed field of characteristic p. Let $d \geq 1$ and let

$$\underline{a} = (a_i)_{1 \leq i \leq d}$$

be a sequence of integers at least 1. A complete intersection V of dimension n and multidegree $\underline{a}$ in the projective space $\mathbb{P}^{n+d}$ over k will be called generic if, after an extension of k, it is projectively isomorphic to a complete intersection defined by equations whose coefficients are algebraically independent over the prime field. In what follows, unless expressly stated otherwise, we suppose that $d \geq 1$ and $a_i \geq 2$.

Theorem 1.2.

Let $V \subset \mathbb{P}^{2+d}$ be a generic complete-intersection surface of multidegree $\underline{a}$. Suppose that

$$\underline{a} \neq (2), (3), (2,2).$$

Then every (reduced) curve on V is the scheme-theoretic intersection of V with a hypersurface of $\mathbb{P}^{2+d}$.

Let us show that the theorem follows from its corollary

$$\mathrm{Pic}(\mathbb{P}^{2+d}) = \mathbb{Z} \xrightarrow{\sim} \mathrm{Pic}(V). \tag{1.2.1}$$

If C is a curve on V, (1.2.1) implies that $\mathcal{O}(C) \simeq \mathcal{O}(n)$ for a suitable n. If H is a hyperplane section, then

$$n = \left(\textstyle\prod_i a_i\right)^{-1}(C, H) > 0.$$

It remains to prove that the linear system induced on V by $|nH|$ is complete. Write V as the intersection of hypersurfaces H_i of degree a_i, and put $V_i = H_1 \cap \cdots \cap H_i$. Induction on i shows that the linear system induced by $|nH|$ on V_i is complete: in the exact sequence

$$H^0(V_i, \mathcal{O}(n)) \longrightarrow H^0(V_{i+1}, \mathcal{O}(n)) \longrightarrow H^1(V_i, \mathcal{O}(n - a_{i+1})),$$

the last group vanishes (FAC).

Theorem 1.3.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of nonzero even dimension $2n$ and multidegree $\underline{a}$. Suppose that one of the following conditions holds:

(a) k is not of characteristic 2, and $(2n; \underline{a})$ is distinct from $(2n; 2)$, $(2n; 2, 2)$, and $(2; 3)$;

(b) the Hodge number $h^{2n,0}$ is nonzero, i.e. $\sum_i (a_i - 1) > 2n$.

Then every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of the class η^n of a linear section.

Let us deduce (1.2.1) from 1.3. The exceptional cases in 1.2 are exactly those for which $\sum_i (a_i - 1) \leq 2$; hence 1.3(b) applies to the surface V of 1.2.[83] We know that $\mathrm{Pic}^\tau(V) = 0$ (XI 1.8), so $\mathrm{Pic}(V)$ injects into $H^2(V, \mathbb{Q}_\ell(1))$. By 1.3 every element of $\mathrm{Pic}(V)$ is a rational multiple of the class η of $\mathcal{O}(1)$, and (XI 1.8) gives the conclusion.

It is clear that the validity of 1.2 and 1.3 does not depend on the chosen generic complete intersection, but only on p, n, and $\underline{a}$.

Theorem 1.4.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of dimension $2n+1$ over k. Let $\ell \subset \check{\mathbb{P}}$ be a Lefschetz pencil of hyperplane sections of X. If k has characteristic 2, suppose in addition that ℓ is a generic Lefschetz pencil.

Suppose that X satisfies condition (LV) (XVIII 5.2.2), and let V be the generic hyperplane section in the pencil ℓ. One of the following two conditions holds:

(a) no nonzero cohomology class in the vanishing part $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ of $H^{2n}(V, \mathbb{Q}_\ell(n))$ is algebraic;

(b) the algebraic cohomology classes generate $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$.

Let us make precise the meaning of "generic" in 1.4.

(a) Let G be the Grassmannian of lines in the projective space $\check{\mathbb{P}}$ dual to $\mathbb{P}$. By a "geometric point" of G we mean a morphism from the spectrum of an algebraically closed field to G. Every geometric point $\bar{s} \to G$ defines a pencil ℓ of hyperplane sections of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call ℓ generic if $\bar{s}$ lies over the generic point of G.

(b) Let ℓ be a pencil of hyperplane sections (a line in $\check{\mathbb{P}}$). Every geometric point $\bar{s} \to \ell$ defines a hyperplane section of the variety obtained from X by extending scalars from k to $k(\bar{s})$. We call this section generic if $\bar{s}$ lies over the generic point of ℓ.

Let $\bar{s}$ be as above, with image the generic point s of ℓ, and let $S \subset \ell$ be the finite set of points giving singular hyperplane sections. The monodromy group $\pi_1(\ell - S, \bar{s})$, a quotient of $\mathrm{Gal}(k(\bar{s})/k(s))$, acts on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$. If k is not of characteristic 2, this action is irreducible (XVIII 6.7); in Section 2 below we shall show that the assertion remains true without a characteristic hypothesis for a generic pencil. The subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by algebraic cohomology classes is stable under monodromy (that is, under Galois). It is therefore either zero or the whole space, which proves 1.4.

[83] The print writes d_i in this sum, although the multidegree has been denoted throughout by (a_i).

Corollary 1.5.

Let $V \subset \mathbb{P}^{2n+d}$ be a generic complete intersection of even dimension $2n$ and multidegree $\underline{a}$. Then either

(a) every algebraic cohomology class in $H^{2n}(V, \mathbb{Q}_\ell(n))$ is a rational multiple of η^n; or

(b) $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic cohomology classes.

If (b) holds, then for every complete intersection of dimension $2n$ and multidegree $\underline{a}$ over an algebraically closed field of characteristic p, the group $H^{2n}(V, \mathbb{Q}_\ell(n))$ is generated by algebraic classes.

Apply 1.4 with X a generic complete intersection of dimension $2n+1$ and multidegree $(a_1, \dots, a_{d-1})$, and with ℓ a generic pencil of sections of X by hypersurfaces of degree a_d (cf. XVII 2.4). One could equally take X of multidegree $(a_1, \dots, a_d)$ and ℓ a generic pencil of hyperplane sections. The final assertion follows because an algebraic cohomology class remains algebraic under specialization.

2. A complement to XVIII

Proposition 2.1.

Let $X \subset \mathbb{P}$ be a smooth projective variety, pure of odd dimension $2n+1$ over an algebraically closed field k, and let $\ell \subset \check{\mathbb{P}}$ be a generic Lefschetz pencil of hyperplane sections of X. Let s be a geometric point of ℓ, let H_s be the corresponding hyperplane section, let $S \subset \ell$ be the set of exceptional values of the pencil, and let $\pi_1(\ell - S, s)$ be the monodromy group. Suppose that X satisfies condition (LV). Then the representation of $\pi_1(\ell - S, s)$ on the vanishing cohomology

$$\mathrm{Ev}(H_s, \mathbb{Q}_\ell) \subset H^{2n}(H_s, \mathbb{Q}_\ell)$$

is absolutely irreducible.[84]

Recall that $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$, and that this image equals the subgroup of monodromy invariants (XVIII 5.6.10).

Proof.

The projective line $\mathbb{P}^1$ is simply connected. Hence $\pi_1(\ell - S, s)$ is topologically generated by the inertia subgroups at the points of S and their conjugates. Its image in $\mathrm{GL}(H^{2n}(H_s, \mathbb{Q}_\ell))$ is therefore topologically generated by the transformations

$$x \longmapsto x - (-1)^n (x \cdot \delta)\delta,$$

where δ is conjugate to a vanishing cycle.[85] Consequently, the image of $H^{2n}(X, \mathbb{Q}_\ell)$ in $H^{2n}(H_s, \mathbb{Q}_\ell)$ is the orthogonal complement of the subspace generated by the vanishing cycles and their conjugates; thus $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ is generated by those cycles and their conjugates.

For a generic Lefschetz pencil, the arguments of XVIII ğ6 show that the vanishing cycles attached to the different points of S are conjugate to one another.

Let $W \subset \mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ be a nonzero invariant subspace. There exist a conjugate δ of a vanishing cycle and an element $v \in W$ such that $(v, \delta) \neq 0$; this is where condition (LV) is used. Since

$$v - (-1)^n (v \cdot \delta)\delta \in W,$$

[84] The print says that X has even dimension $2n$. The displayed degree $H^{2n}(H_s)$, the reflection formula, Remark 2.2, and the application to Theorem 1.4 all force odd dimension $2n+1$.

[85] The print gives $H^n(H_s)$ at this point, although every surrounding occurrence and the statement of the proposition use the middle degree $H^{2n}(H_s)$.

we have $\delta \in W$.[86] The subspace W, containing one vanishing cycle and all its conjugates, is therefore $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$. The argument remains valid after extending the coefficient field $\mathbb{Q}_\ell$, proving absolute irreducibility.

Remark 2.2.

If condition (LV) is omitted, the preceding arguments still show that every invariant subspace of $H^{2n}(H_s, \mathbb{Q}_\ell)$ either contains $\mathrm{Ev}(H_s, \mathbb{Q}_\ell)$ or is contained in the image of $H^{2n}(X, \mathbb{Q}_\ell)$. The same assertion holds for X of even dimension (and for every Lefschetz pencil).

3. Proof of 1.3

3.1.

Under the hypotheses of the theorem, it remains to prove that 1.5(b) does not hold. The case $h^{0,2n} \neq 0$ will be treated by Katz in XXI 4.1 (the condition $h^{0,2n} \neq 0$ is equivalent to

$$\sum_i a_i - (2n + r + 1) \geq 0,$$

this number being the integer m such that $\Omega_V^{2n} \simeq \mathcal{O}(m)$). [87] The results of the present exposé will not be used again in the remainder of the seminar, so there is no vicious circle.

3.2.

When $k = \mathbb{C}$, one proves 1.3 by observing that if 1.5(b) holds, the cohomology of V is purely of type (n, n), since algebraic cohomology is of type (n, n). This occurs only in the exceptional cases excluded in 1.3 (XI 2.9). By the Lefschetz principle, 1.3 is therefore true in characteristic zero.

Proposition 3.3.

Assume the hypotheses of 1.4, and suppose that (b) holds. Let $\mathrm{Ev}(V, \mathbb{Q})$ be the $\mathbb{Q}$-vector subspace of $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ generated by the algebraic cohomology classes.

(a) There is an isomorphism

$$\mathrm{Ev}(V, \mathbb{Q}) \otimes \mathbb{Q}_\ell \xrightarrow{\sim} \mathrm{Ev}(V, \mathbb{Q}_\ell(n)).$$

On $\mathrm{Ev}(V, \mathbb{Q})$ the intersection form is rational and definite, of sign $(-1)^n$.

(b) The vanishing cycles and their conjugates belong to $\mathrm{Ev}(V, \mathbb{Q})$; there they form an irreducible root system of type A, D, or E.

(c) The monodromy group (the image of $\pi_1(\ell - S, \bar{s})$ in $\mathrm{GL}(\mathrm{Ev}(V, \mathbb{Q}))$) is finite; it is the Weyl group of the root system in (b).[88]

Apart from definiteness of the intersection form, (a) follows because the intersection form is nondegenerate on $\mathrm{Ev}(V, \mathbb{Q}_\ell(n))$ (by (LV)) and rational on $\mathrm{Ev}(V, \mathbb{Q})$.

Condition 1.4(b) says equivalently that there are algebraic cycles on V, modulo numerical equivalence, which are not invariant under monodromy; this condition is therefore independent

[86] The print uses the undefined exponent m in this second occurrence; the immediately preceding reflection and the Picard–Lefschetz formula give n.

[87] The print writes d_i in this sum, although the multidegree is denoted by (a_i).

[88] The print repeatedly says “groupe de Weil”; the group attached to a root system is its Weyl group.

of ℓ. Let x be such an algebraic cycle, and let δ be the vanishing cycle associated with $s_0 \in S$ and with a path c from $\bar{s}$ (which defines V) to s_0. For a suitable c, $(x, \delta) \neq 0$. The Galois transform

$$\sigma(x) = x - (-1)^n (x, \delta)\delta$$

is again algebraic; hence $(x, \delta)\delta$ is algebraic, the image of a cycle y independent of ℓ. The rational number

$$\frac{(-1)^n}{2}(y, y) = (x, \delta)^2$$

is consequently independent of ℓ. It is a square in every $\mathbb{Q}_\ell$, hence a square in $\mathbb{Q}$; thus, up to sign, (x, δ) belongs to $\mathbb{Q}$ and is independent of ℓ. This shows not only that $\delta \in \mathrm{Ev}(V, \mathbb{Q})$, but also that the algebraic cycle modulo numerical equivalence

$$\pm\delta = \frac{\sigma(x) - x}{(x, \delta)}$$

is independent of ℓ.

An algebraic cycle on a variety V is always defined over a finite extension of any field of definition of V. It follows that $\pi_1(\ell - S, s)$ acts on $\mathrm{Ev}(V, \mathbb{Q})$ through a finite group. This group is generated by the reflections associated with the vanishing cycles δ and their conjugates (XVIII 6.6.1 and Section 2). The cycles δ and their conjugates therefore form a root system whose Weyl group is the monodromy group. Moreover, all roots of this root system are conjugate under the Weyl group; it is of type A, D, or E.

Up to a scalar, the intersection form is the unique invariant form under the Weyl group. It is therefore definite. Its sign is determined by computing

$$(\delta, \delta) = 2(-1)^n.$$

Proposition 3.4.

If $k = \mathbb{C}$, then under the hypotheses of 1.4 the following conditions are equivalent:

(i) the monodromy group is finite;

(ii) the vanishing cohomology $\mathrm{Ev}(V)$ is purely of type (n, n).

Let $\widetilde{X} \to \ell$ be the family whose fibers are the hyperplane sections in the pencil ℓ, and let V_s be the fiber at $s \in \ell(\mathbb{C})$. The $\mathrm{Ev}(V_s)$ form an algebraic variation of Hodge structures on ℓ. By the fixed-part theorem [2, 4.1.2], applied to a finite covering of ℓ (cf. [2, 4.1.3.3]), (i) implies that the Hodge structure on $\mathrm{Ev}(V_s)$ is locally constant. The monodromy group therefore acts by automorphisms of the Hodge structure. For a Picard–Lefschetz transformation T, $\mathrm{Im}(T - 1)$ is a rank-one Hodge substructure, hence is of type (n, n) and is generated by a vanishing cycle δ. Since the vanishing cycles are of type (n, n), all of $\mathrm{Ev}(V_s)$ is of that type.

Conversely, if $\mathrm{Ev}(V)$ is purely of type (n, n), the intersection form on $\mathrm{Ev}(V)$ is definite and the monodromy group is finite (cf. [2, 4.1.3.3]).

Remark 3.5.

If the equivalent conditions of 3.3 hold, the argument of 3.3 shows that the vanishing cycles and their conjugates form a root system of type A, D, or E, whose Weyl group is the monodromy group.

3.6.

Let us prove 1.3 when k has finite characteristic different from 2. Let $S = \mathrm{Spec}(\mathcal{V})$ be the spectrum of a discrete valuation ring of mixed characteristic with residue field k, and let $X \subset \mathbb{P}_S^{2n+d}$ be a complete intersection of relative dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$ over S. Let ℓ be a Lefschetz pencil of sections of X by hypersurfaces of degree a_d (in the special fiber, ℓ_s is a Lefschetz pencil for X_s). Let

$$f : \widetilde{X} \longrightarrow \ell \simeq \mathbb{P}_S^1$$

be the family whose fibers are the sections in the pencil. The exceptional locus $T \subset \ell$ is ètale over S (XVII 6; here one uses $p \neq 2$). The sheaf $R^{2n} f_* \mathbb{Q}_\ell$ is locally constant on $\ell - T$; it is therefore tamely ramified, and its monodromy is the same in characteristic p as in characteristic zero. If 1.5(b) holds, it follows from 3.3(c) and 3.4 that the cohomology group $H^{2n}(V)$ of complex complete intersections of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n, n). This occurs only in the cases excluded by 1.3 (XI 2.9).

4. A second complement to XVIII

In this section we work over $\mathbb{C}$ and with integral, classical cohomology. Using methods closer to Lefschetz's, we revisit some of the results of XVIII. There they were proved only after tensoring with $\mathbb{Q}$ (or rather with $\mathbb{Q}_\ell$, which amounts to the same thing).

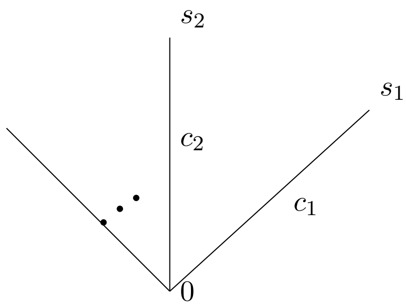
4.1.

Let $X \subset \mathbb{P}$ be a smooth complex projective variety, pure of dimension $n+1$, and let ℓ be a Lefschetz pencil of hyperplane sections of X, with axis A. We also write ℓ for the parameter space of the pencil, which is isomorphic to $\mathbb{P}^1$. Let $S \subset \ell$ be the set of exceptional values. For $s \in \ell$, write H_s for the corresponding hyperplane section of X (of dimension n).

Let $T(A)$ be a closed tubular neighborhood of $A \cap X$ in X, whose boundary is transverse to every H_s.

Choose two points $0, \infty \in \ell - S$, and let $\alpha : \ell \to \mathbb{P}^1(\mathbb{C})$ be an orientation-preserving diffeomorphism that sends 0 to 0, ∞ to ∞, and S to the circle of radius one centered at 0. [89] The composite αf of α with the projection to ℓ makes $X - T(A)^\circ$ a singular fibration in manifolds with boundary over $\mathbb{P}^1(\mathbb{C})$ (one singularity above each point of $\alpha(S)$). Write c_s for the straight segment joining 0 to $\alpha(s)$.

Suppressing α from the notation, the picture is



Near $s \in S$ and the singular point x_s of $H_s \subset X$, suitable local coordinates put us in the situation studied in XIV 3.2. If α is chosen holomorphic near s, one can even choose local coordinates on X and ℓ in which f is written $\sum z_i^2$ and $\alpha^{-1}(c_s)$ is given by "the z_i are real and positive." Above the intersection of a neighborhood of s with $\alpha^{-1}(c_s) - \{s\}$, consider the bundle

[89] The print gives $\ell - S$ as the domain of α, although the same sentence evaluates α on S; the required domain is ℓ.

of real spheres $\Delta'_s \subset X$ defined by "the z_i are real." As $t \to s$, the fiber Δ'_{st} of Δ'_s at t tends to x_s.

Let Δ_s denote the union of $\{x_s\}$ and a sphere bundle over $\alpha^{-1}(c_s)-\{s\}$, contained in $X-T(A)$ and extending Δ'_s. Then Δ_s is an embedded disk in X whose boundary $H_0 \cap \Delta_s$ is a sphere representing the vanishing cycle δ_s at s, transported to H_0 along the path $\alpha^{-1}(c_s)$.

Apply Morse theory to the function $|\alpha f|^2$ to study how the homotopy type of

$$\{x \in X \mid x \in T(A) \text{ or } |\alpha f(x)|^2 \leq r\}$$

changes as r increases. Put

$$X^- = \{x \in X \mid x \in T(A) \text{ or } \alpha f(x) \neq \infty\}.$$

One finds that the inclusions

$$H_0 \cup \bigcup_{s\in S} \Delta_s \hookrightarrow T(A) \cup H_0 \cup \bigcup_{s \in S} \Delta_s \hookrightarrow X^-$$

are homotopy equivalences. Morse theory also shows that $X - X^-$ can be deformation-retracted onto a subcomplex of real dimension at most n. Altogether this gives the following result.

Theorem 4.2.

There exists a compact subcomplex Σ of real dimension at most n in $H_\infty^{\mathrm{aff}} = H_\infty - A$ such that the inclusion

$$H_0 \cup \bigcup_{s\in S} \Delta_s \longrightarrow X - \Sigma$$

is a homotopy equivalence.

4.3.

Removing Σ has no effect on the cohomology, homology, or homotopy groups of X in degrees $i \leq n$. Since $H_0 \cup \bigcup_{s\in S} \Delta_s$ is obtained from H_0 by attaching $(n+1)$-disks along spheres representing the vanishing cycles, homology gives[90]

$$\mathbb{Z}^S \longrightarrow H_n(H_0) \longrightarrow H_n(X) \longrightarrow 0.$$

The analogous exact sequence in cohomology, together with the sequence deduced from it by Poincaré duality, forms the commutative diagram

(4.3.1)
$$\begin{tikzcd}
 & \mathbb{Z}^S & & \\
\mathbb{Z}^S \arrow[r, "\sum n_s\delta_s"] & H^n(H_0) \arrow[u, "(\ ,\delta_s)"'] \arrow[r] & H^{n+2}(X) \arrow[r] & 0 \\
 & H^n(X) \arrow[u] \arrow[ur, "\eta\wedge"'] & & \\
 & 0 \arrow[u] & &
\end{tikzcd}$$

[90]The print writes V_0 in this sequence and in the diagram below, but Section 4 defines the fiber as H_0.

5. The exceptional cases of the Noether theorem

In this section we continue to work over $\mathbb{C}$, with integral cohomology.

5.1.

Let $2n$, d, and $\underline{a} = (a_1, \ldots, a_d)$, with $d \geq 1$ and $a_i \geq 2$, be such that the middle cohomology of smooth complete intersections V of dimension $2n$ and multidegree $\underline{a}$ is purely of type (n, n). In other words, either $n = 0$, or $(2n; \underline{a})$ is one of $(2n; 2)$, $(2n; 2, 2)$, and $(2; 3)$ (XI 2.9).

Suppose that V is obtained from a complete intersection $X \subset \mathbb{P}^{2n+d}$ (respectively, $X \subset \mathbb{P}^{2n+d+1}$) of dimension $2n+1$ and multidegree $(a_1, \ldots, a_{d-1})$ (respectively, $(a_1, \ldots, a_d)$) by taking a section by a hypersurface of degree a_d (respectively, a hyperplane section) belonging to a Lefschetz pencil ℓ.

By (4.3.1), the vanishing part $\mathrm{Ev}(V)$ of $H^{2n}(V)$, which equals its primitive part, is generated over $\mathbb{Z}$ by the vanishing cycles. It is the orthogonal complement of the power η^n of the cohomology class η of a hyperplane section.

Since the intersection form on $H^{2n}(V)$ is unimodular and η^{n}[91] is not by, with $b > 1$, for another cohomology class y (XI 1.6(iv)), the discriminant of the intersection form on $\mathrm{Ev}(V)$ is

$$(\eta^n, \eta^n) = \prod_i a_i.$$

Let R be the root system in $H^{2n}(V)$ formed by the vanishing cycles and their conjugates. We have the following information about R:

(a) R is empty or is of type A, D, or E;

(b) the rank of R is given in XI 1.10;

(c) its discriminant is $\prod_i a_i$. In the tables, this discriminant is the order of the center of the corresponding simply connected complex simple group. The argument by which we computed the discriminant shows that this center, dual to the quotient of the weight lattice by the root lattice, is cyclic.

In every case this information determines R completely if $n \neq 0$. One finds:

Proposition 5.2.

(i) If V is an even-dimensional quadric, R is of type A_1.

(ii) If V is a cubic surface, R is of type E_6.

(iii) If V has dimension $2n$ and is the intersection of two quadrics, R is of type D_{2n+3}.

5.3.

Suppose that $n \neq 0$. The set R^1 of vectors of length ± 2 in $\mathrm{Ev}(V)$ is also a root system and has the same properties (a), (b), and (c) as R. Hence $R = R^1$. Let π_1 be the fundamental group of the parameter space of all complete intersections $V \subset \mathbb{P}^{2n+d}$ of the type under consideration. The image of π_1 in $\mathrm{GL}(\mathrm{Ev}(V))$ is the Weyl group of R. In every case it is the automorphism group of the triple

$$\big(H^{2n}(V), \eta^n, \text{intersection form}\big).$$

[91] The print has η^{2n} in the primitivity assertion, but that class lies in degree $4n$; both the surrounding discussion and the next display require $\eta^n \in H^{2n}(V)$.

5.4.

With R replaced by R^1 from 5.3, assertion 5.2(ii) is well known [1], and for small values of n, 5.2(iii) was known, at least to Manin ($n = 0$ is trivial; a surface that is the intersection of two quadrics is a del Pezzo surface).[92] A detailed study of the intersection of two quadrics was made by M. Reid [4]. He deduces 5.2(iii) from a study of the geometry of the n-dimensional linear subspaces on an intersection of two quadrics $V \subset \mathbb{P}^{2n+2}$.

Bibliography

[1] E. Cartan, "Quelques remarques sur les 28 bitangentes d'une quartique plane et les 27 droites d'une surface cubique," *Bull. Sci. Math.* **70** (1946), 42–45; *Oeuvres*, part I, vol. 2, p. 1353.

[2] P. Deligne, "Théorie de Hodge II," *Publ. Math. IHES* **40** (1971), 5–58.

[3] S. Lefschetz, *L'analysis situs et la géométrie algébrique*, Gauthier–Villars, 1924 (reprinted in *Selected Papers*, Chelsea).

[4] M. Reid, Ph.D. thesis, Cambridge, 1972.

[5] J.-P. Serre, "Faisceaux algébriques cohérents," *Ann. of Math.* **61** (1955), 197–278 (cited as FAC).

[92] The print refers here to 4.2(ii) and 4.2(iii), but the assertions cited are the immediately preceding cases of Proposition 5.2.

SGA 7 — EXPOSÉ XX

The Griffiths Theorem

by N. Katz

0. Introduction

We prove Griffiths's theorem [1], which gives a systematic (though nonconstructive) way of "finding" algebraic cycles that are $\mathbb{Q}_\ell$-cohomologous to zero, but no multiple of which is algebraically equivalent to zero. The proof given here (due to Grothendieck) is the translation into purely algebraic terms of Griffiths's original, more or less transcendental, proof. In both proofs, the irreducibility theorem XVIII 6.7 is a key point (cf. 3.1.6 and 3.2).

1. The formalism of primitive classes

Throughout what follows, a prime number ℓ is fixed. All schemes under consideration are assumed to have characteristic different from ℓ. Consider the situation

(1.0.1)
$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

in which we assume

(1.0.3) $$f,\ g \quad \text{smooth of relative dimensions} \quad d_1 \text{ and } d_2,$$

and

(1.0.4) $$\pi \quad \text{proper.}$$

Since X and Y are smooth over S, the properness of π gives a Gysin morphism (SGA 5, Exposé IV; the further citation is left blank in the print)

(1.0.5) $$\pi_* : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q-2(d_1-d_2)}(Y, \mathbb{Q}_\ell(i - d_1 + d_2)).$$

Localizing on S in the ètale topology, we likewise obtain a morphism of sheaves on S[93]

(1.0.6) $$\pi_* : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2),$$

and, on applying the functor $H^p(S, -)$, a morphism

(1.0.7) $$\pi_* : H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q-2(d_1-d_2)} g_* \mathbb{Q}_\ell(i - d_1 + d_2)).$$

Definition 1.0.8.

Let $(E_r^{p,q}, d_r^{p,q})$ and $({}'E_r^{p,q}, {}'d_r^{p,q})$ be two spectral sequences in a given abelian category, and let $(a, b) \in \mathbb{Z} \times \mathbb{Z}$. The shift of $({}'E_r^{p,q}, {}'d_r^{p,q})$ by (a, b) is the spectral sequence $({}''E_r^{p,q}, {}''d_r^{p,q})$ given by[94]

(1.0.9) $$\begin{cases} {}''E_r^{p,q} = {}'E_r^{p+a,q+b}, \\ {}''d_r^{p,q} = {}'d_r^{p+a,q+b}. \end{cases}$$

[93] In (1.0.6) the print has an apostrophe in place of the closing parenthesis in the exponent and has $i - d_1 - d_2$ in the target twist. Formulae (1.0.5) and (1.0.7), as well as the Gysin degree, give the reading displayed here.

[94] The print omits the prime on the right-hand differential in (1.0.9). The definition of the shift of the primed spectral sequence requires the primed differential.

1.0.10.

A morphism of spectral sequences from $(E_r^{p,q}, d_r^{p,q})$ to $(''E_r^{p,q}, ''d_r^{p,q})$ is called a morphism of bidegree (a,b) from $(E_r^{p,q}, d_r^{p,q})$ to $('E_r^{p,q}, 'd_r^{p,q})$.

The following lemma is an immediate consequence of the theory of the Gysin morphism (SGA 5, Exposé IV).

Lemma 1.1.

There is a canonical morphism π_* of bidegree $(0, -2(d_1 - d_2))$[95] between the Leray spectral sequences

(1.1.0) $$R^p\alpha_* R^q f_* \mathbb{Q}_\ell(i) \Longrightarrow R^{p+q}(\alpha f)_* \mathbb{Q}_\ell(i)$$

and

(1.1.1) $$R^p\alpha_* R^q g_* \mathbb{Q}_\ell(i - d_1 + d_2) \Longrightarrow R^{p+q}(\alpha g)_* \mathbb{Q}_\ell(i - d_1 + d_2),$$

which is given on the E_2 term by (1.0.7), and on the abutment by (1.0.5).

1.2.

Now consider the situation

(1.2.1) $$\begin{tikzcd} & X \times_S Y \arrow[dl, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] & \\ X \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S \arrow[d, "\alpha"] & \\ & T & \end{tikzcd}$$

in which S, X, f, Y, g are as in (1.0.1)–(1.0.4), but now f is also assumed proper.[96] Let

(1.2.2) $$z \in H^a(X \times_S Y, \mathbb{Q}_\ell(b)).$$

Then z induces a map, again denoted by z,

(1.2.3) $$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i + b - d_1))$$

by the formula

(1.2.4) $$z(x) = \mathrm{pr}_{2*}\big(z \wedge \mathrm{pr}_1^*(x)\big),$$

where pr_{2*} is the Gysin homomorphism. By ètale localization on S, the class z likewise defines

(1.2.5) $$z : R^q f_* \mathbb{Q}_\ell(i) \longrightarrow R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1),$$

and, after applying $H^p(S, -)$, maps

(1.2.6) $$H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \longrightarrow H^p(S, R^{q+a-2d_1} g_* \mathbb{Q}_\ell(i + b - d_1)).$$

[95] The print has $(a, -2(d_1 - d_2))$. The map (1.0.7) leaves the first Leray index p unchanged, so the first component of the bidegree is 0.

[96] The print begins this range at the nonexistent (1.0.2); the displayed situation is (1.0.1).

Lemma 1.3.

The class z defines a morphism of bidegree $(0, a-2d_1)$[97] between the Leray spectral sequences

$$H^p(S, R^q f_* \mathbb{Q}_\ell(i)) \Longrightarrow H^{p+q}(X, \mathbb{Q}_\ell(i)) \tag{1.2.7}$$

and

$$H^p(S, R^q g_* \mathbb{Q}_\ell(i+b-d_1)) \Longrightarrow H^{p+q}(Y, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.2.8}$$

which is given on the E_2 terms by (1.2.6), and on the abutments by (1.2.3).
Proof. In view of (1.2.4), and since "multiplication by z" defines a morphism of bidegree $(0, a)$ between the Leray spectral sequences

$$H^p\big(S, R^q(f \circ \mathrm{pr}_1)_* \mathbb{Q}_\ell(i)\big) \Longrightarrow H^{p+q}(X \times_S Y, \mathbb{Q}_\ell(i)) \tag{1.2.9}$$

$$\Big\downarrow z$$

$$H^p\big(S, R^{q+a}(f \circ \mathrm{pr}_1)_* \mathbb{Q}_\ell(i+b)\big) \Longrightarrow H^{p+q+a}(X \times_S Y, \mathbb{Q}_\ell(i+b)), \tag{1.2.10}$$

it suffices to apply Lemma 1.1 to the situation[98]

$$\begin{tikzcd} X \times_S Y \arrow[rr, "\mathrm{pr}_2"] \arrow[dr, "f \circ \mathrm{pr}_1"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd} \quad . \tag{1.2.11}$$

Corollary 1.4.

In the situation (1.2.1)–(1.2.2), the morphism (1.2.3)

$$z : H^q(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1}(Y, \mathbb{Q}_\ell(i+b-d_1)) \tag{1.4.1}$$

takes primitive classes (XVIII 5.8.1) to primitive classes, and the following diagram is commutative:

$$\begin{tikzcd} \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i)) \arrow[r, "z"] \arrow[d, "u_f"] & \mathrm{Prim}^{q+a-2d_1}(Y/S, \mathbb{Q}_\ell(i+b-d_1)) \arrow[d, "u_g"] \\ H^1(S, R^{q-1} f_* \mathbb{Q}_\ell(i)) \arrow[r, "z"] & H^1(S, R^{q+a-2d_1-1} g_* \mathbb{Q}_\ell(i+b-d_1)). \end{tikzcd} \tag{1.4.2}$$

Here u_f and u_g are the canonical homomorphisms of XVIII 5.8.1.

1.5.

Let $\bar\eta$ be a geometric point of S (lying over a point η of S), and let

$$z_{\bar\eta} \in H^a\big(X_{\bar\eta} \times_{k(\bar\eta)} Y_{\bar\eta}, \mathbb{Q}_\ell(b)\big) \tag{1.5.1}$$

be the class induced by z on the corresponding geometric fiber of

$$X \times_S Y \longrightarrow S. \tag{1.5.2}$$

[97] The print has $(0, b-2d_1)$. The maps (1.2.3), (1.2.5), and (1.2.6) change cohomological degree by $a-2d_1$; b controls the Tate twist.

[98] In (1.2.9) and (1.2.10) the print writes $(\mathrm{pr}_1 f)_*$, whereas the map to S is $f \circ \mathrm{pr}_1$, as the printed diagram (1.2.11) itself shows.

Proposition 1.5.3.

Suppose that f and g are proper, that S is connected,[99] and let $q \in \mathbb{Z}$. If the class $z_{\bar\eta}$ induces the zero homomorphism

$$0 = z_{\bar\eta} : H^{q-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+a-2d_1-1}(Y_{\bar\eta}, \mathbb{Q}_\ell(i+b-d_1)), \tag{1.5.3}$$

then, for every $x \in \mathrm{Prim}^q(X/S, \mathbb{Q}_\ell(i))$, one has

$$u_g(z(x)) = 0 \quad \text{in} \quad H^1\big(S, R^{q+a-2d_1-1} g_* \mathbb{Q}_\ell(i+b-d_1)\big). \tag{1.5.4}$$

Proof. By the commutativity of (1.4.2), it is enough to see that z induces the zero homomorphism

$$z : R^{q-1} f_* \mathbb{Q}_\ell(i) \xrightarrow{0} R^{q+a-2d_1-1} g_* \mathbb{Q}_\ell(i+b-d_1). \tag{1.5.5}$$

Indeed, since f and g are proper and smooth, the sheaves $R^\bullet f_* \mathbb{Q}_\ell(\cdot)$ and $R^\bullet g_* \mathbb{Q}_\ell(\cdot)$ are twisted constant sheaves (SGA 4, XVI 2.2), hence are equivalent to their values at $\bar\eta$, regarded as modules under $\pi_1(S, \bar\eta)$.

1.6. A construction of Manin

Suppose that two sections of $f : X \longrightarrow S$ are given:

$$\begin{tikzcd} S \arrow[r, bend left, "t_1"] \arrow[r, bend right, "t_2"'] & X \arrow[l, "f"'] \end{tikzcd} , \tag{1.6.0}$$

and associate with each its cohomology class

$$\mathrm{cl}(t_i) = t_{i*}(1) \in H^{2d_1}(X, \mathbb{Q}_\ell(d_1)). \tag{1.6.1}$$

If the fibers of $f : X \longrightarrow S$ are geometrically connected (more generally, if, for $s \in S$, $t_1(s)$ and $t_2(s)$ belong to the same connected component of X_s), it is clear that[100]

$$\mathrm{cl}(t_1) - \mathrm{cl}(t_2) \in \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)). \tag{1.6.2}$$

We denote by

$$u_f(t_1 - t_2) \in H^1(S, R^{2d_1-1} f_* \mathbb{Q}_\ell(d_1)) \tag{1.6.3}$$

its image under

$$u_f : \mathrm{Prim}^{2d_1}(X/S, \mathbb{Q}_\ell(d_1)) \longrightarrow H^1(S, R^{2d_1-1} f_* \mathbb{Q}_\ell(d_1)). \tag{1.6.4}$$

(It is an analogue $\mu(t_1 - t_2)$ of this element that Manin studies in [5].)

[99] The print has the nonsensical adjective "annexe" in place of "connexe", and uses the capital X for the element in the conclusion. The proof requires connectedness of S, while (1.5.4) itself uses the element x.
[100] The print says $f(s)$ and $g(s)$ here, although the two maps under discussion are the sections t_1 and t_2.

1.6.5.

Now let z be an algebraic cycle on $X \times_S Y$, of codimension at least b on each fiber of $X \times_S Y$,[101] which determines a cohomology class (SGA 5, Exposé IV; the further citation is blank in the print)

$$\mathrm{cl}(z) \in H^{2b}(X \times_S Y, \mathbb{Q}_\ell(b)). \tag{1.6.6}$$

This class defines a map (cf. (1.2.3))

$$\mathrm{cl}(z) \cdot H^{2d_1}(X, \mathbb{Q}_\ell(d_1)) \longrightarrow H^{2b}(Y, \mathbb{Q}_\ell(b)), \tag{1.6.7}$$

and, if the cycles $z(t_1)$ and $z(t_2)$ are defined, the compatibility of the cup product with intersections of cycles (SGA 5, Exposé IV; the further citation is blank in the print) gives

$$\mathrm{cl}(z)\big(\mathrm{cl}(t_1) - \mathrm{cl}(t_2)\big) = \mathrm{cl}\big(z(t_1) - z(t_2)\big). \tag{1.6.8}$$

The commutativity of (1.4.2) gives us

$$\mathrm{cl}(z)\big(u_f(t_1 - t_2)\big) = u_g\big(\mathrm{cl}(z(t_1) - z(t_2))\big). \tag{1.6.9}$$

As a corollary of Proposition 1.5.3, we therefore have:

Proposition 1.7.

Let t_1 and t_2 be two sections of $f : X \longrightarrow S$, with f assumed to have connected fibers, and let z be an algebraic cycle on $X \times_S Y$ of codimension b, such that the algebraic cycle

$$z(t_1) - z(t_2) \tag{1.7.1}$$

is defined. Then its cohomology class is primitive (XVIII 5.8.1). If the algebraic cycle $z_{\bar\eta}$ on $X_{\bar\eta} \times Y_{\bar\eta}$ induces the zero homomorphism

$$z_{\bar\eta} : H^{2d_1-1}(X_{\bar\eta}, \mathbb{Q}_\ell(d_1)) \xrightarrow{0} H^{2b-1}(Y_{\bar\eta}, \mathbb{Q}_\ell(b)), \tag{1.7.2}$$

then

$$u_g\big(z(t_1) - z(t_2)\big) = 0 \quad \text{in} \quad H^1(S, R^{2b-1}g_*\mathbb{Q}_\ell(b)), \tag{1.7.3}$$

where u_g is the homomorphism of XVIII 5.8.1 occurring in 1.4. [102]

2. A reminder on level

(cf. [2], 9.7 and 10.1).

2.0.1.

Let X be proper and smooth over an algebraically closed field k. We saw in (1.2.3) that, for every

$$T \quad \text{proper, smooth, and connected over } k\text{, of dimension } d, \tag{2.0.2}$$

and every

$$\text{algebraic cycle } z \text{ of codimension } b \text{ on } T \times X, \tag{2.0.3}$$

[101] If S is smooth over a scheme T (for example $T = \operatorname{Spec} k$, with k a field), it would be enough to suppose that z has codimension at least b on each fiber over T.

[102] The sentence in the print uses an upper-case $Z_{\bar\eta}$, whereas (1.7.2) and the cycle used throughout the section are denoted by $z_{\bar\eta}$.

the cohomology class

$$\mathrm{cl}(z) \in H^{2b}(T \times X, \mathbb{Q}_\ell(b)) \tag{2.0.4}$$

gives rise to maps

$$\mathrm{cl}(z) : H^q(T, \mathbb{Q}_\ell(i)) \longrightarrow H^{q+2b-2d}(X, \mathbb{Q}_\ell(i+b-d)). \tag{2.0.5}$$

The <u>coniveau filtration</u> (N^j) on $H^q(X, \mathbb{Q}_\ell(i))$ is defined by

$$N^j H^q(X, \mathbb{Q}_\ell(i)) = \sum_{(T,z)} \mathrm{Im}\big(\mathrm{cl}(z) : H^{q-2j}(T, \mathbb{Q}_\ell(i-j)) \longrightarrow H^q(X, \mathbb{Q}_\ell(i))\big)\,, \tag{2.0.6}$$

where the sum extends over the pairs

$$(T, z), \quad \begin{array}{l} T/k \text{ proper, smooth, and connected, of dimension } d, \\ z \text{ an algebraic cycle on } T \times X \text{ of codimension } j+d. \end{array} \tag{2.0.7}$$

It should be noted that one sometimes speaks of the increasing filtration F^j "by level", defined by

$$F^j H^q(X, \mathbb{Q}_\ell(i)) = N^{\left[\frac{q-j}{2}\right]} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.8}$$

Replacing (T, z) by $(\mathbb{P}^1 \times T, \{\mathrm{pt}\} \times z)$, one sees that

$$N^j H^q(X, \mathbb{Q}_\ell(i)) \subset N^{j-1} H^q(X, \mathbb{Q}_\ell(i)). \tag{2.0.9}$$

Obviously, one has:

Proposition 2.1.

Let X be proper, smooth, and geometrically connected over a field k, let $\bar{k}$ be an algebraic closure of k, and put $X_{\bar{k}} = X \otimes_k \bar{k}$. Then $\mathrm{Gal}(\bar{k}/k)$, acting on $H^q(X_{\bar{k}}, \mathbb{Q}_\ell(i))$, preserves the coniveau filtration.

Standard arguments (cf. [2]) prove the following statements.

Proposition 2.2.

The level filtration is invariant under an extension k'/k of algebraically closed fields.

Proposition 2.3.

Let S be a reduced and irreducible scheme, with generic point η, and let

$$\pi : X \longrightarrow S$$

be a proper and smooth morphism with geometrically connected fibers. If, for given integers j and q, one has on the geometric generic fiber $X_{\bar{\eta}}$

$$H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{\eta}}, \mathbb{Q}_\ell), \tag{2.3.1}$$

then there exists a nonempty open subset $U \subset S$ such that, at every point $u \in U$, one has on the geometric fiber $X_{\bar{u}}$

$$H^q(X_{\bar{u}}, \mathbb{Q}_\ell) = N^j H^q(X_{\bar{u}}, \mathbb{Q}_\ell). \tag{2.3.2}$$

3. Application to Lefschetz pencils

3.1.

From now on, we fix

$$k, \quad \text{an algebraically closed field,} \tag{3.1.0}$$

and

$$X \subset \mathbb{P}^r, \quad \begin{array}{l}\text{a projective, smooth, connected } k\text{-scheme} \\ \text{of even dimension } n, \text{ satisfying (LV) (XVIII 5.2.2).}\end{array} \tag{3.1.1}$$

Take

$$D = \{H_t\}_{t \in \mathbb{P}^1}, \quad \text{a Lefschetz pencil of hypersurfaces of degree } d, \tag{3.1.2}$$

with generic fiber

$$X_\eta, \qquad \eta = \text{the generic point of } \mathbb{P}^1. \tag{3.1.3}$$

Put (cf. XVIII 5.4.4)

$$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta), \quad \text{the inclusion,} \tag{3.1.4}$$

and

$$E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = \big(m^* H^{n-1}(X, \mathbb{Q}_\ell(i))\big)^\perp \subset H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)). \tag{3.1.5}$$

We saw in XVIII 6.7.2 that the Picard–Lefschetz formula implies[103]

$$\begin{array}{c}\text{If } n \text{ is even, every finite-index subgroup of } \operatorname{Gal}(k(\bar\eta)/k(\eta)) \\ \text{acts irreducibly on } E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).\end{array} \tag{3.1.6}$$

Proposition 3.2.

Let $D = \{H_t\}$ be a Lefschetz pencil, and suppose that $n = 2m$ is even. Then either

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \subset m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1} H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)), \tag{3.2.1}$$

or else:

(3.2.2)

For every curve C proper and smooth over $k(\bar\eta)$ and every algebraic cycle z on $C \times X_{\bar\eta}$ of codimension m, the induced homomorphism (1.2.3)

$$\operatorname{cl}(z) : H^1(C, \mathbb{Q}_\ell(1 + i - m)) \longrightarrow H^{2m-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \tag{3.2.3}$$

has its image in $m^* H^{2m-1}(X, \mathbb{Q}_\ell(i))$.

Proof. Suppose that the image of such a homomorphism (3.2.3) is not contained in $m^* H^{n-1}(X, \mathbb{Q}_\ell(i))$, or, equivalently, that its image in $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$ is nonzero. This image is evidently stable under the action of a finite-index subgroup of $\operatorname{Gal}(k(\bar\eta)/k(\eta))$. The irreducibility (3.1.6) of $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i))$ therefore says that

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + \operatorname{Im}(3.2.3).$$

[103] The print has E^{2n-1} in (3.1.6). The space defined in (3.1.5), and used throughout the proof of Proposition 3.2, is E^{n-1}.

By definition,

$$\operatorname{Im}(3.2.3) = \operatorname{cl}(z)\big(H^1(C, \mathbb{Q}_\ell(1+i-m))\big) \subset N^{m-1}H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)).$$

Consequently,

$$H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) = m^* H^{n-1}(X, \mathbb{Q}_\ell(i)) + N^{m-1}H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)),$$

which is precisely the first alternative.

3.3. Condition (B)

3.3.1. We shall say that the inclusion

$$m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$$

satisfies condition (B) if there exists

$$V_{\bar\eta}, \quad \text{an algebraic cycle of codimension } n-1 \text{ on } X_{\bar\eta} \times \big(X \times_k k(\bar\eta)\big), \tag{3.3.2}$$

such that the map it defines

$$V_{\bar\eta} : H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X, \mathbb{Q}_\ell(i)) \tag{3.3.3}$$

is a left inverse of

$$m^* : H^{n-1}(X, \mathbb{Q}_\ell(i)) \longrightarrow H^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell(i)), \tag{3.3.4}$$

that is, such that

$$V_{\bar\eta}\, m^* = \mathrm{id}. \tag{3.3.5}$$

3.4.

Condition (B) is satisfied if the operation $\wedge$ of [4, 1.4.2.1] is algebraic (cf. [4, 2.12(ii)]). It is conjectured that $\wedge$ is always algebraic, and hence that condition (B) always holds; this is known in many cases [4]. It holds trivially when X is a complete intersection of even dimension, since in that case we know (IX 1.6) that

$$H^{n-1}(X, \mathbb{Q}_\ell(i)) = 0. \tag{3.4.1}$$

4. Griffiths's theorem

Theorem 4.1.

Let X be projective, smooth, and connected over k, of even dimension $n = 2m \geq 4$. Let D be a Lefschetz pencil of sections by hypersurfaces of degree d. Assume that:

(4.1.0) X satisfies (LV) (XVIII 5.2.2);
(4.1.1) the pencil D satisfies hypothesis (A) (XVIII 5.3);
(4.1.2) the pencil D satisfies alternative (3.2.2), that is, in the setting of Proposition 3.2, the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ is not of level ≤ 1;
(4.1.3) the inclusion $m : X_{\bar\eta} \hookrightarrow X \times_k k(\bar\eta)$ satisfies hypothesis (B) (3.3).[104]

[104] The print writes m^* before this geometric inclusion. Condition (B) in 3.3 is imposed on m; m^* denotes its induced map in cohomology.

Then, if an algebraic cycle z of codimension m on X has a restriction to $X_{\bar\eta}$ which, as an algebraic cycle, satisfies

$$z_{\bar\eta} = z|_{X_{\bar\eta}} \quad \text{is algebraically equivalent to zero,} \tag{4.1.4}$$

then

$$\mathrm{cl}(z) = 0 \quad \text{in } H^{2m}(X, \mathbb{Q}_\ell(m)); \tag{4.1.5}$$

in other words, z is $\mathbb{Q}_\ell$-homologous to zero.

4.2. A remark on the case where X is a complete intersection

As already noted (cf. XVIII 5.2.5 and (3.4)), hypotheses (4.1.0) and (4.1.3) then hold automatically. We saw in XVIII 6.3.4 and 6.4 that hypothesis (4.1.1) holds for a pencil of hypersurfaces of degree $d > 2n$. Moreover, in the next exposé we shall prove condition (4.1.2) for a "general" pencil of hypersurfaces of degree $d > 2n$ (XXI 5.2). Consequently, a "general" pencil of hypersurfaces of degree $d > 2n$ satisfies all the hypotheses of Theorem 4.1.

4.3.

Returning to the case of arbitrary X, it seems very plausible that, for d sufficiently large, all the conditions of 4.1 hold for D, at least when D is sufficiently general. We have already seen this for (4.1.1) (XVIII 6.4), and we noted the conjecture that (4.1.0) and (4.1.3) always hold (with (4.1.0) known in every case when $\mathrm{char}(k) = 0$). As for condition (4.1.2), Hodge theory proves it for large d when $\mathrm{char}(k) = 0$,[105] while heuristic arguments, given in the appendix to the next exposé, make the same conclusion plausible when $\mathrm{char}(k) > 0$.

More generally, these arguments—Hodge theory and, respectively, crystalline cohomology—prove, and respectively make plausible, that for a smooth projective variety X of dimension n, and for every Lefschetz pencil D of hypersurfaces of sufficiently large degree d (respectively, every sufficiently general pencil D of hypersurfaces of degree d), the vanishing cohomology $E^{n-1}(X_{\bar\eta}, \mathbb{Q}_\ell)$ has level exactly $n-1$; that is, it is not of level $\leq n-3$, or equivalently is not of coniveau 1.

[105]The print has $\mathrm{char}(k) \neq 0$ in this Hodge-theoretic clause. The following clause treats positive characteristic, so the required reading here is $\mathrm{char}(k) = 0$.